

Model Testing Report

Experiment: bank_aml_tm_50k_local

Generated: 2026-04-08 19:11:05

Segmentation Column: value_date

Segments Defined: 4

Segment	Rows	Status
value_date_quarter=2024Q1	10801	completed
value_date_quarter=2024Q2	12206	completed
value_date_quarter=2024Q3	13077	completed
value_date_quarter=2024Q4	13916	completed

Overall - 1) Model Effectiveness (1/3)

Metric	Figure	Explanation
auc_roc	0.8470	Measures ranking power across all thresholds. 0.8470 indicates strong separation. Next: compare AUC-ROC across segments and keep threshold tuning separate from ranking validation.
auc_pr	0.5343	Measures precision-recall performance under class imbalance; the observed event rate is 16.00%. Next: use this together with alert capacity to decide whether the high-score queue is practically useful.
precision	0.6121	Shows the share of flagged cases that are true positives at the current threshold. Next: if this is below review expectations, raise the threshold or improve exclusion features.
recall	0.3701	Shows the share of true positives captured at the current threshold. Next: if missed cases are too high, lower the threshold or add stronger risk features.
f1	0.4613	Balances precision and recall in one score. Next: use it as a threshold-selection aid when both missed cases and false alerts matter.
ks_statistic	0.5528	Measures score separation between positive and negative classes. Next: low KS suggests the model needs better features or a different specification.

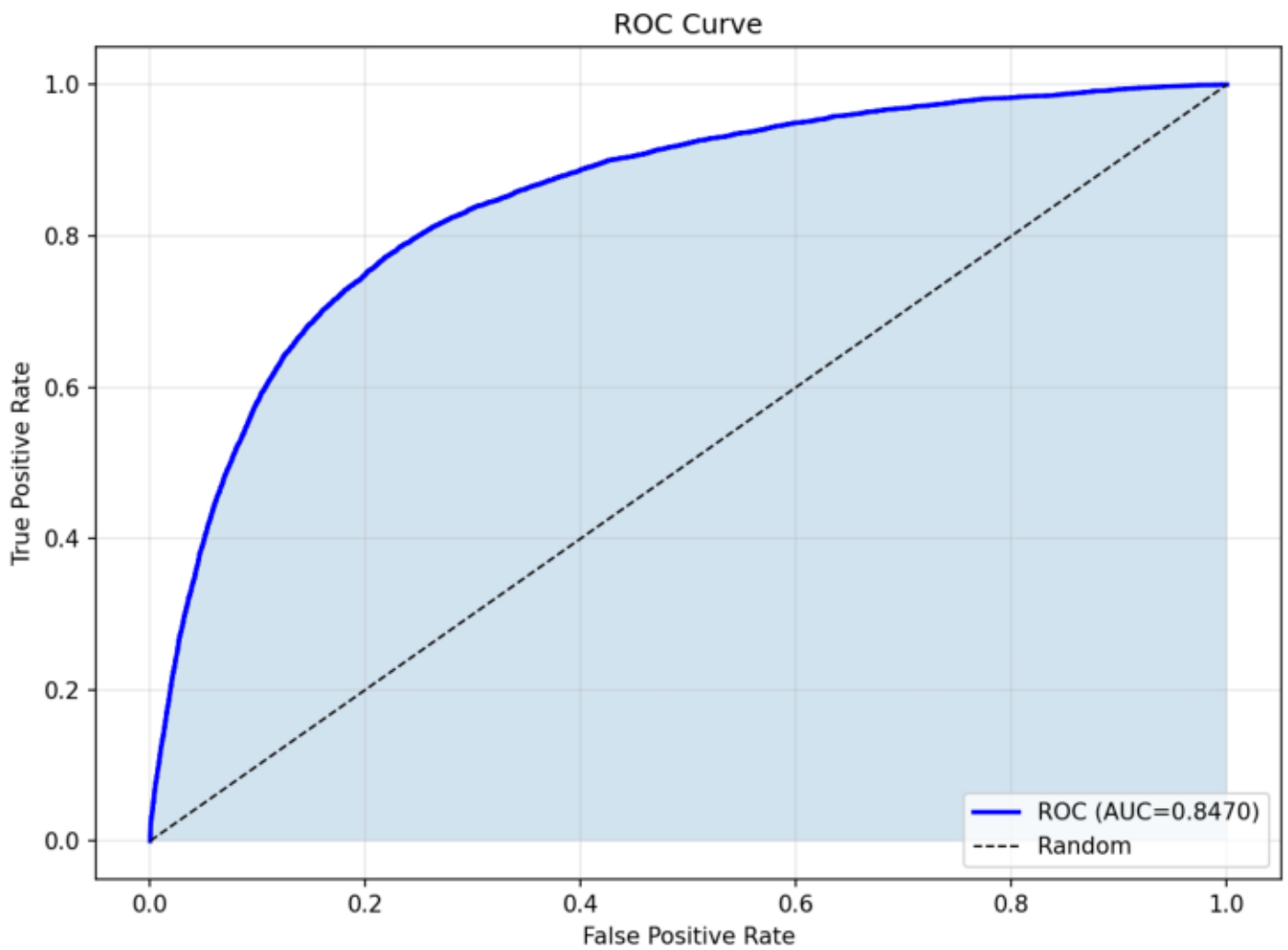
Overall - 1) Model Effectiveness (2/3)

Metric	Figure	Explanation
ks_threshold	0.1500	This is the threshold that maximizes KS separation. Next: treat it as a candidate operating point, then compare it with business capacity and policy limits.
confusion_matrix[TN]	40126	True negatives correctly cleared. Next: preserve this base while tuning the alert threshold.
confusion_matrix[FP]	1876	False positives create review cost without SAR conversion. Next: reduce this count with threshold tuning or sharper risk features.
confusion_matrix[FN]	5038	False negatives are missed SAR cases. Next: if this count is unacceptable, lower the threshold or add features that capture suspicious behavior earlier.
confusion_matrix[TP]	2960	True positives are correctly escalated cases. Next: assess whether this volume is sufficient relative to expected SAR capture goals.
precision_at_k[10]	0.9000	Shows how accurate the top-10 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[50]	0.9200	Shows how accurate the top-50 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[100]	0.9000	Shows how accurate the top-100 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.

Overall - 1) Model Effectiveness (3/3)

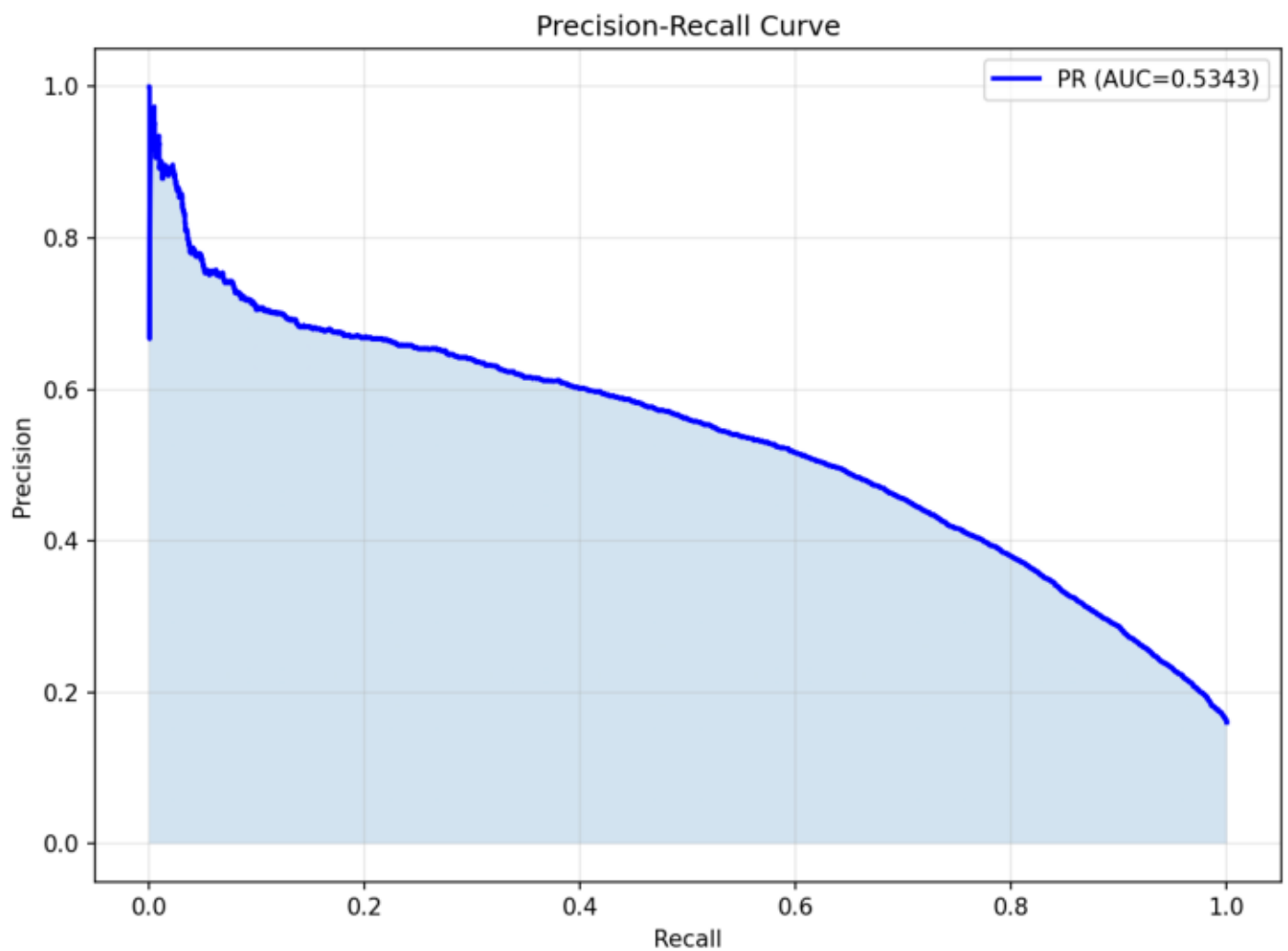
Metric	Figure	Explanation
precision_at_k[200]	0.8900	Shows how accurate the top-200 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[500]	0.7720	Shows how accurate the top-500 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
recall_at_k[10]	0.0011	Shows how much of the positive population is captured within the top-10 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[50]	0.0058	Shows how much of the positive population is captured within the top-50 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[100]	0.0113	Shows how much of the positive population is captured within the top-100 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[200]	0.0223	Shows how much of the positive population is captured within the top-200 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[500]	0.0483	Shows how much of the positive population is captured within the top-500 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.

Overall - 1) Model Effectiveness



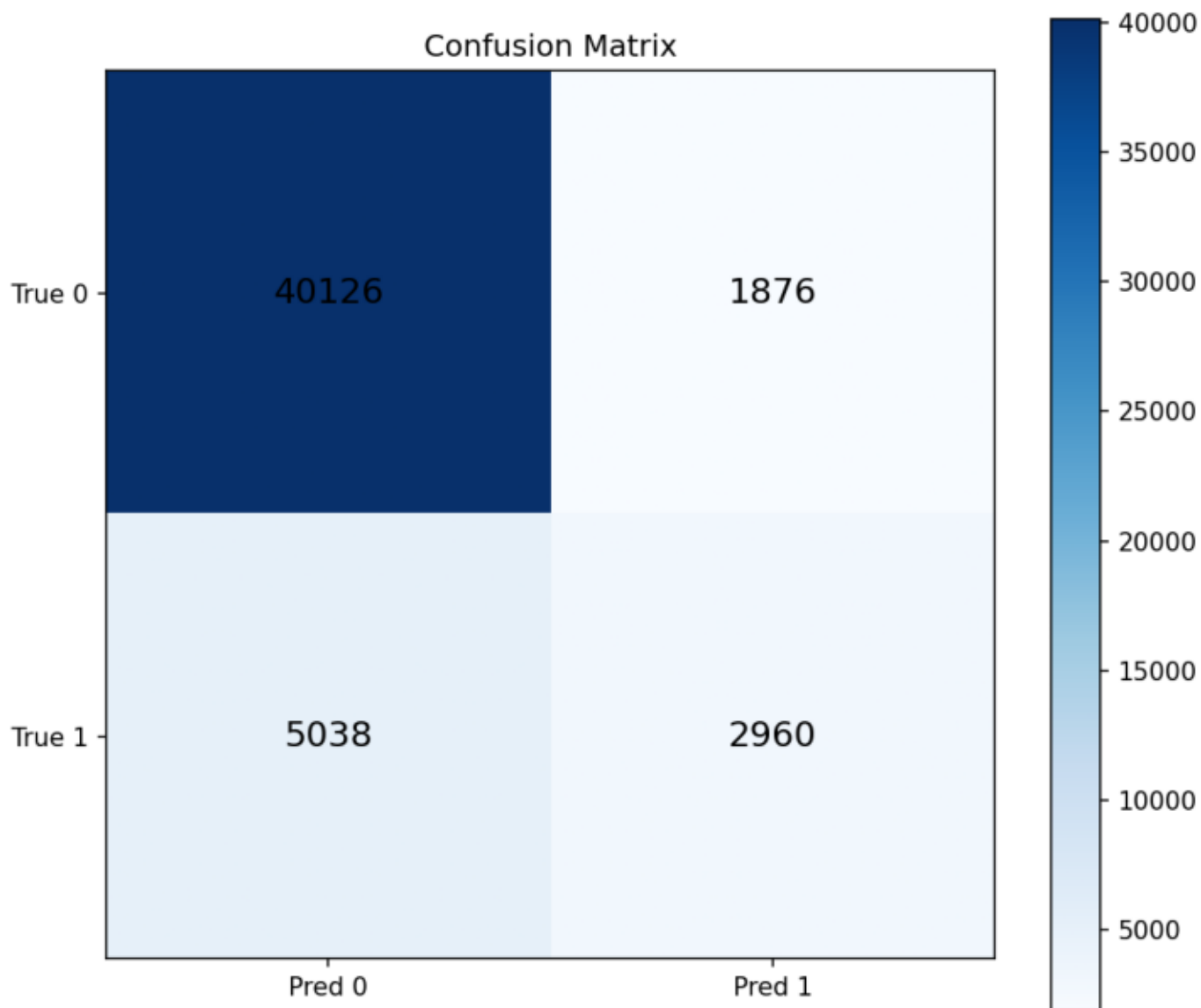
roc_curve: AUC-ROC=0.8470; curve above diagonal indicates strong separation.

Overall - 1) Model Effectiveness



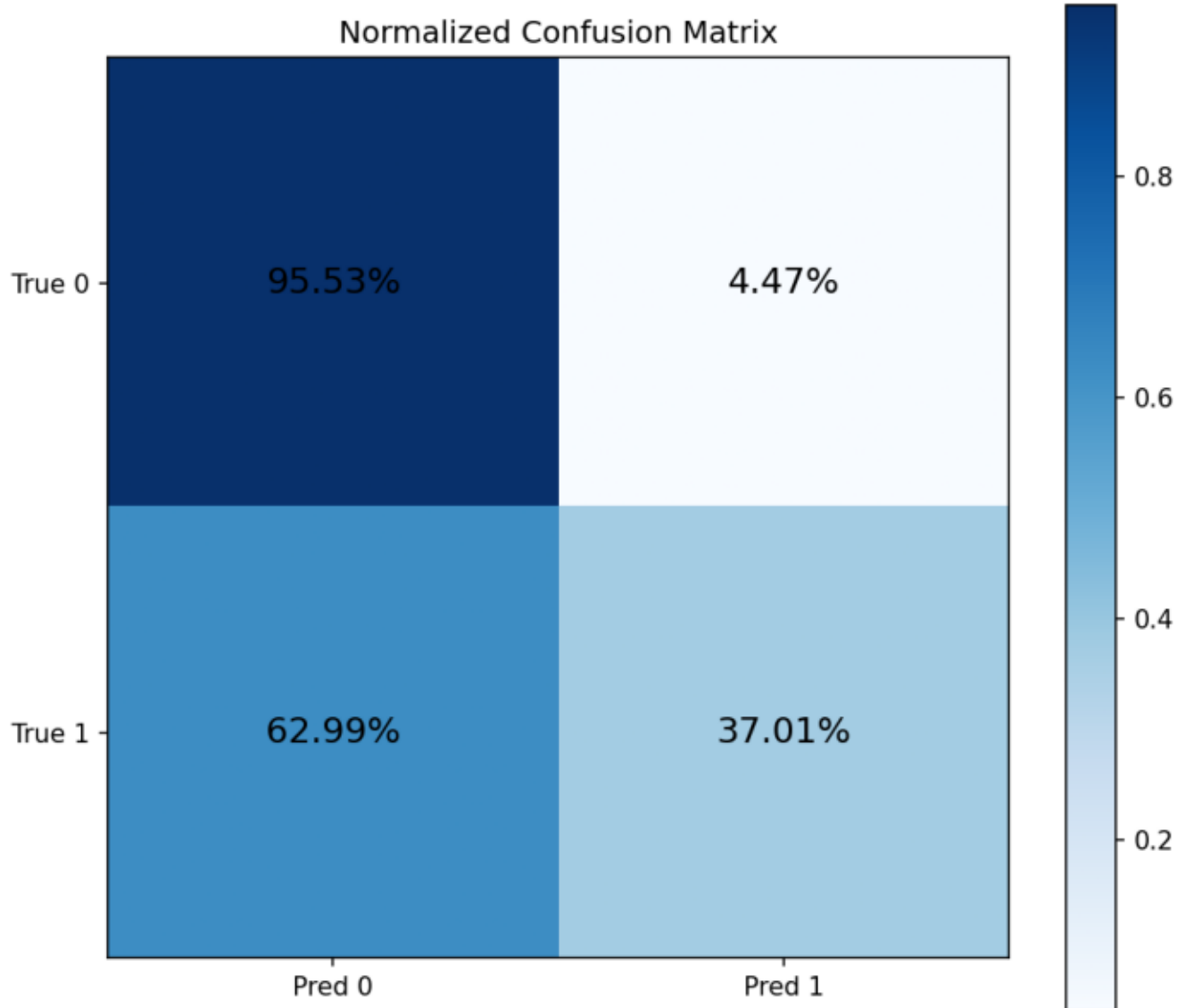
pr_curve: AUC-PR=0.5343 vs baseline 16.00%; higher curve implies better precision at recall.

Overall - 1) Model Effectiveness



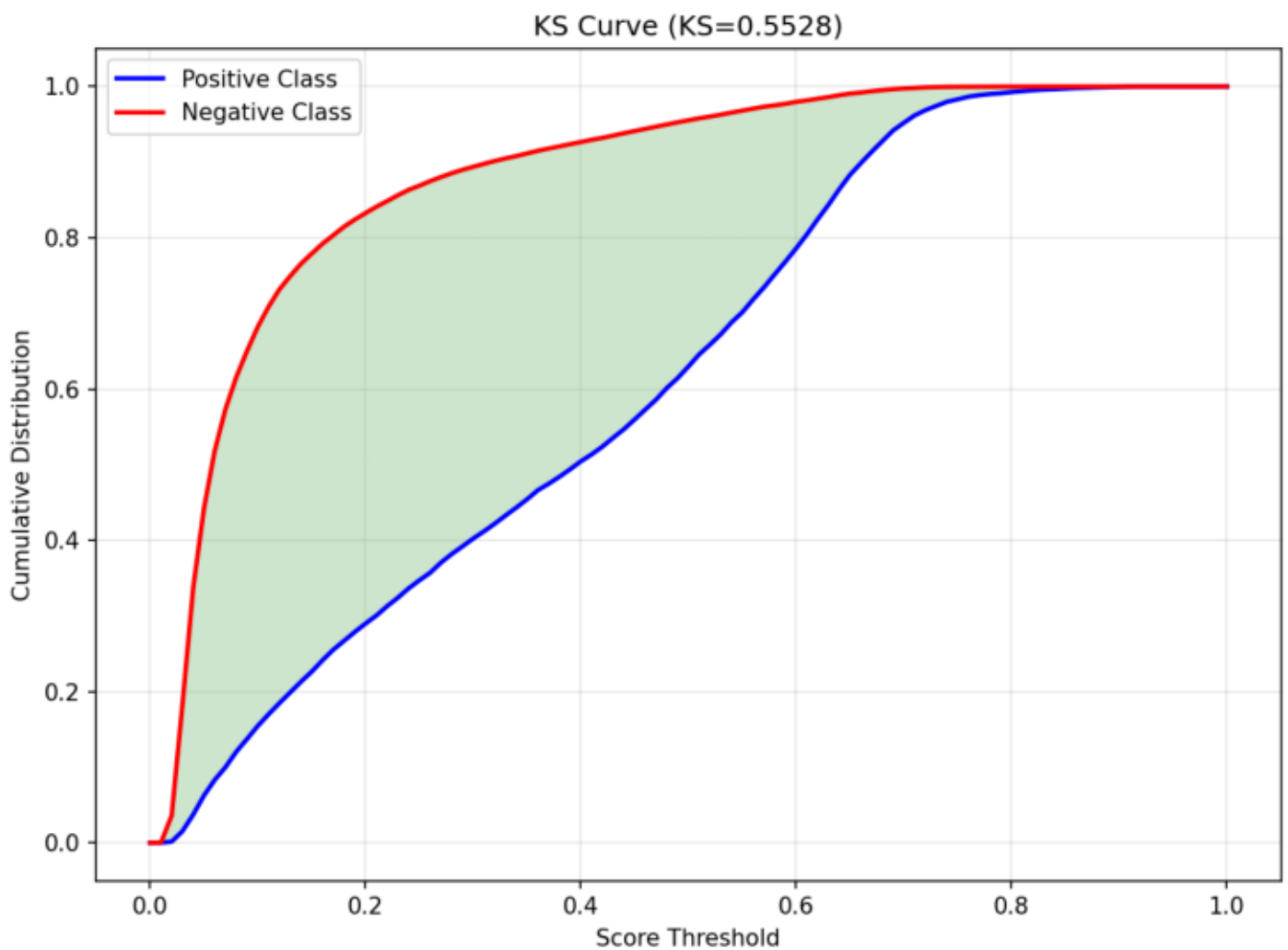
confusion_matrix: At threshold 0.50: TP=2960, FP=1876, FN=5038, TN=40126.

Overall - 1) Model Effectiveness



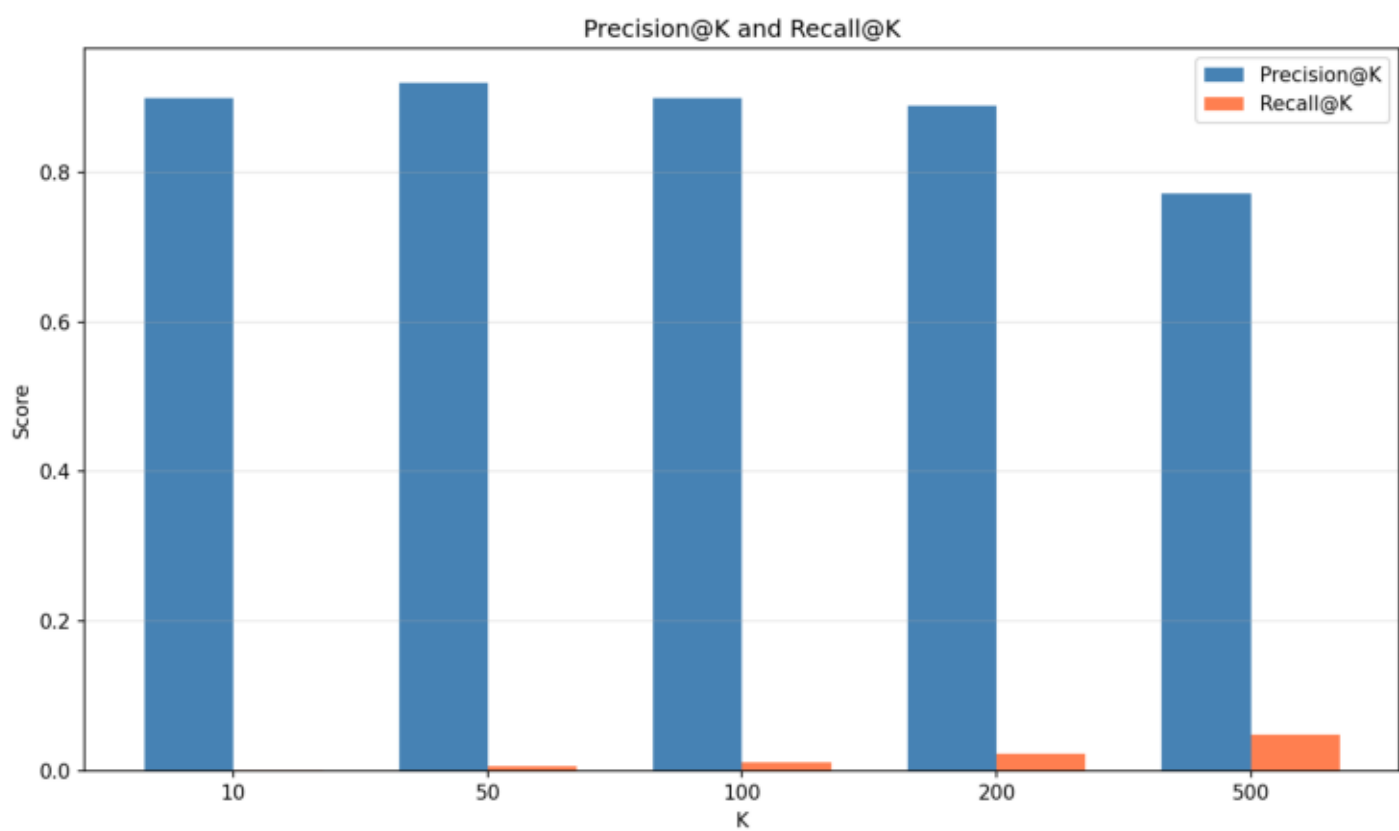
confusion_matrix_norm: Normalized rates: TPR=37.01%, TNR=95.53%.

Overall - 1) Model Effectiveness



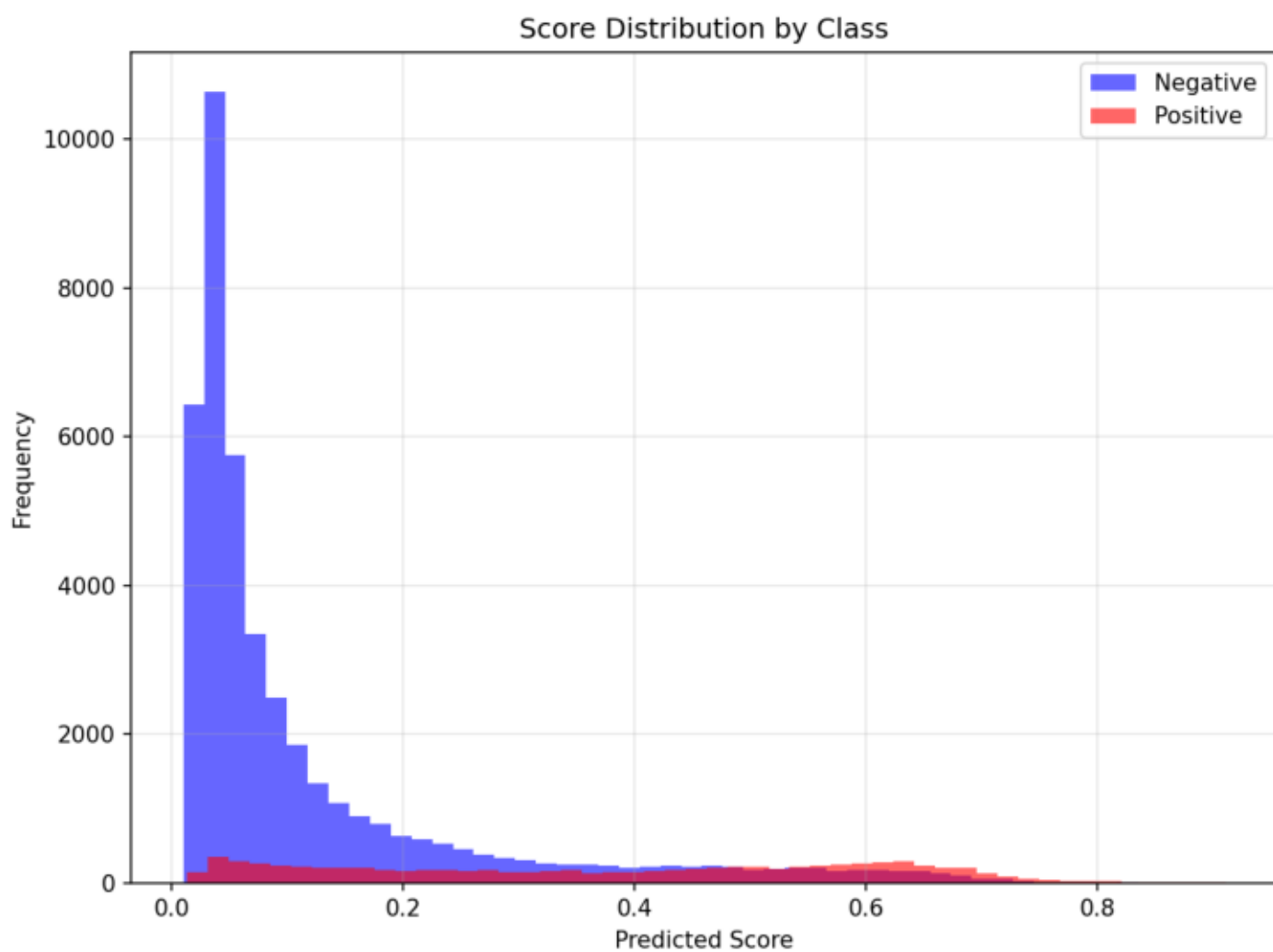
ks_curve: Maximum separation KS=0.5528 at threshold 0.15.

Overall - 1) Model Effectiveness



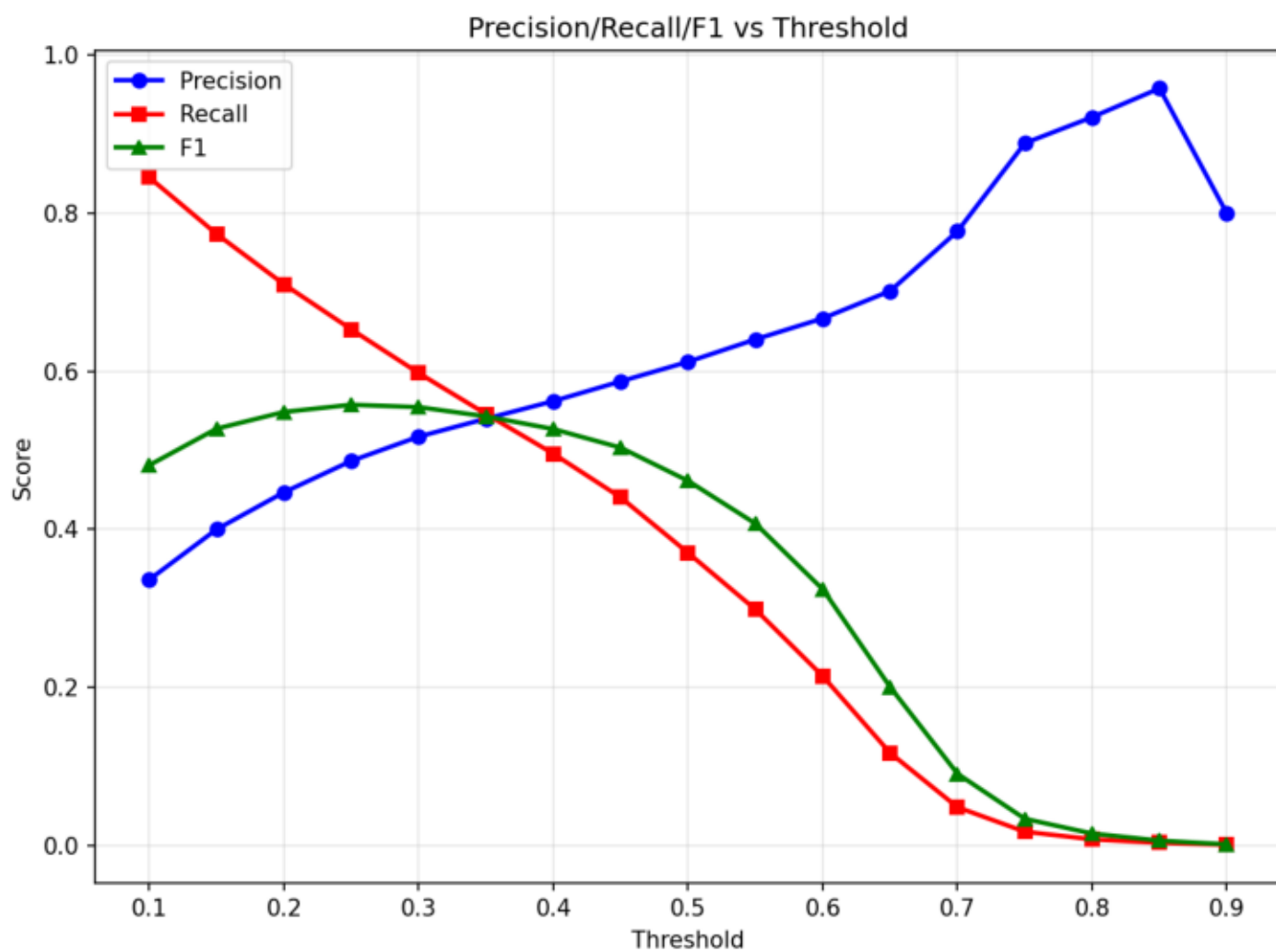
precision_recall_at_k: Bars show precision/recall as you expand the review queue.

Overall - 1) Model Effectiveness



score_distribution: Mean score: positive=0.379, negative=0.118 (larger gap is better).

Overall - 1) Model Effectiveness



threshold_analysis: Best F1 in tested grid is 0.5577 at threshold 0.25.

Overall - 2) Model Efficiency (1/4)

Metric	Figure	Explanation
fpr	0.0447	Measures the share of negatives incorrectly flagged; 0.0447 is low. Next: if review demand is too high, raise the threshold or add features that better suppress false alerts.
tn	40126	Counts negatives correctly cleared without manual review. Next: preserve this clearance rate while checking that recall does not fall below policy expectations.
fp	1876	Counts non-events that would still be sent to review. Next: use this with staffing capacity to decide whether threshold or feature refinement is required.
threshold	0.5000	This is the active operating cutoff used to flag alerts. Next: tune it jointly with recall, precision, and review capacity rather than in isolation.
fpr_at_thresholds[thresholds.t_0.05]	0.5579	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.05. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.10]	0.3177	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.10. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

Overall - 2) Model Efficiency (2/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.15]	0.2210	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.15. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.20]	0.1674	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.20. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.25]	0.1312	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.25. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.30]	0.1062	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.30. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.35]	0.0885	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.35. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.40]	0.0735	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.40. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

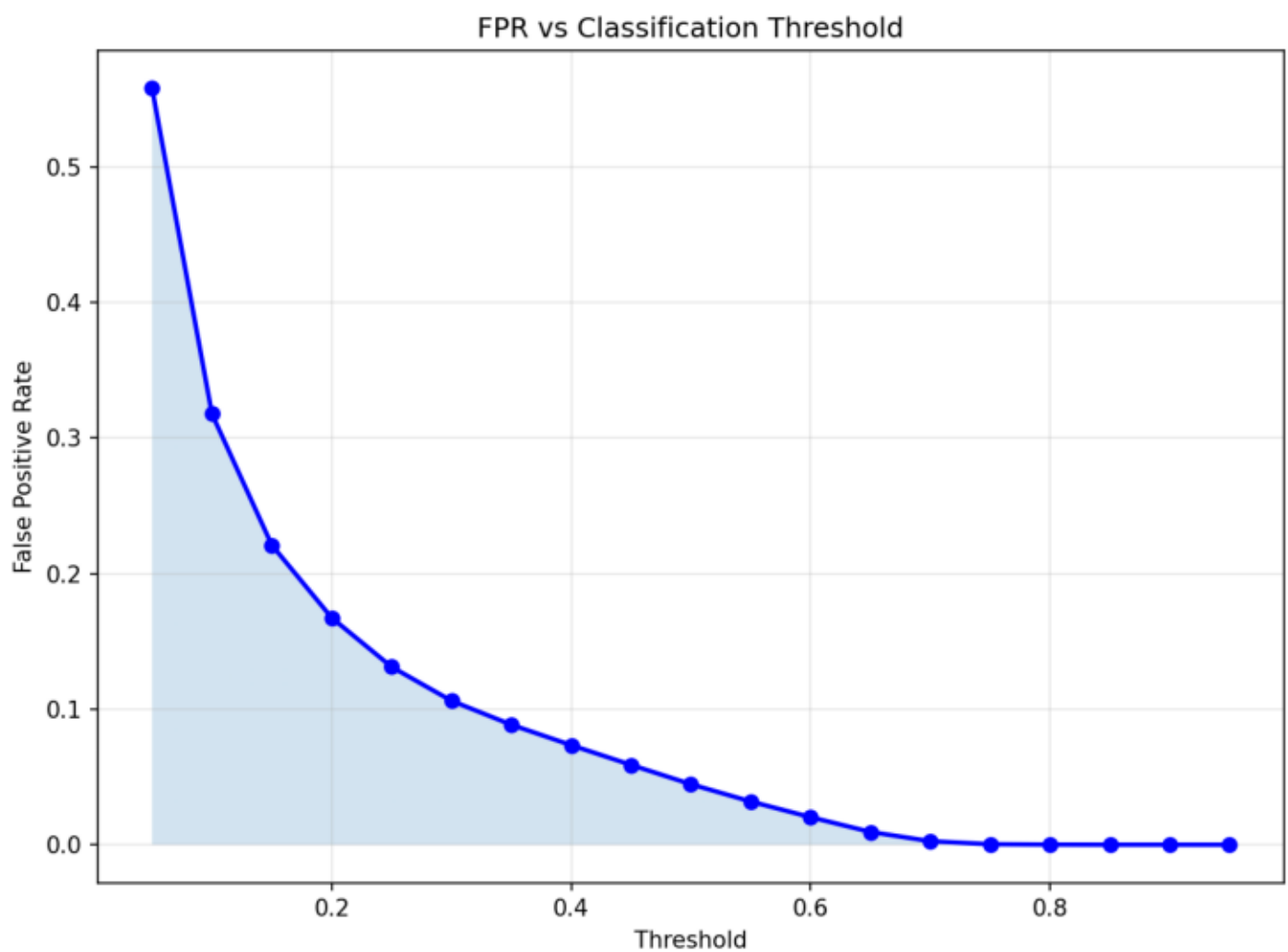
Overall - 2) Model Efficiency (3/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.45]	0.0589	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.45. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.50]	0.0447	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.50. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.55]	0.0319	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.55. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.60]	0.0203	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.60. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.65]	0.0095	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.65. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.70]	0.0026	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.70. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

Overall - 2) Model Efficiency (4/4)

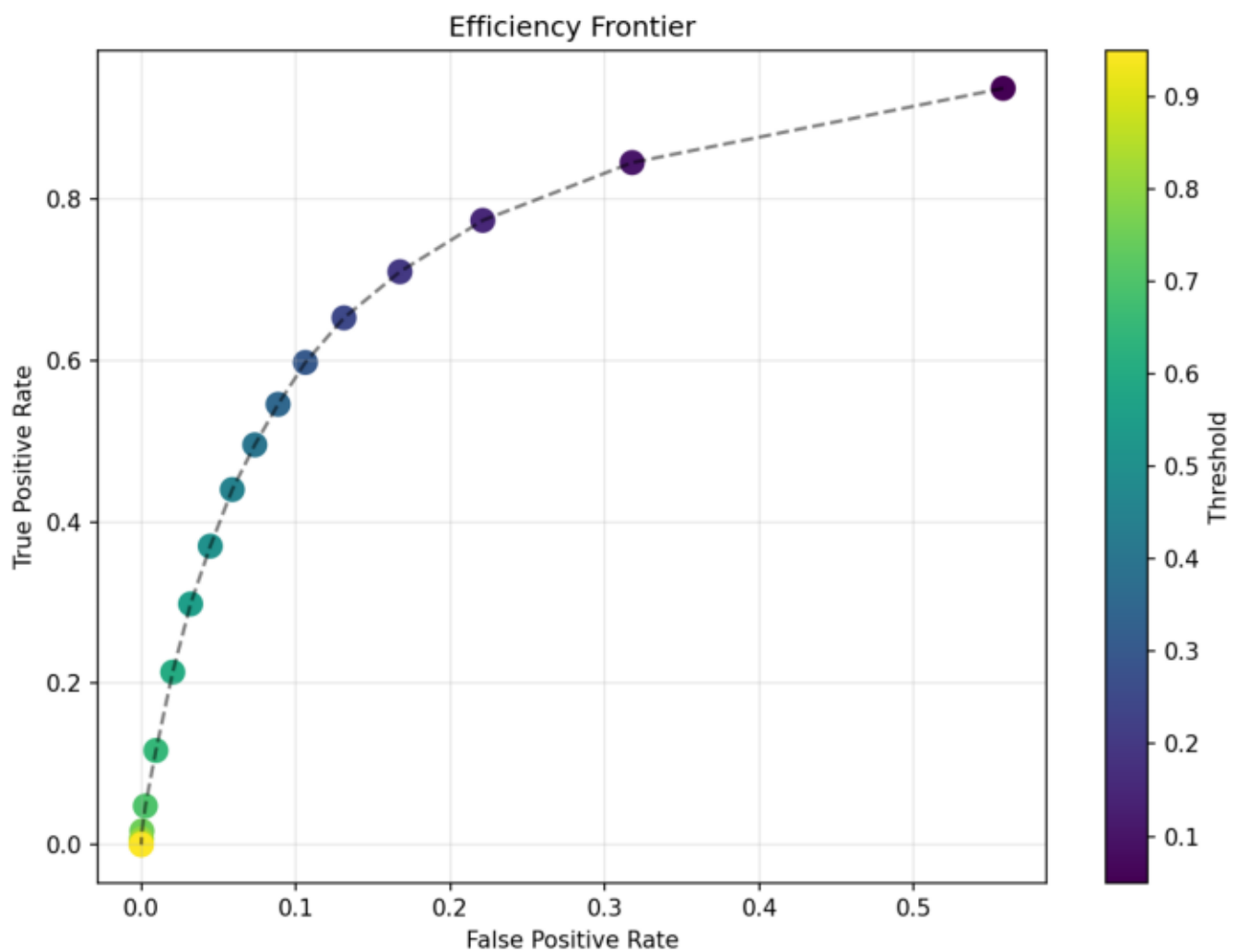
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.75]	0.0004	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.75. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.80]	0.0001	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.80. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.85]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.85. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.90]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.90. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.95]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.95. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

Overall - 2) Model Efficiency



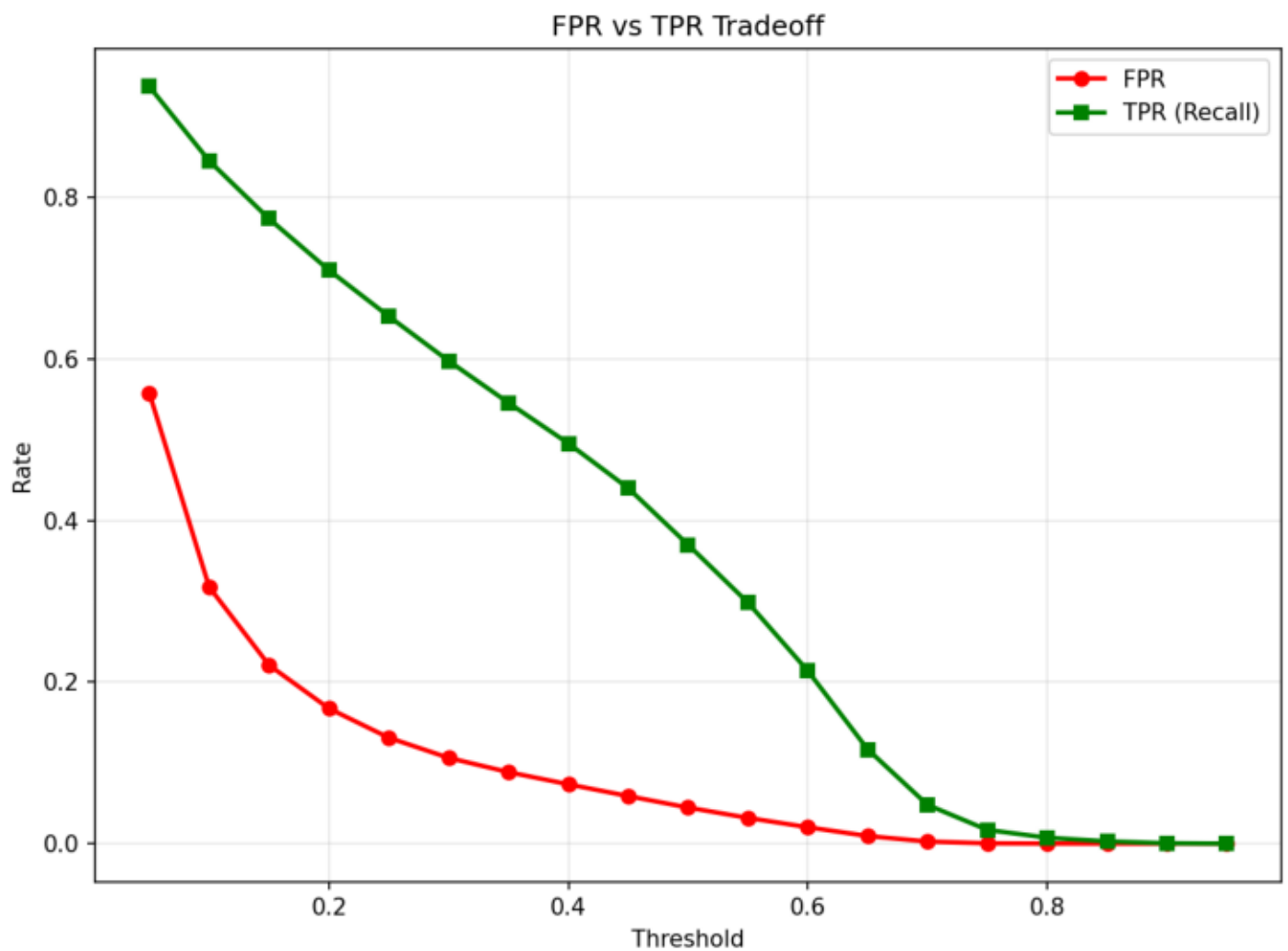
fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.

Overall - 2) Model Efficiency



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

Overall - 2) Model Efficiency



fpr_tpr_tradeoff: FPR and TPR curves highlight how recall drops as you reduce false positives.

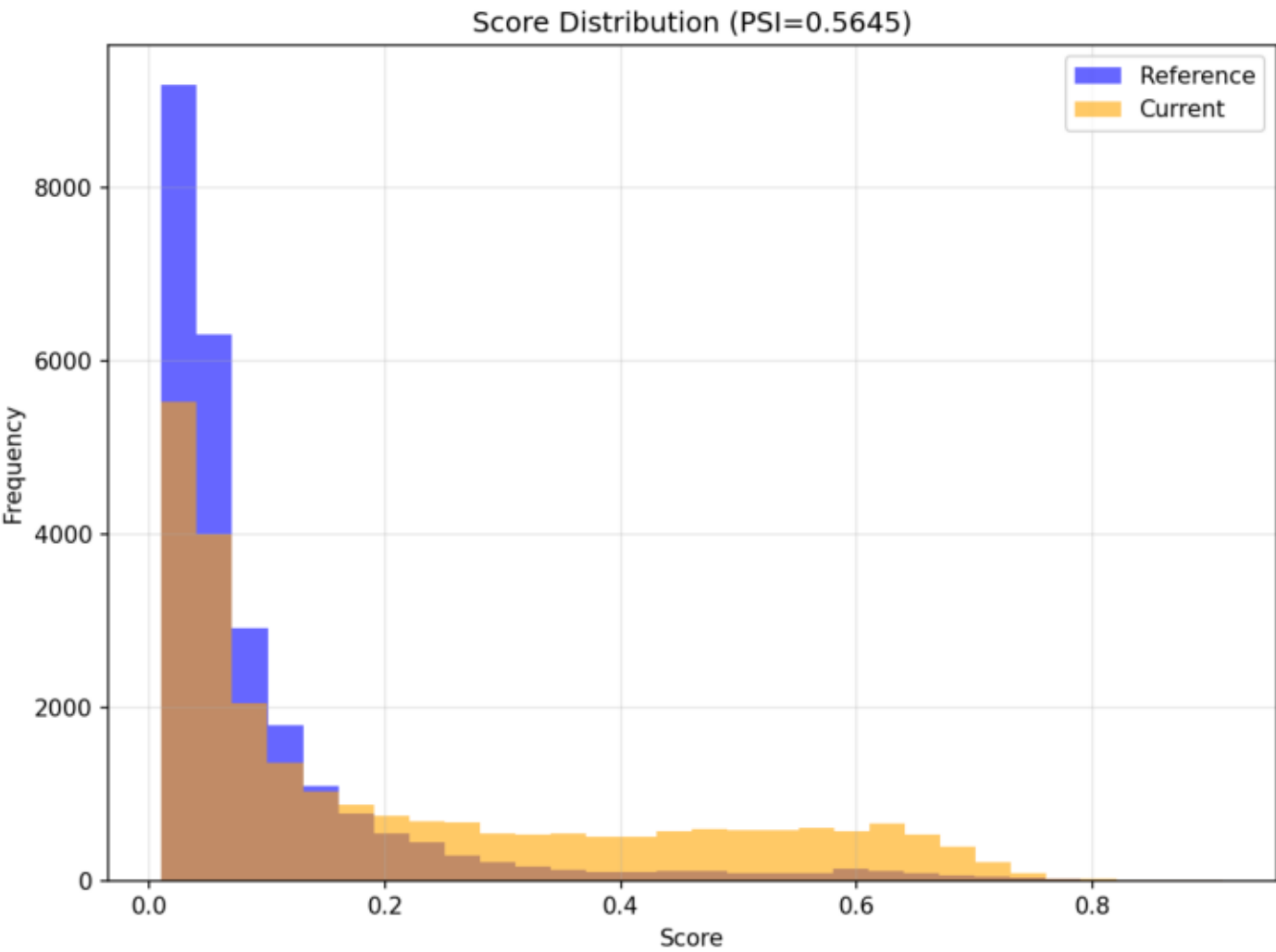
Overall - 3) Model Stability (1/2)

Metric	Figure	Explanation
psi	0.5645	Measures score-distribution shift between reference and current populations; 0.5645 is material. Next: investigate segment mix, data drift, and recalibration needs when PSI is material.
cv_auc_roc_mean	0.8471	Average AUC-ROC across validation folds. Next: compare it with overall AUC-ROC; a large gap can indicate overfitting or unstable sampling.
cv_auc_roc_std	0.0082	Standard deviation of AUC-ROC across folds. Next: if variability is high, simplify the model or increase training stability before deployment.
cv_auc_pr_mean	0.5350	Average AUC-PR across validation folds. Next: confirm that precision-recall performance remains acceptable under resampling.
cv_auc_pr_std	0.0155	Standard deviation of AUC-PR across folds. Next: high dispersion suggests unstable alert quality and warrants deeper validation.
bootstrap_auc_roc_mean	0.8219	Bootstrap mean AUC-ROC estimates expected ranking performance under repeated sampling. Next: compare it with the point estimate to judge optimism in the current test run.
bootstrap_auc_roc_ci_lower	0.8154	Lower confidence bound for bootstrap AUC-ROC. Next: use it as a conservative performance floor in model-risk discussions.
bootstrap_auc_roc_ci_upper	0.8279	Upper confidence bound for bootstrap AUC-ROC. Next: review the interval width with the lower bound to assess uncertainty.

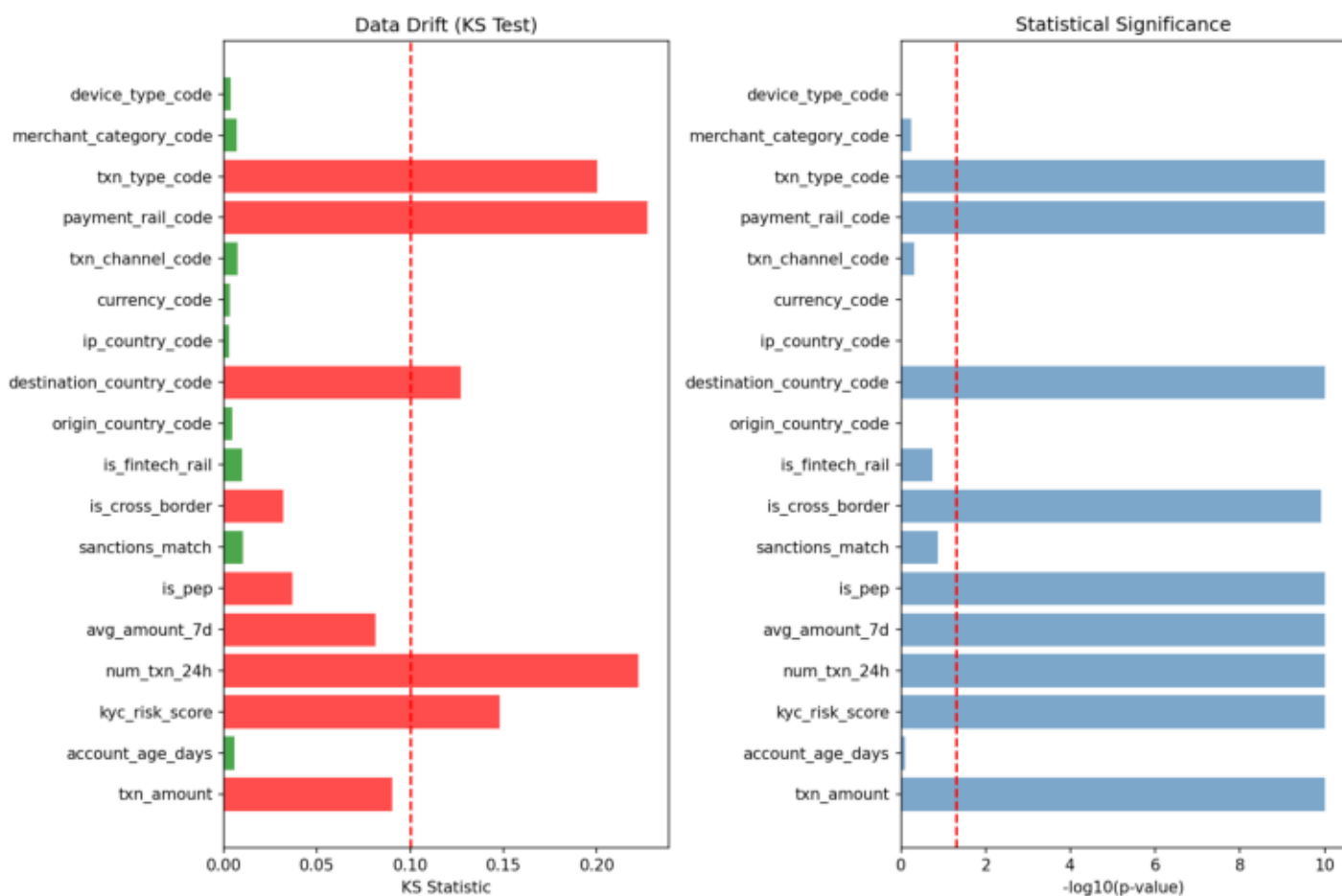
Overall - 3) Model Stability (2/2)

Metric	Figure	Explanation
concept_drift_detected	False	No material concept-drift signal was detected in the evaluated slices. Next: retain ongoing monitoring because stable historical results do not guarantee future stability.
concept_drift_score	0.0058	Quantifies the magnitude of performance change across ordered slices. Next: rising scores should trigger deeper review of segment behavior and model refresh timing.

Overall - 3) Model Stability

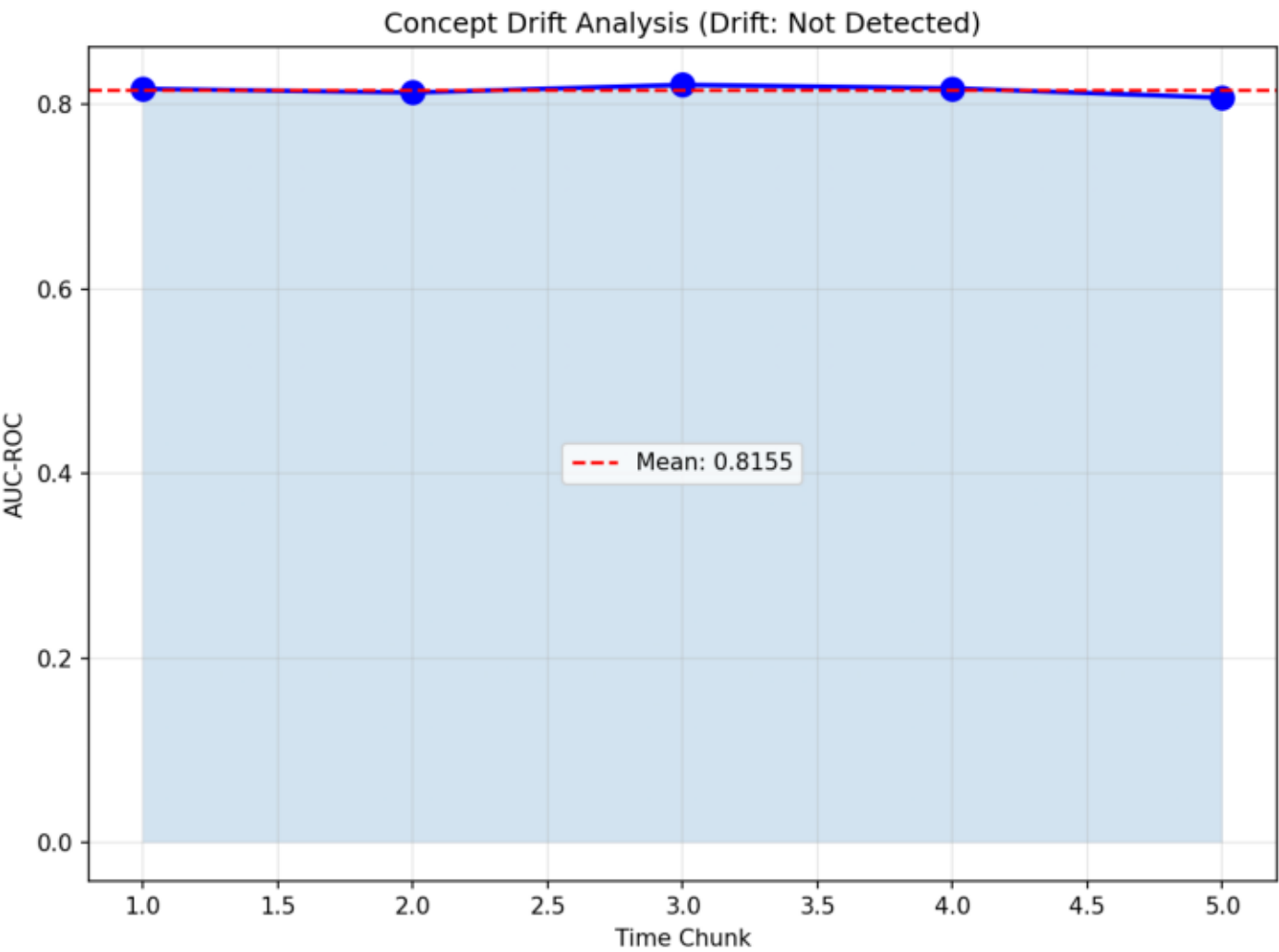


Overall - 3) Model Stability



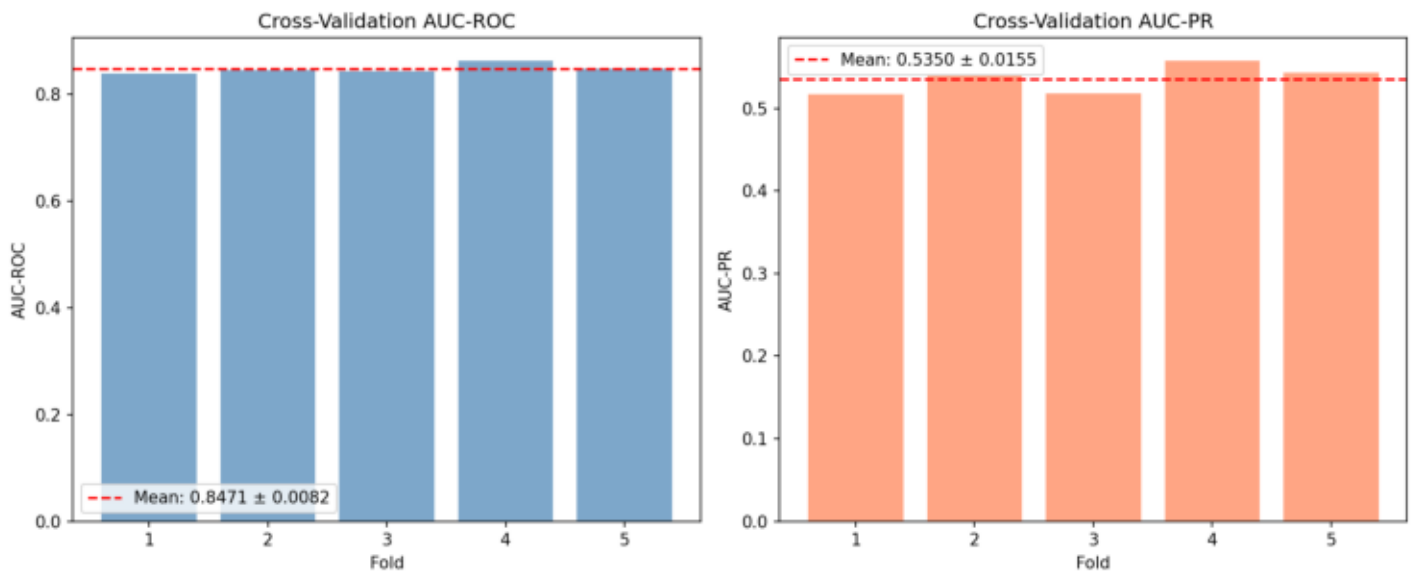
data_drift_heatmap: 9/18 features show drift; red bars highlight flagged features.

Overall - 3) Model Stability



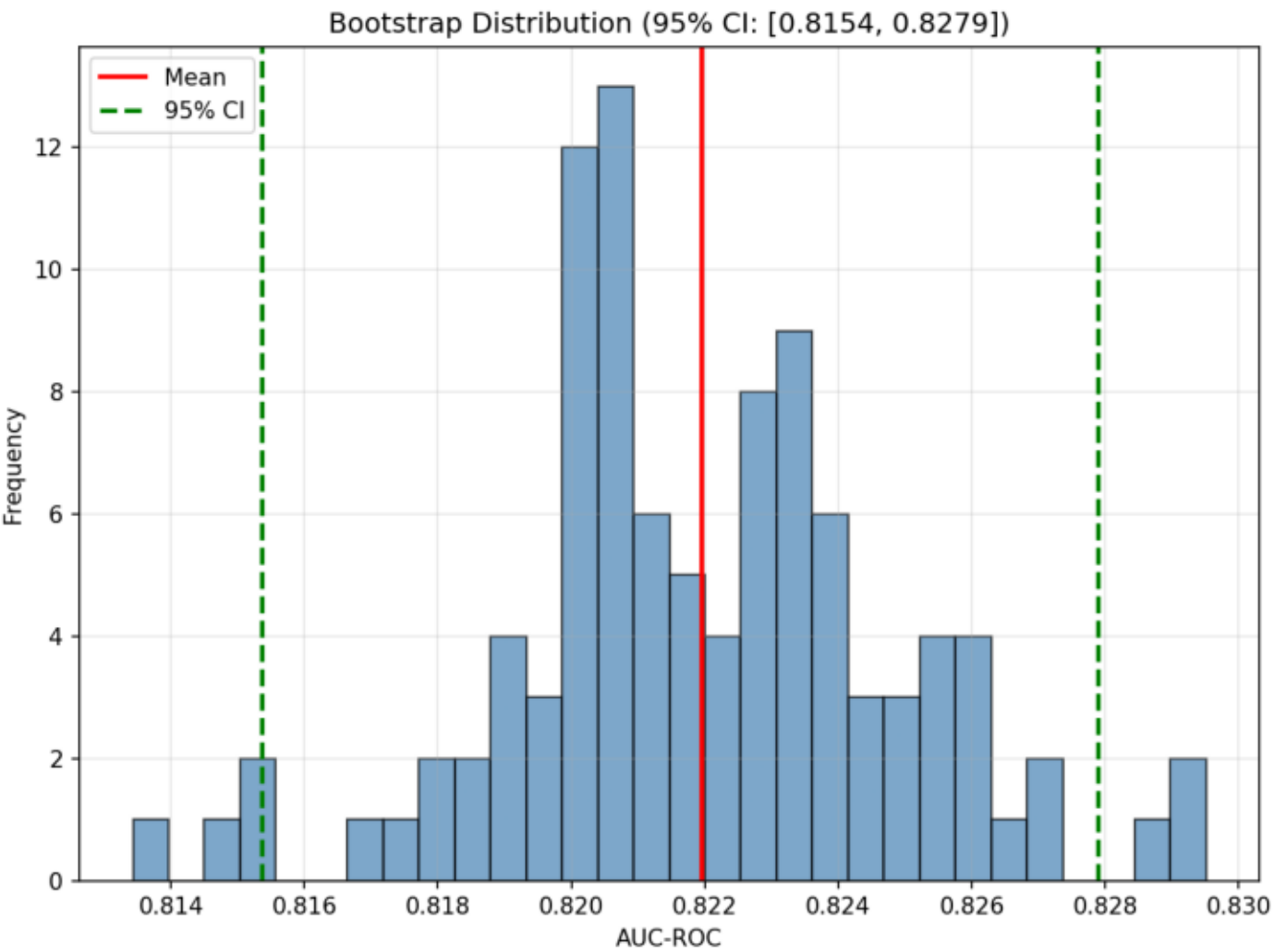
concept_drift: AUC across time chunks shows whether performance is stable over time.

Overall - 3) Model Stability

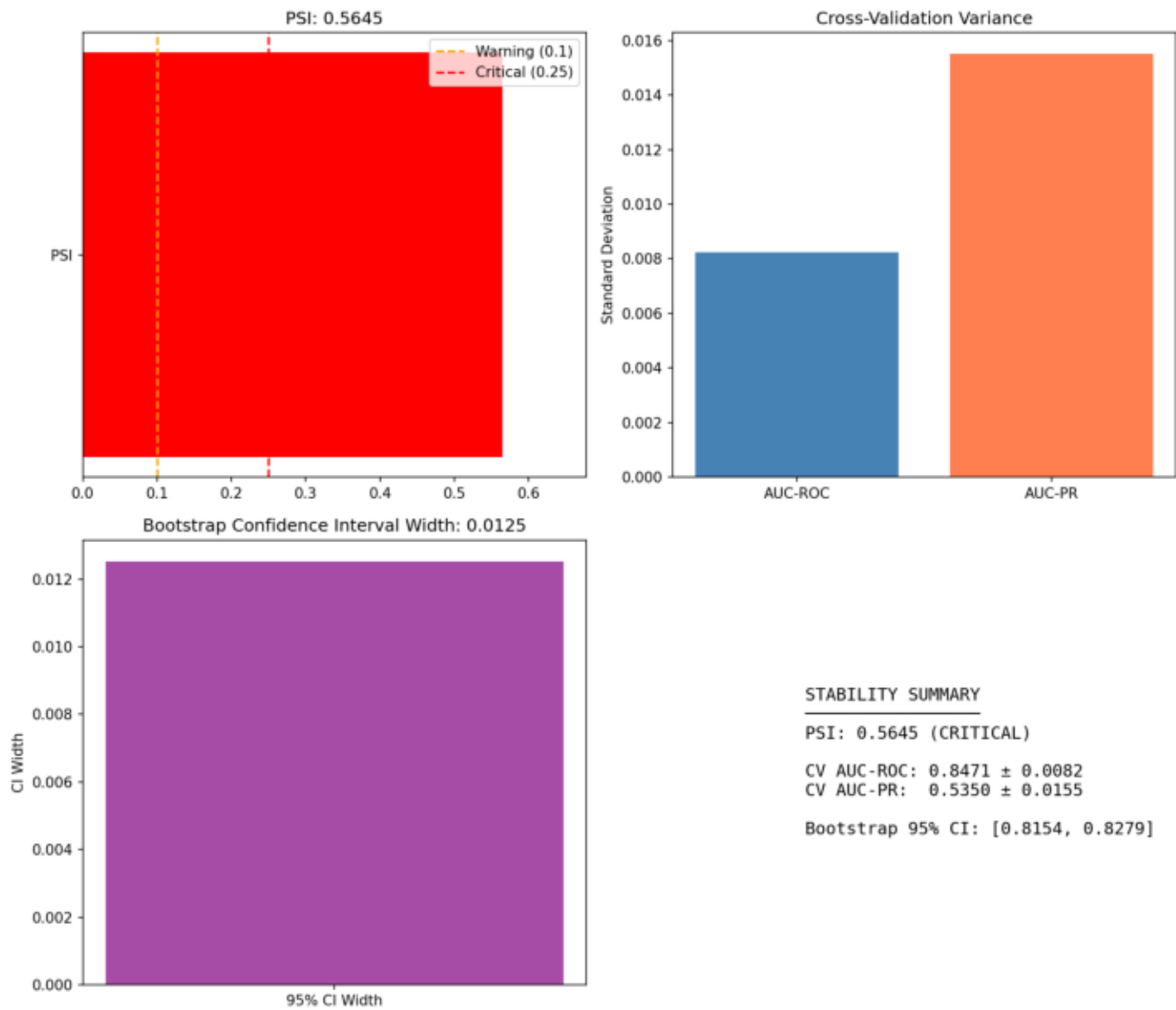


cv_results: Fold scores cluster around ROC 0.8471 and PR 0.5350.

Overall - 3) Model Stability



Overall - 3) Model Stability



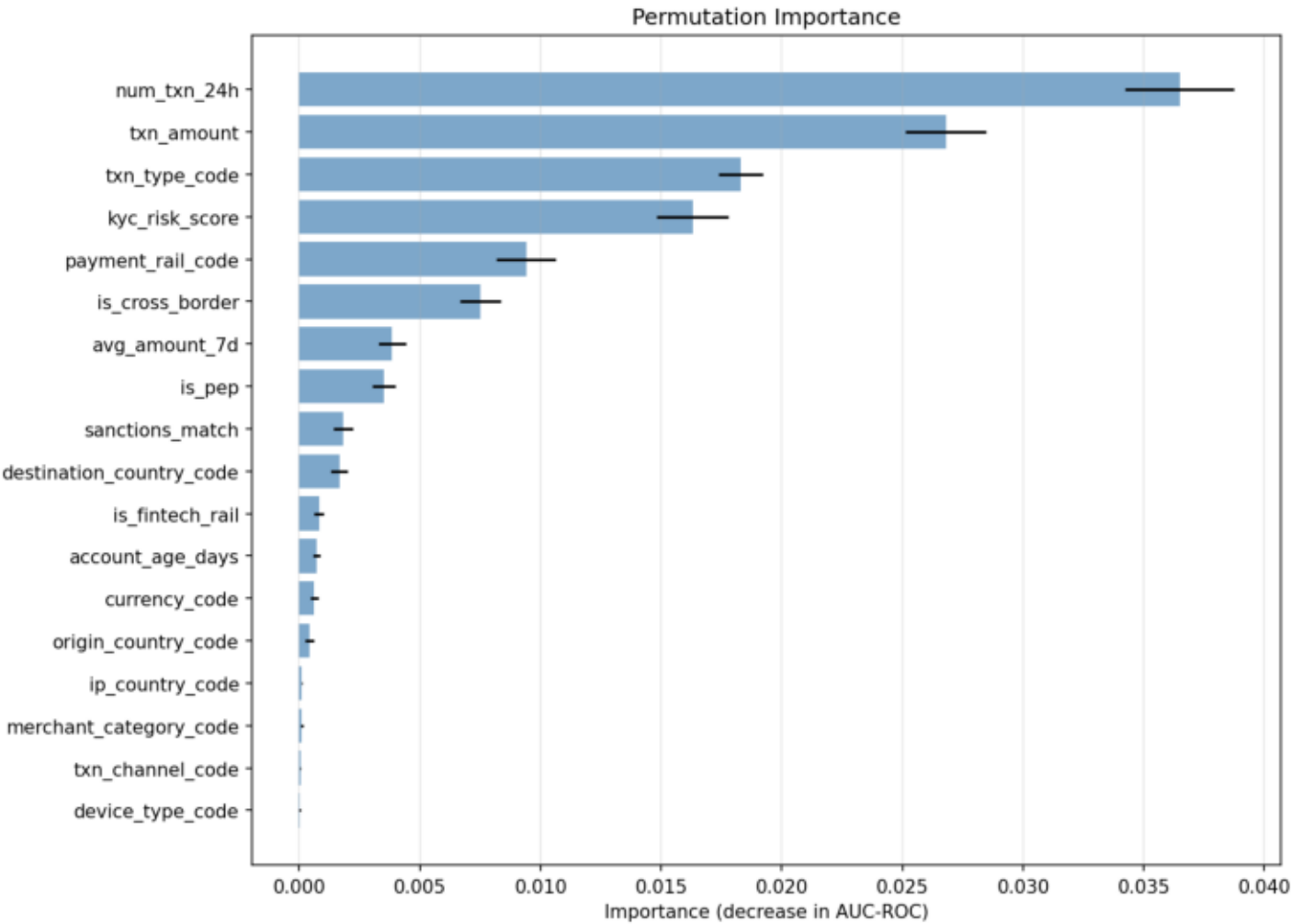
Overall - 4) Model Interpretability (1/2)

Metric	Figure	Explanation
model_type	tree	Identifies the estimator family so explanation methods can be interpreted in the right context. Next: confirm the chosen explanation methods are appropriate for this model class.
methods_used	permutation, pdp, ice, shap, lime	Lists the interpretability methods produced in this run. Next: compare signals across methods before using them in documentation or model challenge.
perm_top_features	num_txn_24h, txn_amount, txn_type_code, kyc_risk_score, payment_rail_code ...	Shows features with the largest performance drop when permuted, which highlights global importance. Next: validate that the top drivers align with business logic, feature governance, and policy expectations.
shap_top_features	num_txn_24h, txn_amount, txn_type_code, kyc_risk_score, payment_rail_code ...	Shows the most influential features under SHAP-based contribution analysis. Next: inspect whether directionality and ranking remain sensible across key segments.

Overall - 4) Model Interpretability (2/2)

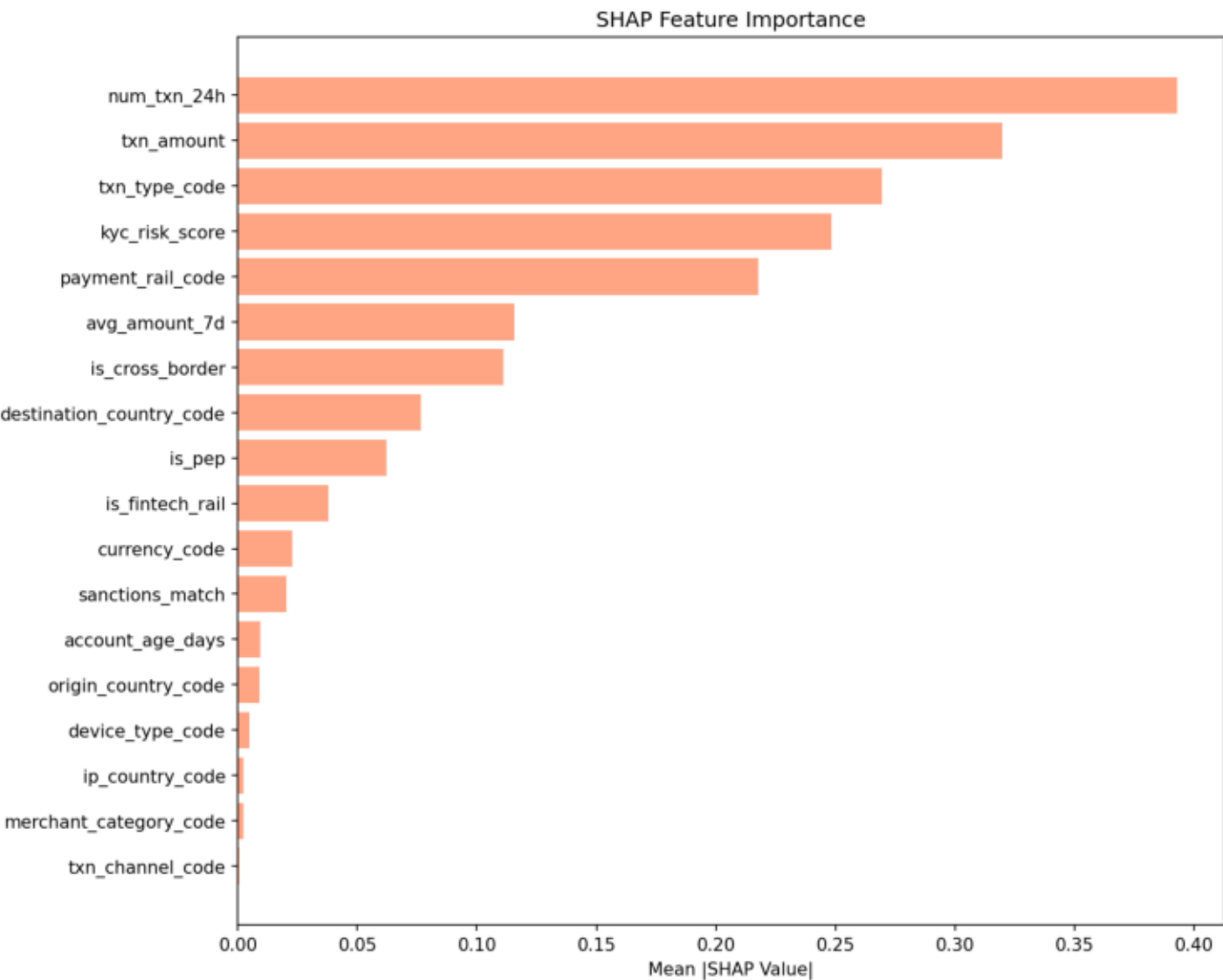
Metric	Figure	Explanation
lime_instances	3	Counts the locally explained observations reviewed with LIME. Next: examine representative true-positive and false-positive cases to understand local decision logic.
pdp_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score, num_txn_24h ...	Lists features with partial-dependence plots. Next: review whether average model response is monotonic, stable, and operationally sensible.
ice_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score	Lists features with individual conditional expectation plots. Next: inspect heterogeneity and interactions that may be hidden in average-only views.

Overall - 4) Model Interpretability



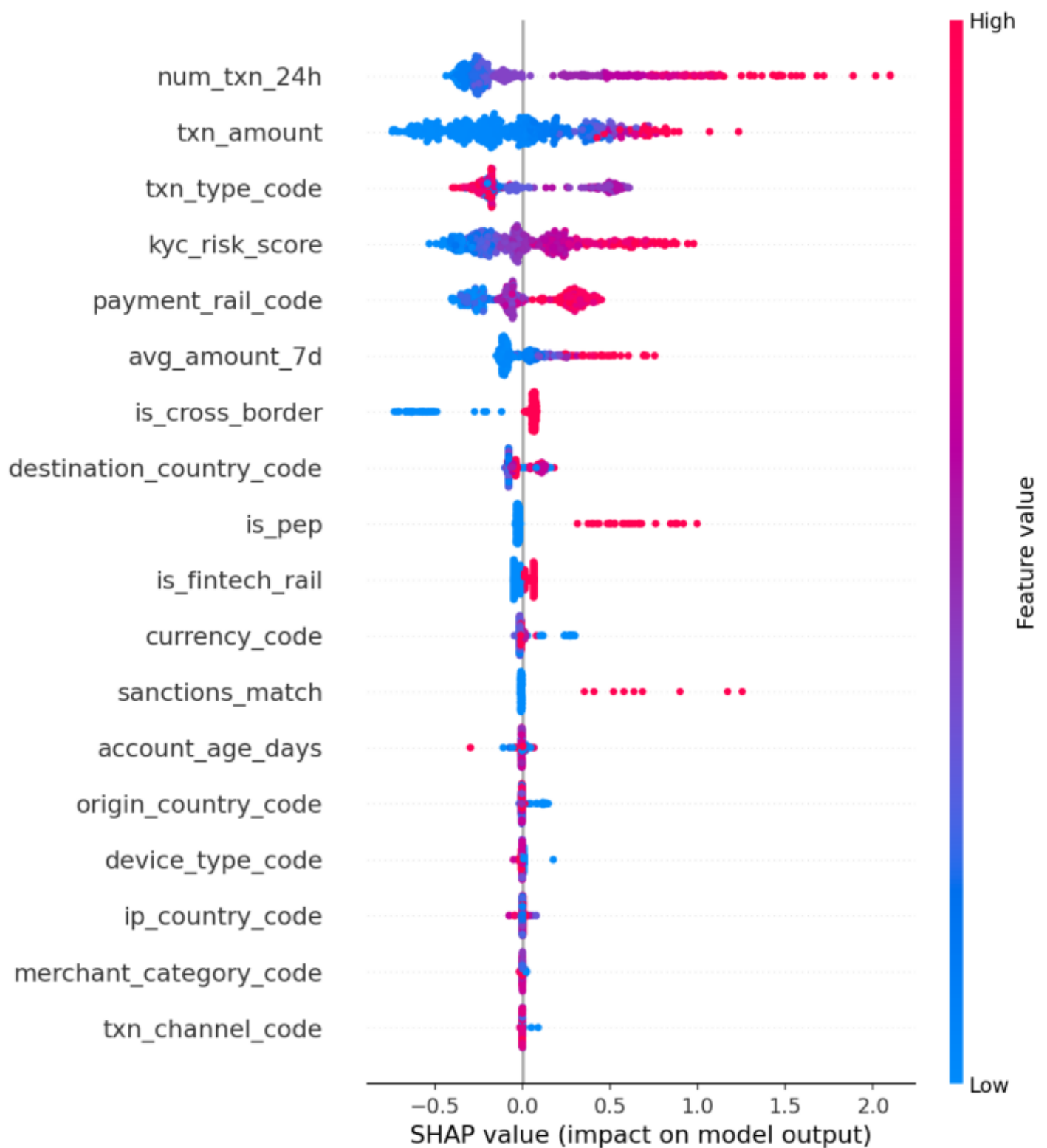
permutation_importance: Bars show how much AUC drops when each feature is permuted.

Overall - 4) Model Interpretability

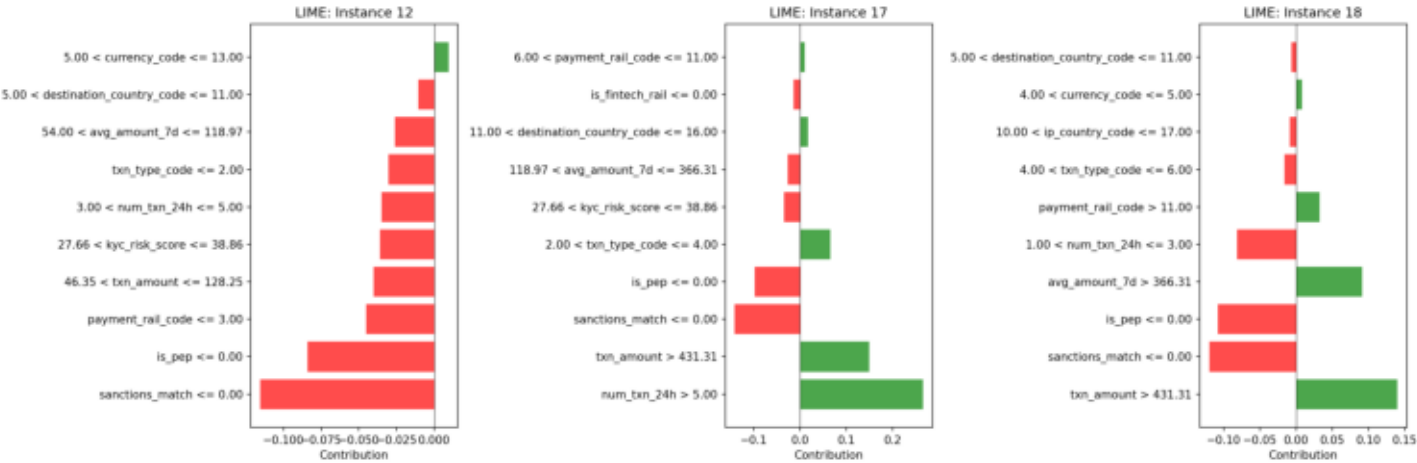


shap_bar: SHAP bar plot ranks global feature impact by mean absolute SHAP values.

Overall - 4) Model Interpretability



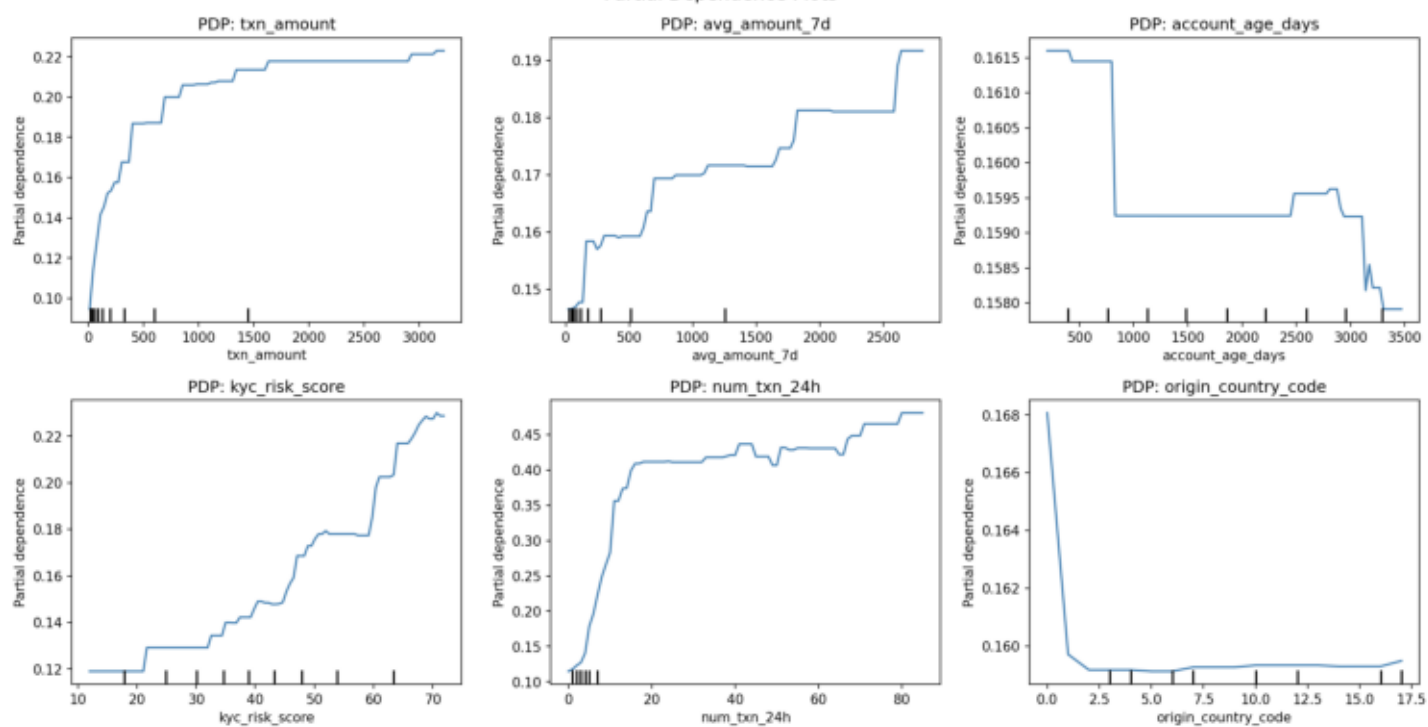
Overall - 4) Model Interpretability



lime_explanation: LIME bars show local feature contributions for selected instances.

Overall - 4) Model Interpretability

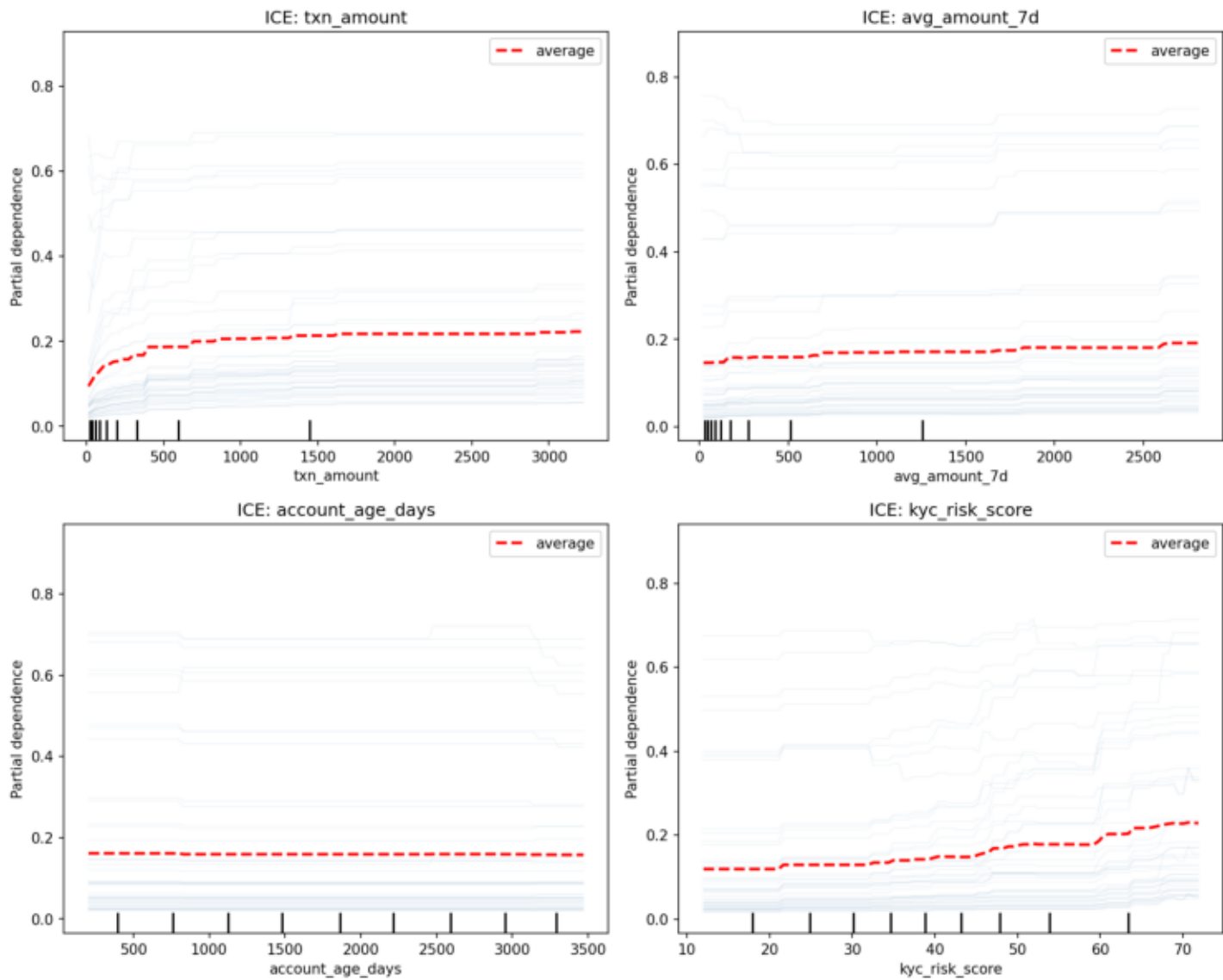
Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Overall - 4) Model Interpretability

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q1 - 1) Model Effectiveness (1/3)

Metric	Figure	Explanation
auc_roc	0.8265	Measures ranking power across all thresholds. 0.8265 indicates strong separation. Next: compare AUC-ROC across segments and keep threshold tuning separate from ranking validation.
auc_pr	0.4177	Measures precision-recall performance under class imbalance; the observed event rate is 7.04%. Next: use this together with alert capacity to decide whether the high-score queue is practically useful.
precision	0.7004	Shows the share of flagged cases that are true positives at the current threshold. Next: if this is below review expectations, raise the threshold or improve exclusion features.
recall	0.2276	Shows the share of true positives captured at the current threshold. Next: if missed cases are too high, lower the threshold or add stronger risk features.
f1	0.3436	Balances precision and recall in one score. Next: use it as a threshold-selection aid when both missed cases and false alerts matter.
ks_statistic	0.5010	Measures score separation between positive and negative classes. Next: low KS suggests the model needs better features or a different specification.

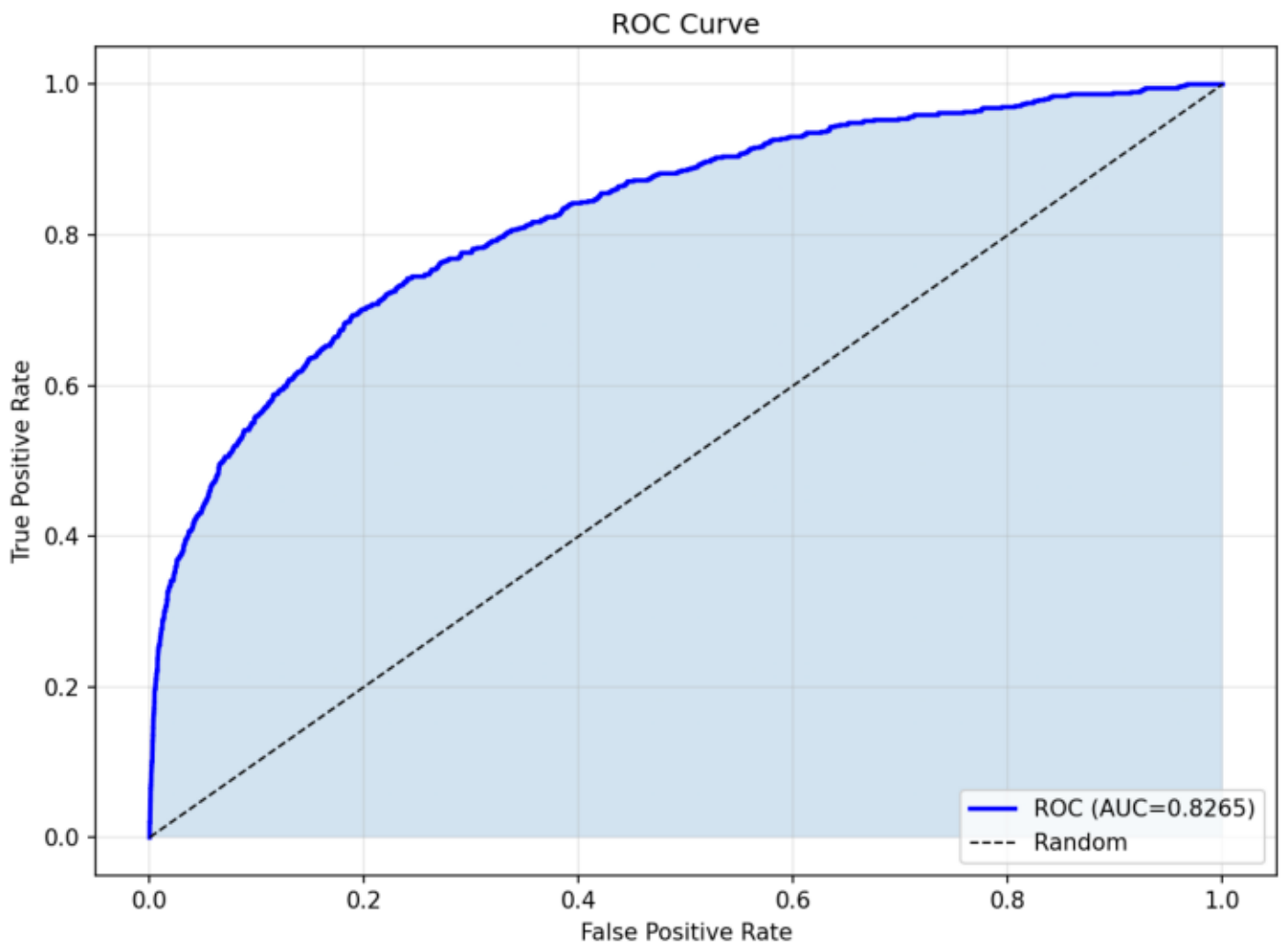
value_date_quarter=2024Q1 - 1) Model Effectiveness (2/3)

Metric	Figure	Explanation
ks_threshold	0.1100	This is the threshold that maximizes KS separation. Next: treat it as a candidate operating point, then compare it with business capacity and policy limits.
confusion_matrix[TN]	9967	True negatives correctly cleared. Next: preserve this base while tuning the alert threshold.
confusion_matrix[FP]	74	False positives create review cost without SAR conversion. Next: reduce this count with threshold tuning or sharper risk features.
confusion_matrix[FN]	587	False negatives are missed SAR cases. Next: if this count is unacceptable, lower the threshold or add features that capture suspicious behavior earlier.
confusion_matrix[TP]	173	True positives are correctly escalated cases. Next: assess whether this volume is sufficient relative to expected SAR capture goals.
precision_at_k[10]	0.9000	Shows how accurate the top-10 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[50]	0.8400	Shows how accurate the top-50 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[100]	0.7700	Shows how accurate the top-100 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.

value_date_quarter=2024Q1 - 1) Model Effectiveness (3/3)

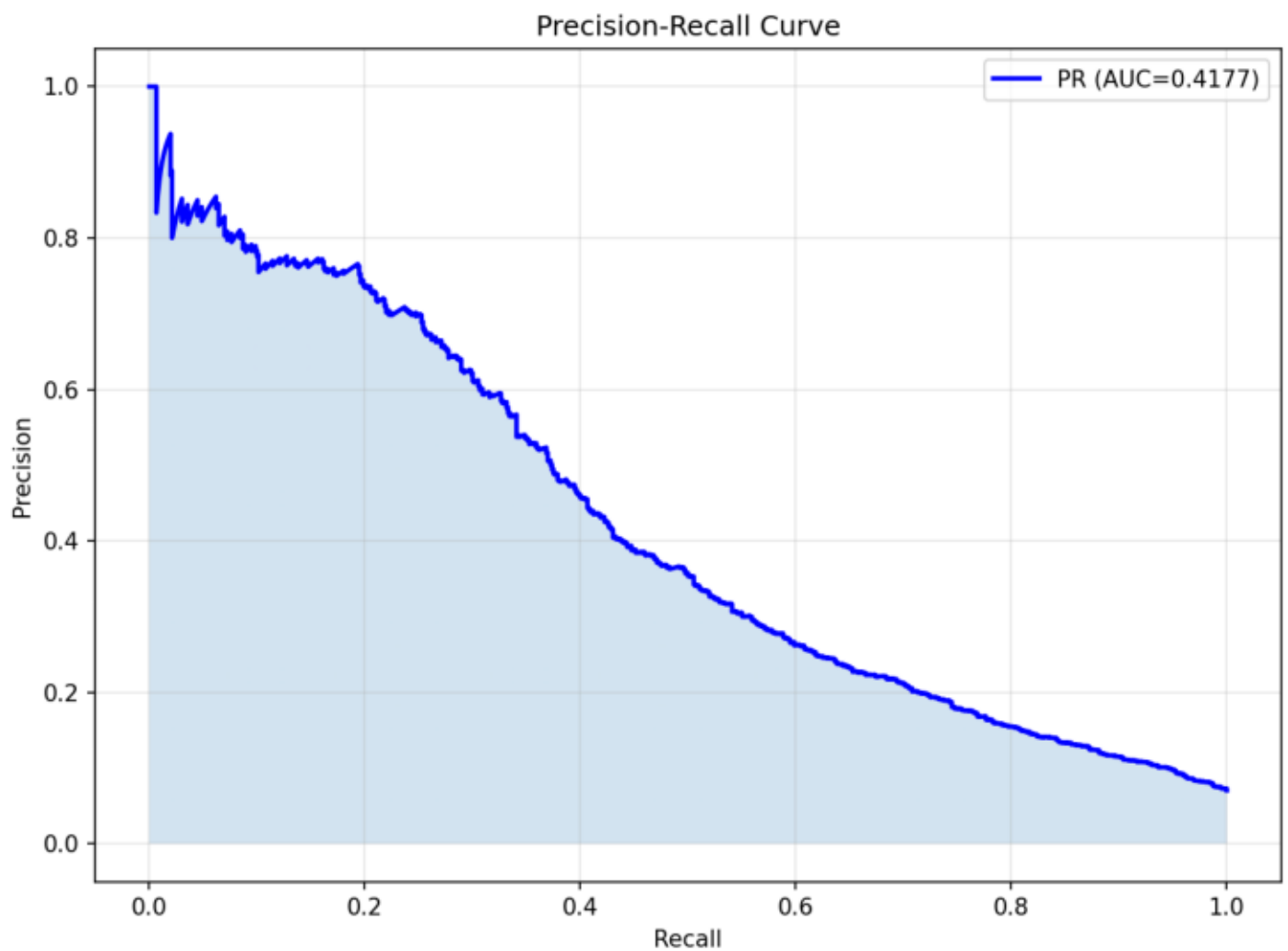
Metric	Figure	Explanation
precision_at_k[200]	0.7450	Shows how accurate the top-200 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[500]	0.5340	Shows how accurate the top-500 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
recall_at_k[10]	0.0118	Shows how much of the positive population is captured within the top-10 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[50]	0.0553	Shows how much of the positive population is captured within the top-50 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[100]	0.1013	Shows how much of the positive population is captured within the top-100 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[200]	0.1961	Shows how much of the positive population is captured within the top-200 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[500]	0.3513	Shows how much of the positive population is captured within the top-500 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.

value_date_quarter=2024Q1 - 1) Model Effectiveness



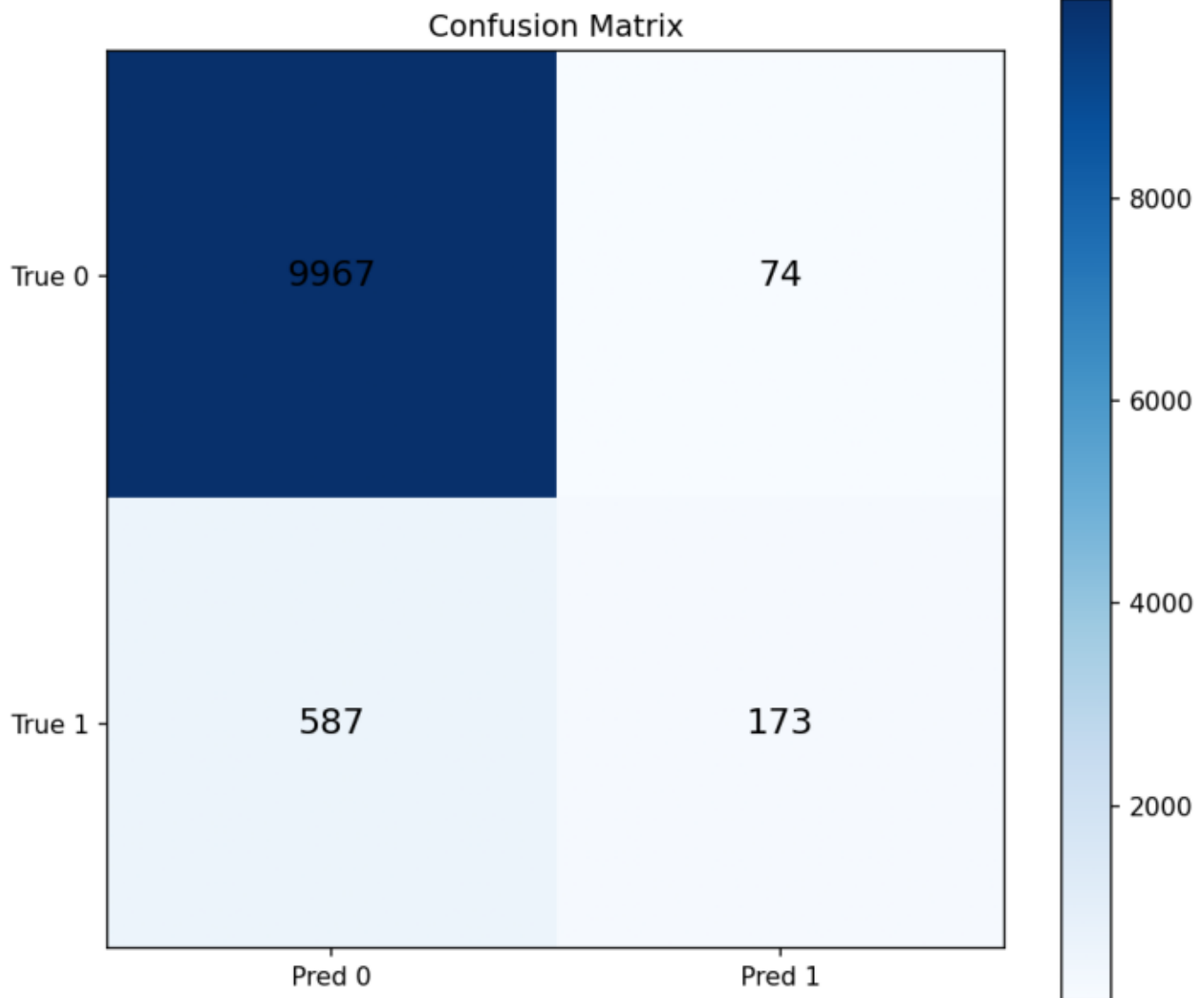
roc_curve: AUC-ROC=0.8265; curve above diagonal indicates strong separation.

value_date_quarter=2024Q1 - 1) Model Effectiveness



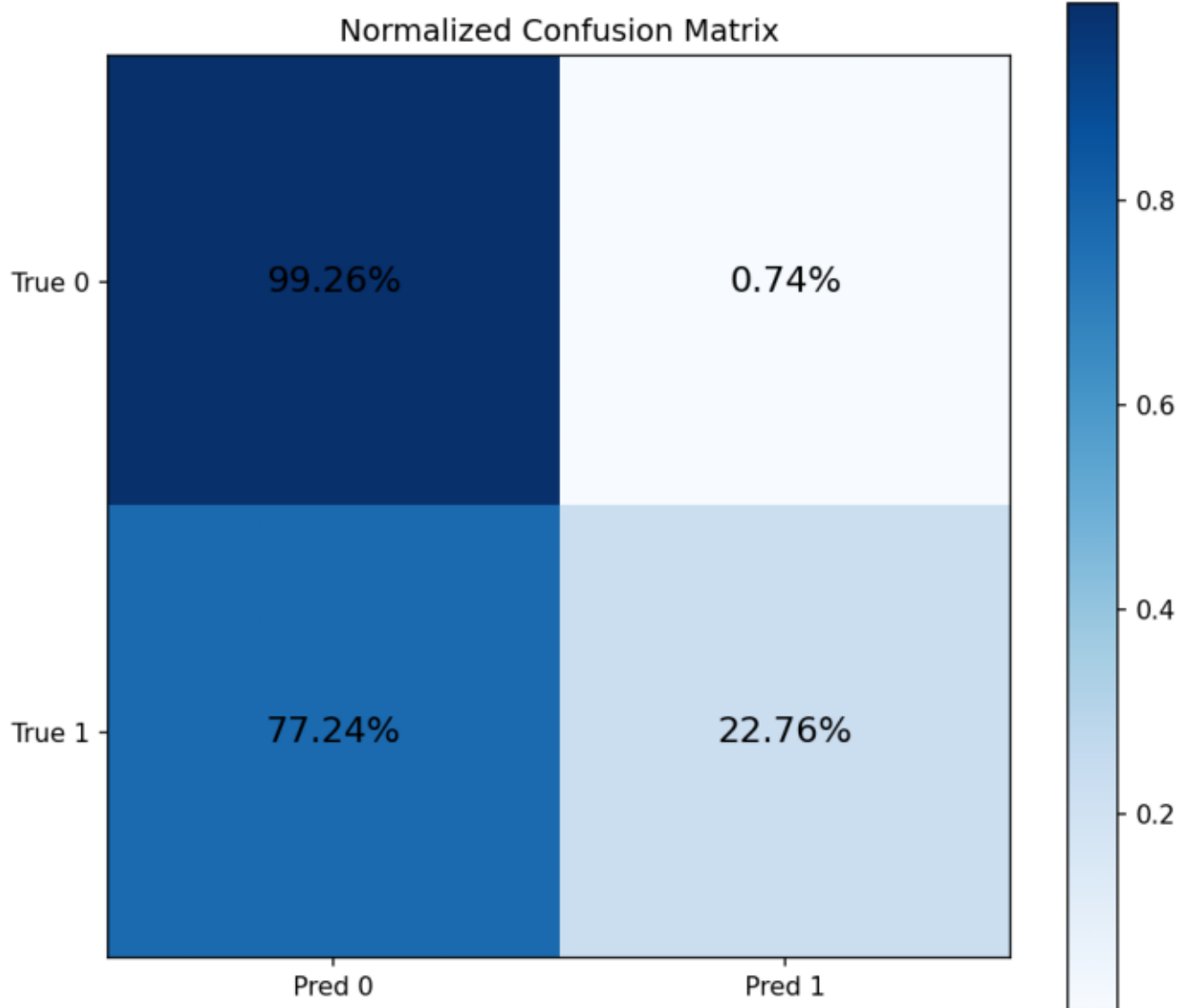
pr_curve: AUC-PR=0.4177 vs baseline 7.04%; higher curve implies better precision at recall.

value_date_quarter=2024Q1 - 1) Model Effectiveness



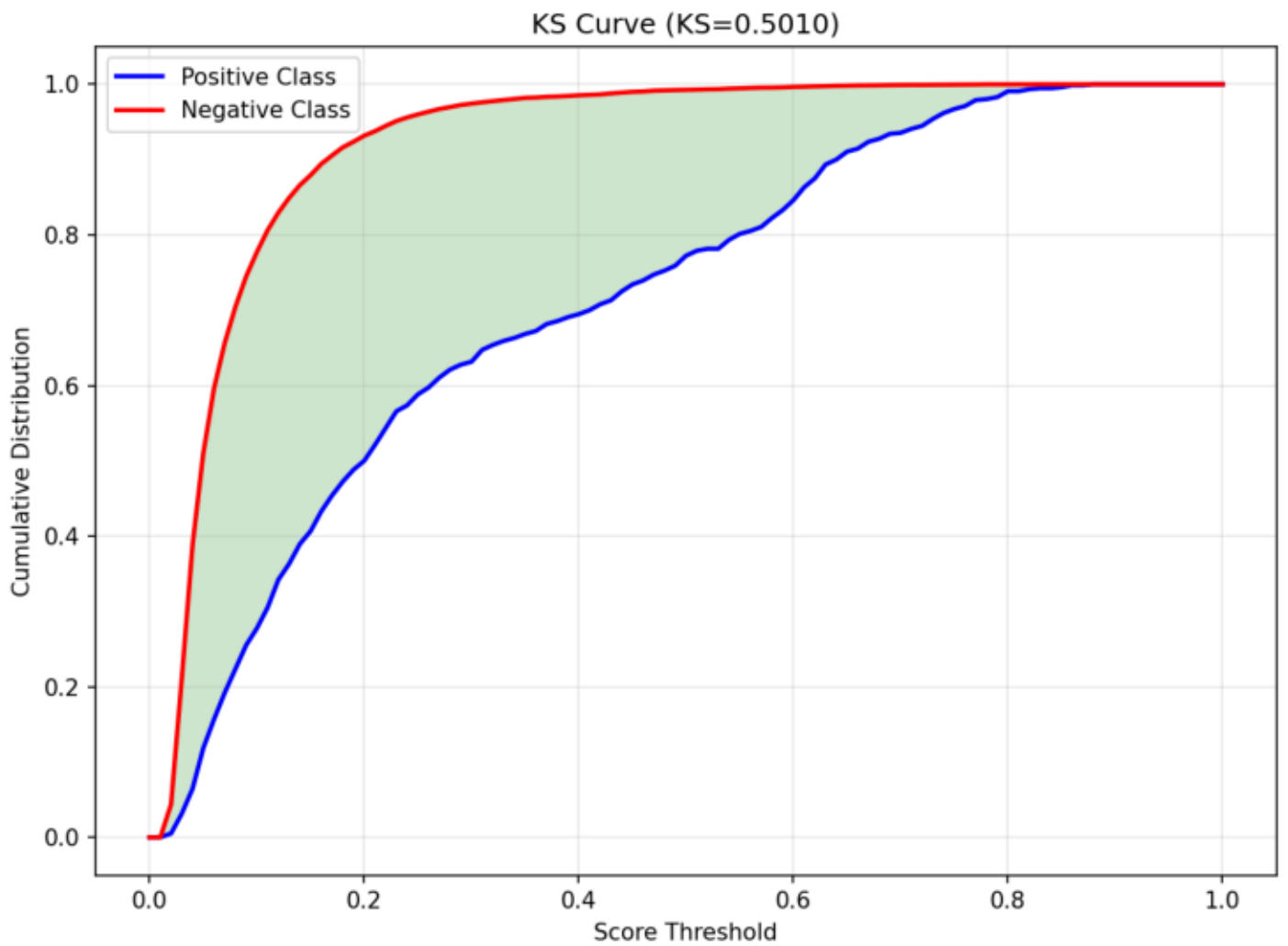
confusion_matrix: At threshold 0.50: TP=173, FP=74, FN=587, TN=9967.

value_date_quarter=2024Q1 - 1) Model Effectiveness



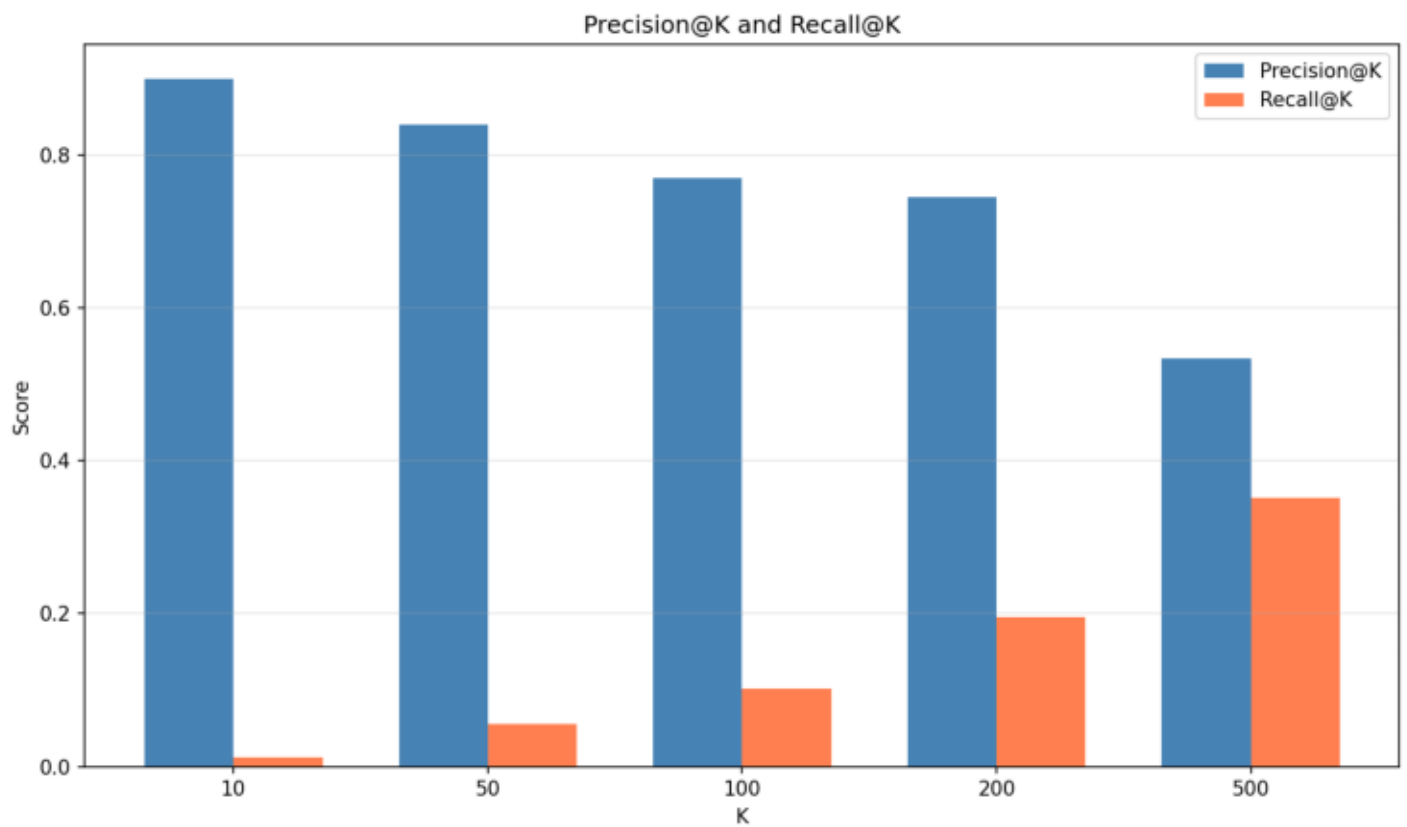
confusion_matrix_norm: Normalized rates: TPR=22.76%, TNR=99.26%.

value_date_quarter=2024Q1 - 1) Model Effectiveness



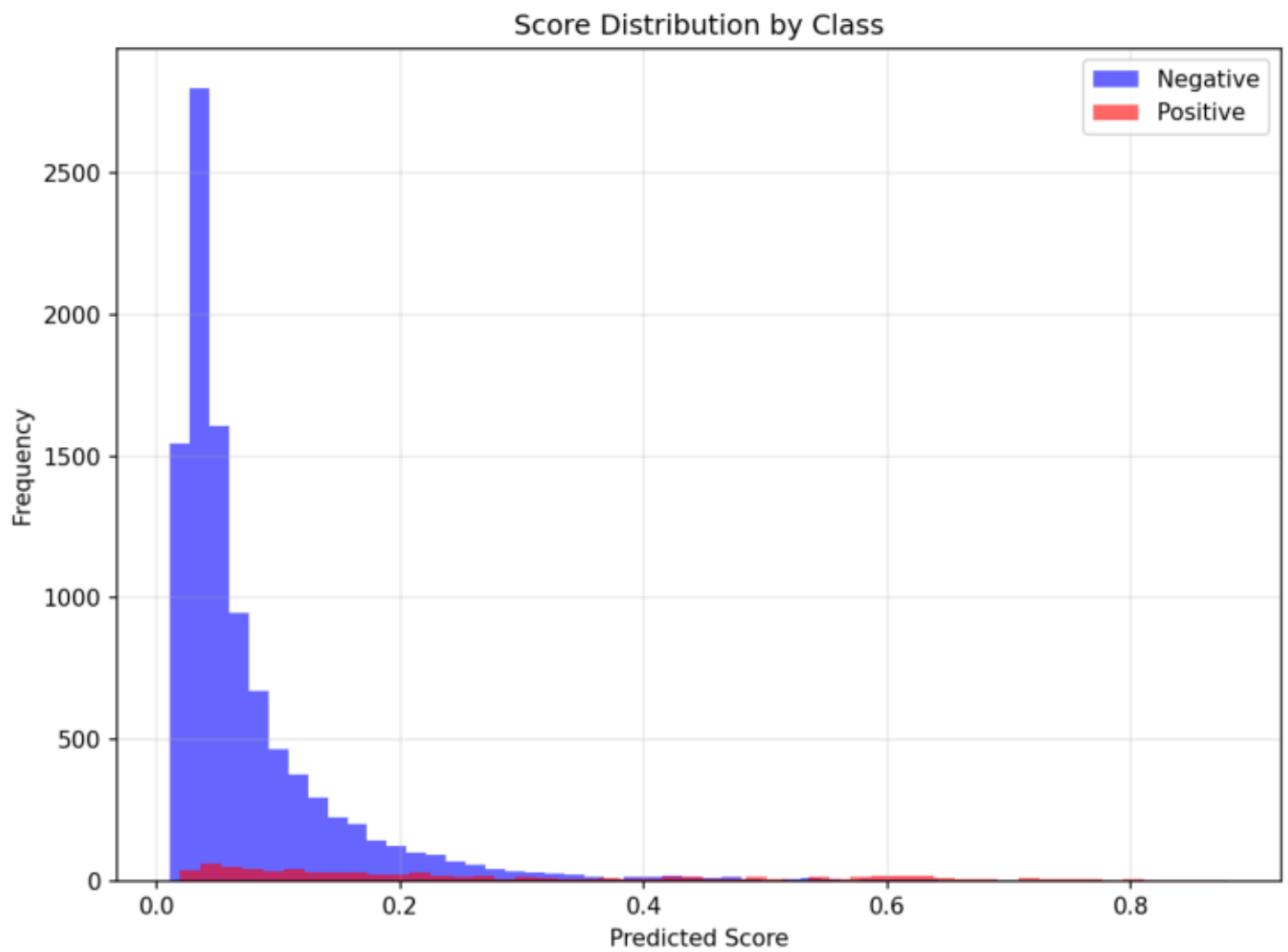
ks_curve: Maximum separation KS=0.5010 at threshold 0.11.

value_date_quarter=2024Q1 - 1) Model Effectiveness



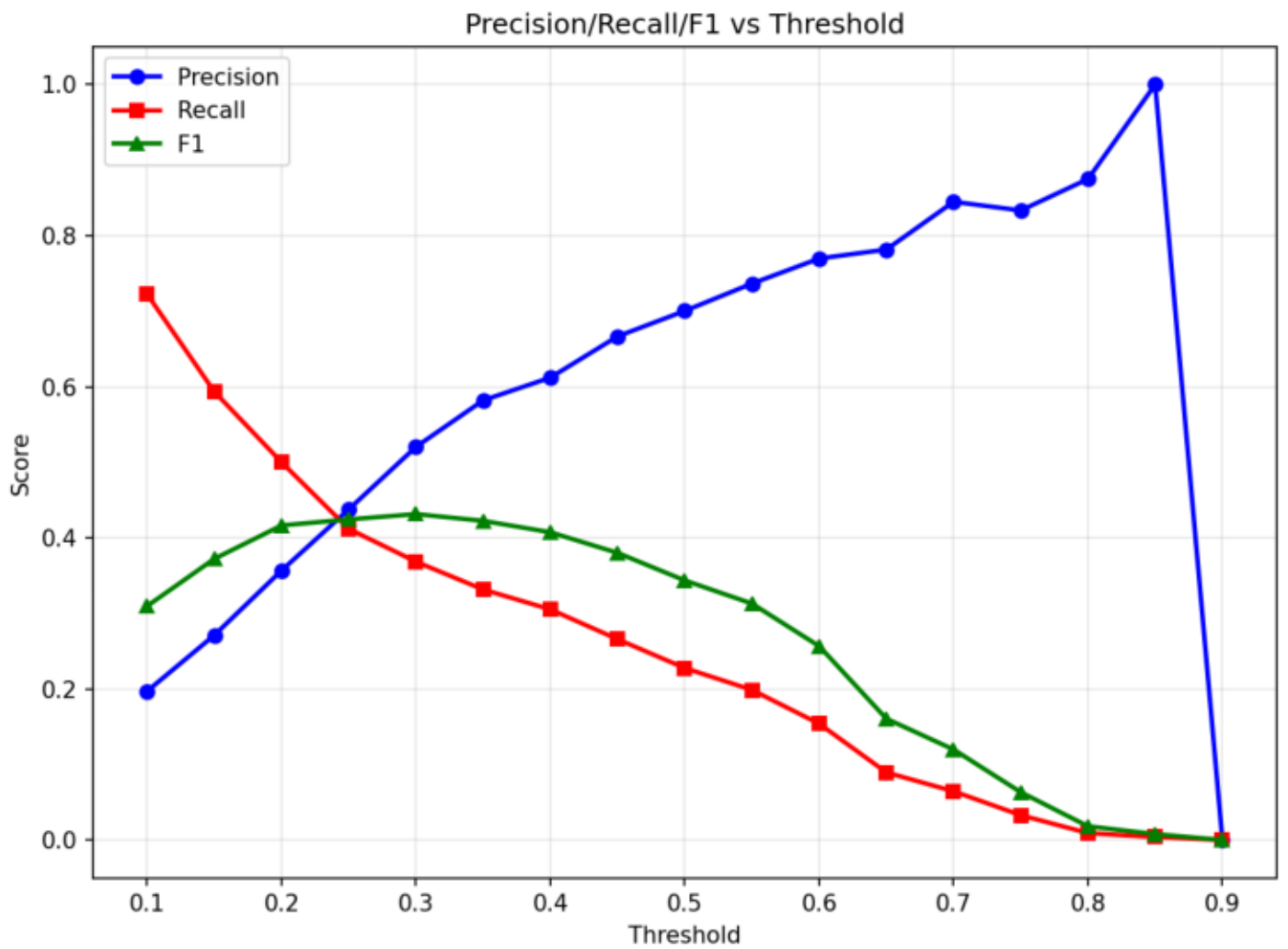
precision_recall_at_k: Bars show precision/recall as you expand the review queue.

value_date_quarter=2024Q1 - 1) Model Effectiveness



score_distribution: Mean score: positive=0.283, negative=0.078 (larger gap is better).

value_date_quarter=2024Q1 - 1) Model Effectiveness



threshold_analysis: Best F1 in tested grid is 0.4314 at threshold 0.30.

value_date_quarter=2024Q1 - 2) Model Efficiency (1/4)

Metric	Figure	Explanation
fpr	0.0074	Measures the share of negatives incorrectly flagged; 0.0074 is low. Next: if review demand is too high, raise the threshold or add features that better suppress false alerts.
tn	9967	Counts negatives correctly cleared without manual review. Next: preserve this clearance rate while checking that recall does not fall below policy expectations.
fp	74	Counts non-events that would still be sent to review. Next: use this with staffing capacity to decide whether threshold or feature refinement is required.
threshold	0.5000	This is the active operating cutoff used to flag alerts. Next: tune it jointly with recall, precision, and review capacity rather than in isolation.
fpr_at_thresholds[thresholds.t_0.05]	0.4902	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.05. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.10]	0.2227	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.10. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q1 - 2) Model Efficiency (2/4)

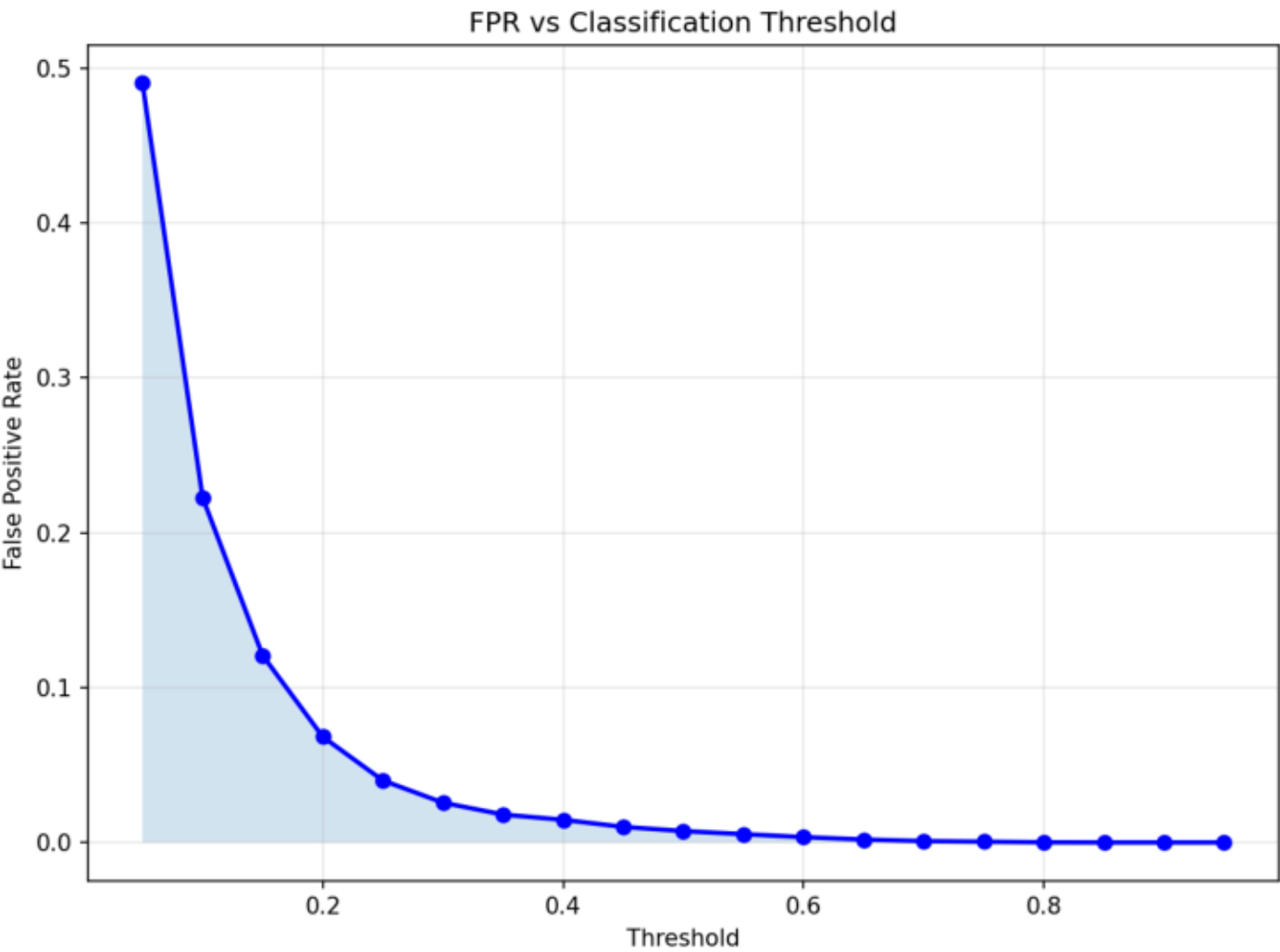
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.15]	0.1207	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.15. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.20]	0.0682	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.20. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.25]	0.0400	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.25. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.30]	0.0257	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.30. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.35]	0.0180	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.35. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.40]	0.0146	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.40. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q1 - 2) Model Efficiency (3/4)

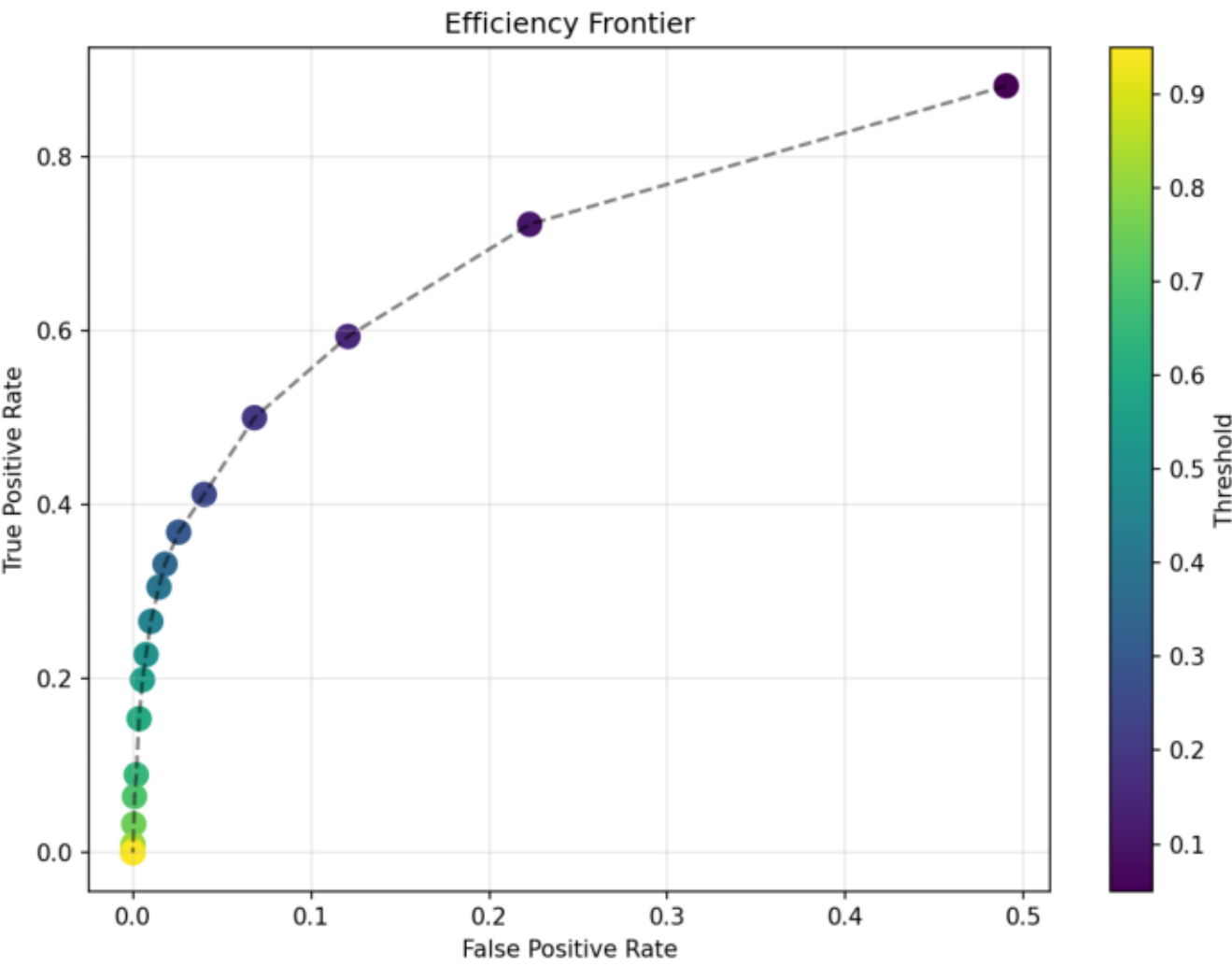
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.45]	0.0101	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.45. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.50]	0.0074	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.50. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.55]	0.0054	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.55. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.60]	0.0035	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.60. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.65]	0.0019	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.65. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.70]	0.0009	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.70. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q1 - 2) Model Efficiency (4/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.75]	0.0005	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.75. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.80]	0.0001	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.80. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.85]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.85. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.90]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.90. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.95]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.95. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

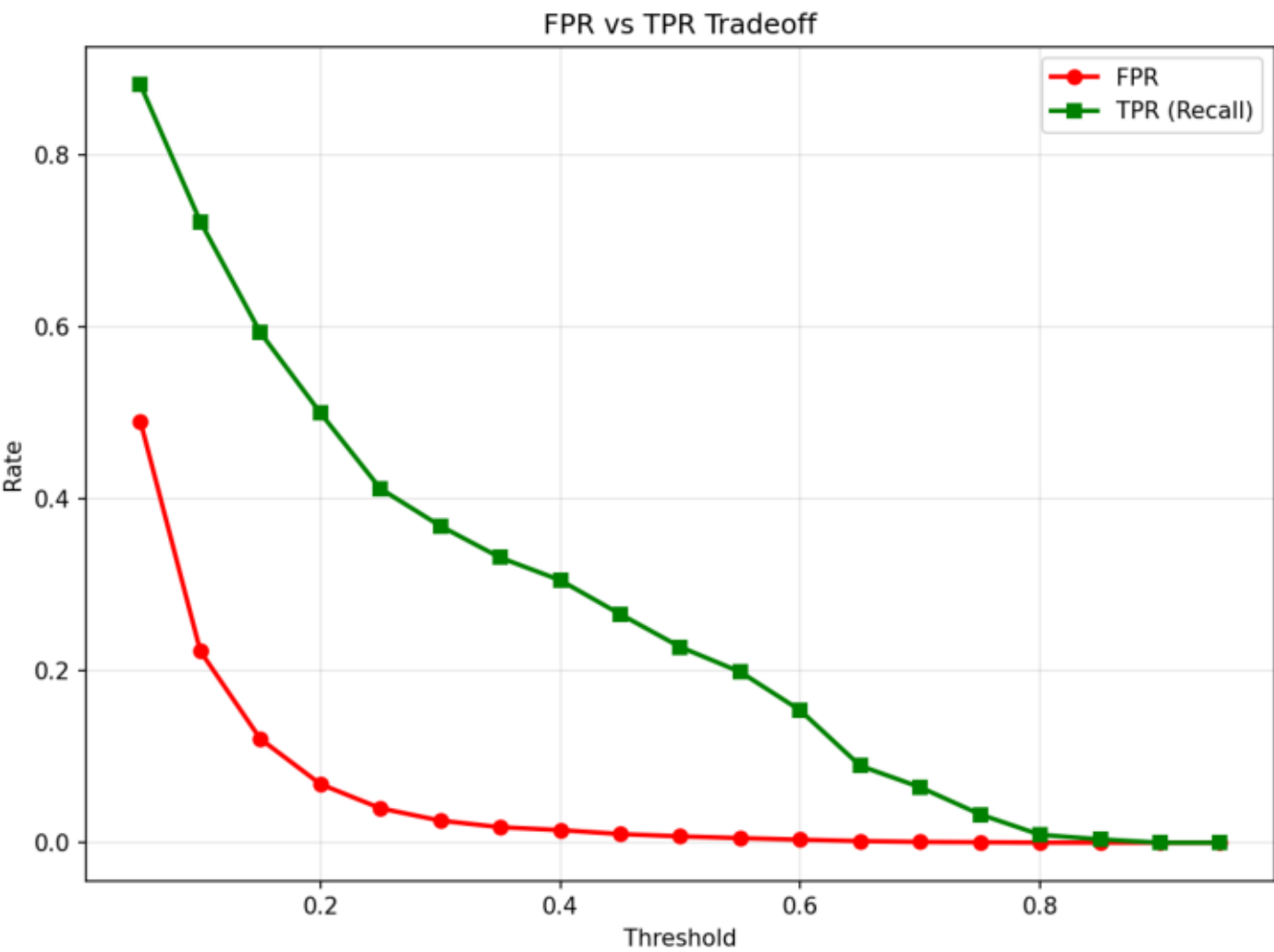


value_date_quarter=2024Q1 - 2) Model Efficiency



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

value_date_quarter=2024Q1 - 2) Model Efficiency



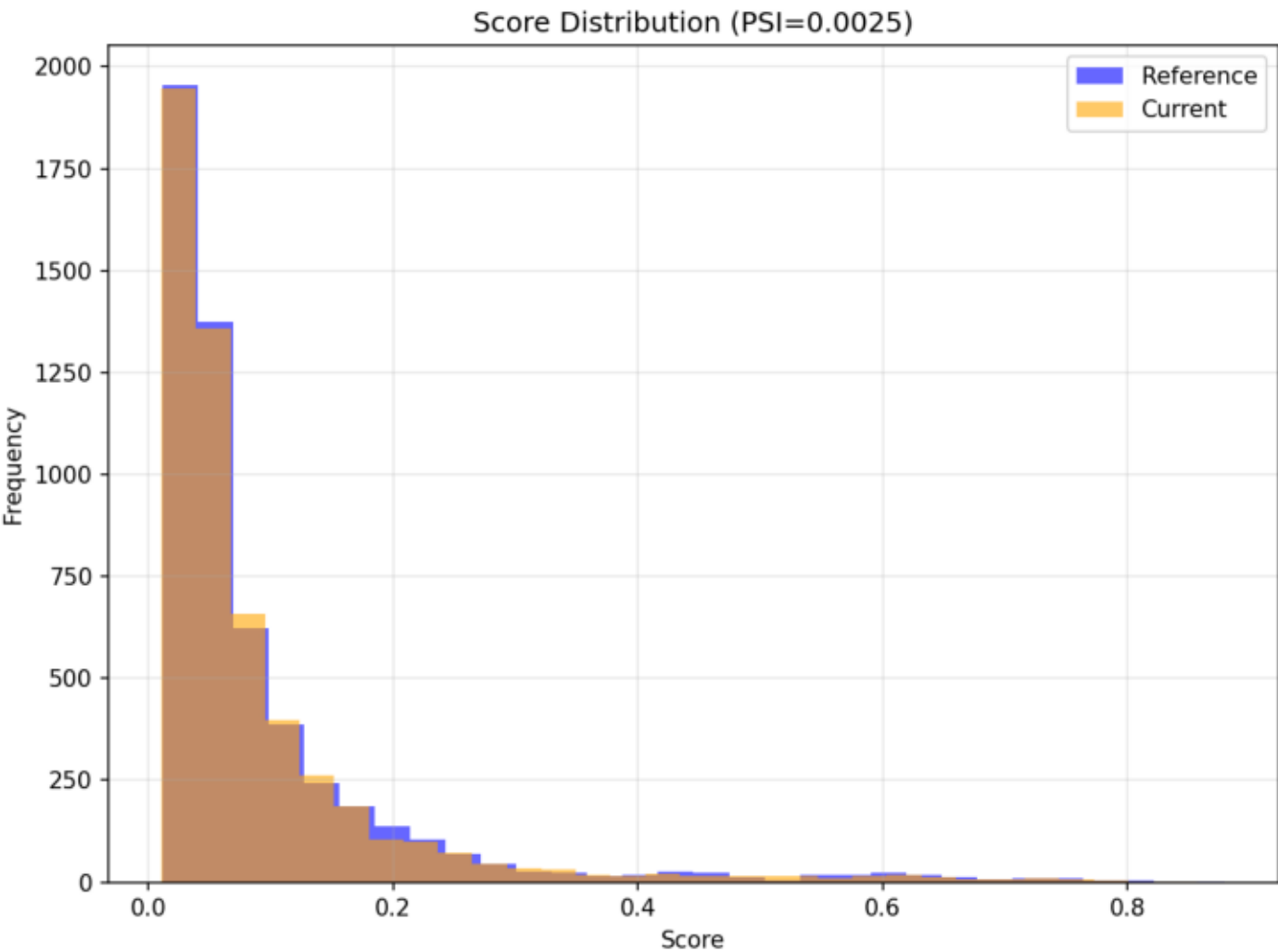
value_date_quarter=2024Q1 - 3) Model Stability (1/2)

Metric	Figure	Explanation
psi	0.0025	Measures score-distribution shift between reference and current populations; 0.0025 is negligible. Next: investigate segment mix, data drift, and recalibration needs when PSI is material.
cv_auc_roc_mean	0.8262	Average AUC-ROC across validation folds. Next: compare it with overall AUC-ROC; a large gap can indicate overfitting or unstable sampling.
cv_auc_roc_std	0.0077	Standard deviation of AUC-ROC across folds. Next: if variability is high, simplify the model or increase training stability before deployment.
cv_auc_pr_mean	0.4196	Average AUC-PR across validation folds. Next: confirm that precision-recall performance remains acceptable under resampling.
cv_auc_pr_std	0.0141	Standard deviation of AUC-PR across folds. Next: high dispersion suggests unstable alert quality and warrants deeper validation.
bootstrap_auc_roc_mean	0.8177	Bootstrap mean AUC-ROC estimates expected ranking performance under repeated sampling. Next: compare it with the point estimate to judge optimism in the current test run.
bootstrap_auc_roc_ci_lower	0.7946	Lower confidence bound for bootstrap AUC-ROC. Next: use it as a conservative performance floor in model-risk discussions.
bootstrap_auc_roc_ci_upper	0.8393	Upper confidence bound for bootstrap AUC-ROC. Next: review the interval width with the lower bound to assess uncertainty.

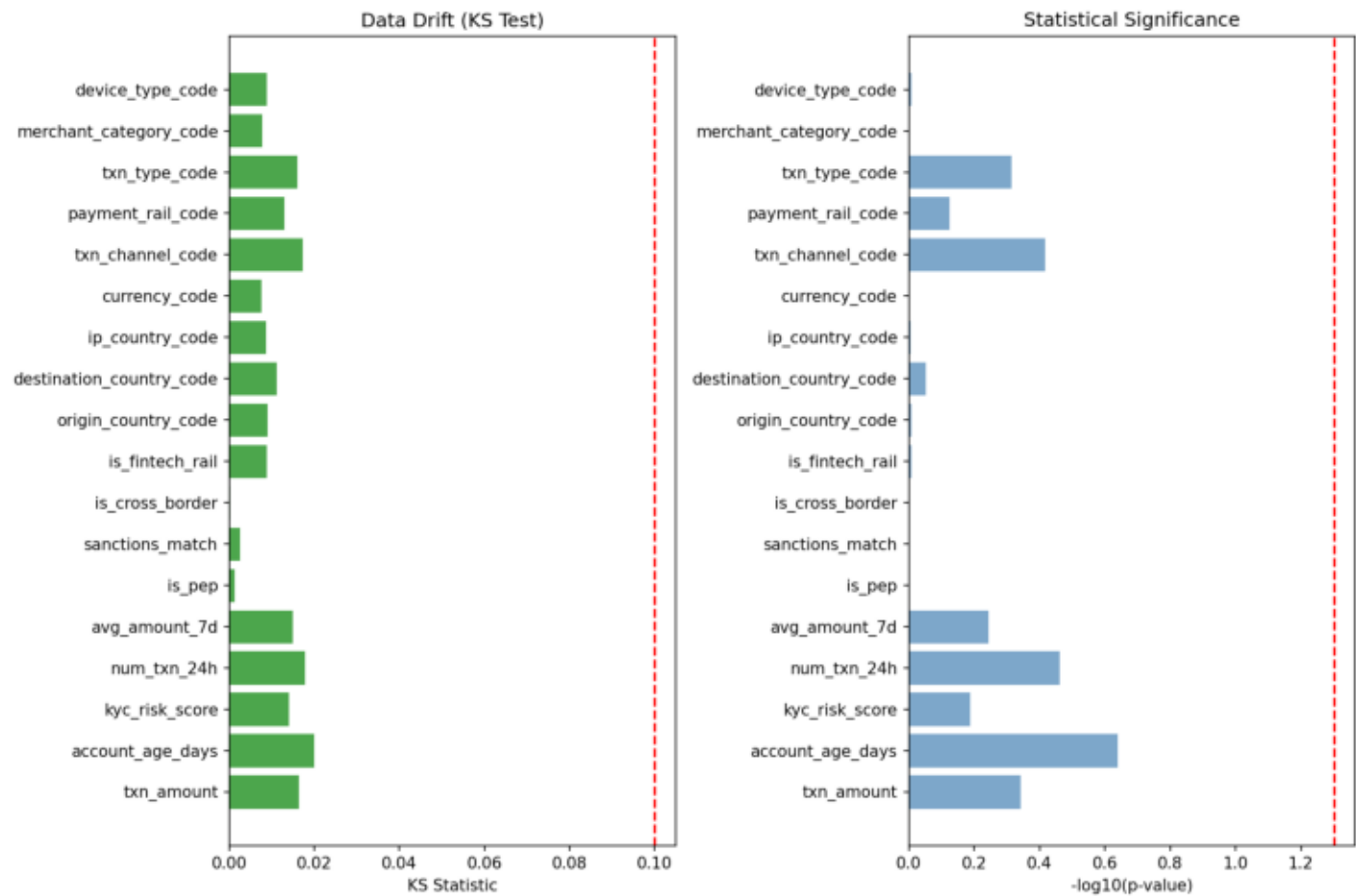
value_date_quarter=2024Q1 - 3) Model Stability (2/2)

Metric	Figure	Explanation
concept_drift_detected	False	No material concept-drift signal was detected in the evaluated slices. Next: retain ongoing monitoring because stable historical results do not guarantee future stability.
concept_drift_score	0.0308	Quantifies the magnitude of performance change across ordered slices. Next: rising scores should trigger deeper review of segment behavior and model refresh timing.

value_date_quarter=2024Q1 - 3) Model Stability

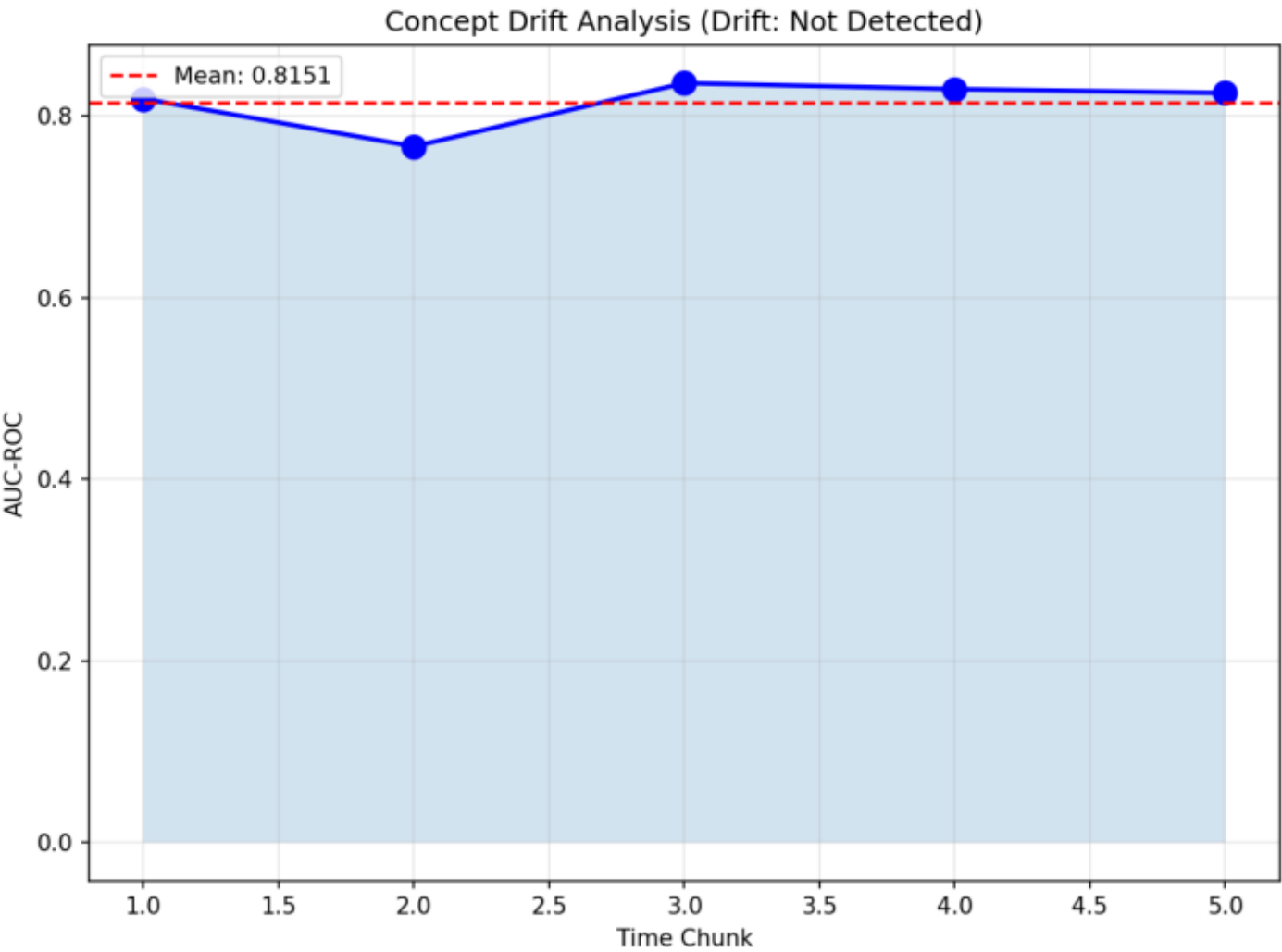


value_date_quarter=2024Q1 - 3) Model Stability



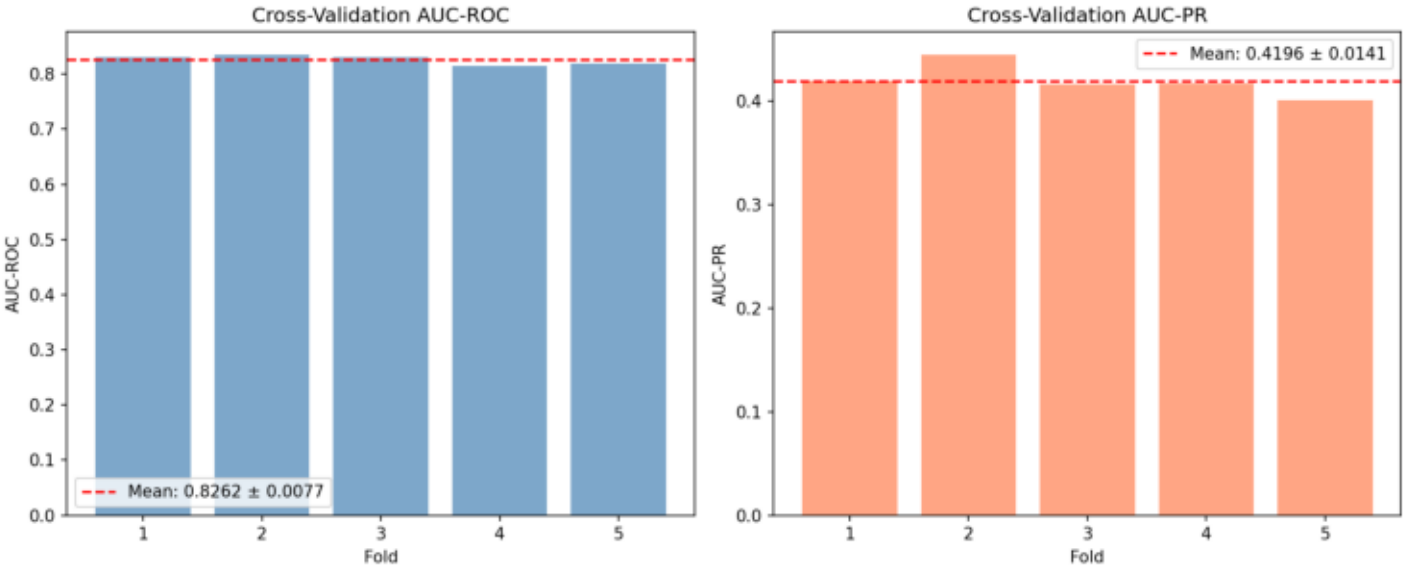
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.

value_date_quarter=2024Q1 - 3) Model Stability



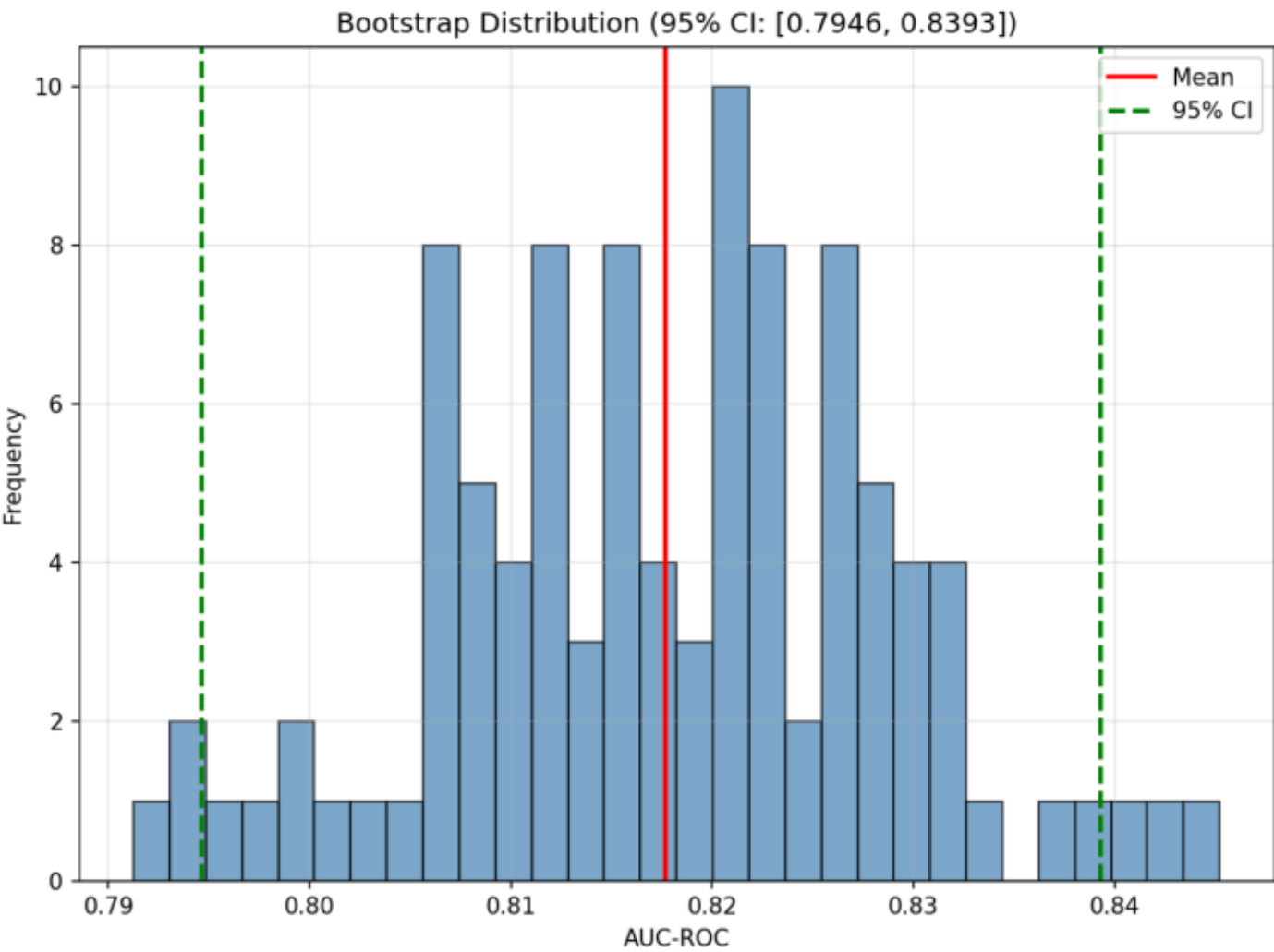
concept_drift: AUC across time chunks shows whether performance is stable over time.

value_date_quarter=2024Q1 - 3) Model Stability



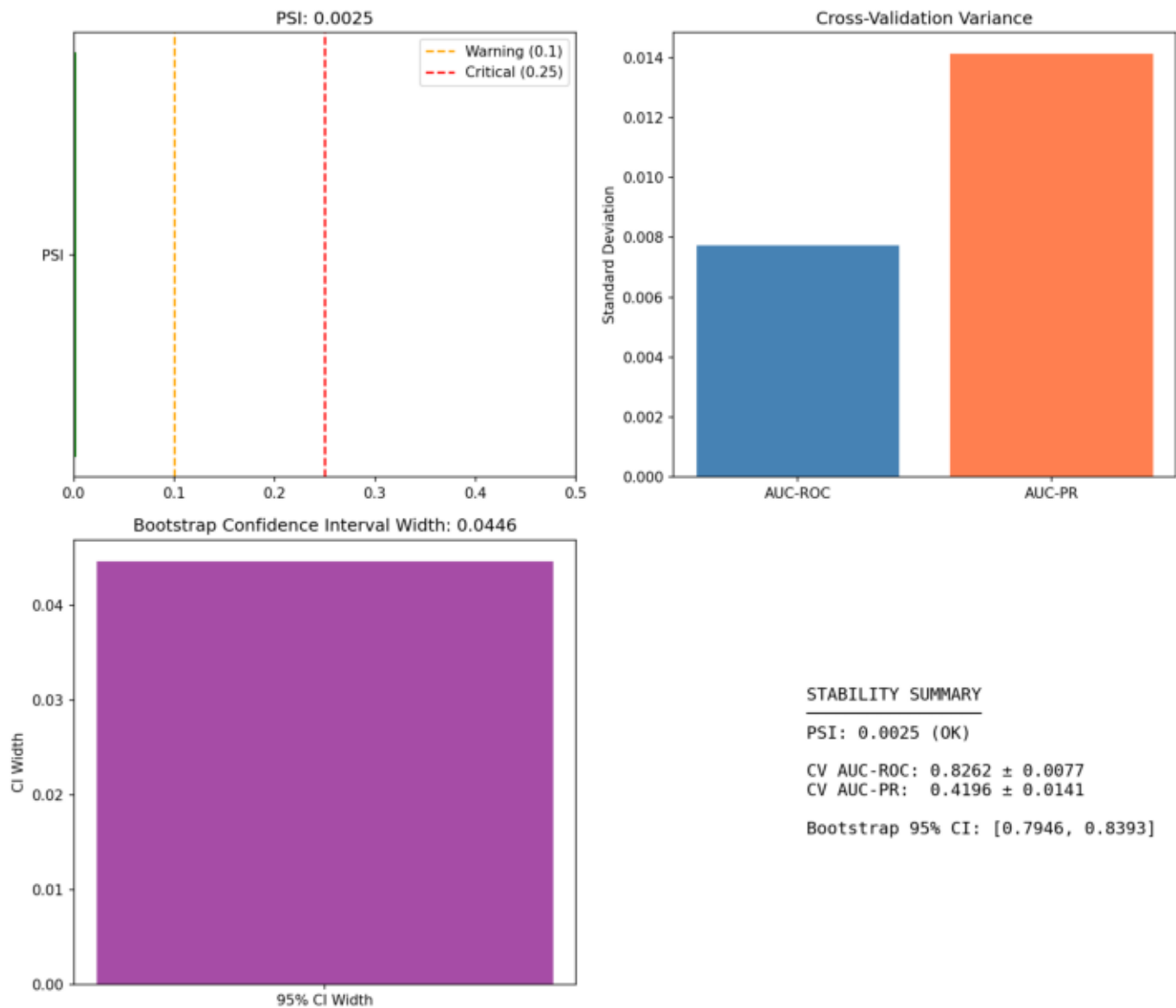
cv_results: Fold scores cluster around ROC 0.8262 and PR 0.4196.

value_date_quarter=2024Q1 - 3) Model Stability



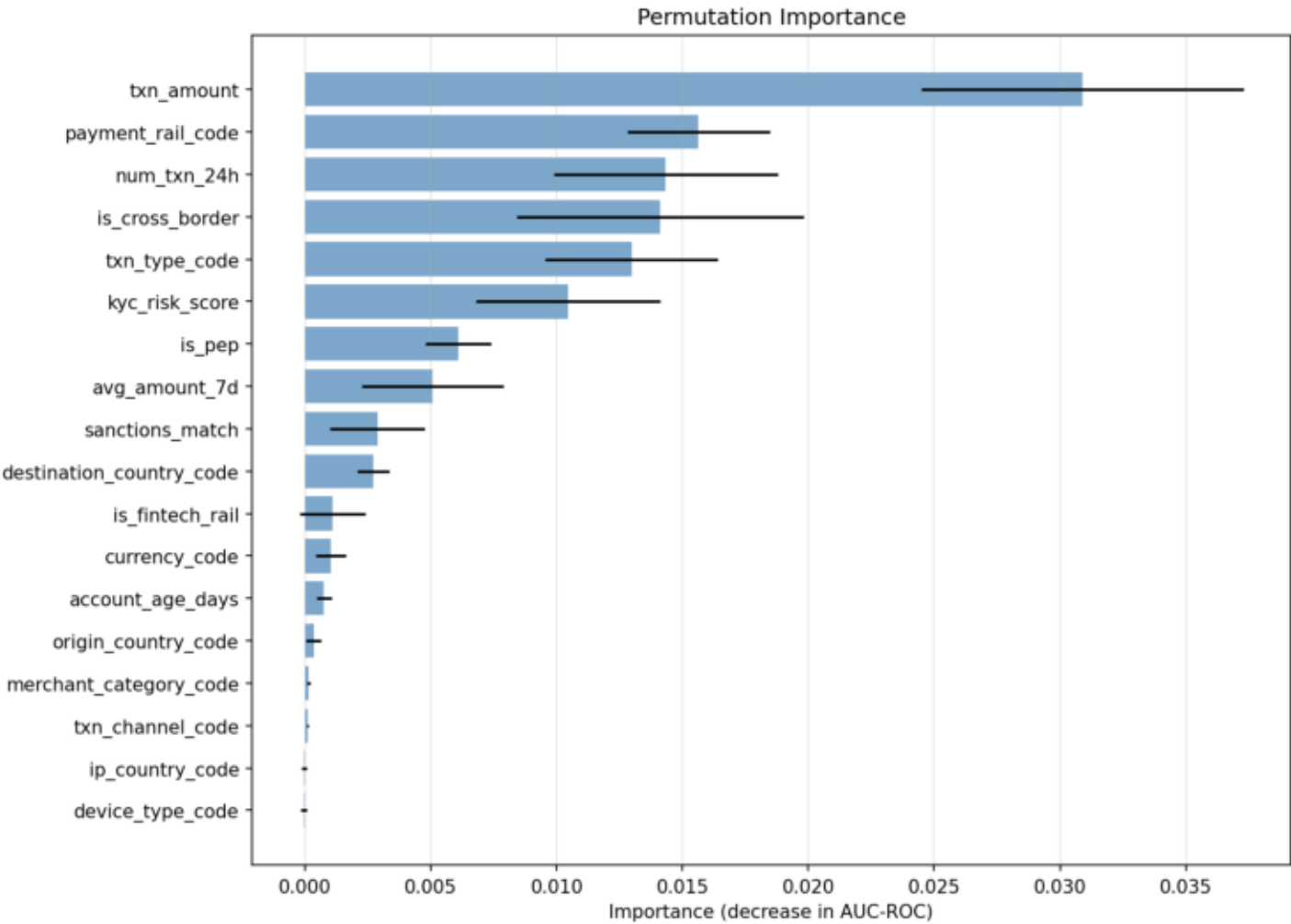
bootstrap_distribution: Bootstrap CI [0.7946, 0.8393] reflects performance stability.

value_date_quarter=2024Q1 - 3) Model Stability

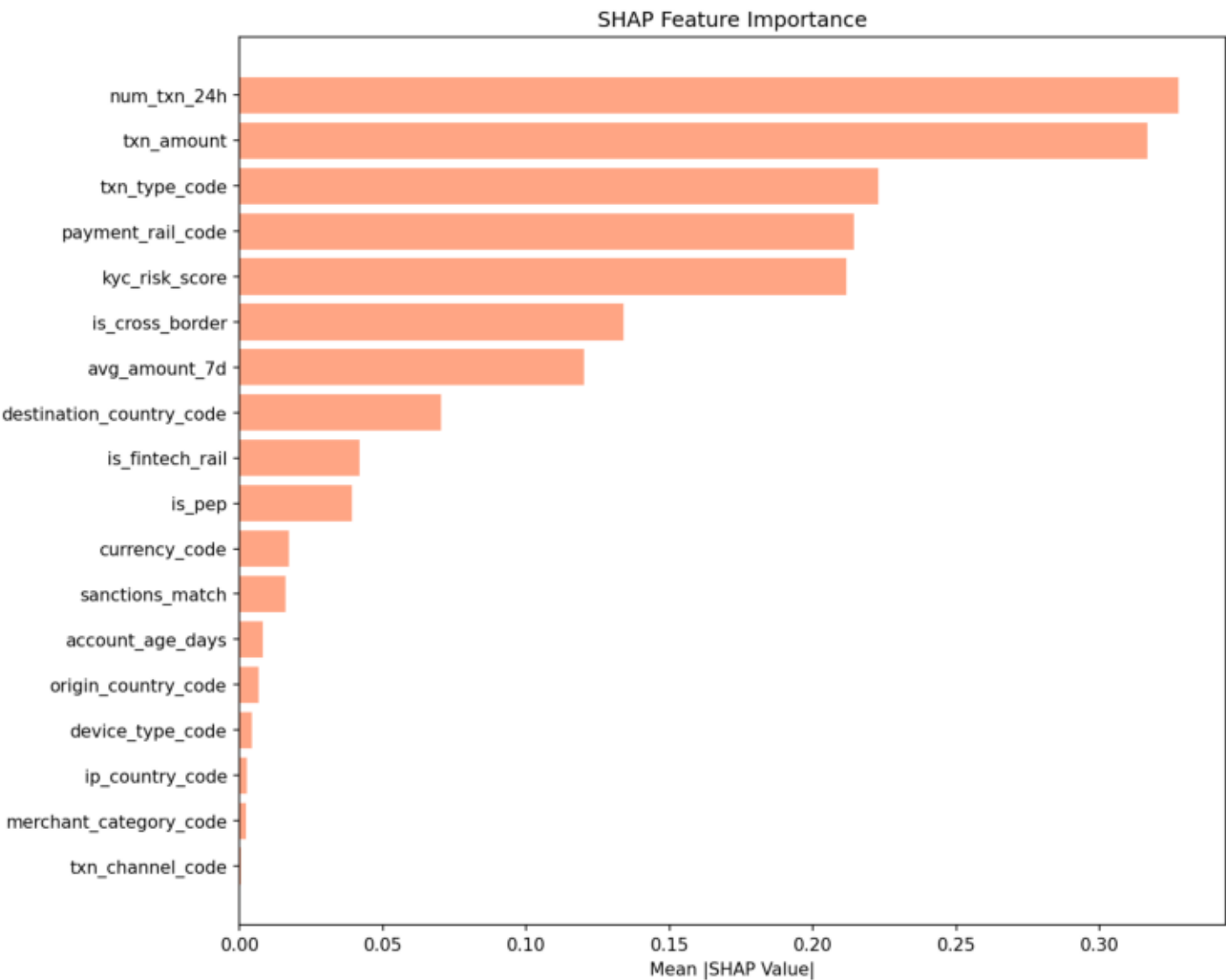


Metric	Figure	Explanation
model_type	tree	Identifies the estimator family so explanation methods can be interpreted in the right context. Next: confirm the chosen explanation methods are appropriate for this model class.
methods_used	permutation, pdp, ice, shap, lime	Lists the interpretability methods produced in this run. Next: compare signals across methods before using them in documentation or model challenge.
perm_top_features	txn_amount, payment_rail_code, num_txn_24h, is_cross_border, txn_type_code ...	Shows features with the largest performance drop when permuted, which highlights global importance. Next: validate that the top drivers align with business logic, feature governance, and policy expectations.
shap_top_features	num_txn_24h, txn_amount, txn_type_code, payment_rail_code, kyc_risk_score ...	Shows the most influential features under SHAP-based contribution analysis. Next: inspect whether directionality and ranking remain sensible across key segments.
lime_instances	3	Counts the locally explained observations reviewed with LIME. Next: examine representative true-positive and false-positive cases to understand local decision logic.

Metric	Figure	Explanation
pdp_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score, num_txn_24h ...	Lists features with partial-dependence plots. Next: review whether average model response is monotonic, stable, and operationally sensible.
ice_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score	Lists features with individual conditional expectation plots. Next: inspect heterogeneity and interactions that may be hidden in average-only views.

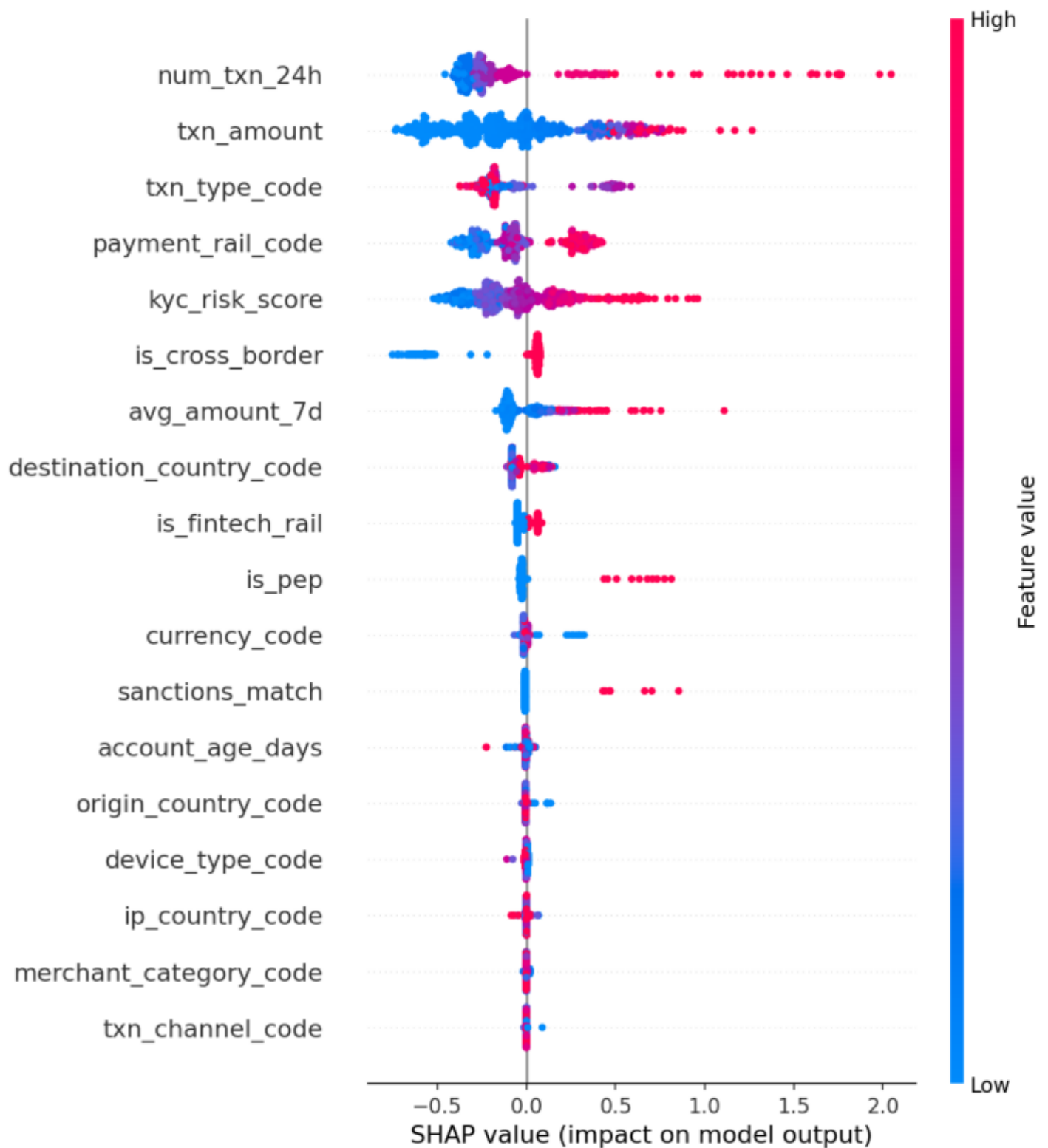


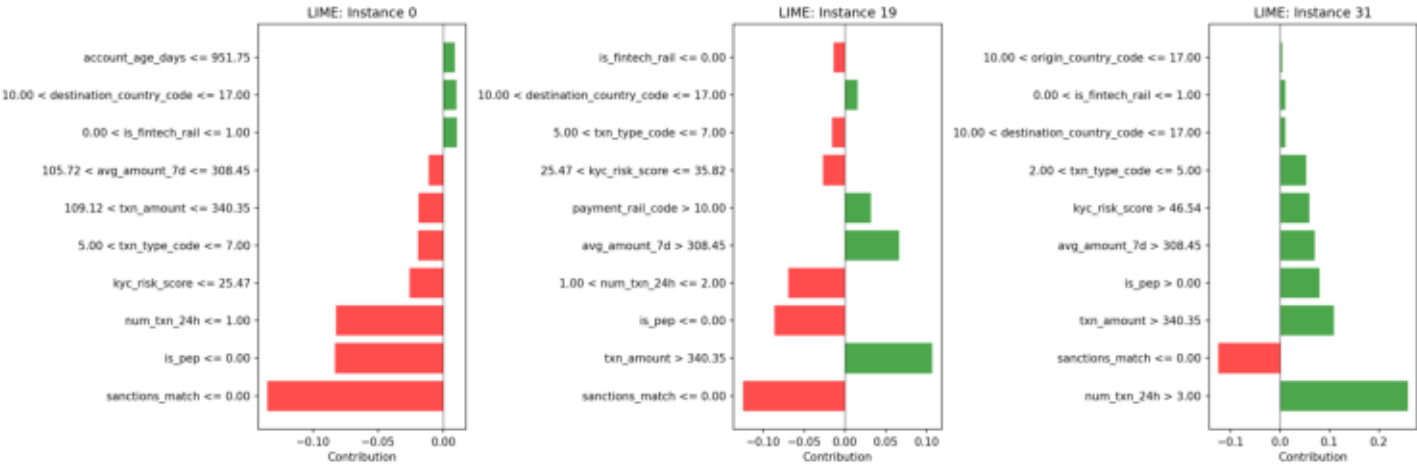
permutation_importance: Bars show how much AUC drops when each feature is permuted.



shap_bar: SHAP bar plot ranks global feature impact by mean absolute SHAP values.

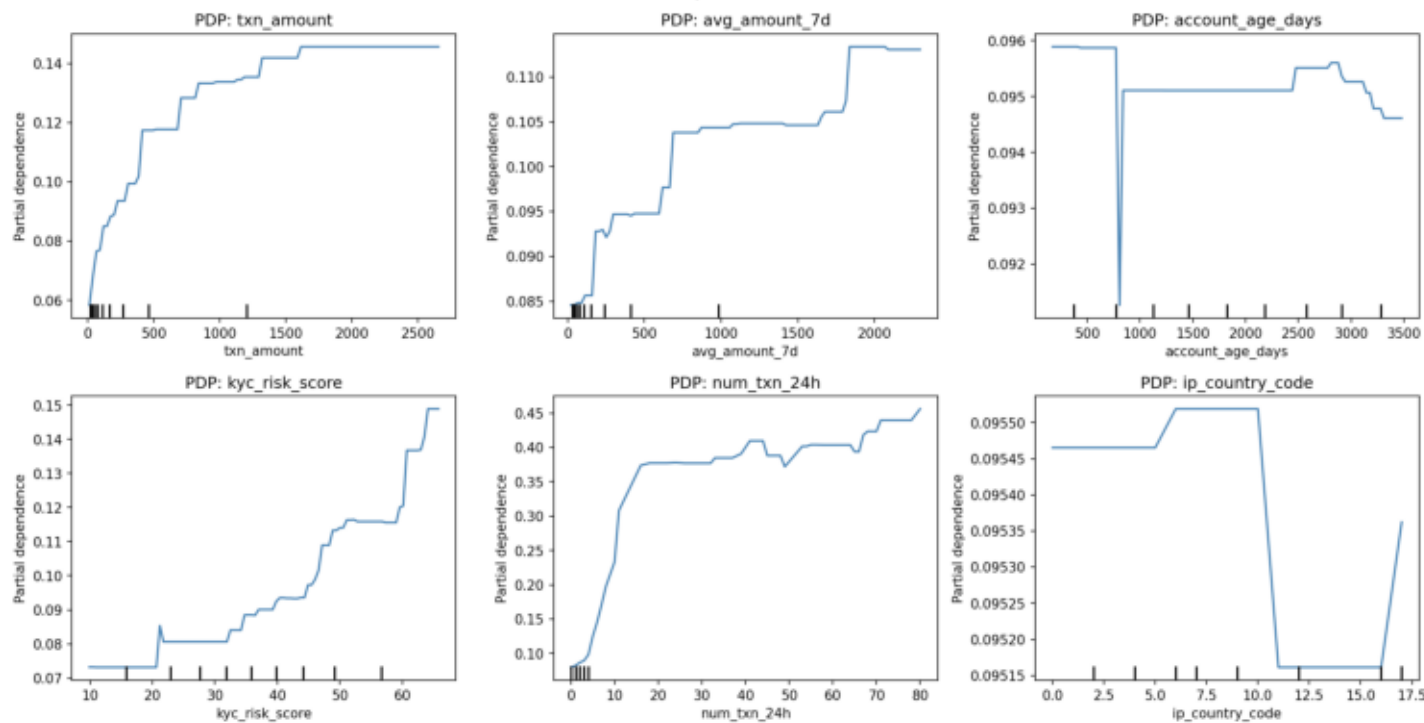
value_date_quarter=2024Q1 - 4) Model Interpretability





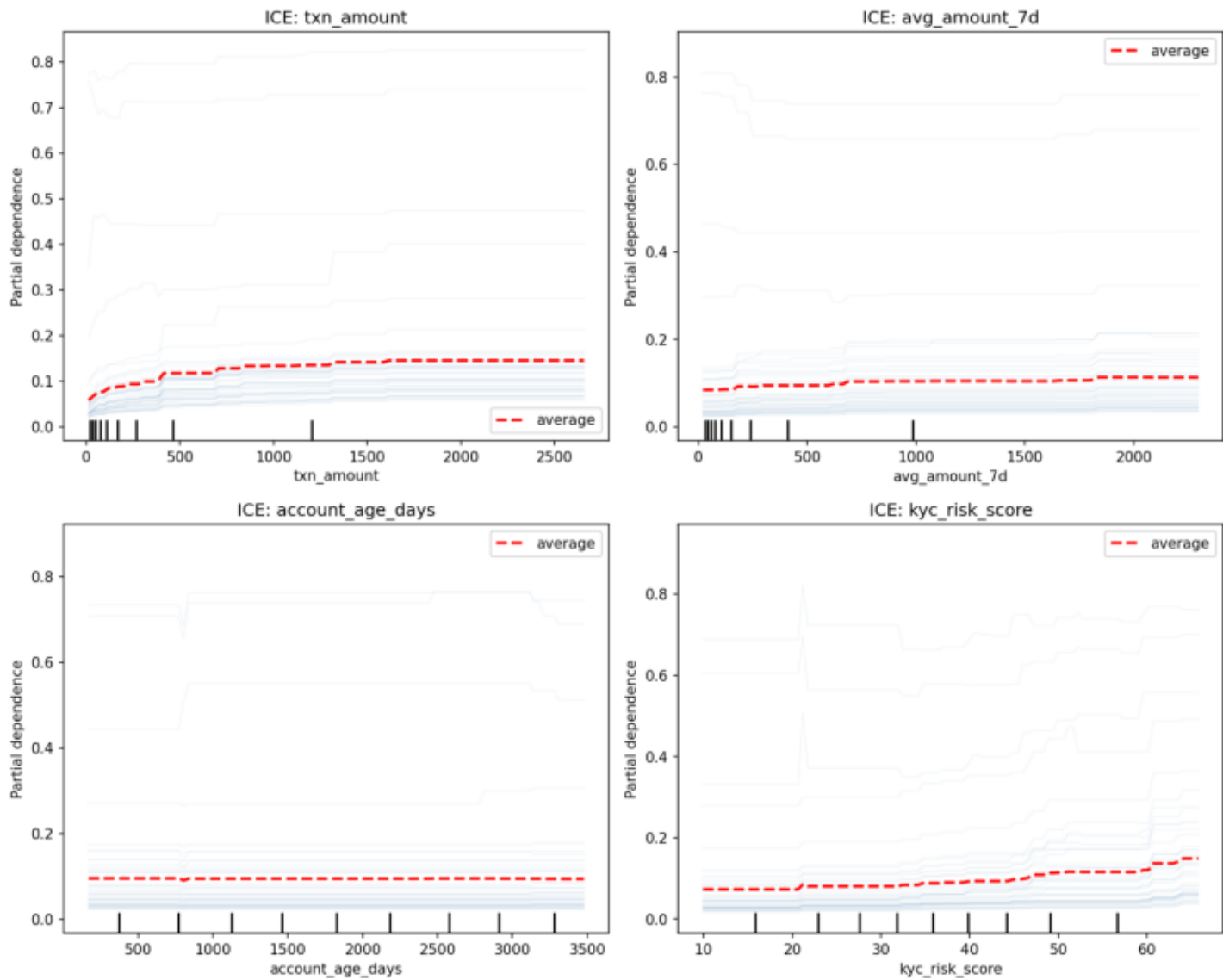
lime_explanation: LIME bars show local feature contributions for selected instances.

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q2 - 1) Model Effectiveness (1/3)

Metric	Figure	Explanation
auc_roc	0.8025	Measures ranking power across all thresholds. 0.8025 indicates strong separation. Next: compare AUC-ROC across segments and keep threshold tuning separate from ranking validation.
auc_pr	0.4067	Measures precision-recall performance under class imbalance; the observed event rate is 7.95%. Next: use this together with alert capacity to decide whether the high-score queue is practically useful.
precision	0.6795	Shows the share of flagged cases that are true positives at the current threshold. Next: if this is below review expectations, raise the threshold or improve exclusion features.
recall	0.2186	Shows the share of true positives captured at the current threshold. Next: if missed cases are too high, lower the threshold or add stronger risk features.
f1	0.3307	Balances precision and recall in one score. Next: use it as a threshold-selection aid when both missed cases and false alerts matter.
ks_statistic	0.4591	Measures score separation between positive and negative classes. Next: low KS suggests the model needs better features or a different specification.

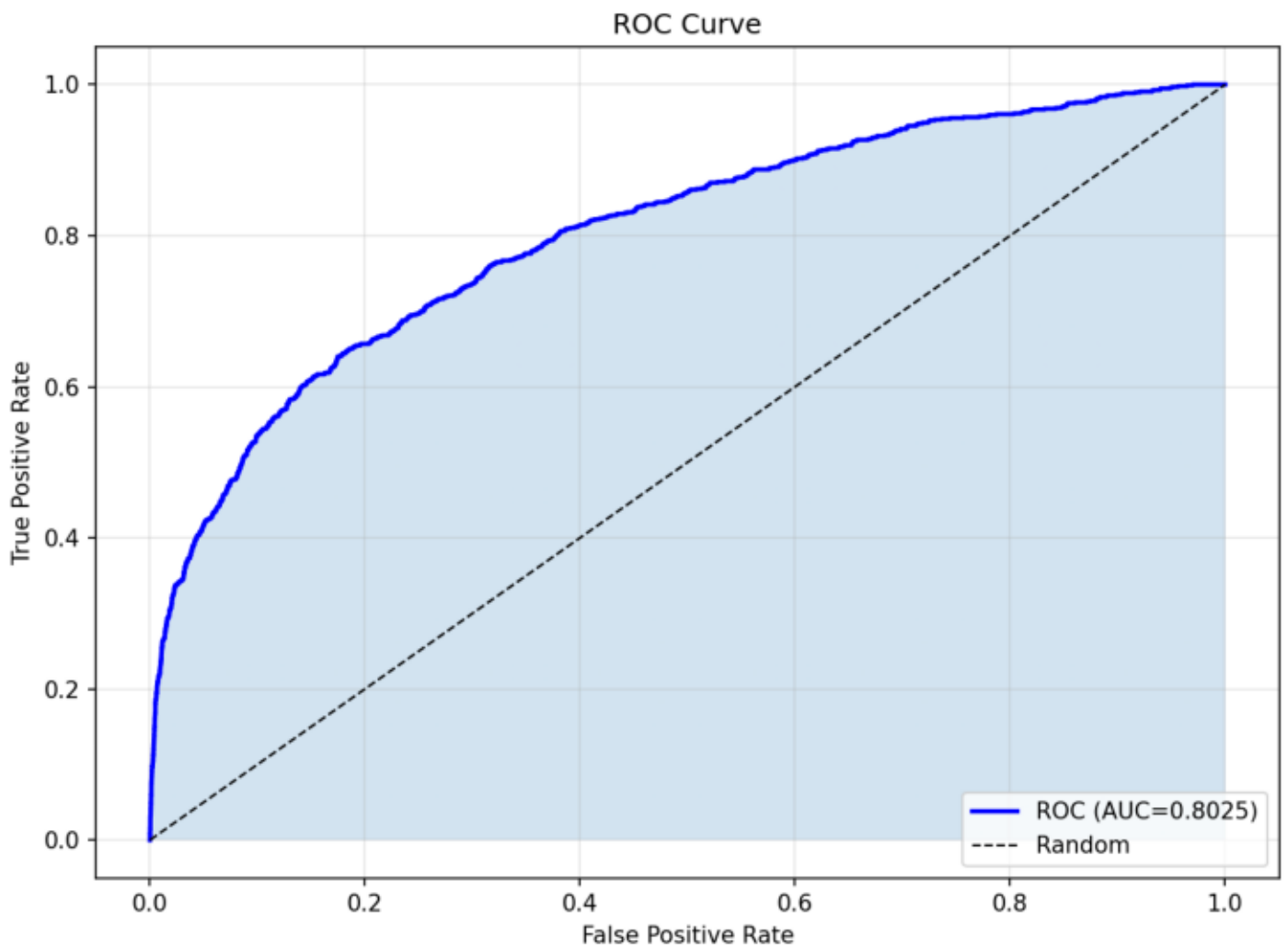
value_date_quarter=2024Q2 - 1) Model Effectiveness (2/3)

Metric	Figure	Explanation
ks_threshold	0.1200	This is the threshold that maximizes KS separation. Next: treat it as a candidate operating point, then compare it with business capacity and policy limits.
confusion_matrix[TN]	11136	True negatives correctly cleared. Next: preserve this base while tuning the alert threshold.
confusion_matrix[FP]	100	False positives create review cost without SAR conversion. Next: reduce this count with threshold tuning or sharper risk features.
confusion_matrix[FN]	758	False negatives are missed SAR cases. Next: if this count is unacceptable, lower the threshold or add features that capture suspicious behavior earlier.
confusion_matrix[TP]	212	True positives are correctly escalated cases. Next: assess whether this volume is sufficient relative to expected SAR capture goals.
precision_at_k[10]	0.8000	Shows how accurate the top-10 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[50]	0.8200	Shows how accurate the top-50 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[100]	0.8100	Shows how accurate the top-100 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.

value_date_quarter=2024Q2 - 1) Model Effectiveness (3/3)

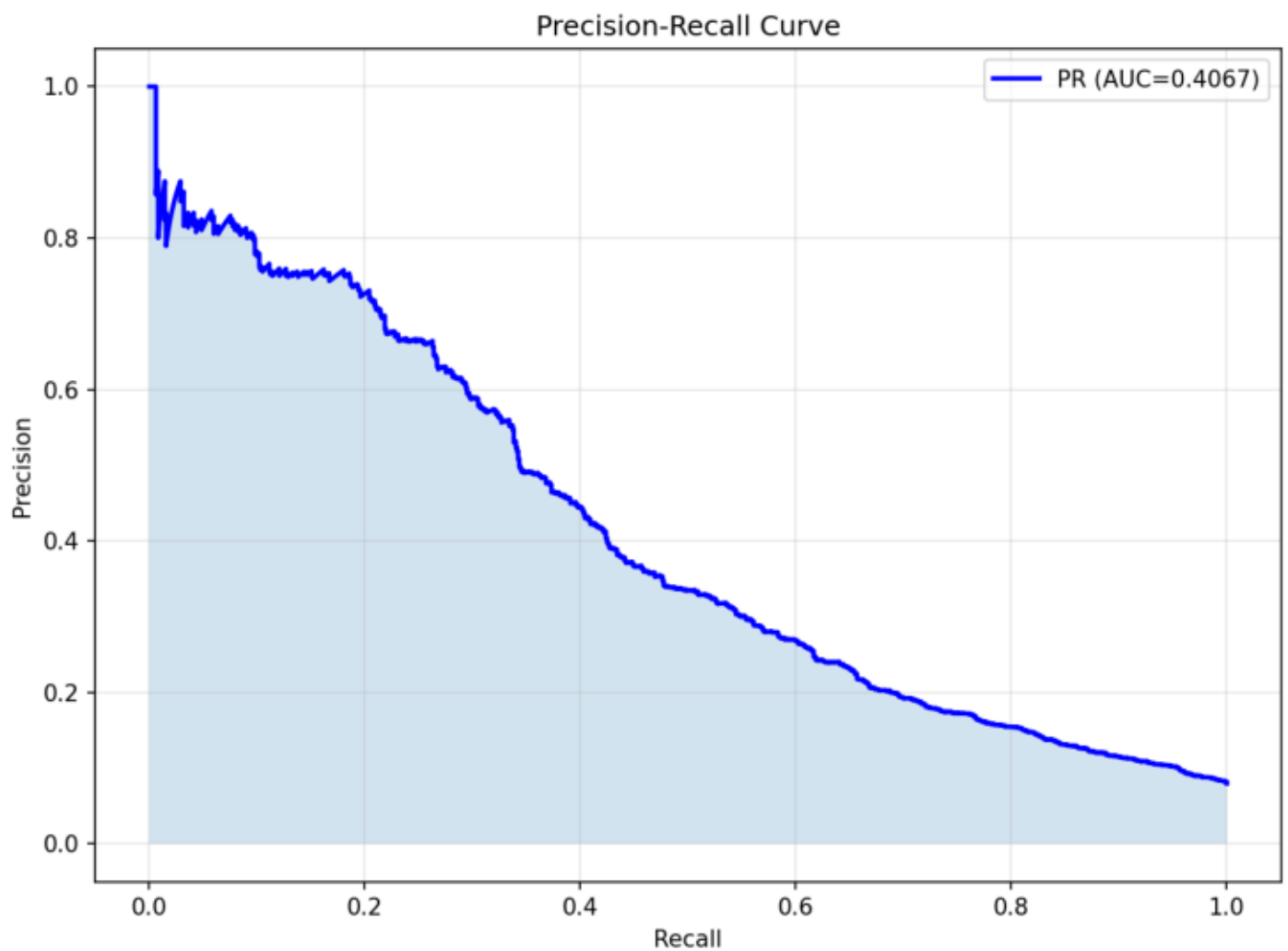
Metric	Figure	Explanation
precision_at_k[200]	0.7500	Shows how accurate the top-200 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[500]	0.5880	Shows how accurate the top-500 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
recall_at_k[10]	0.0082	Shows how much of the positive population is captured within the top-10 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[50]	0.0423	Shows how much of the positive population is captured within the top-50 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[100]	0.0835	Shows how much of the positive population is captured within the top-100 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[200]	0.1546	Shows how much of the positive population is captured within the top-200 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[500]	0.3031	Shows how much of the positive population is captured within the top-500 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.

value_date_quarter=2024Q2 - 1) Model Effectiveness



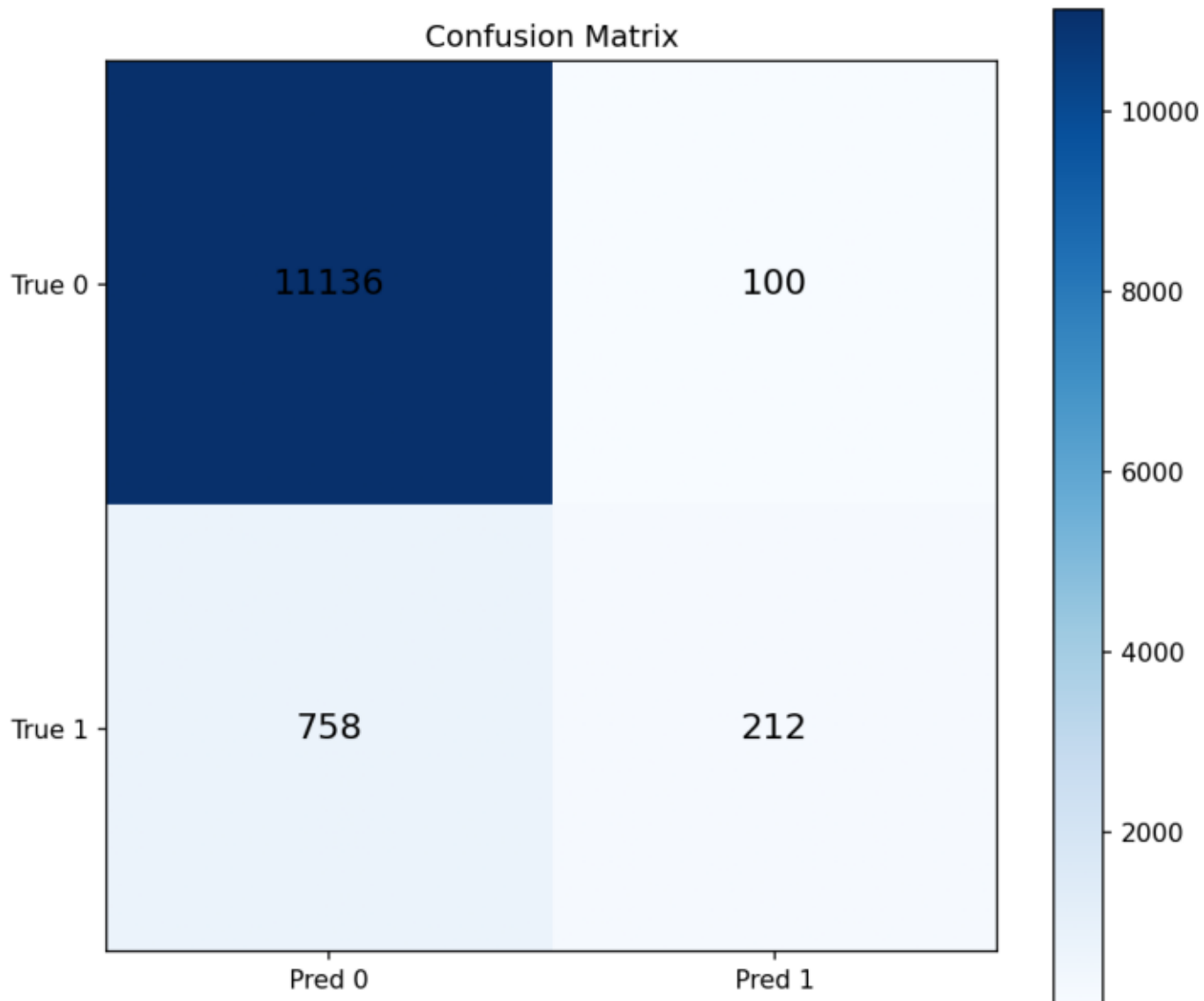
roc_curve: AUC-ROC=0.8025; curve above diagonal indicates strong separation.

value_date_quarter=2024Q2 - 1) Model Effectiveness



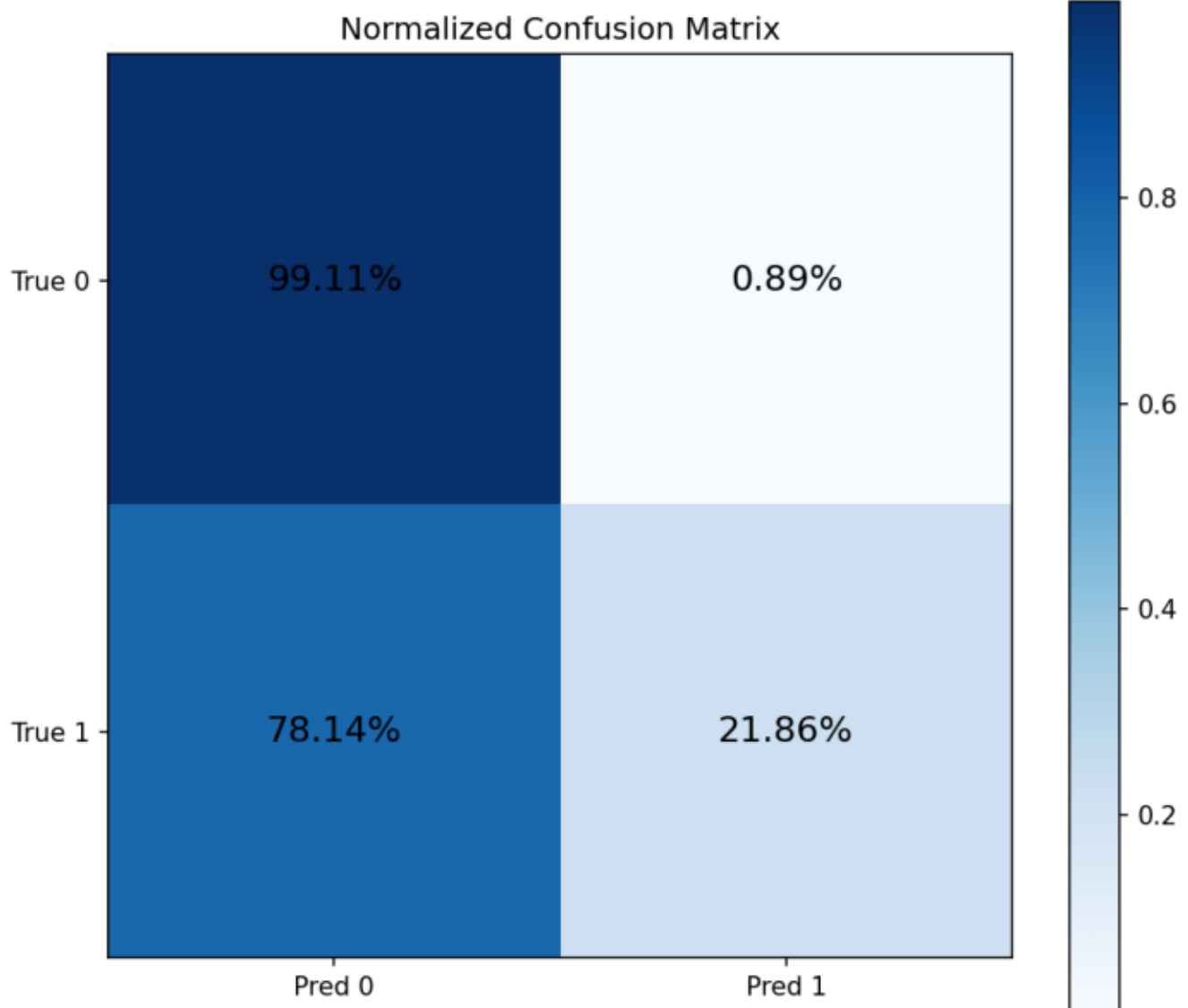
pr_curve: AUC-PR=0.4067 vs baseline 7.95%; higher curve implies better precision at recall.

value_date_quarter=2024Q2 - 1) Model Effectiveness



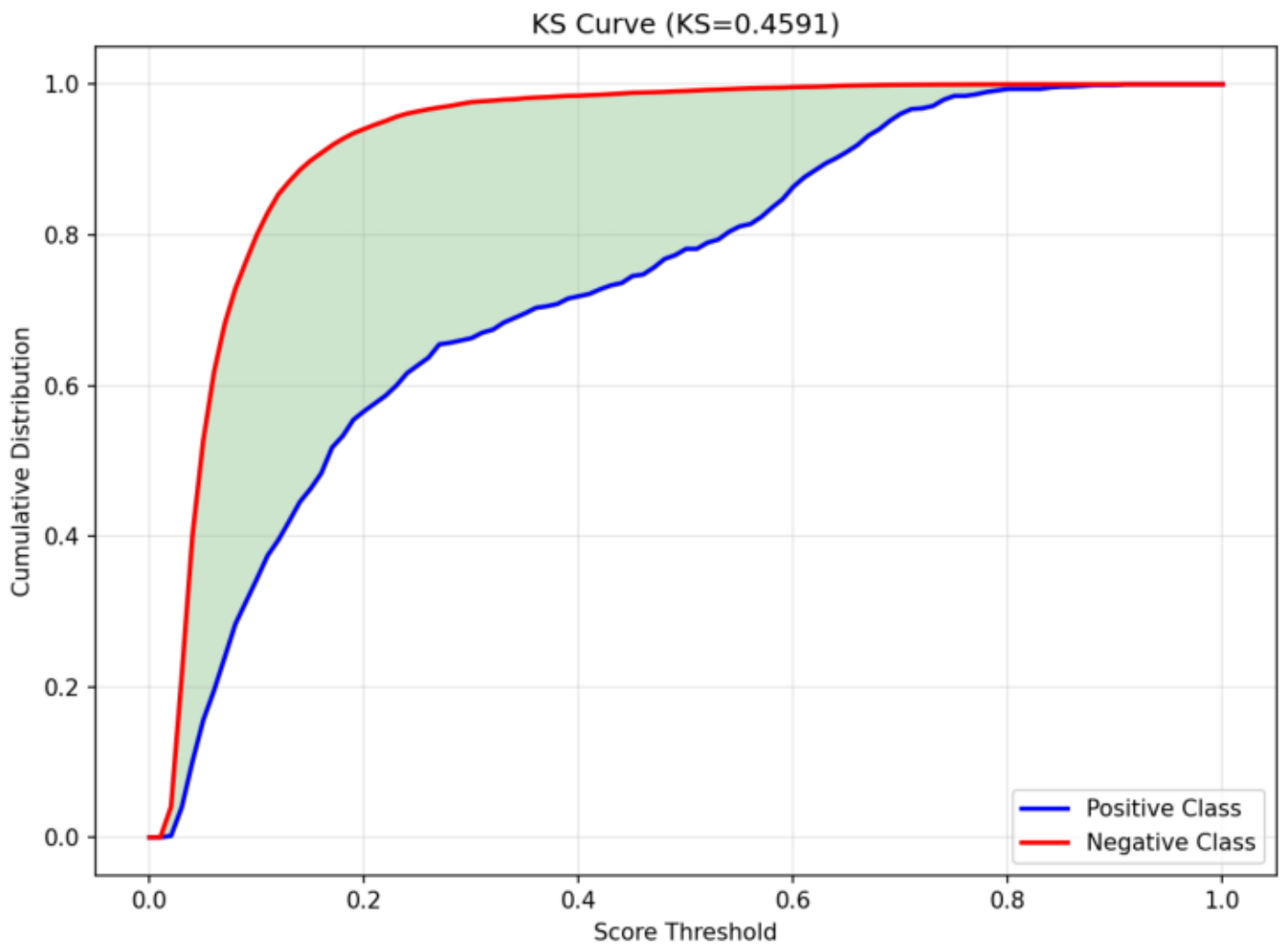
confusion_matrix: At threshold 0.50: TP=212, FP=100, FN=758, TN=11136.

value_date_quarter=2024Q2 - 1) Model Effectiveness



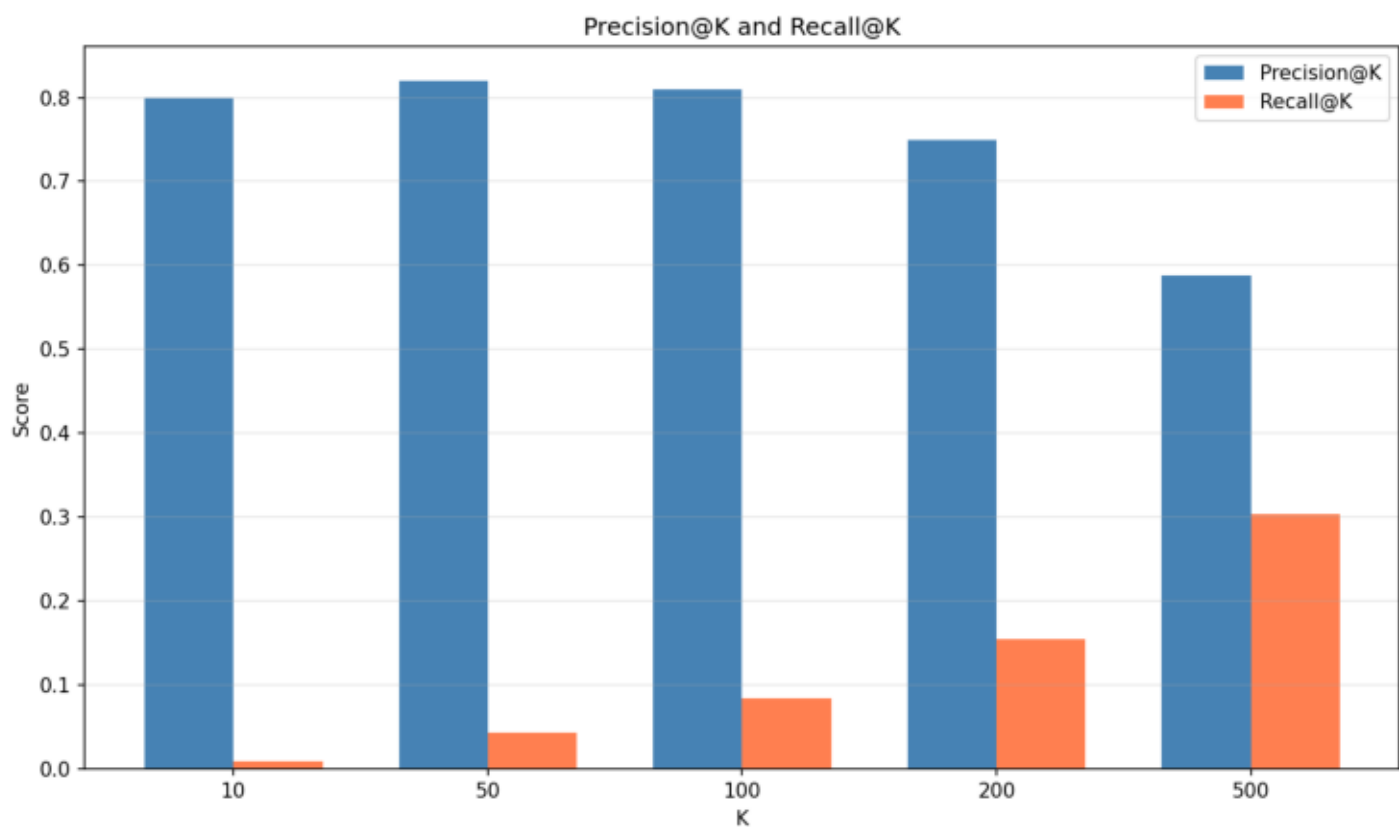
confusion_matrix_norm: Normalized rates: TPR=21.86%, TNR=99.11%.

value_date_quarter=2024Q2 - 1) Model Effectiveness



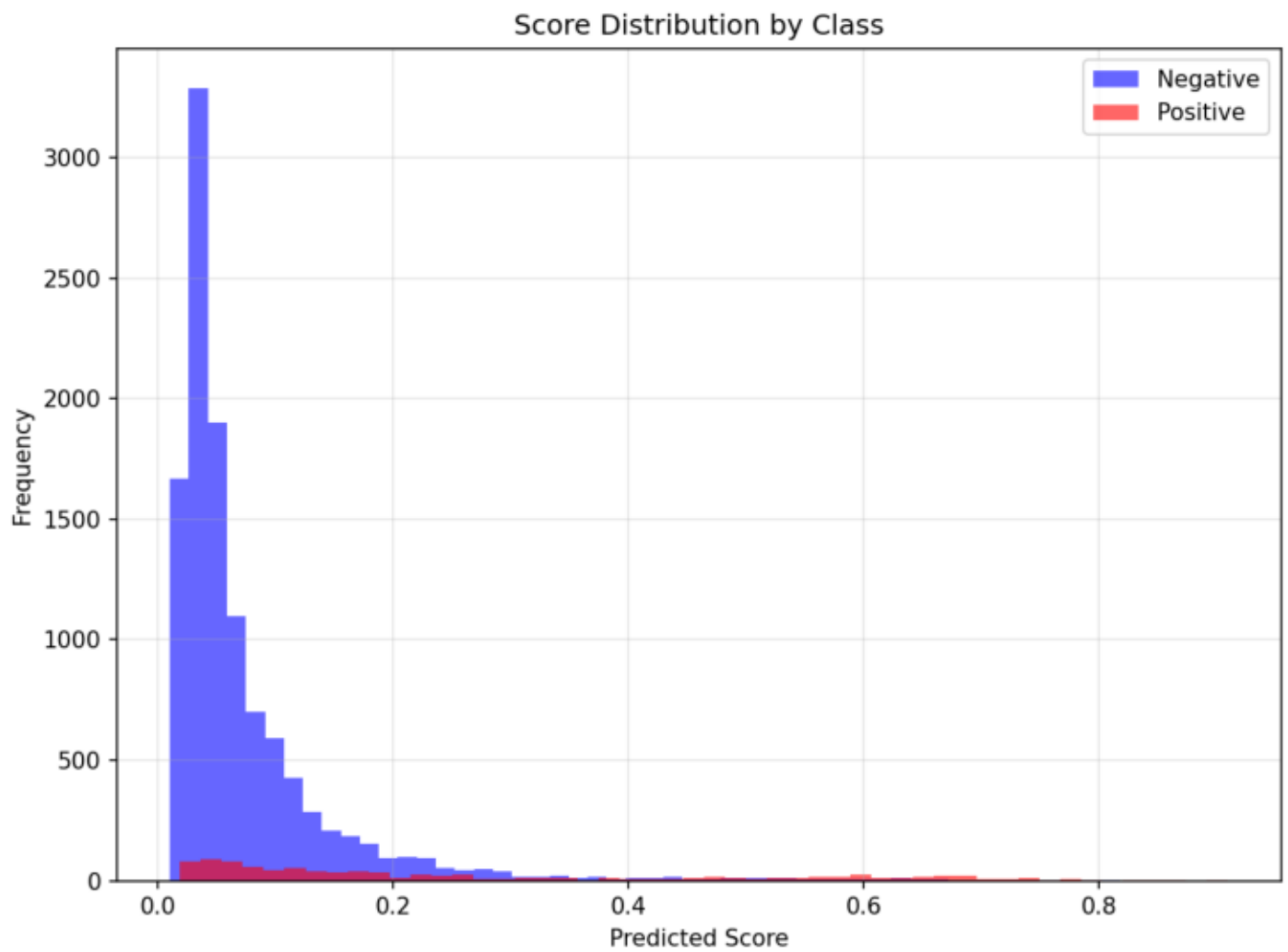
ks_curve: Maximum separation KS=0.4591 at threshold 0.12.

value_date_quarter=2024Q2 - 1) Model Effectiveness



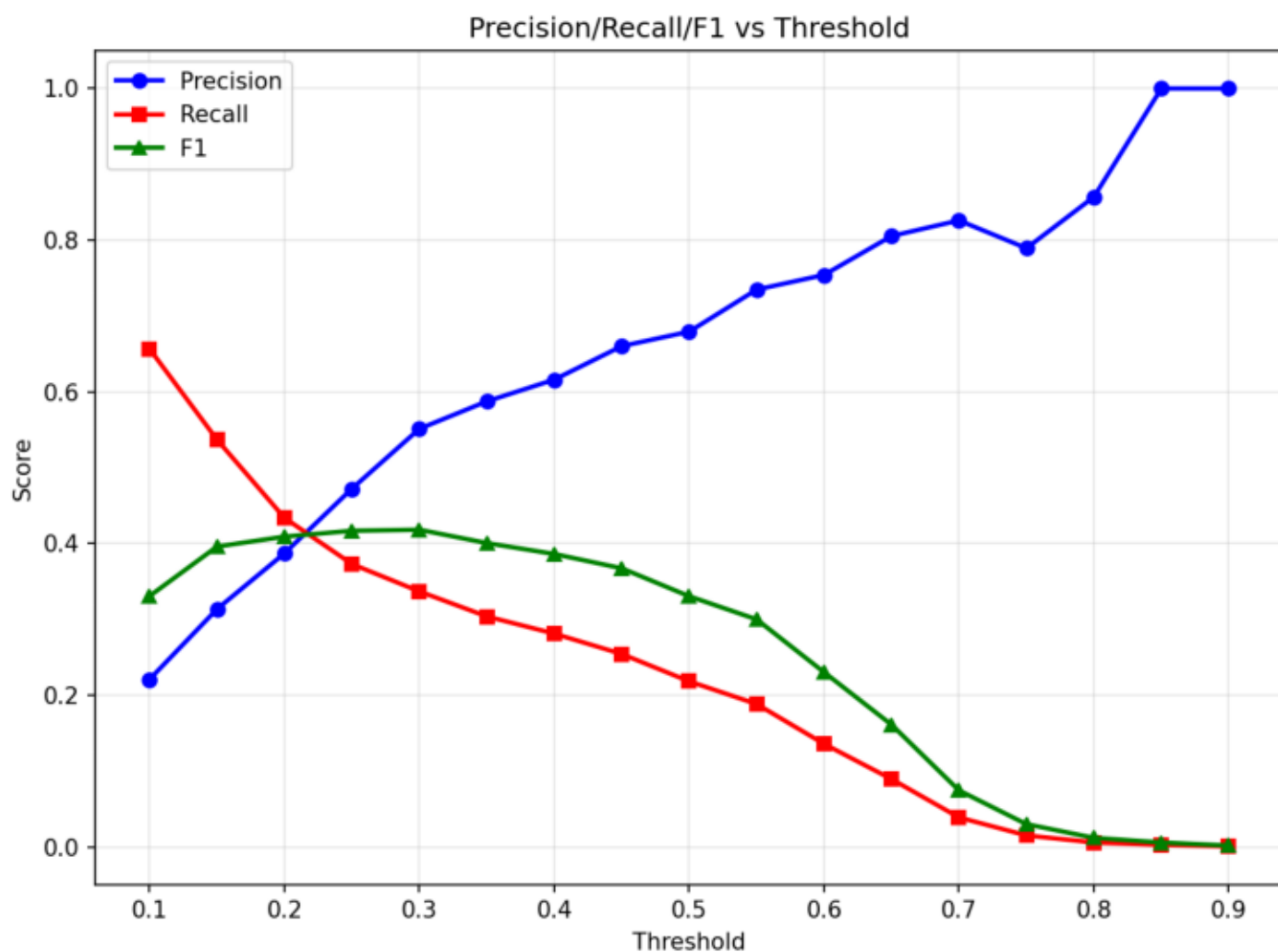
precision_recall_at_k: Bars show precision/recall as you expand the review queue.

value_date_quarter=2024Q2 - 1) Model Effectiveness



score_distribution: Mean score: positive=0.262, negative=0.075 (larger gap is better).

value_date_quarter=2024Q2 - 1) Model Effectiveness



threshold_analysis: Best F1 in tested grid is 0.4184 at threshold 0.30.

value_date_quarter=2024Q2 - 2) Model Efficiency (1/4)

Metric	Figure	Explanation
fpr	0.0089	Measures the share of negatives incorrectly flagged; 0.0089 is low. Next: if review demand is too high, raise the threshold or add features that better suppress false alerts.
tn	11136	Counts negatives correctly cleared without manual review. Next: preserve this clearance rate while checking that recall does not fall below policy expectations.
fp	100	Counts non-events that would still be sent to review. Next: use this with staffing capacity to decide whether threshold or feature refinement is required.
threshold	0.5000	This is the active operating cutoff used to flag alerts. Next: tune it jointly with recall, precision, and review capacity rather than in isolation.
fpr_at_thresholds[thresholds.t_0.05]	0.4726	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.05. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.10]	0.1995	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.10. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q2 - 2) Model Efficiency (2/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.15]	0.1014	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.15. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.20]	0.0593	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.20. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.25]	0.0360	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.25. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.30]	0.0237	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.30. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.35]	0.0184	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.35. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.40]	0.0151	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.40. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

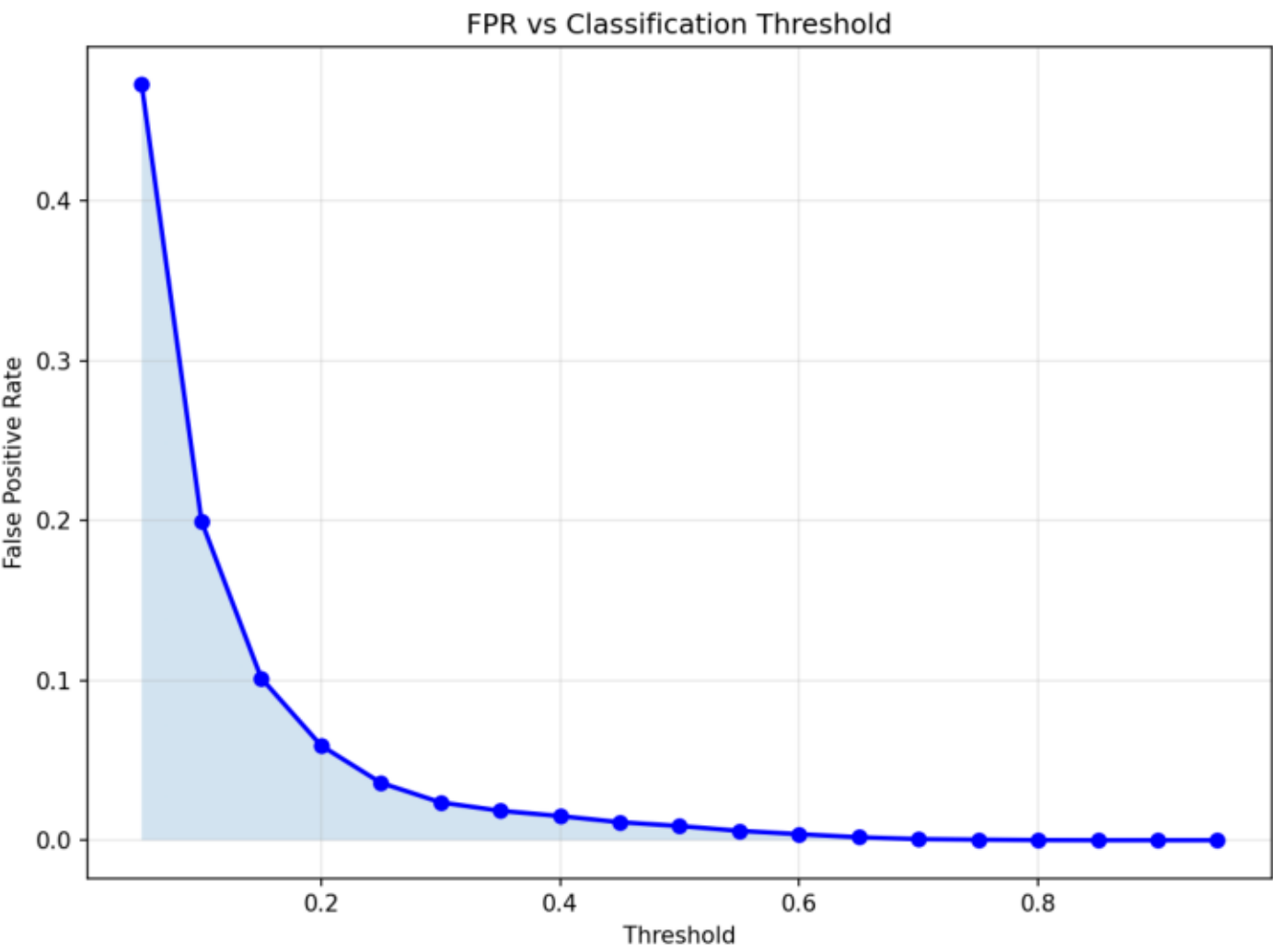
value_date_quarter=2024Q2 - 2) Model Efficiency (3/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.45]	0.0113	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.45. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.50]	0.0089	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.50. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.55]	0.0059	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.55. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.60]	0.0038	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.60. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.65]	0.0019	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.65. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.70]	0.0007	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.70. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q2 - 2) Model Efficiency (4/4)

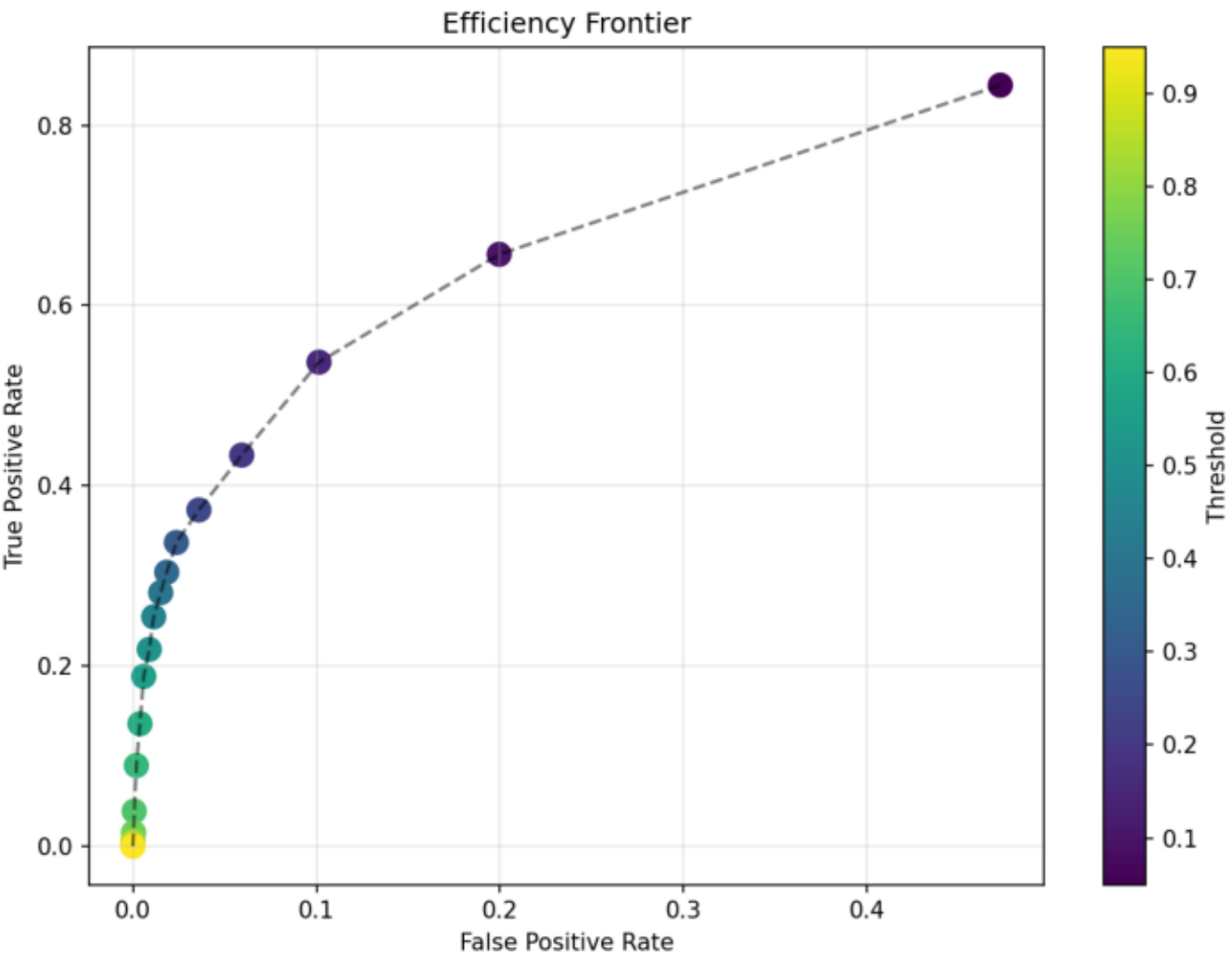
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.75]	0.0004	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.75. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.80]	0.0001	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.80. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.85]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.85. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.90]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.90. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.95]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.95. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q2 - 2) Model Efficiency



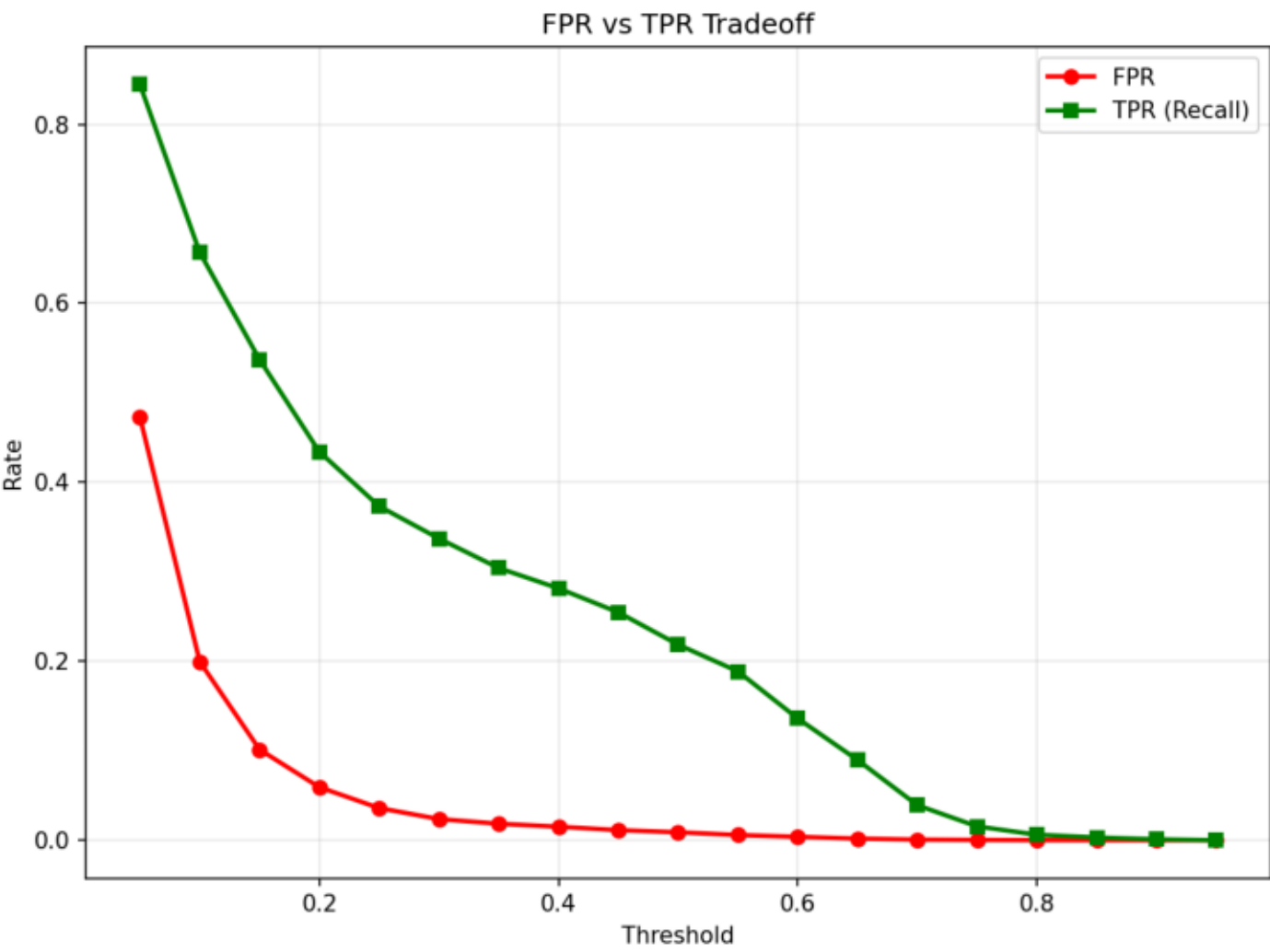
fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.

value_date_quarter=2024Q2 - 2) Model Efficiency



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

value_date_quarter=2024Q2 - 2) Model Efficiency



fpr_tpr_tradeoff: FPR and TPR curves highlight how recall drops as you reduce false positives.

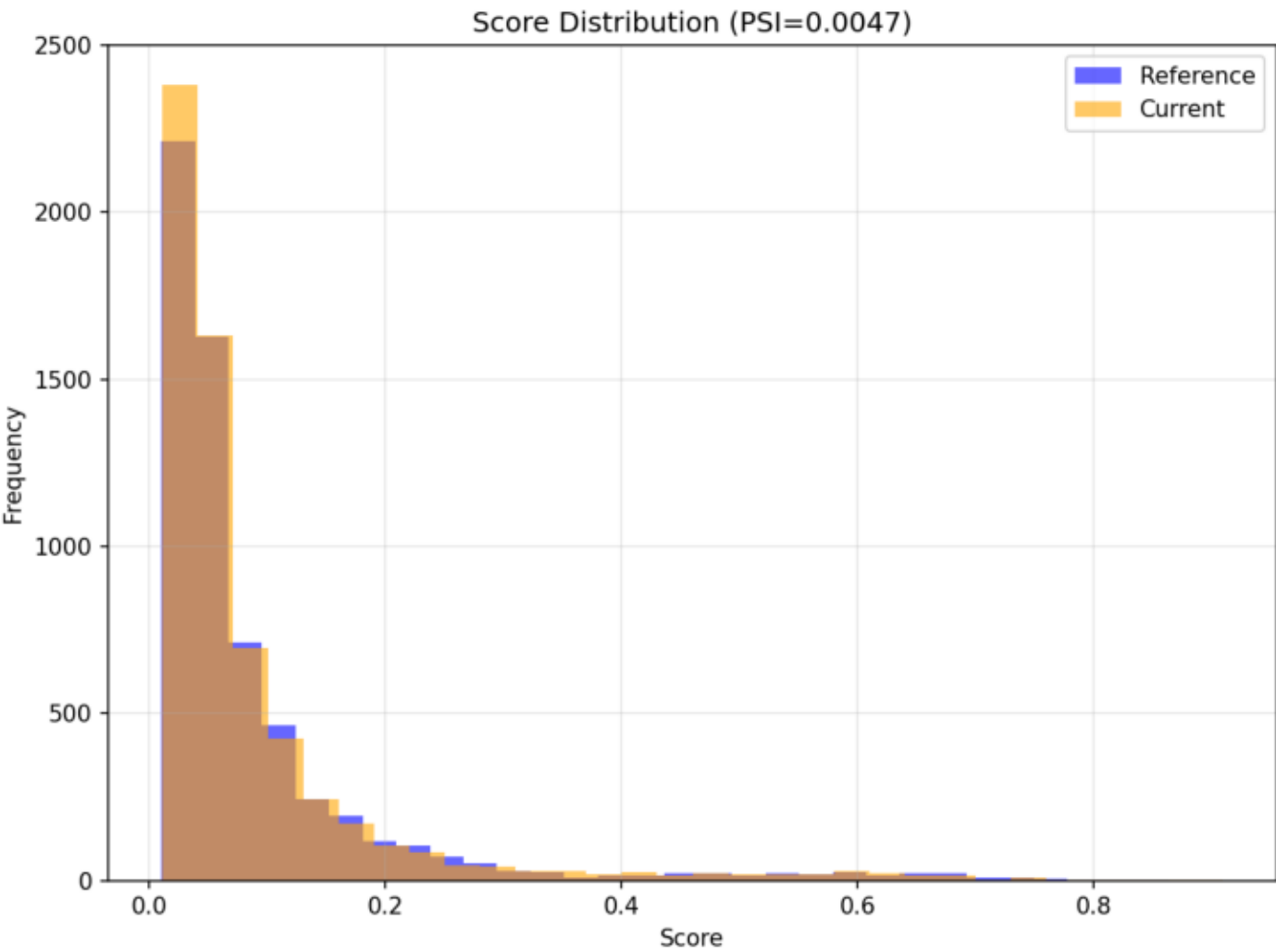
value_date_quarter=2024Q2 - 3) Model Stability (1/2)

Metric	Figure	Explanation
psi	0.0047	Measures score-distribution shift between reference and current populations; 0.0047 is negligible. Next: investigate segment mix, data drift, and recalibration needs when PSI is material.
cv_auc_roc_mean	0.8025	Average AUC-ROC across validation folds. Next: compare it with overall AUC-ROC; a large gap can indicate overfitting or unstable sampling.
cv_auc_roc_std	0.0174	Standard deviation of AUC-ROC across folds. Next: if variability is high, simplify the model or increase training stability before deployment.
cv_auc_pr_mean	0.4091	Average AUC-PR across validation folds. Next: confirm that precision-recall performance remains acceptable under resampling.
cv_auc_pr_std	0.0321	Standard deviation of AUC-PR across folds. Next: high dispersion suggests unstable alert quality and warrants deeper validation.
bootstrap_auc_roc_mean	0.7998	Bootstrap mean AUC-ROC estimates expected ranking performance under repeated sampling. Next: compare it with the point estimate to judge optimism in the current test run.
bootstrap_auc_roc_ci_lower	0.7788	Lower confidence bound for bootstrap AUC-ROC. Next: use it as a conservative performance floor in model-risk discussions.
bootstrap_auc_roc_ci_upper	0.8200	Upper confidence bound for bootstrap AUC-ROC. Next: review the interval width with the lower bound to assess uncertainty.

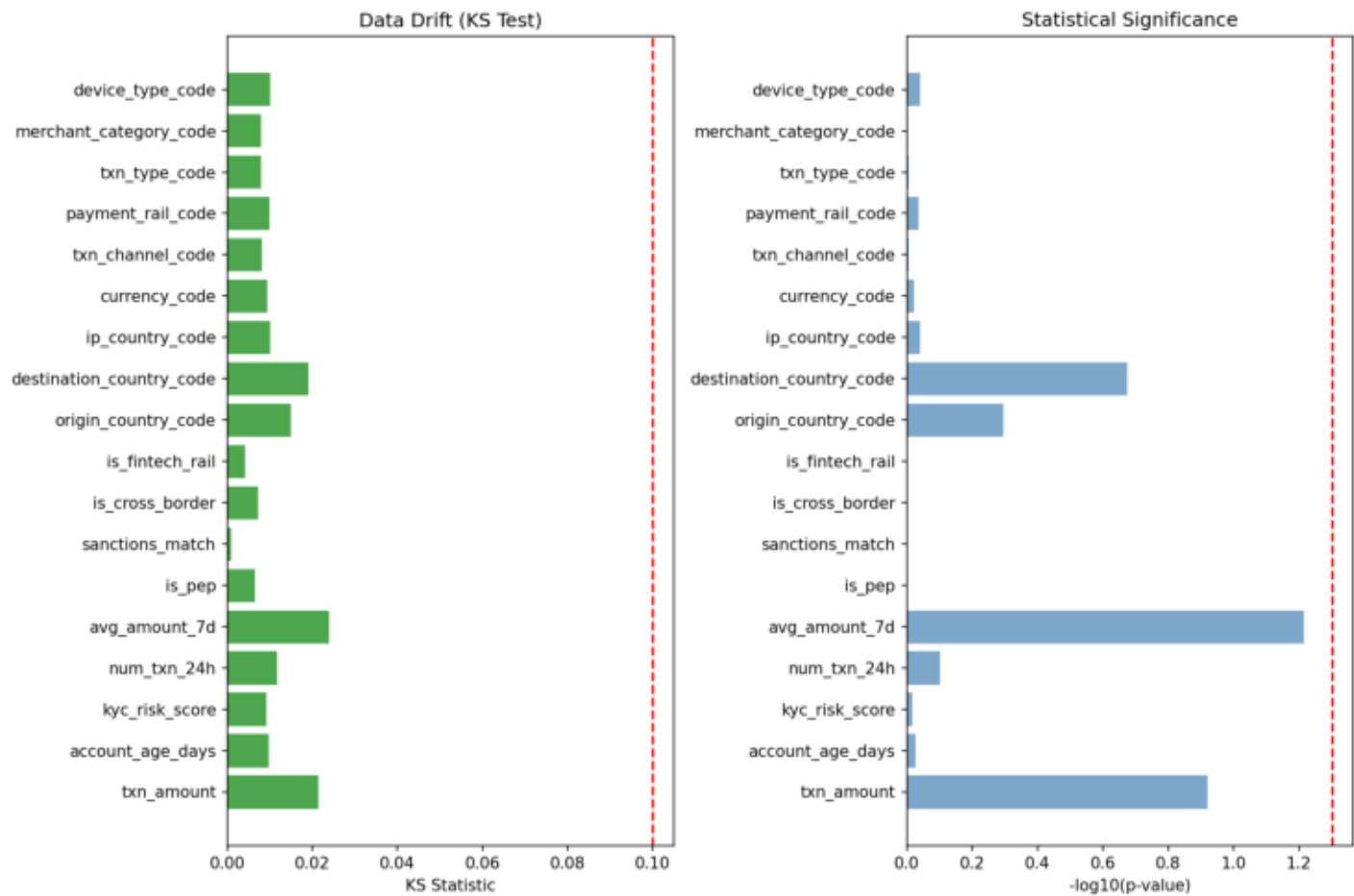
value_date_quarter=2024Q2 - 3) Model Stability (2/2)

Metric	Figure	Explanation
concept_drift_detected	False	No material concept-drift signal was detected in the evaluated slices. Next: retain ongoing monitoring because stable historical results do not guarantee future stability.
concept_drift_score	0.0177	Quantifies the magnitude of performance change across ordered slices. Next: rising scores should trigger deeper review of segment behavior and model refresh timing.

value_date_quarter=2024Q2 - 3) Model Stability

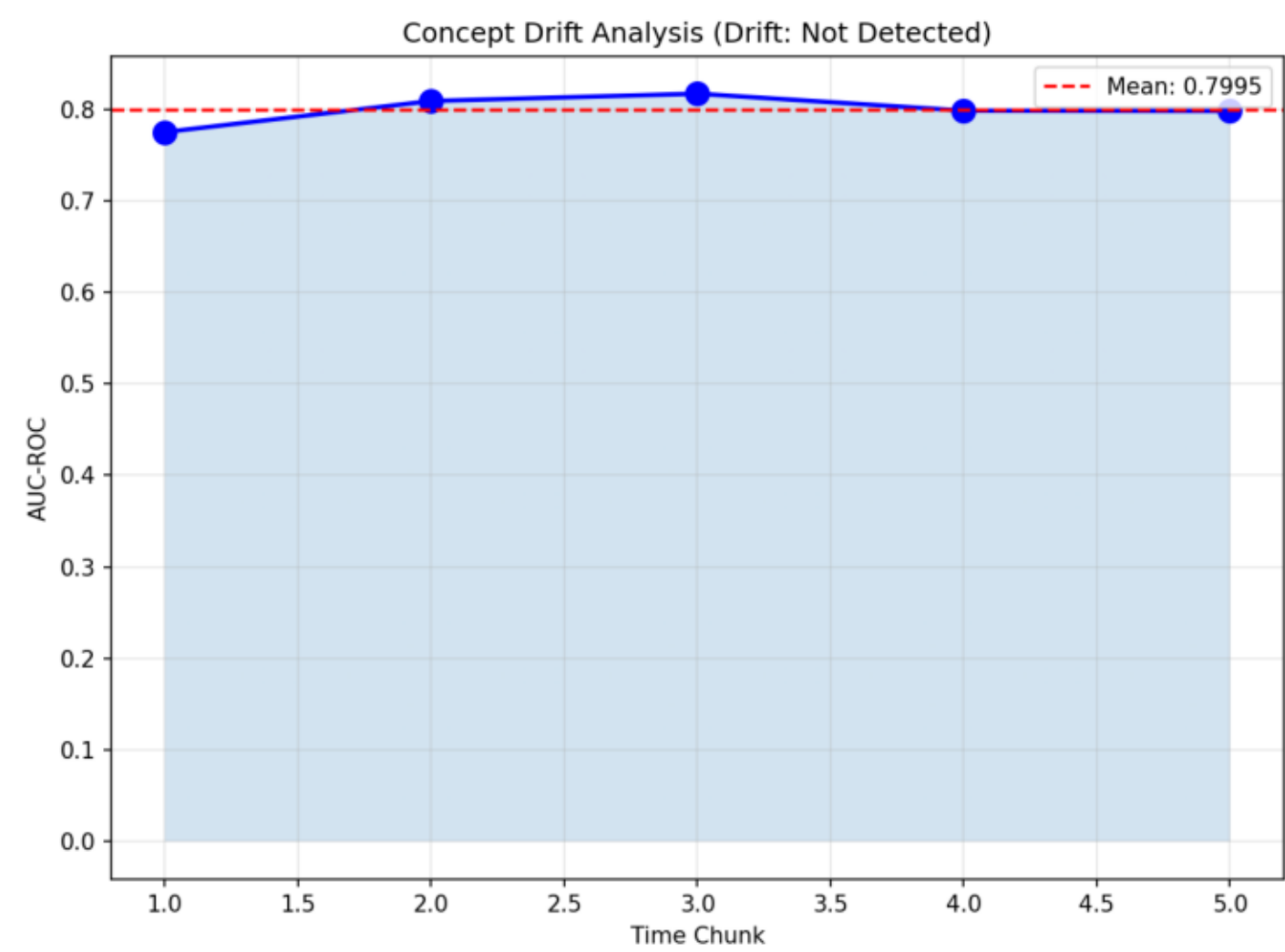


value_date_quarter=2024Q2 - 3) Model Stability



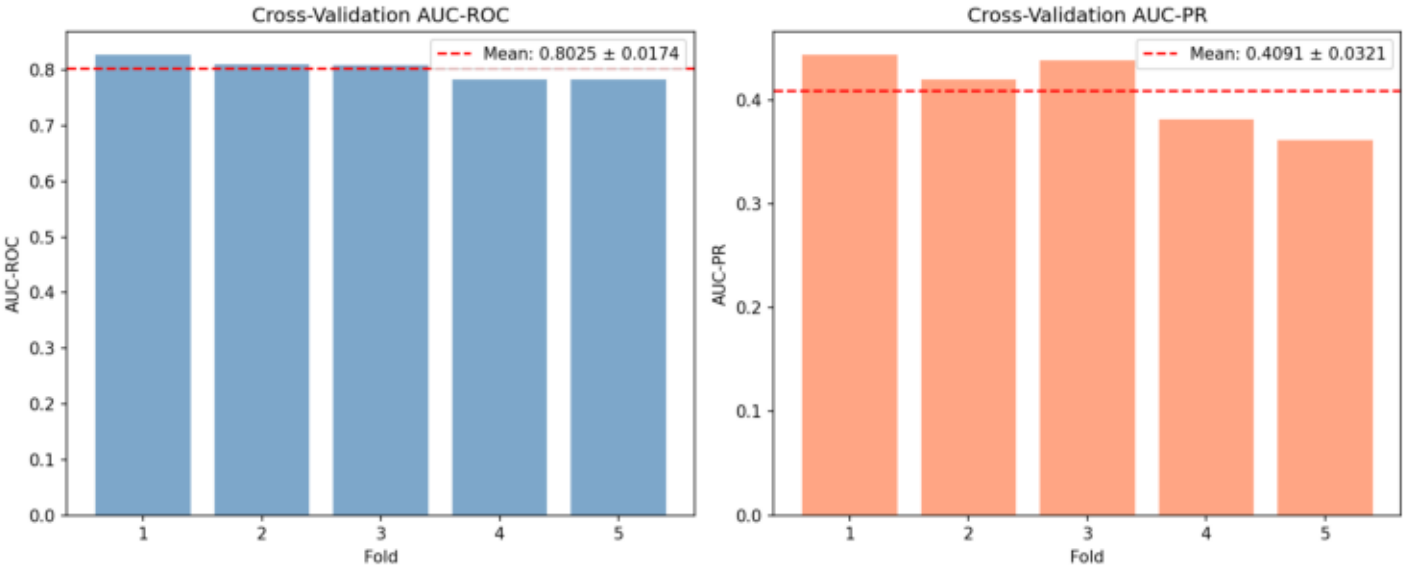
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.

value_date_quarter=2024Q2 - 3) Model Stability



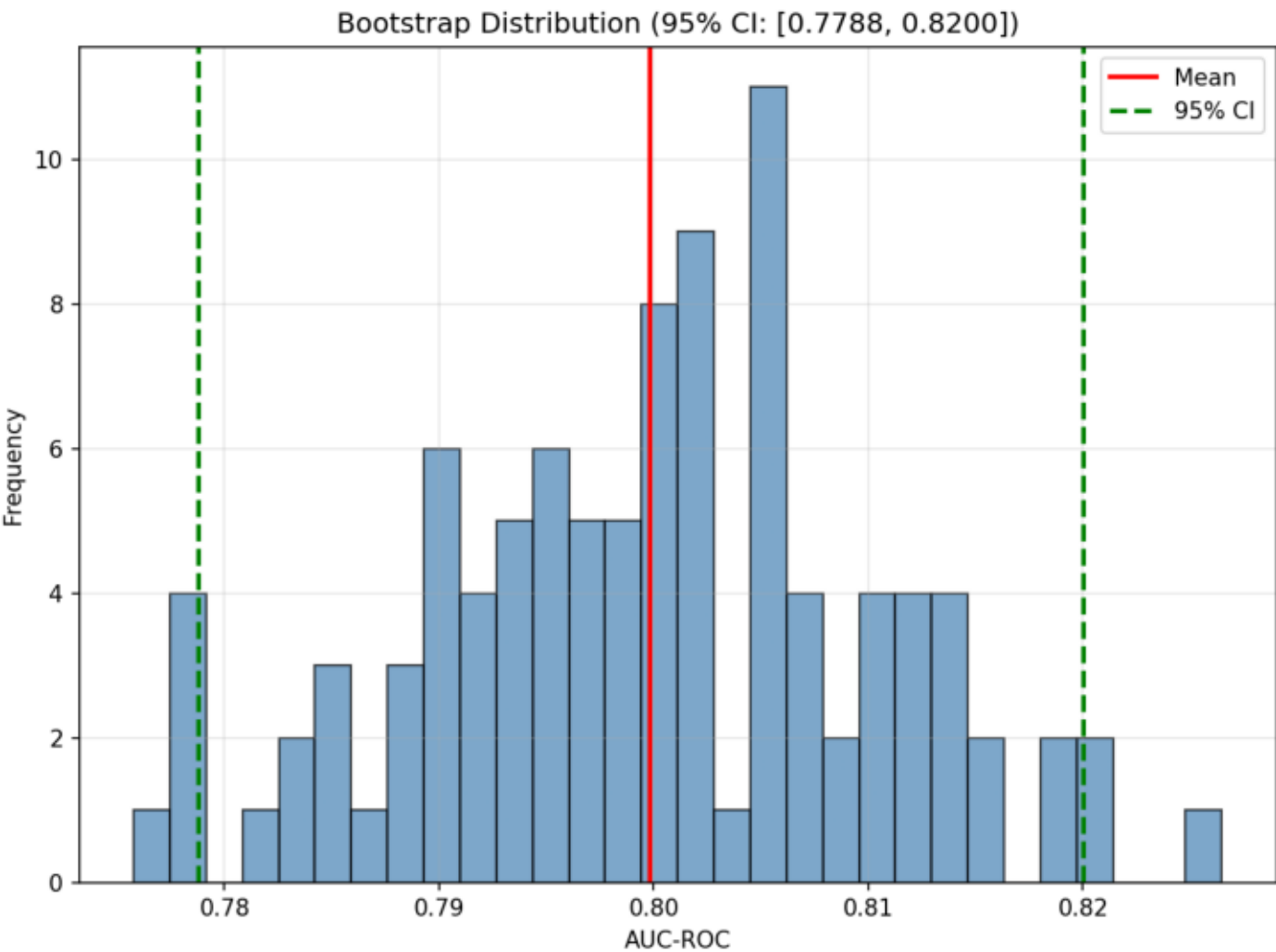
concept_drift: AUC across time chunks shows whether performance is stable over time.

value_date_quarter=2024Q2 - 3) Model Stability

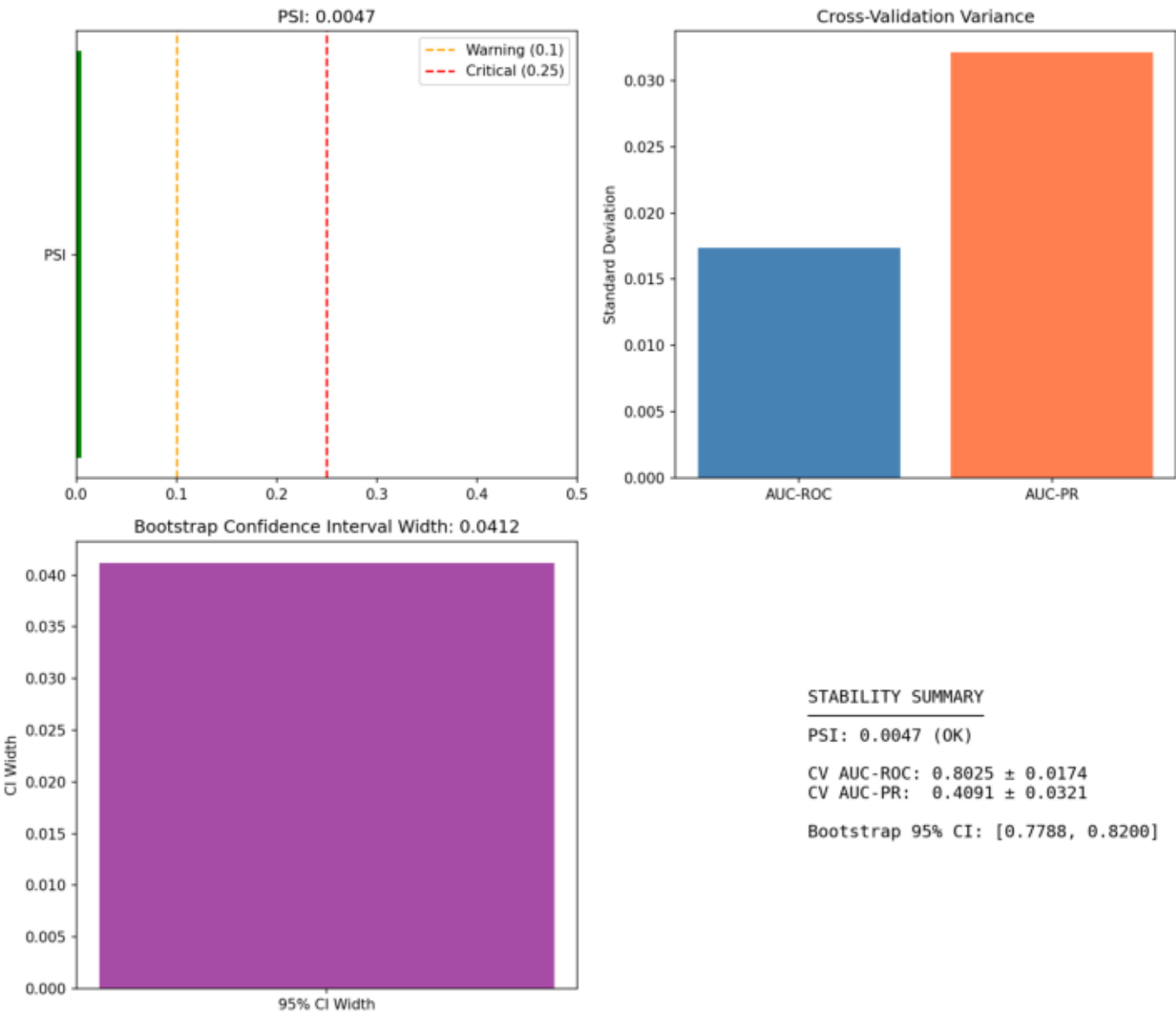


cv_results: Fold scores cluster around ROC 0.8025 and PR 0.4091.

value_date_quarter=2024Q2 - 3) Model Stability



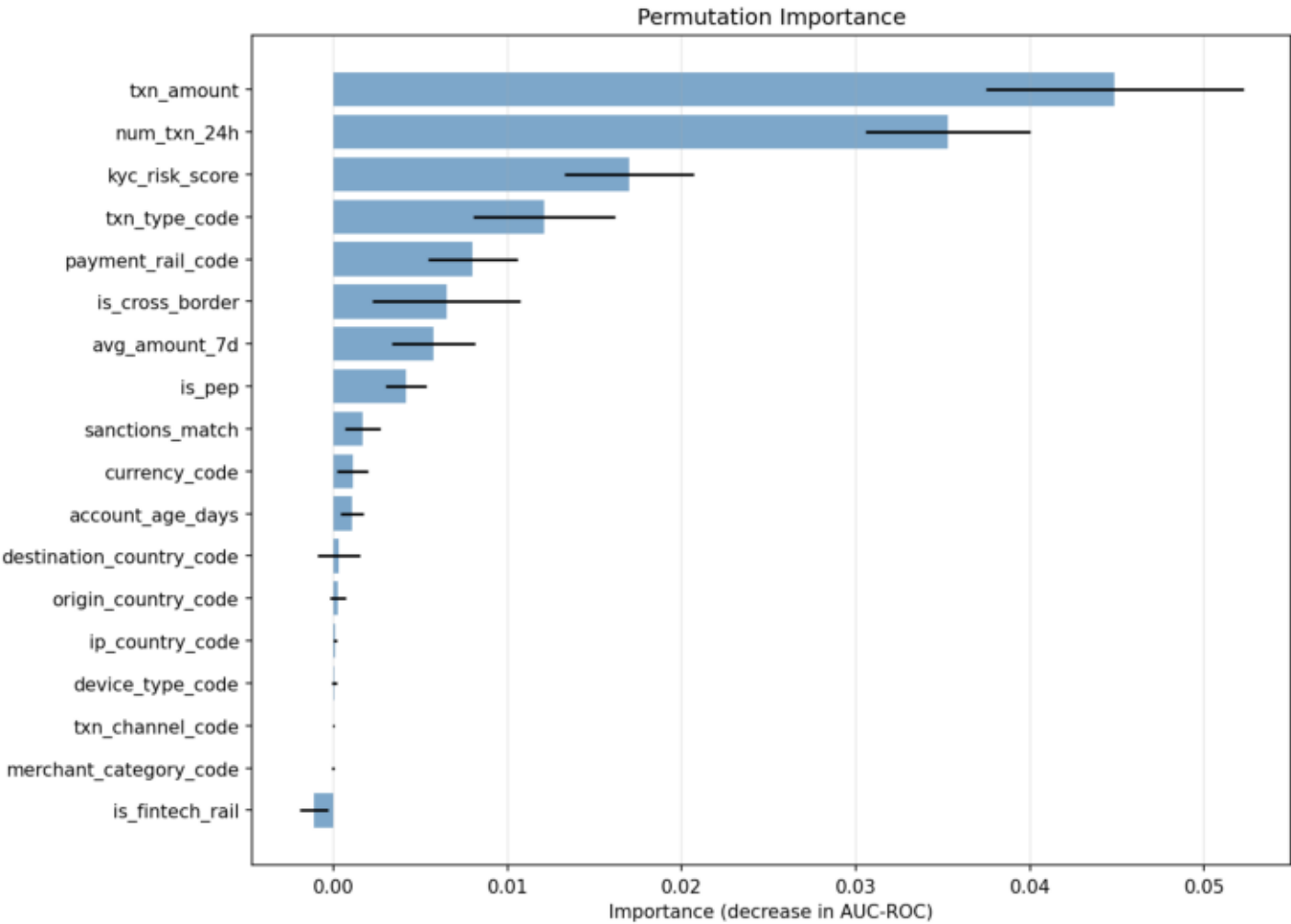
value_date_quarter=2024Q2 - 3) Model Stability



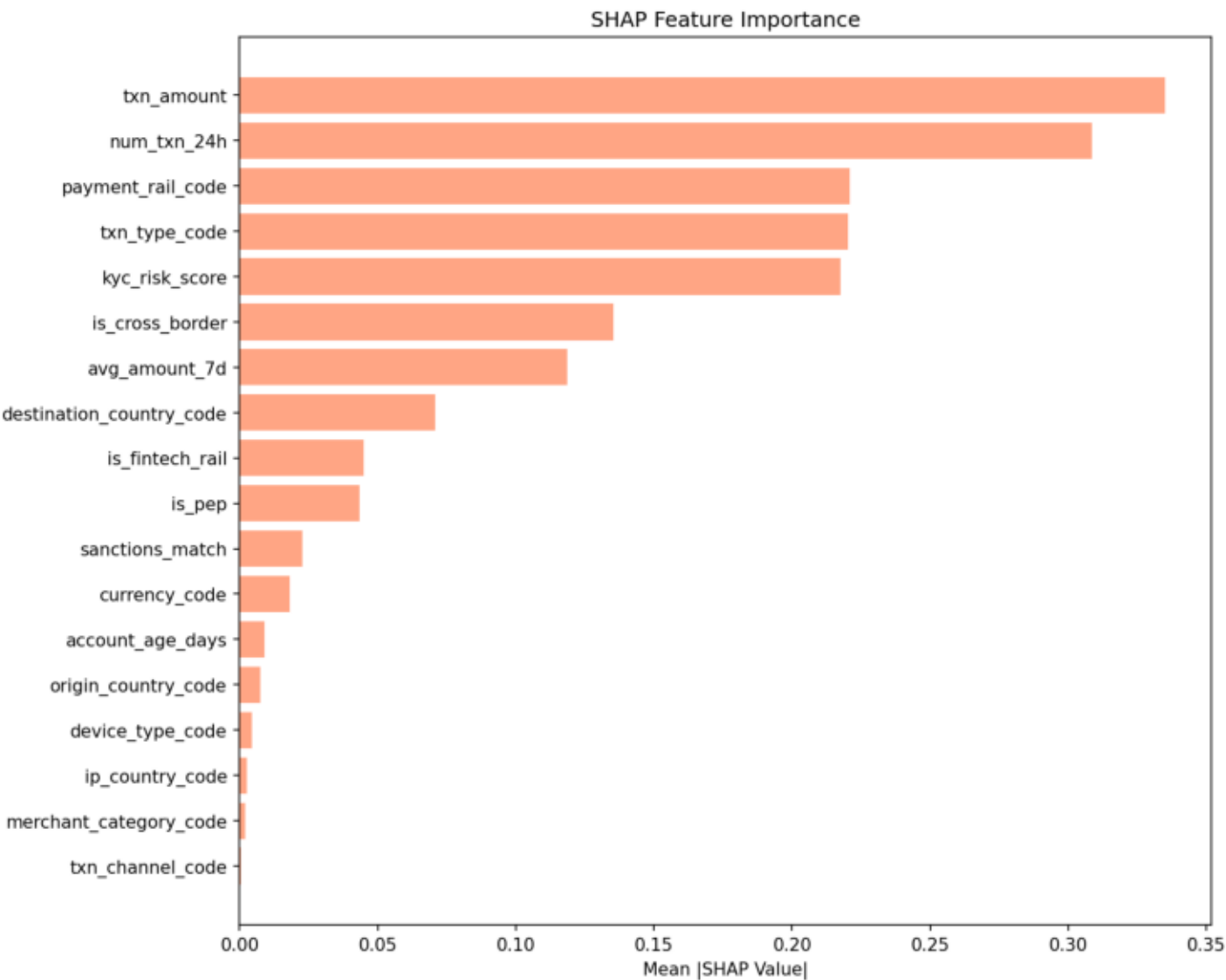
Metric	Figure	Explanation
model_type	tree	Identifies the estimator family so explanation methods can be interpreted in the right context. Next: confirm the chosen explanation methods are appropriate for this model class.
methods_used	permutation, pdp, ice, shap, lime	Lists the interpretability methods produced in this run. Next: compare signals across methods before using them in documentation or model challenge.
perm_top_features	txn_amount, num_txn_24h, kyc_risk_score, txn_type_code, payment_rail_code ...	Shows features with the largest performance drop when permuted, which highlights global importance. Next: validate that the top drivers align with business logic, feature governance, and policy expectations.
shap_top_features	txn_amount, num_txn_24h, payment_rail_code, txn_type_code, kyc_risk_score ...	Shows the most influential features under SHAP-based contribution analysis. Next: inspect whether directionality and ranking remain sensible across key segments.
lime_instances	3	Counts the locally explained observations reviewed with LIME. Next: examine representative true-positive and false-positive cases to understand local decision logic.

Metric	Figure	Explanation
pdp_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score, num_txn_24h ...	Lists features with partial-dependence plots. Next: review whether average model response is monotonic, stable, and operationally sensible.
ice_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score	Lists features with individual conditional expectation plots. Next: inspect heterogeneity and interactions that may be hidden in average-only views.

value_date_quarter=2024Q2 - 4) Model Interpretability

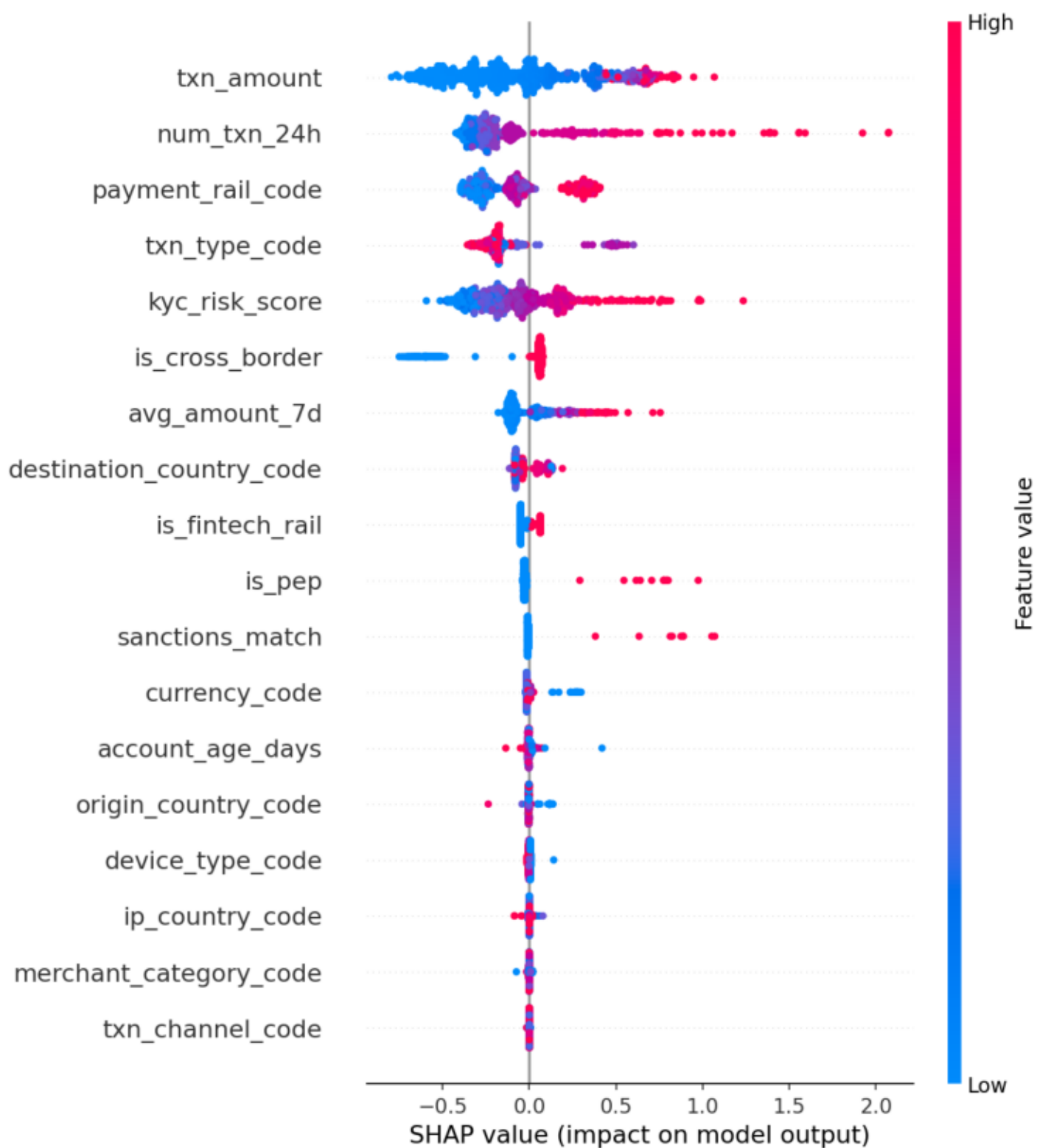


permutation_importance: Bars show how much AUC drops when each feature is permuted.

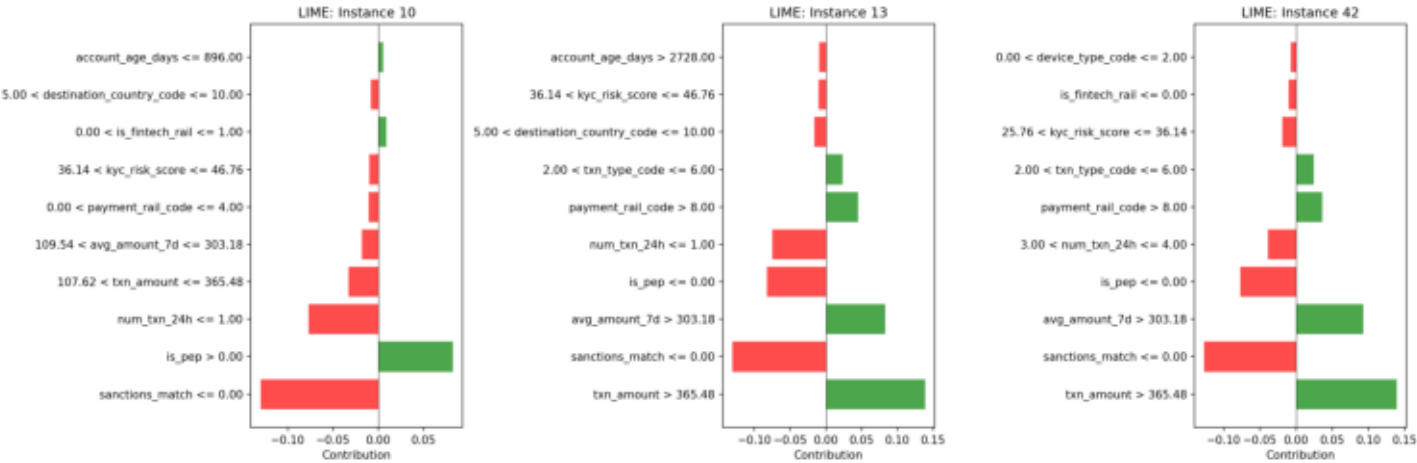


shap_bar: SHAP bar plot ranks global feature impact by mean absolute SHAP values.

value_date_quarter=2024Q2 - 4) Model Interpretability



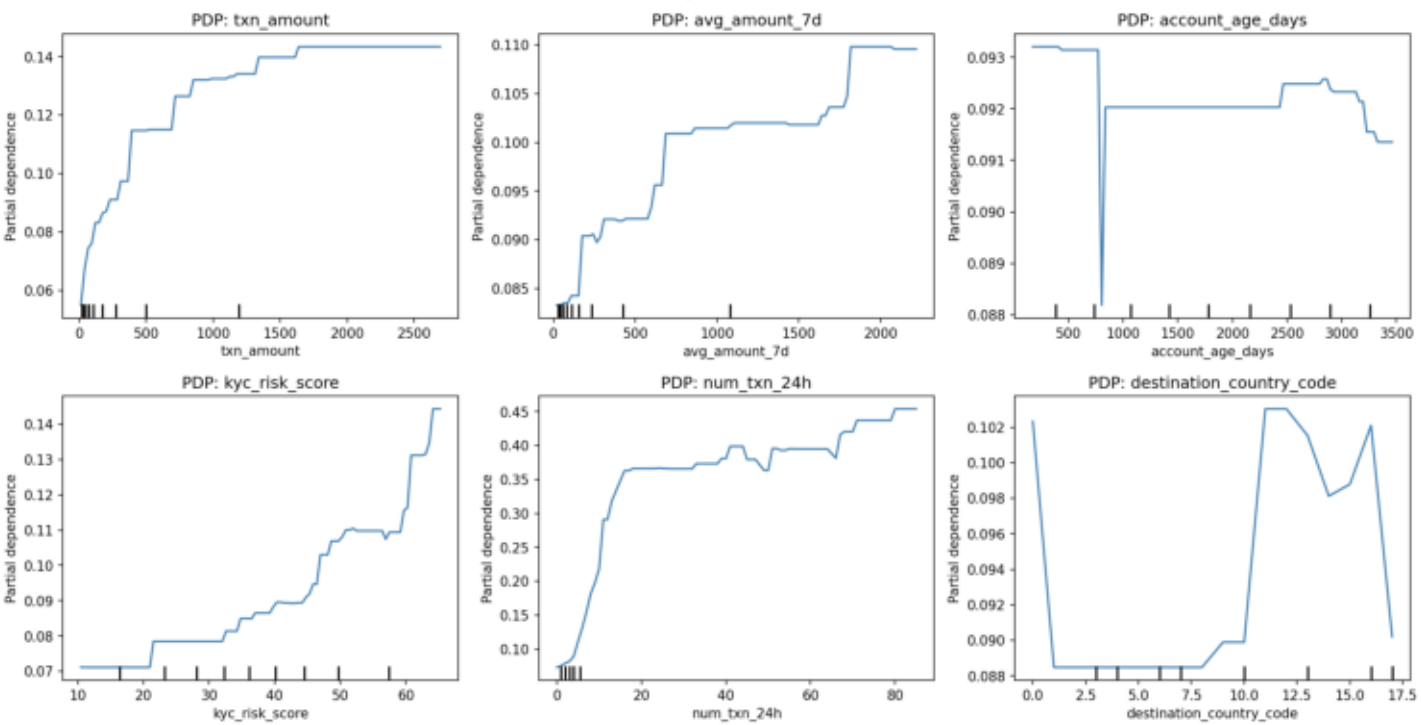
value_date_quarter=2024Q2 - 4) Model Interpretability



lime_explanation: LIME bars show local feature contributions for selected instances.

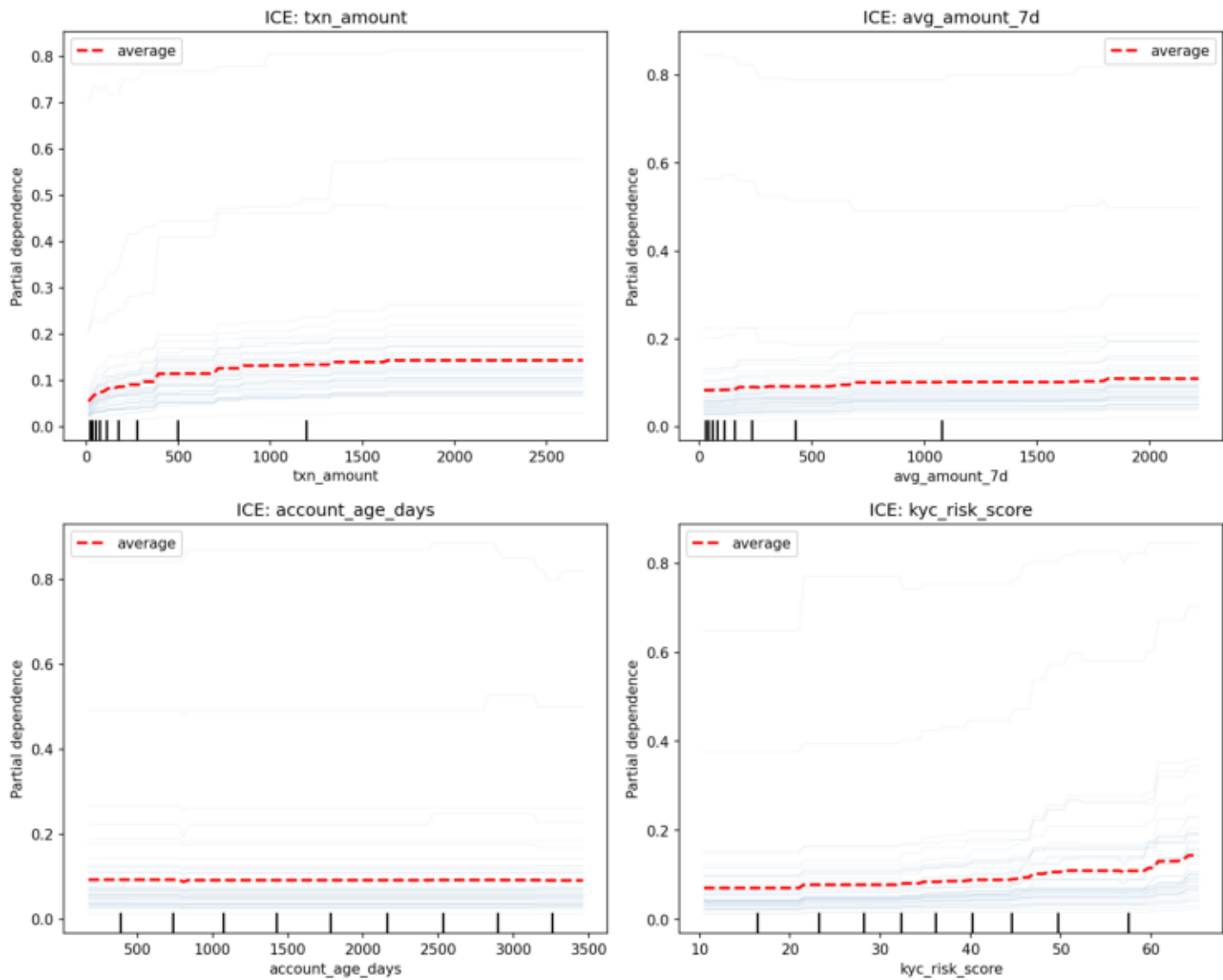
value_date_quarter=2024Q2 - 4) Model Interpretability

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q3 - 1) Model Effectiveness (1/3)

Metric	Figure	Explanation
auc_roc	0.8128	Measures ranking power across all thresholds. 0.8128 indicates strong separation. Next: compare AUC-ROC across segments and keep threshold tuning separate from ranking validation.
auc_pr	0.4926	Measures precision-recall performance under class imbalance; the observed event rate is 17.67%. Next: use this together with alert capacity to decide whether the high-score queue is practically useful.
precision	0.5826	Shows the share of flagged cases that are true positives at the current threshold. Next: if this is below review expectations, raise the threshold or improve exclusion features.
recall	0.2717	Shows the share of true positives captured at the current threshold. Next: if missed cases are too high, lower the threshold or add stronger risk features.
f1	0.3706	Balances precision and recall in one score. Next: use it as a threshold-selection aid when both missed cases and false alerts matter.
ks_statistic	0.4777	Measures score separation between positive and negative classes. Next: low KS suggests the model needs better features or a different specification.

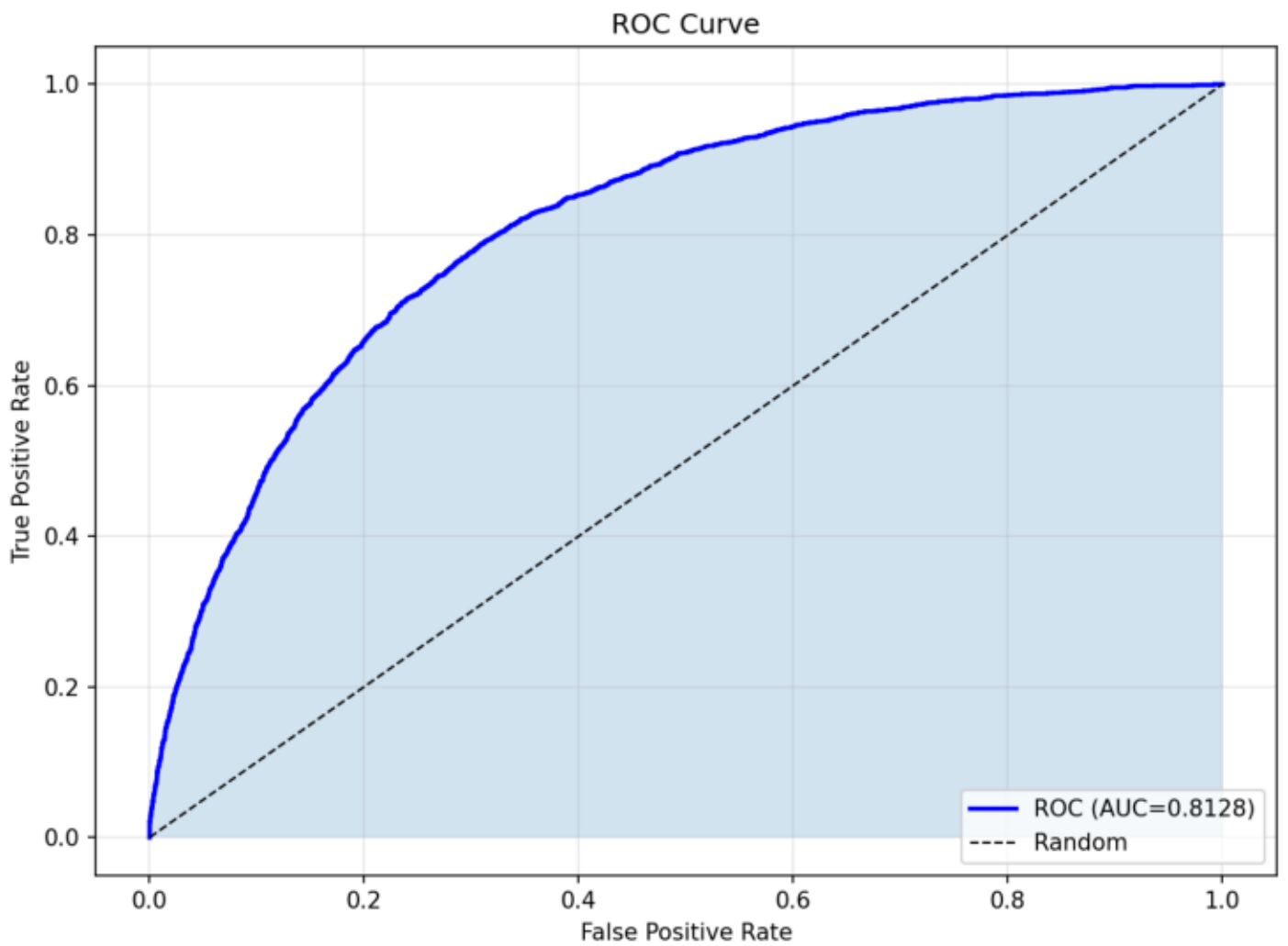
value_date_quarter=2024Q3 - 1) Model Effectiveness (2/3)

Metric	Figure	Explanation
ks_threshold	0.1400	This is the threshold that maximizes KS separation. Next: treat it as a candidate operating point, then compare it with business capacity and policy limits.
confusion_matrix[TN]	10316	True negatives correctly cleared. Next: preserve this base while tuning the alert threshold.
confusion_matrix[FP]	450	False positives create review cost without SAR conversion. Next: reduce this count with threshold tuning or sharper risk features.
confusion_matrix[FN]	1683	False negatives are missed SAR cases. Next: if this count is unacceptable, lower the threshold or add features that capture suspicious behavior earlier.
confusion_matrix[TP]	628	True positives are correctly escalated cases. Next: assess whether this volume is sufficient relative to expected SAR capture goals.
precision_at_k[10]	1.0000	Shows how accurate the top-10 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[50]	0.9600	Shows how accurate the top-50 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[100]	0.8300	Shows how accurate the top-100 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.

value_date_quarter=2024Q3 - 1) Model Effectiveness (3/3)

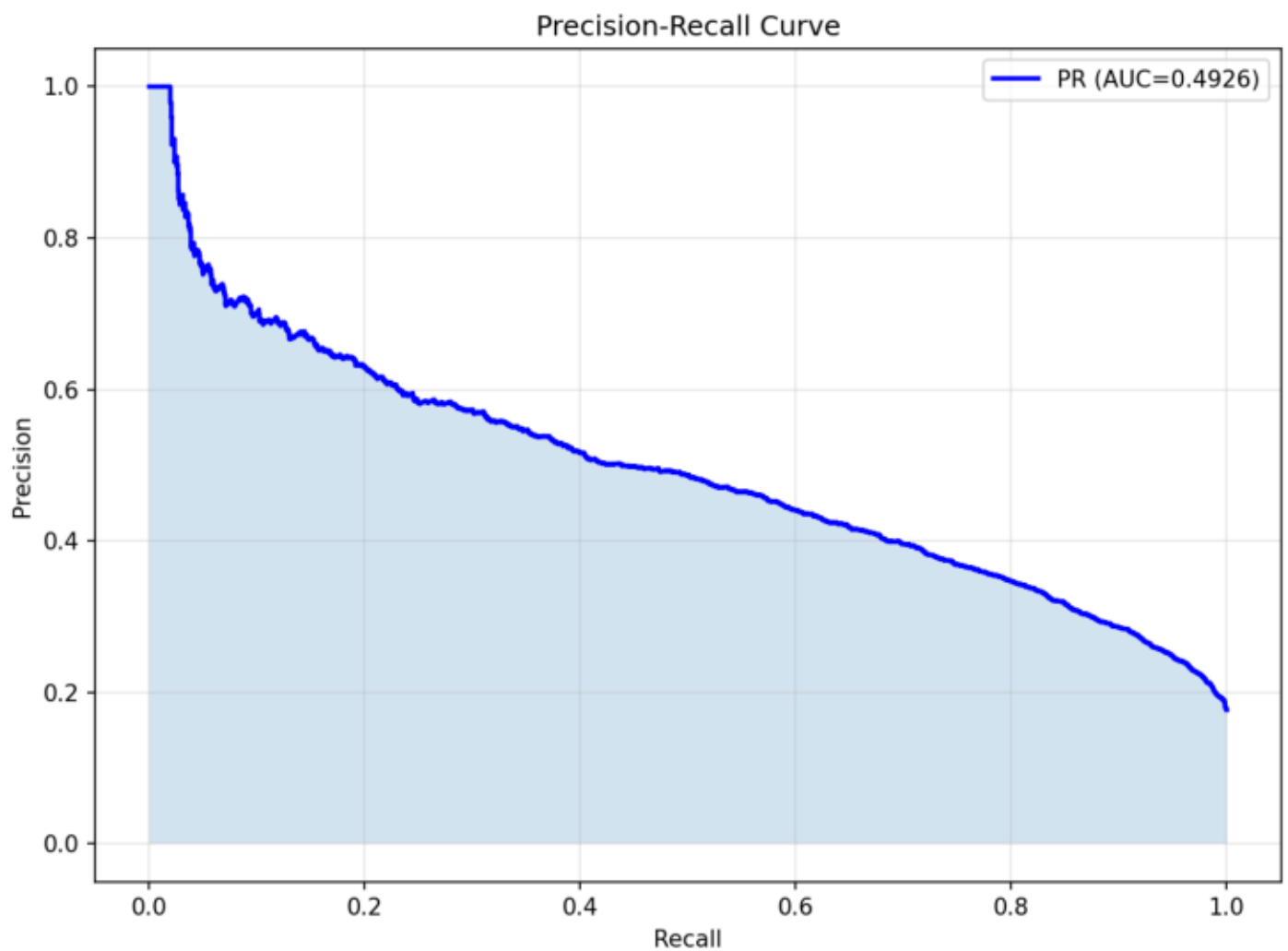
Metric	Figure	Explanation
precision_at_k[200]	0.7350	Shows how accurate the top-200 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[500]	0.6720	Shows how accurate the top-500 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
recall_at_k[10]	0.0043	Shows how much of the positive population is captured within the top-10 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[50]	0.0208	Shows how much of the positive population is captured within the top-50 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[100]	0.0359	Shows how much of the positive population is captured within the top-100 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[200]	0.0636	Shows how much of the positive population is captured within the top-200 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[500]	0.1454	Shows how much of the positive population is captured within the top-500 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.

value_date_quarter=2024Q3 - 1) Model Effectiveness



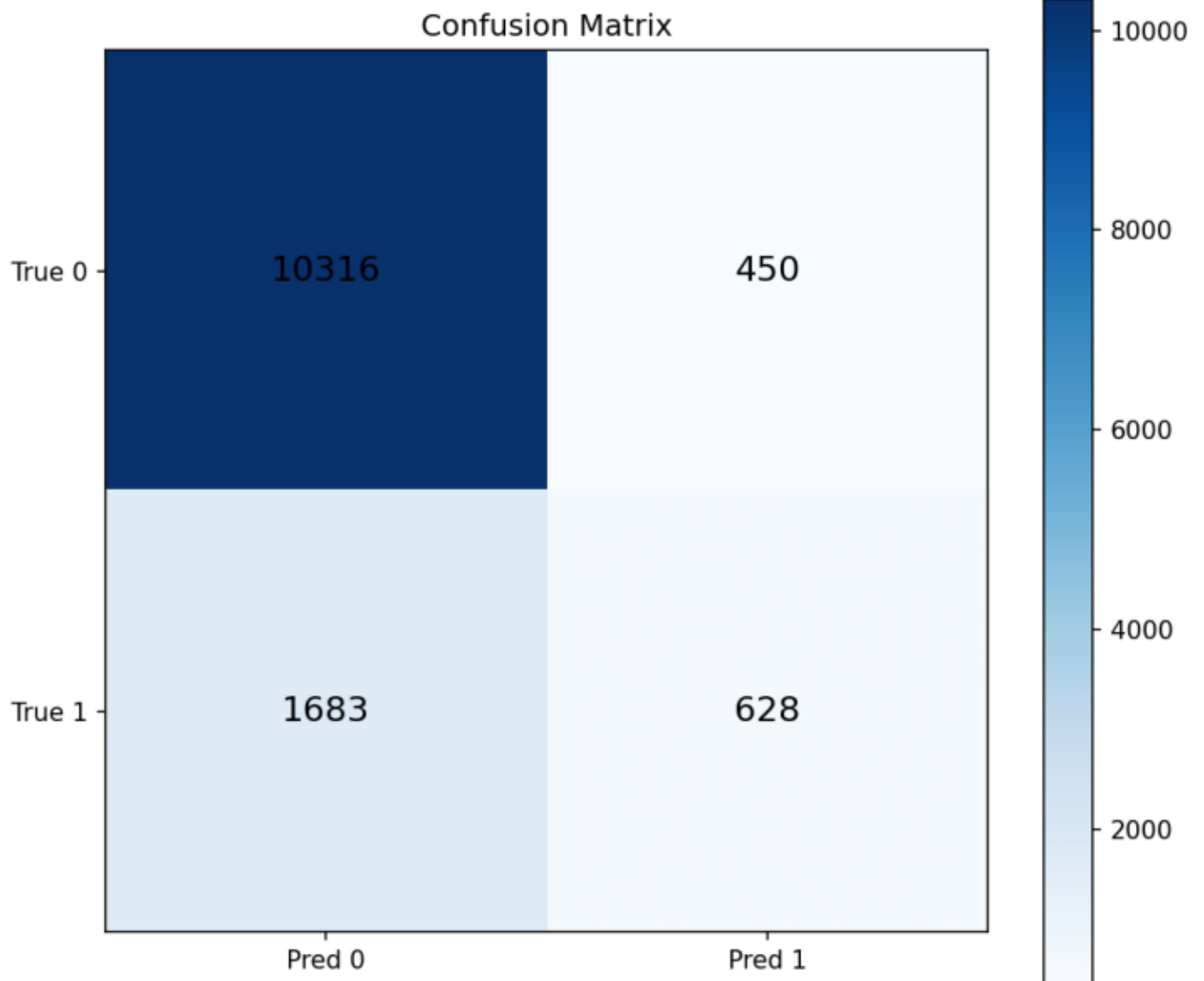
roc_curve: AUC-ROC=0.8128; curve above diagonal indicates strong separation.

value_date_quarter=2024Q3 - 1) Model Effectiveness



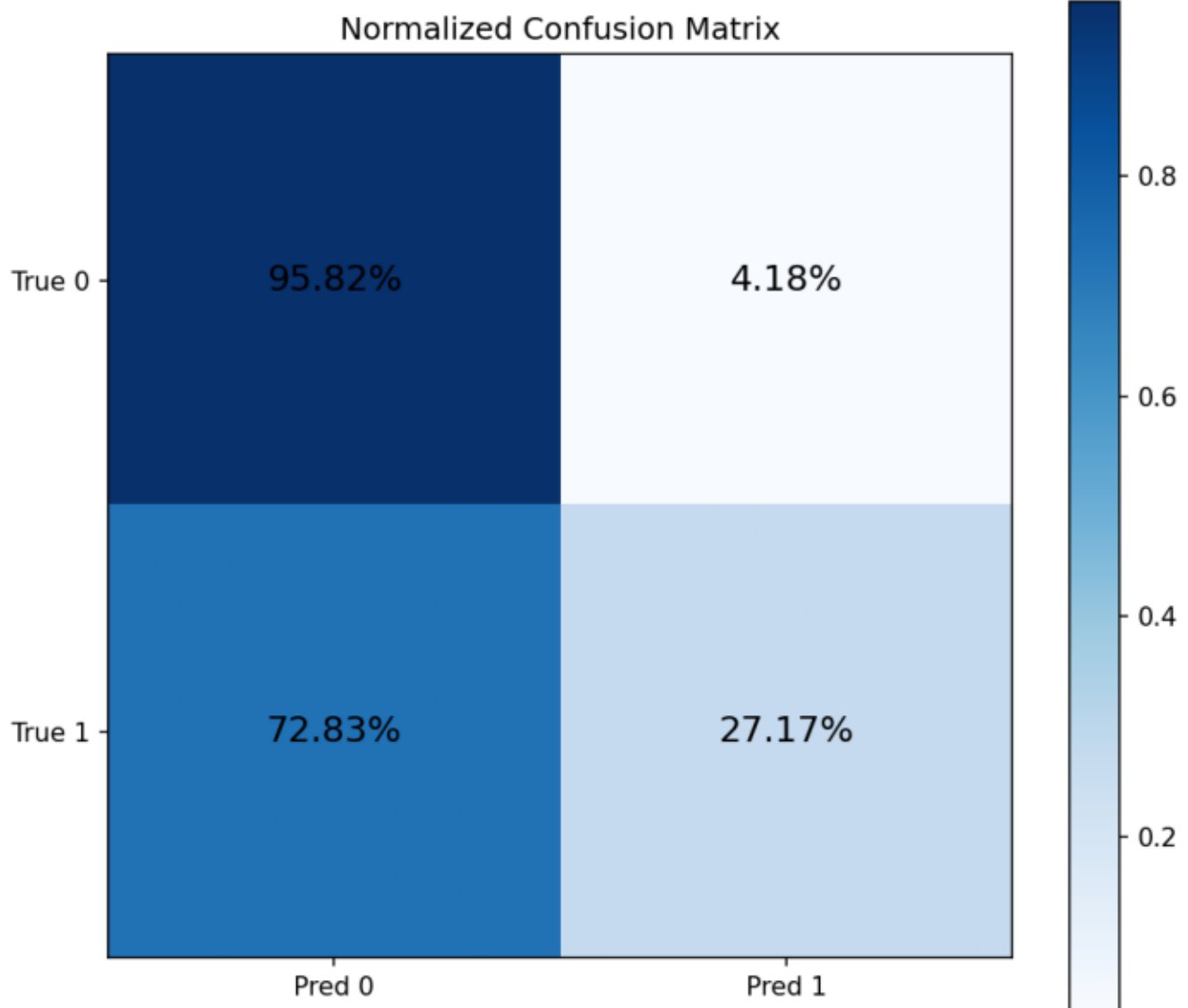
pr_curve: AUC-PR=0.4926 vs baseline 17.67%; higher curve implies better precision at recall.

value_date_quarter=2024Q3 - 1) Model Effectiveness



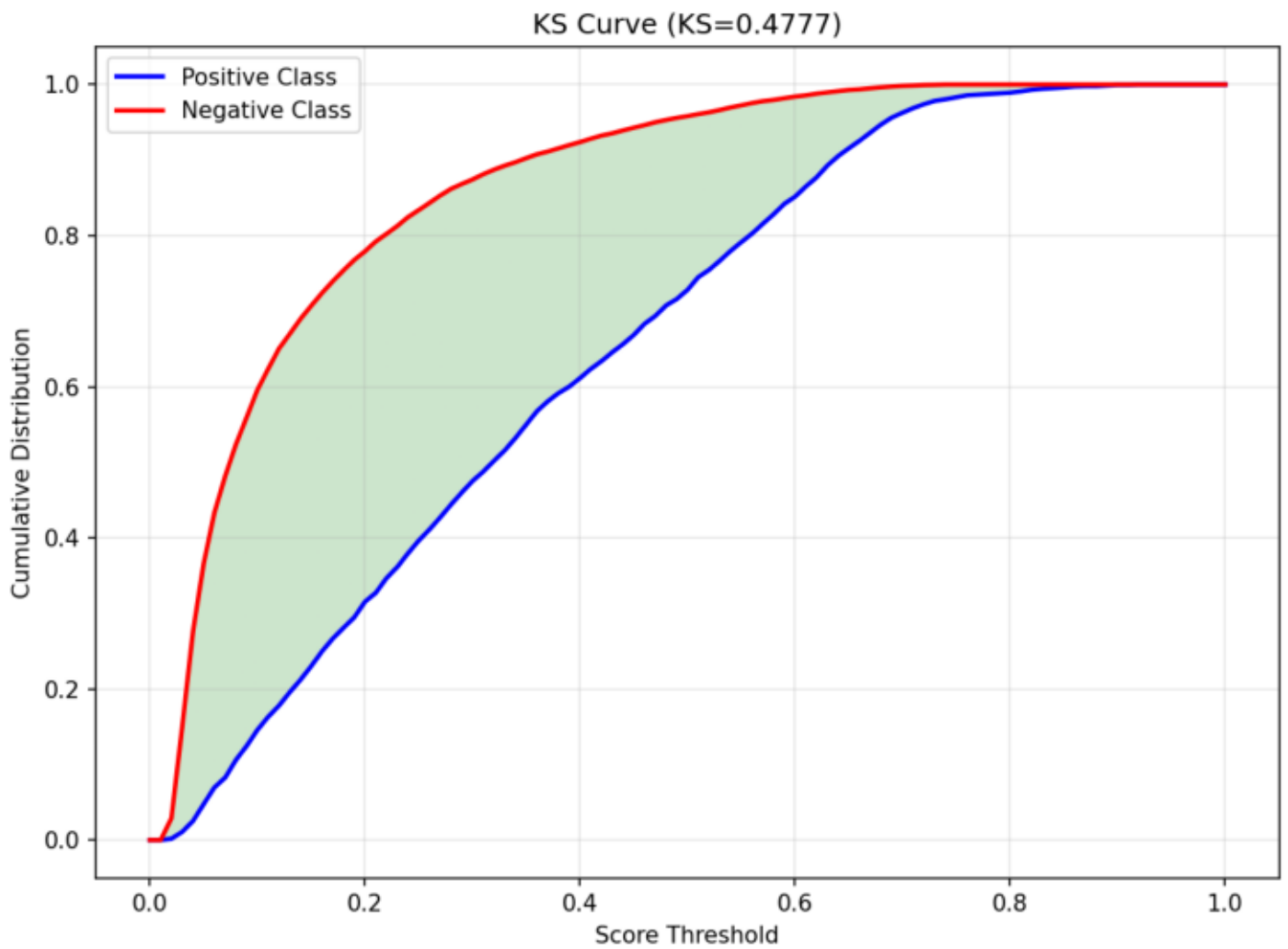
confusion_matrix: At threshold 0.50: TP=628, FP=450, FN=1683, TN=10316.

value_date_quarter=2024Q3 - 1) Model Effectiveness



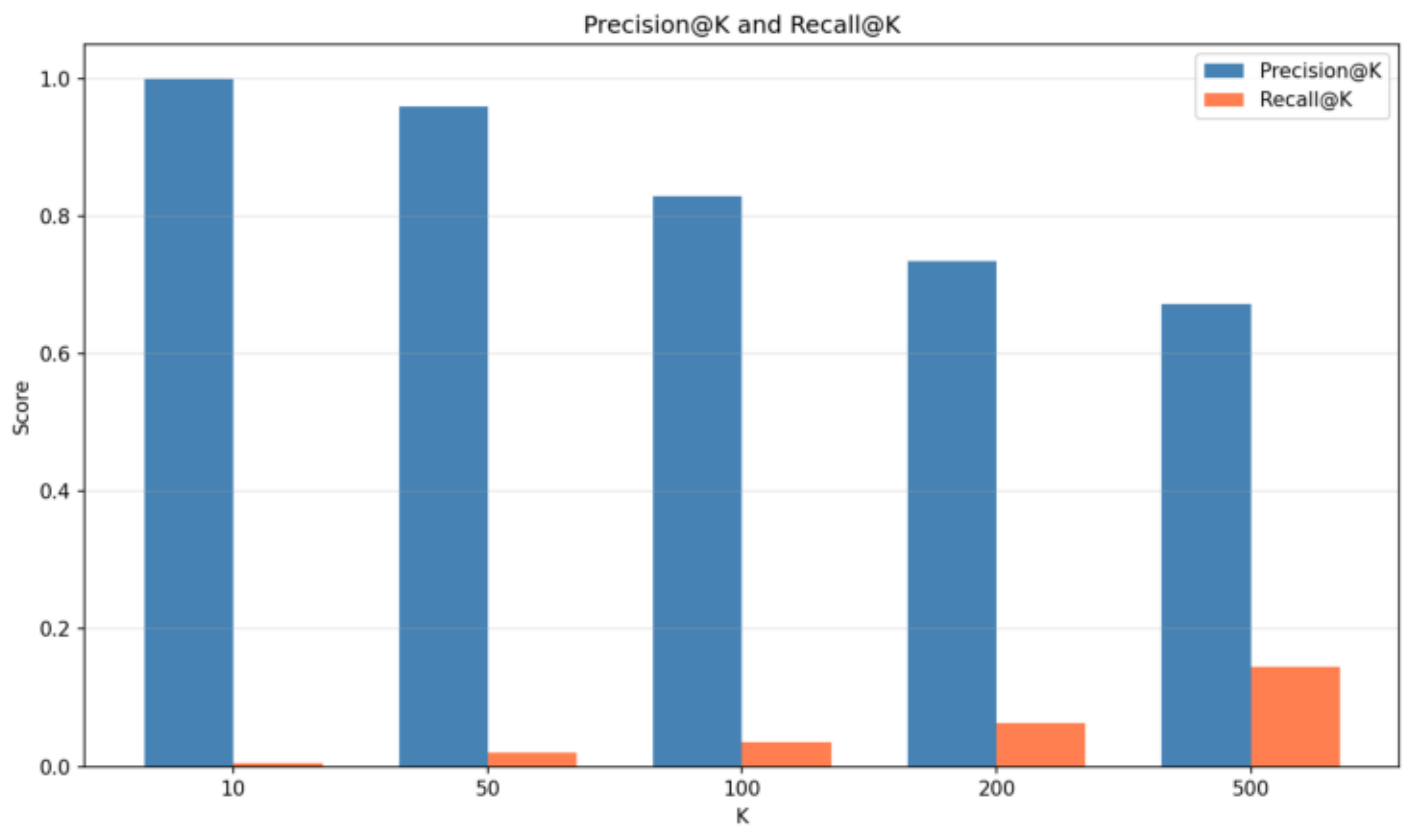
confusion_matrix_norm: Normalized rates: TPR=27.17%, TNR=95.82%.

value_date_quarter=2024Q3 - 1) Model Effectiveness



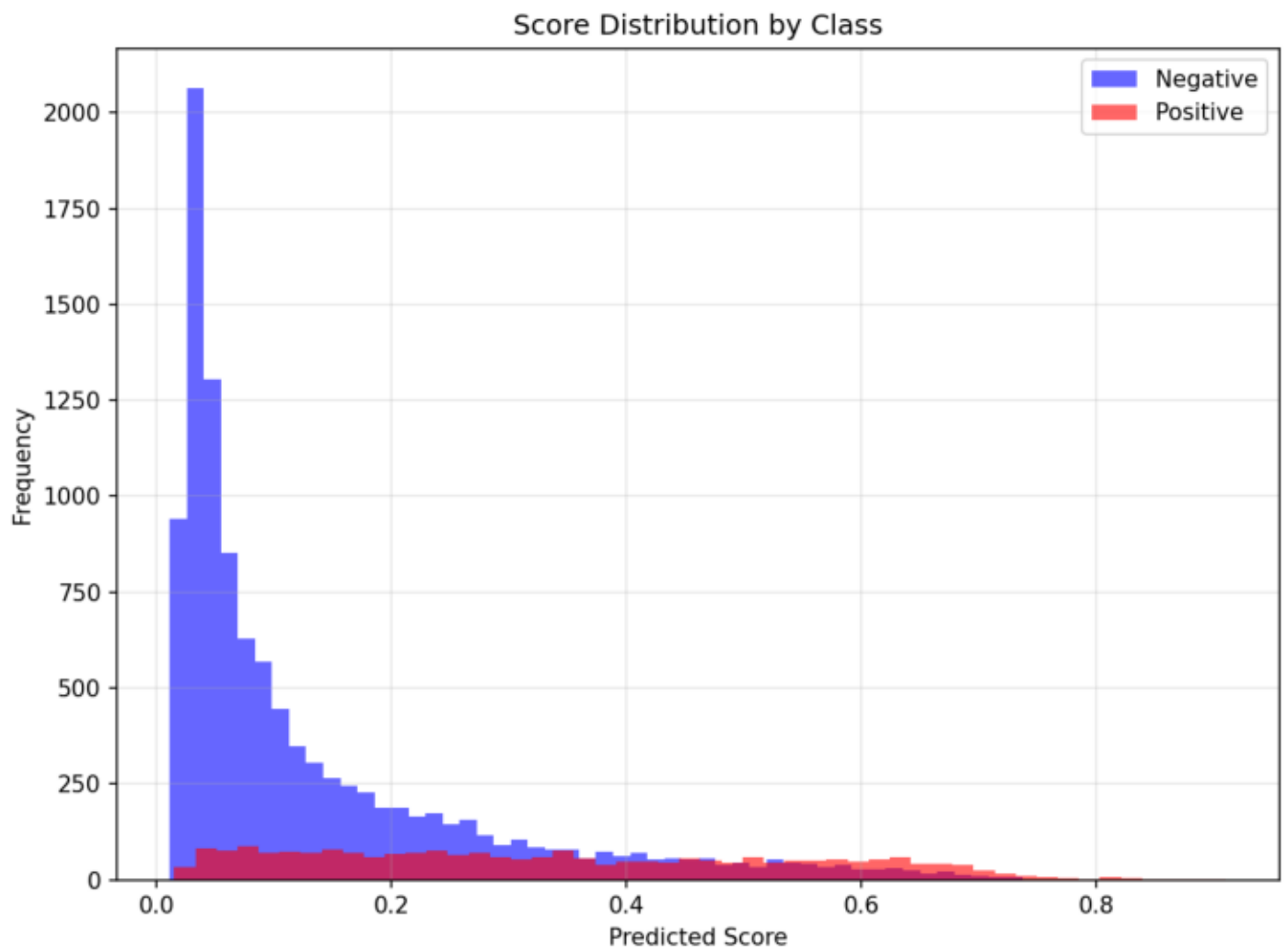
ks_curve: Maximum separation KS=0.4777 at threshold 0.14.

value_date_quarter=2024Q3 - 1) Model Effectiveness



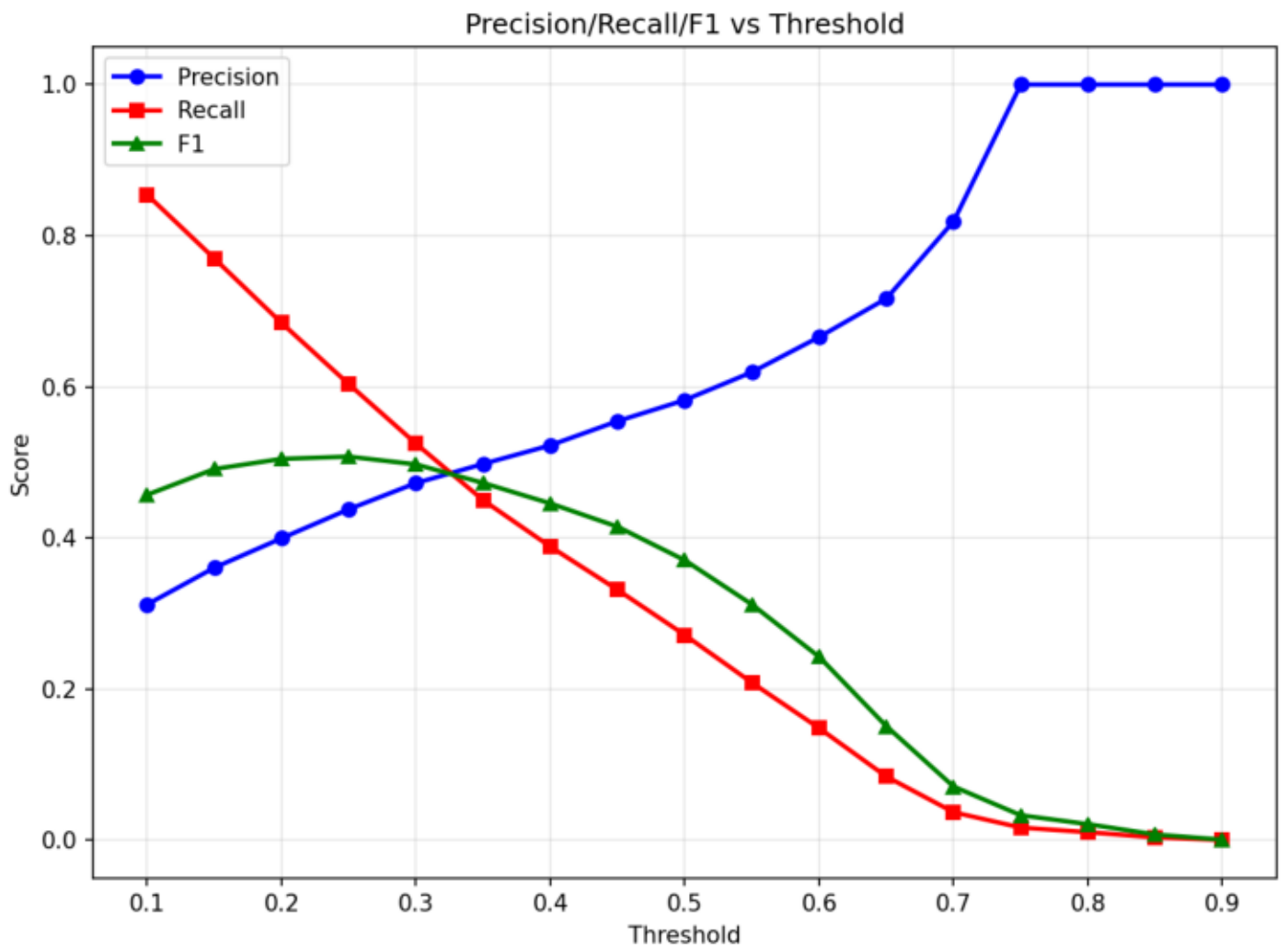
precision_recall_at_k: Bars show precision/recall as you expand the review queue.

value_date_quarter=2024Q3 - 1) Model Effectiveness



score_distribution: Mean score: positive=0.342, negative=0.135 (larger gap is better).

value_date_quarter=2024Q3 - 1) Model Effectiveness



threshold_analysis: Best F1 in tested grid is 0.5077 at threshold 0.25.

value_date_quarter=2024Q3 - 2) Model Efficiency (1/4)

Metric	Figure	Explanation
fpr	0.0418	Measures the share of negatives incorrectly flagged; 0.0418 is low. Next: if review demand is too high, raise the threshold or add features that better suppress false alerts.
tn	10316	Counts negatives correctly cleared without manual review. Next: preserve this clearance rate while checking that recall does not fall below policy expectations.
fp	450	Counts non-events that would still be sent to review. Next: use this with staffing capacity to decide whether threshold or feature refinement is required.
threshold	0.5000	This is the active operating cutoff used to flag alerts. Next: tune it jointly with recall, precision, and review capacity rather than in isolation.
fpr_at_thresholds[thresholds.t_0.05]	0.6339	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.05. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.10]	0.4040	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.10. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q3 - 2) Model Efficiency (2/4)

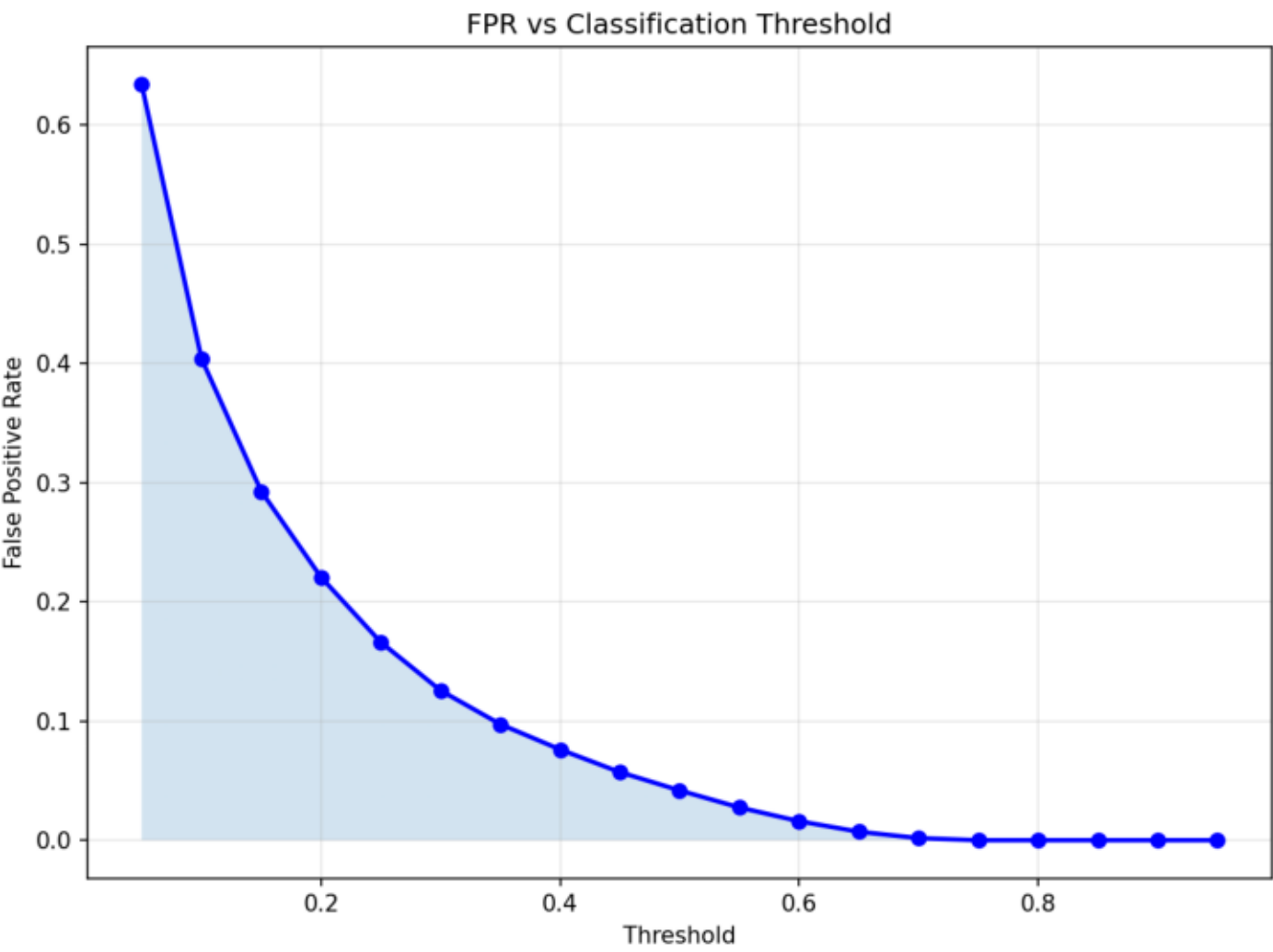
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.15]	0.2928	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.15. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.20]	0.2207	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.20. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.25]	0.1662	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.25. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.30]	0.1257	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.30. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.35]	0.0973	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.35. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.40]	0.0762	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.40. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q3 - 2) Model Efficiency (3/4)

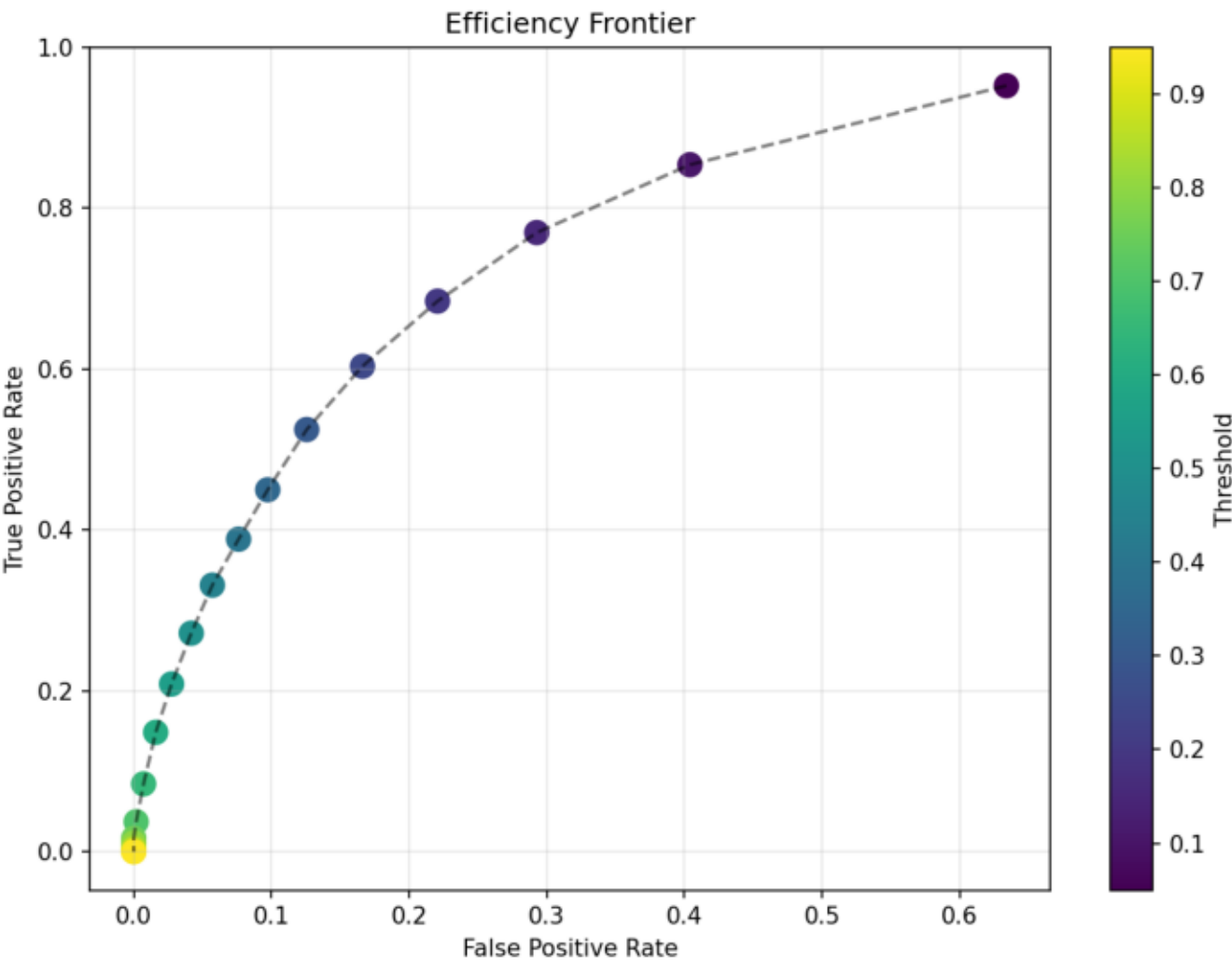
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.45]	0.0571	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.45. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.50]	0.0418	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.50. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.55]	0.0275	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.55. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.60]	0.0160	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.60. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.65]	0.0072	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.65. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.70]	0.0018	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.70. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q3 - 2) Model Efficiency (4/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.75]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.75. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.80]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.80. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.85]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.85. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.90]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.90. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.95]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.95. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

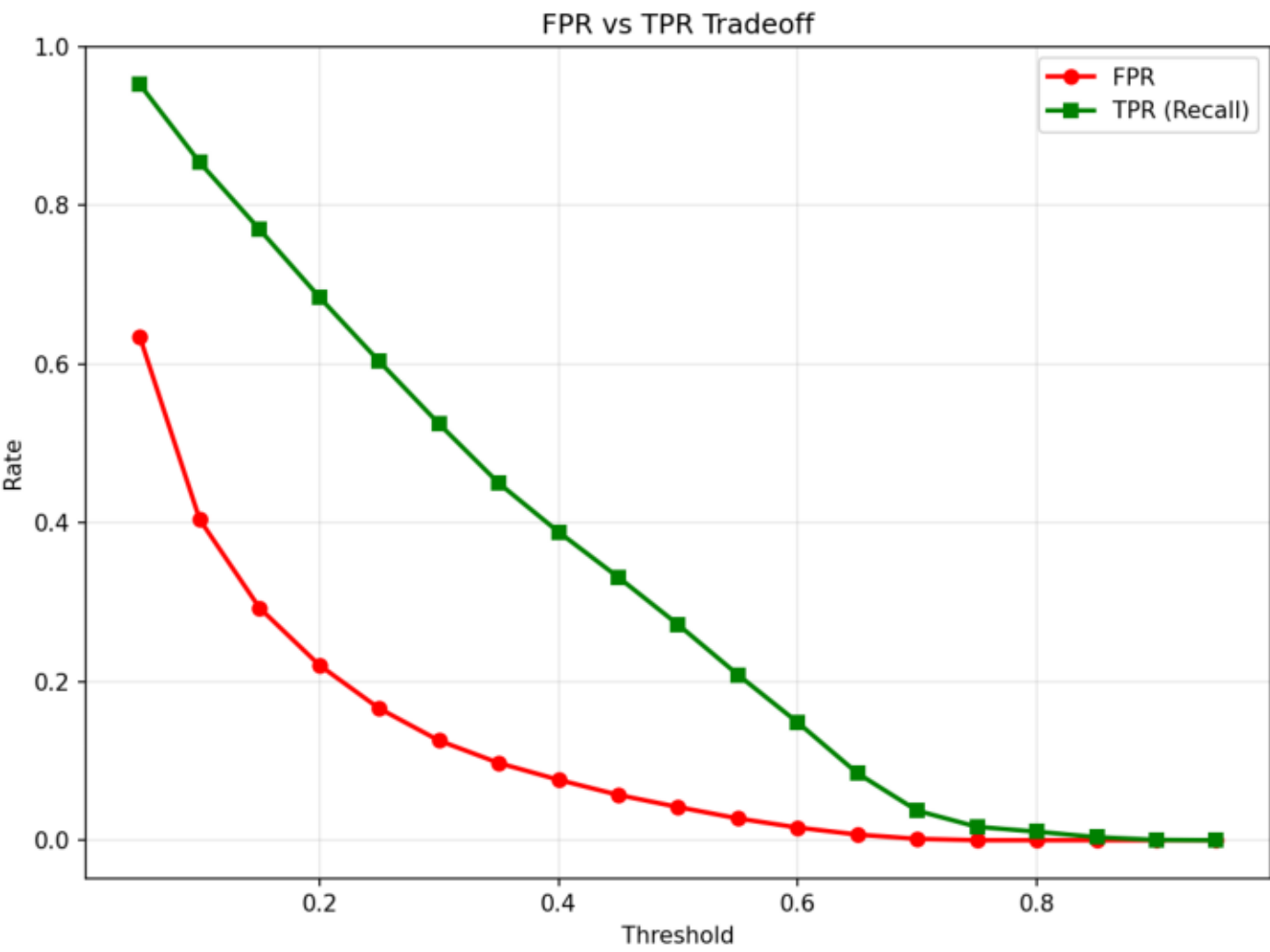


value_date_quarter=2024Q3 - 2) Model Efficiency



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

value_date_quarter=2024Q3 - 2) Model Efficiency



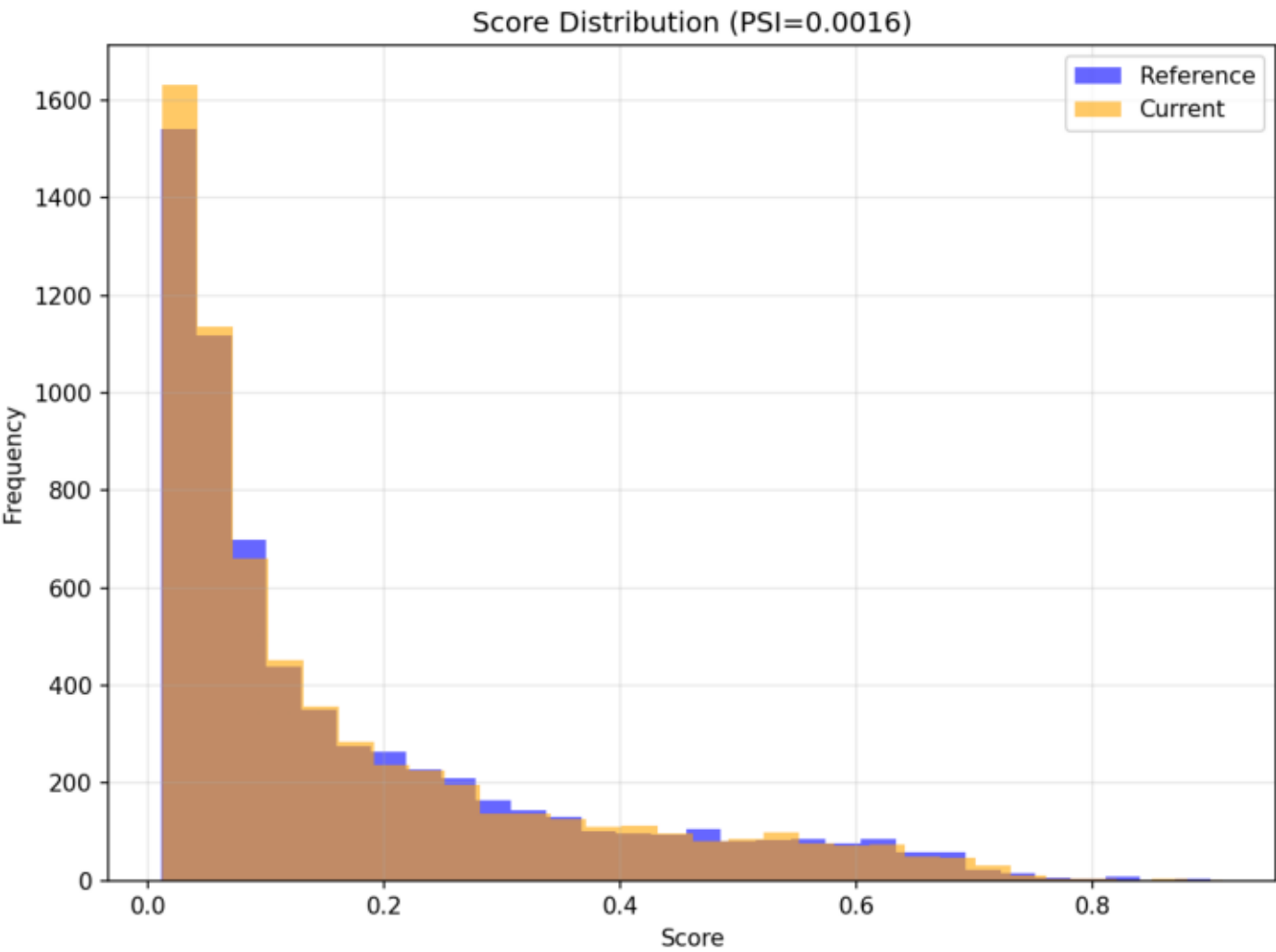
value_date_quarter=2024Q3 - 3) Model Stability (1/2)

Metric	Figure	Explanation
psi	0.0016	Measures score-distribution shift between reference and current populations; 0.0016 is negligible. Next: investigate segment mix, data drift, and recalibration needs when PSI is material.
cv_auc_roc_mean	0.8126	Average AUC-ROC across validation folds. Next: compare it with overall AUC-ROC; a large gap can indicate overfitting or unstable sampling.
cv_auc_roc_std	0.0139	Standard deviation of AUC-ROC across folds. Next: if variability is high, simplify the model or increase training stability before deployment.
cv_auc_pr_mean	0.4935	Average AUC-PR across validation folds. Next: confirm that precision-recall performance remains acceptable under resampling.
cv_auc_pr_std	0.0318	Standard deviation of AUC-PR across folds. Next: high dispersion suggests unstable alert quality and warrants deeper validation.
bootstrap_auc_roc_mean	0.8120	Bootstrap mean AUC-ROC estimates expected ranking performance under repeated sampling. Next: compare it with the point estimate to judge optimism in the current test run.
bootstrap_auc_roc_ci_lower	0.7996	Lower confidence bound for bootstrap AUC-ROC. Next: use it as a conservative performance floor in model-risk discussions.
bootstrap_auc_roc_ci_upper	0.8228	Upper confidence bound for bootstrap AUC-ROC. Next: review the interval width with the lower bound to assess uncertainty.

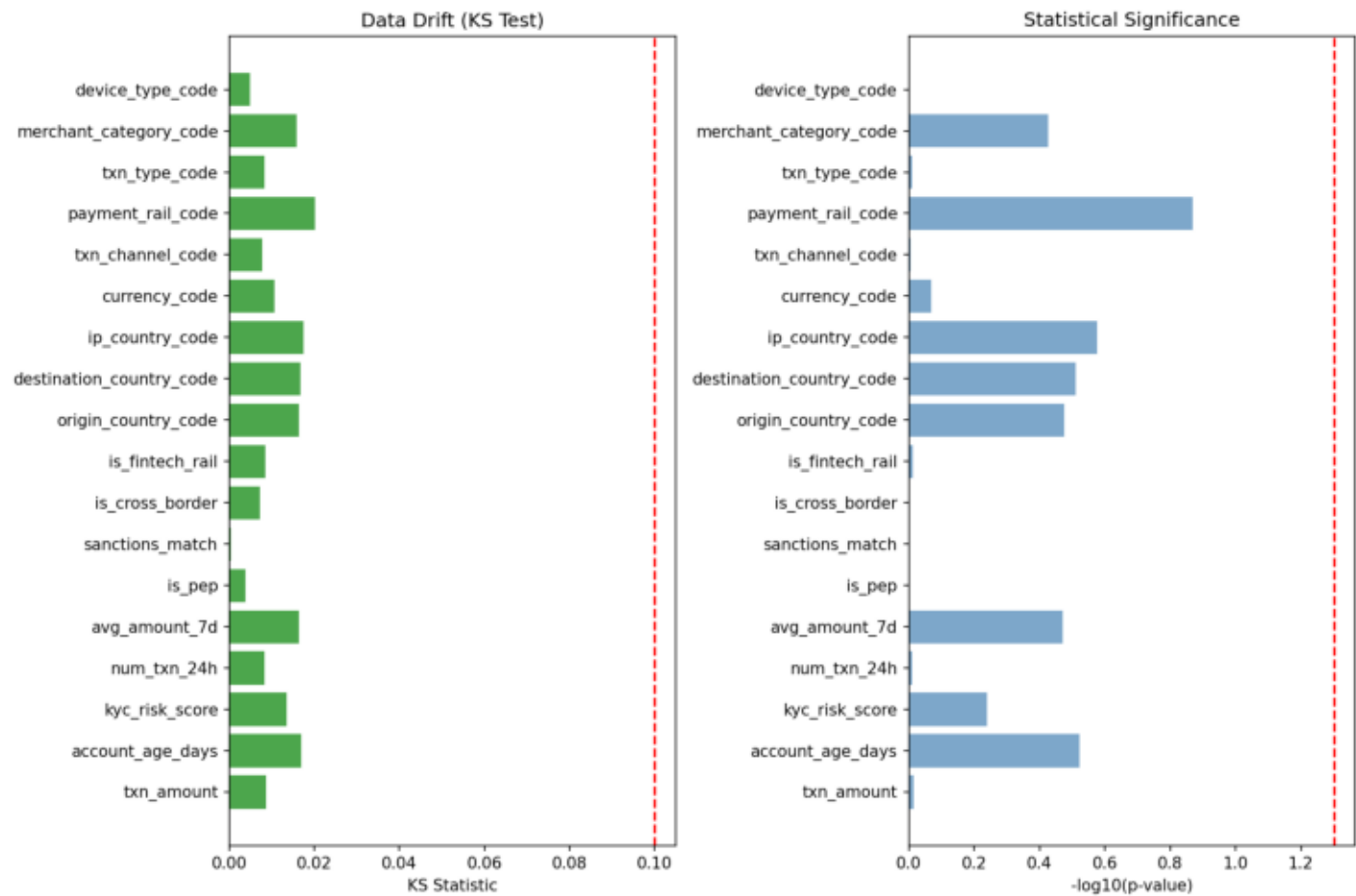
value_date_quarter=2024Q3 - 3) Model Stability (2/2)

Metric	Figure	Explanation
concept_drift_detected	False	No material concept-drift signal was detected in the evaluated slices. Next: retain ongoing monitoring because stable historical results do not guarantee future stability.
concept_drift_score	0.0309	Quantifies the magnitude of performance change across ordered slices. Next: rising scores should trigger deeper review of segment behavior and model refresh timing.

value_date_quarter=2024Q3 - 3) Model Stability

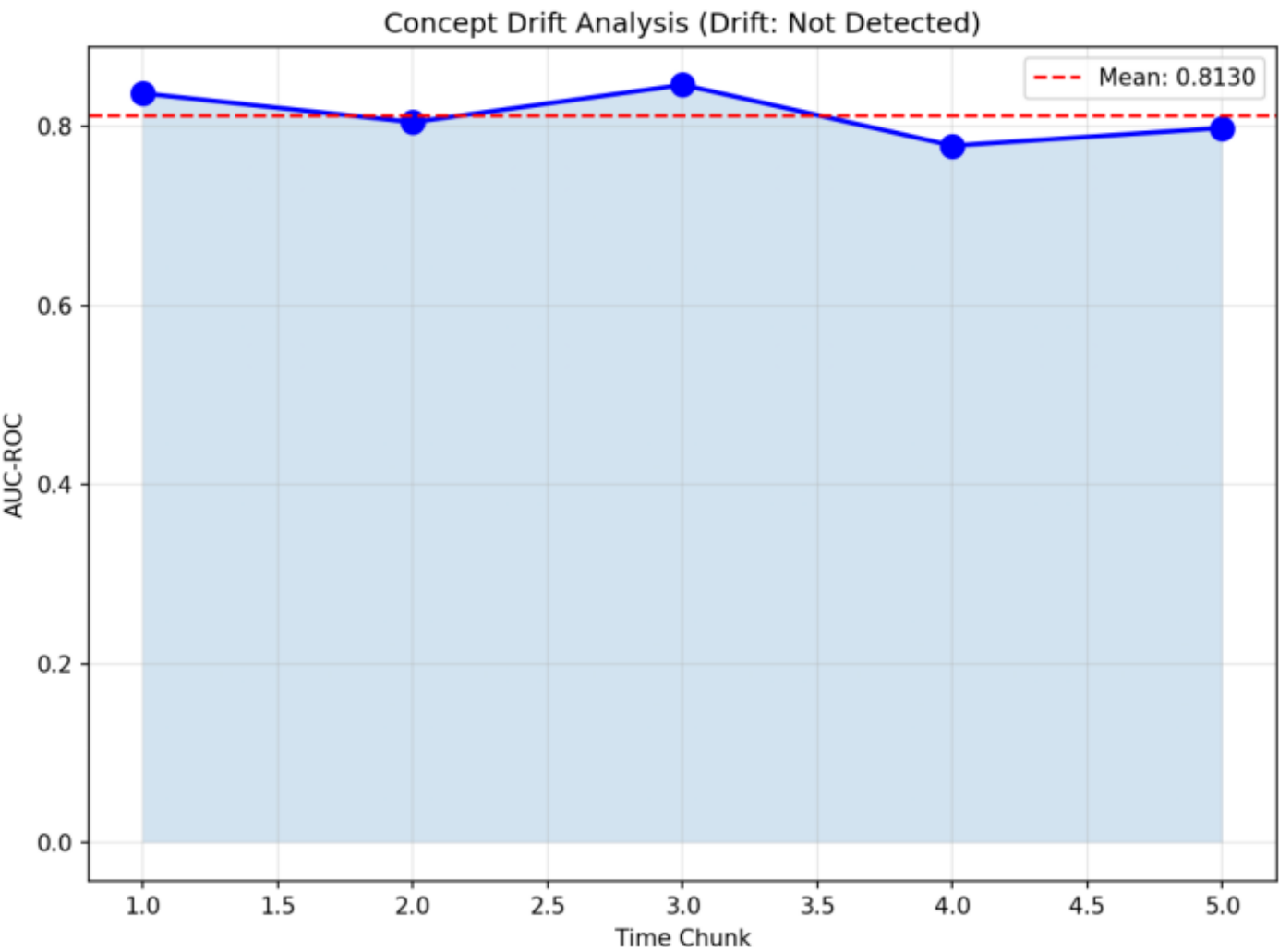


value_date_quarter=2024Q3 - 3) Model Stability



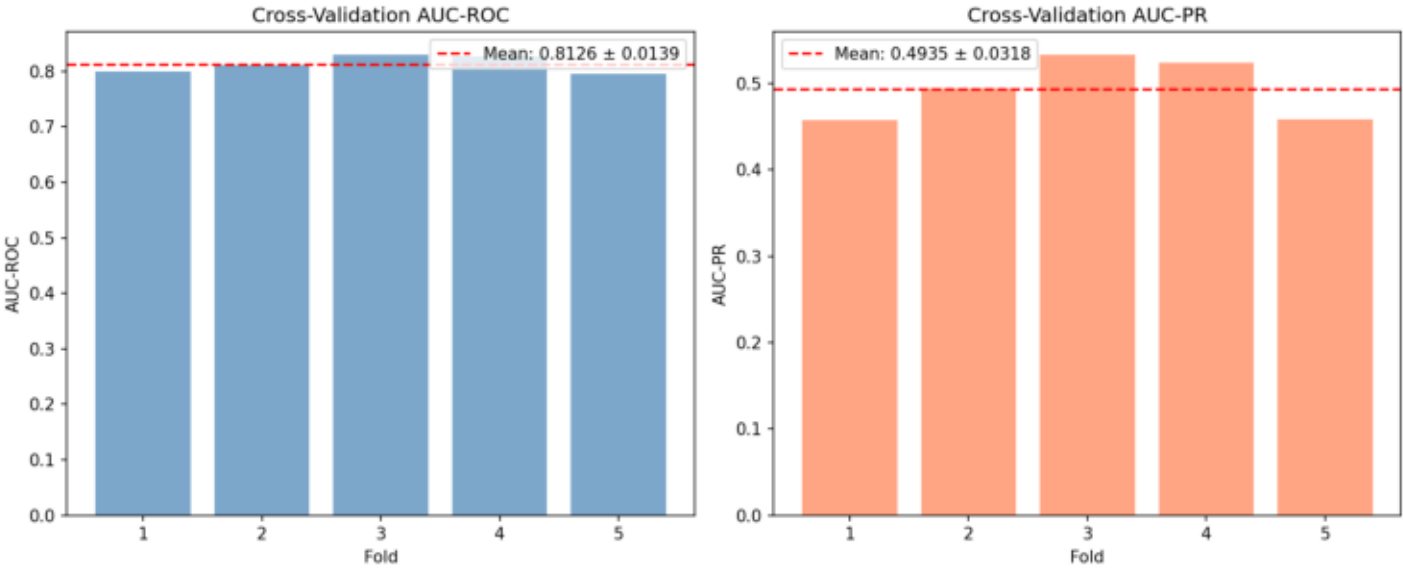
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.

value_date_quarter=2024Q3 - 3) Model Stability



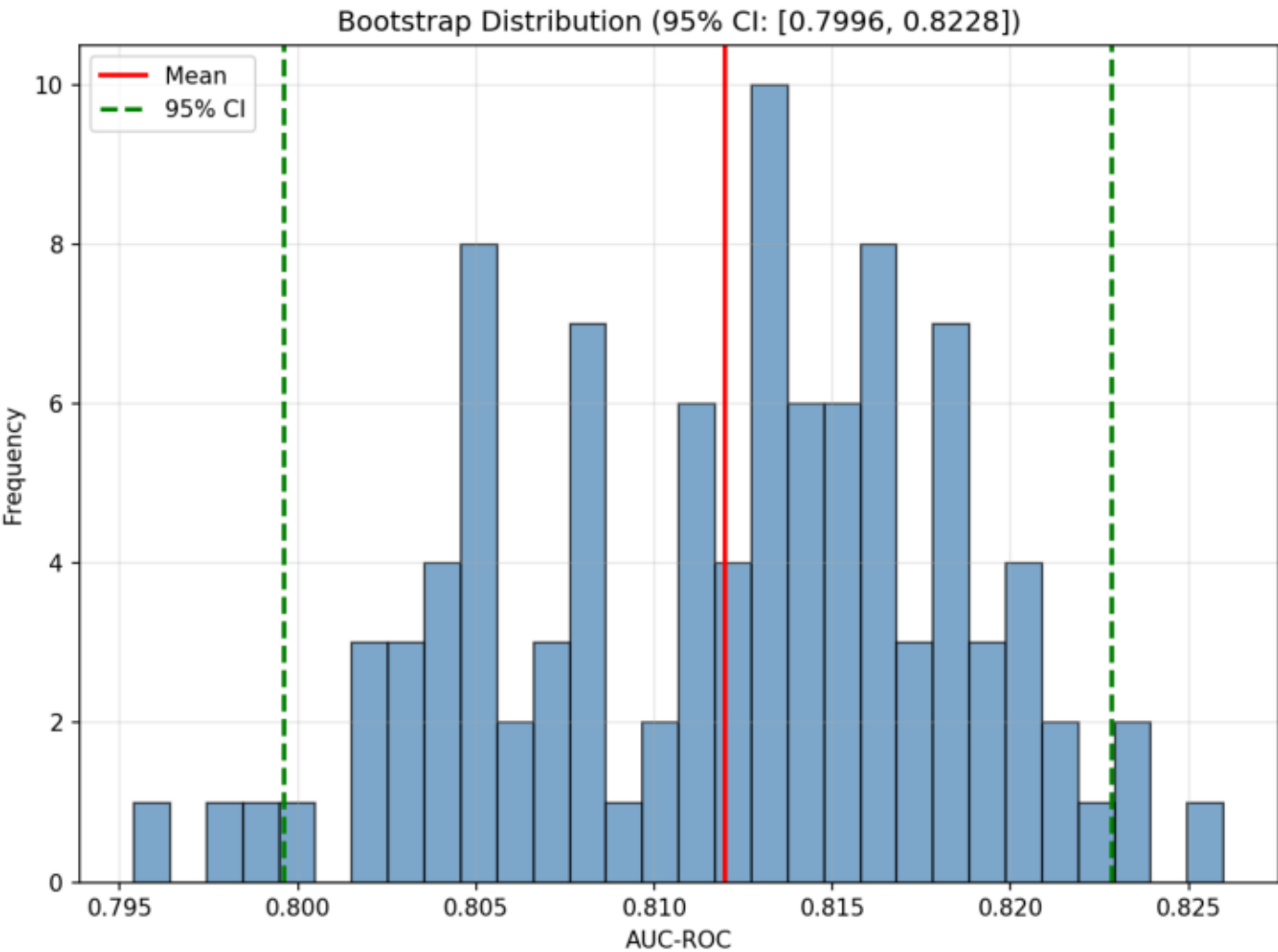
concept_drift: AUC across time chunks shows whether performance is stable over time.

value_date_quarter=2024Q3 - 3) Model Stability

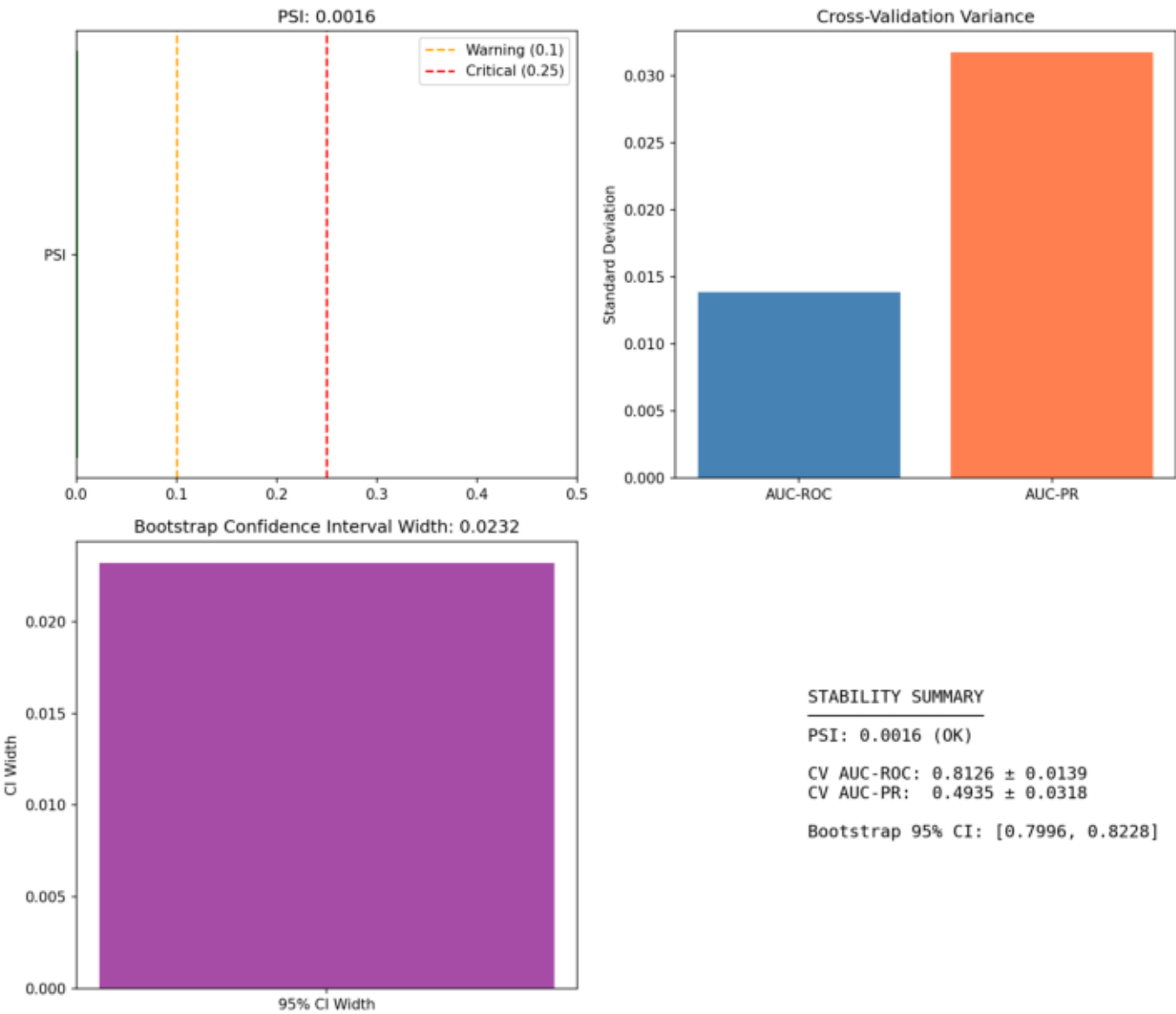


cv_results: Fold scores cluster around ROC 0.8126 and PR 0.4935.

value_date_quarter=2024Q3 - 3) Model Stability

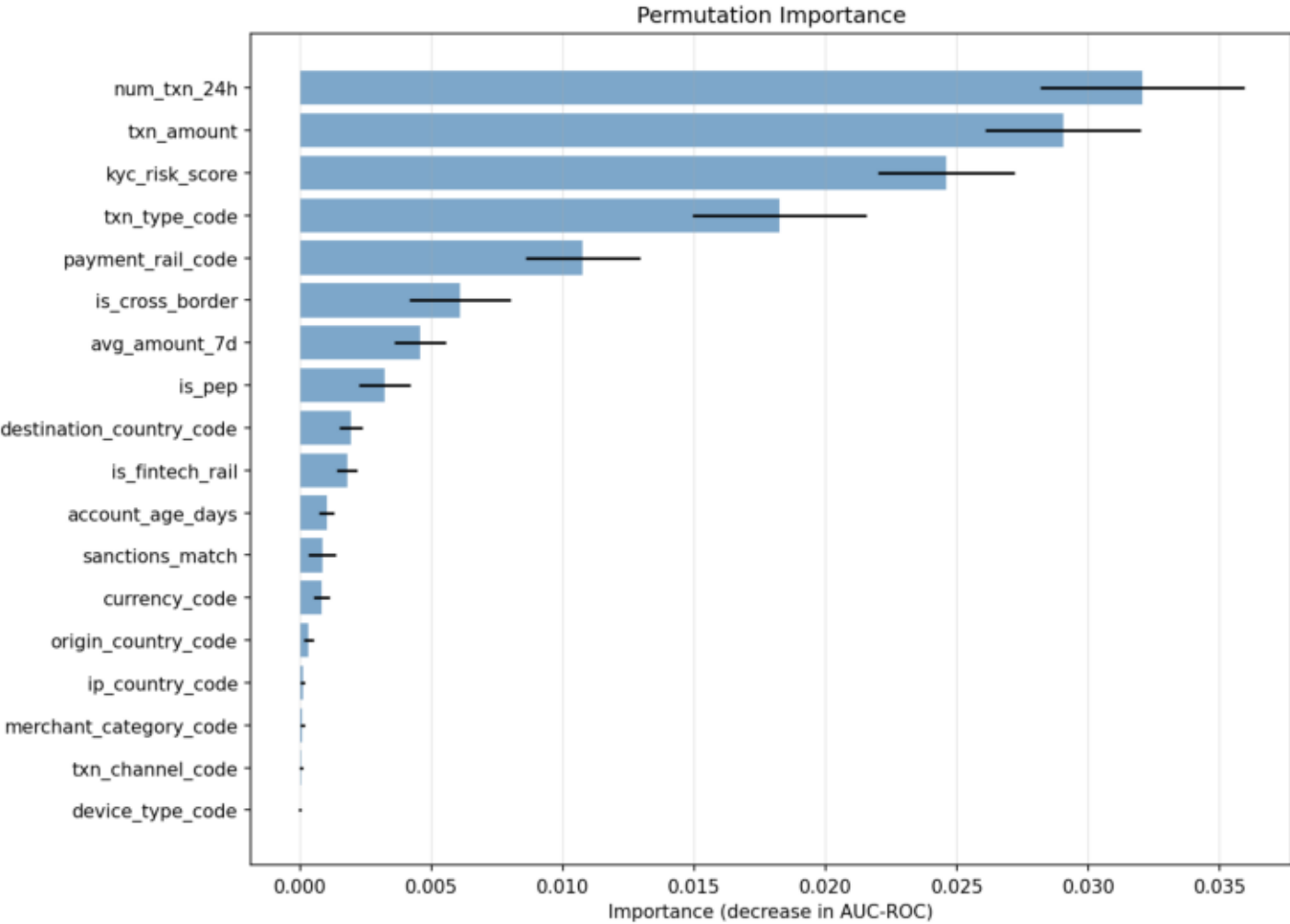


value_date_quarter=2024Q3 - 3) Model Stability

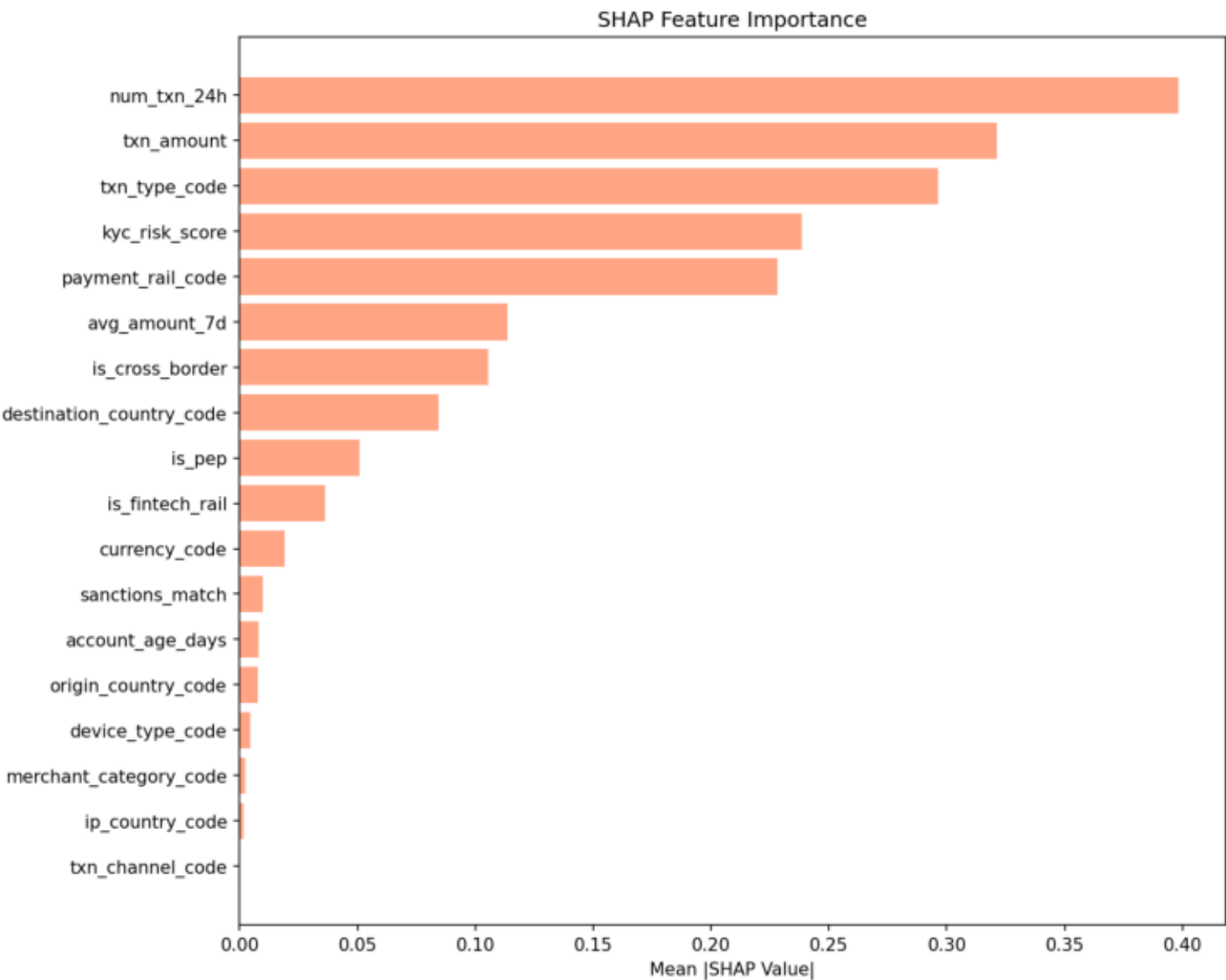


Metric	Figure	Explanation
model_type	tree	Identifies the estimator family so explanation methods can be interpreted in the right context. Next: confirm the chosen explanation methods are appropriate for this model class.
methods_used	permutation, pdp, ice, shap, lime	Lists the interpretability methods produced in this run. Next: compare signals across methods before using them in documentation or model challenge.
perm_top_features	num_txn_24h, txn_amount, kyc_risk_score, txn_type_code, payment_rail_code ...	Shows features with the largest performance drop when permuted, which highlights global importance. Next: validate that the top drivers align with business logic, feature governance, and policy expectations.
shap_top_features	num_txn_24h, txn_amount, txn_type_code, kyc_risk_score, payment_rail_code ...	Shows the most influential features under SHAP-based contribution analysis. Next: inspect whether directionality and ranking remain sensible across key segments.

Metric	Figure	Explanation
lime_instances	3	Counts the locally explained observations reviewed with LIME. Next: examine representative true-positive and false-positive cases to understand local decision logic.
pdp_features	avg_amount_7d, txn_amount, account_age_days, kyc_risk_score, num_txn_24h ...	Lists features with partial-dependence plots. Next: review whether average model response is monotonic, stable, and operationally sensible.
ice_features	avg_amount_7d, txn_amount, account_age_days, kyc_risk_score	Lists features with individual conditional expectation plots. Next: inspect heterogeneity and interactions that may be hidden in average-only views.

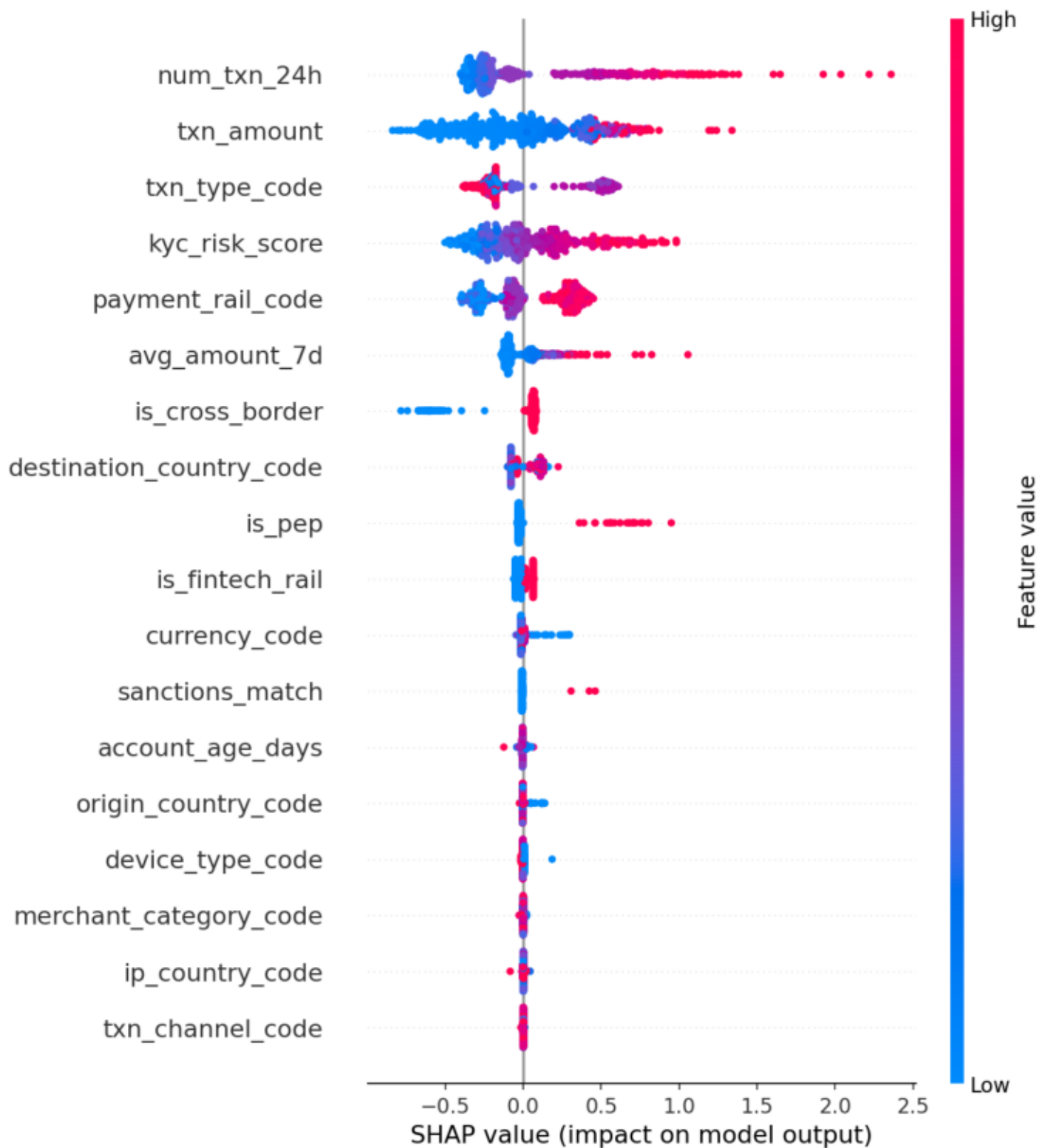


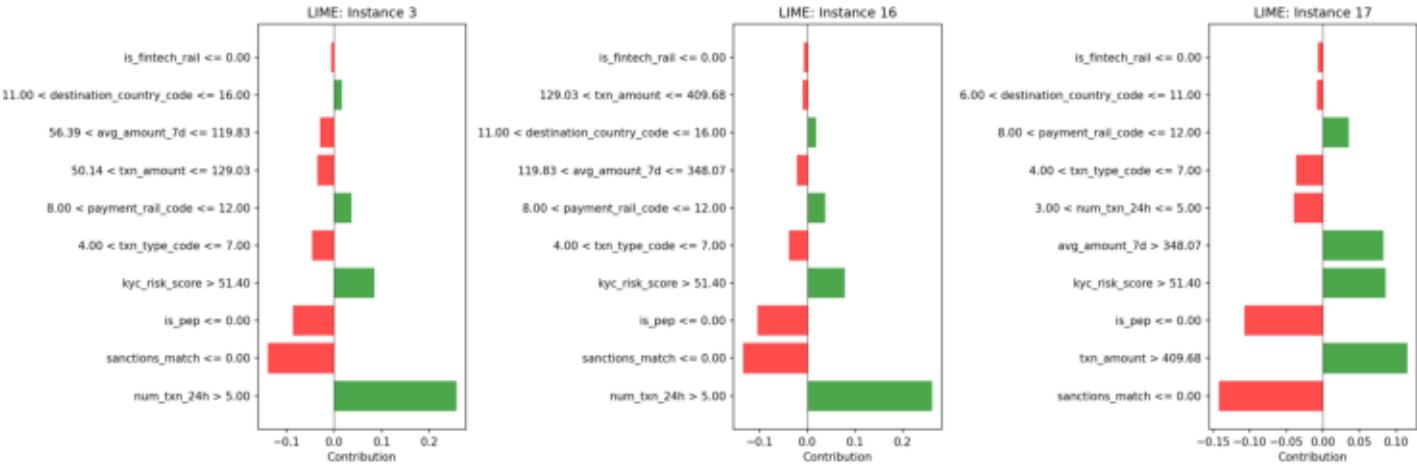
permutation_importance: Bars show how much AUC drops when each feature is permuted.



shap_bar: SHAP bar plot ranks global feature impact by mean absolute SHAP values.

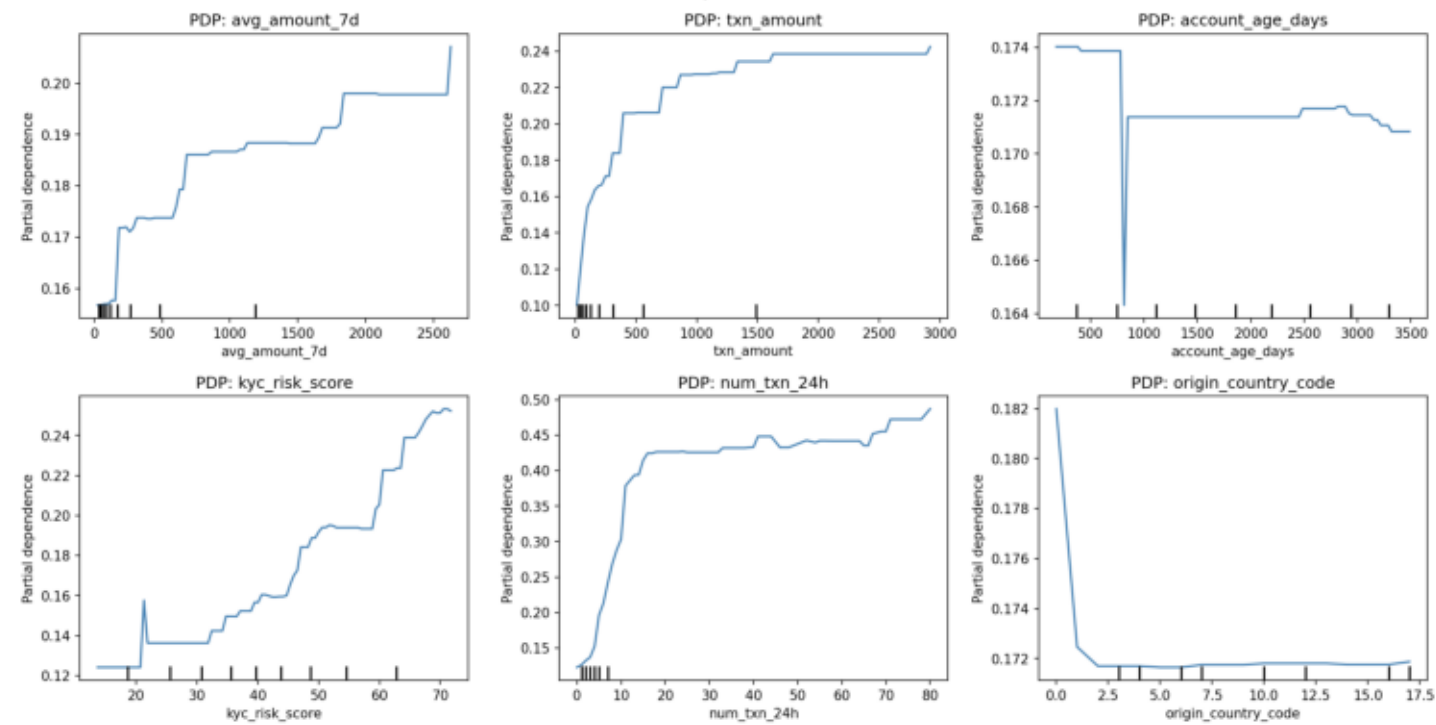
value_date_quarter=2024Q3 - 4) Model Interpretability





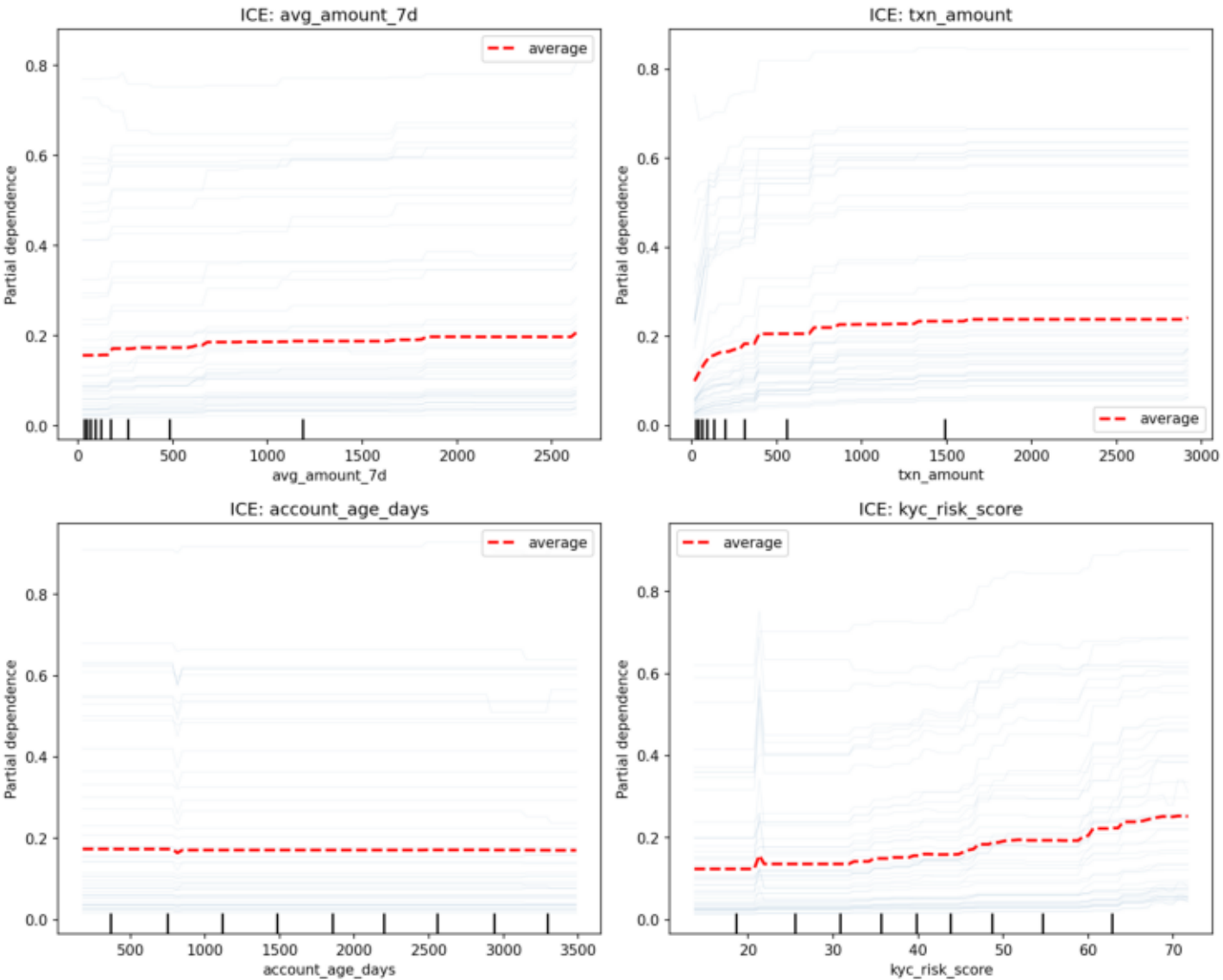
value_date_quarter=2024Q3 - 4) Model Interpretability

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q4 - 1) Model Effectiveness (1/3)

Metric	Figure	Explanation
auc_roc	0.8144	Measures ranking power across all thresholds. 0.8144 indicates strong separation. Next: compare AUC-ROC across segments and keep threshold tuning separate from ranking validation.
auc_pr	0.5921	Measures precision-recall performance under class imbalance; the observed event rate is 28.43%. Next: use this together with alert capacity to decide whether the high-score queue is practically useful.
precision	0.6086	Shows the share of flagged cases that are true positives at the current threshold. Next: if this is below review expectations, raise the threshold or improve exclusion features.
recall	0.4920	Shows the share of true positives captured at the current threshold. Next: if missed cases are too high, lower the threshold or add stronger risk features.
f1	0.5442	Balances precision and recall in one score. Next: use it as a threshold-selection aid when both missed cases and false alerts matter.
ks_statistic	0.5053	Measures score separation between positive and negative classes. Next: low KS suggests the model needs better features or a different specification.

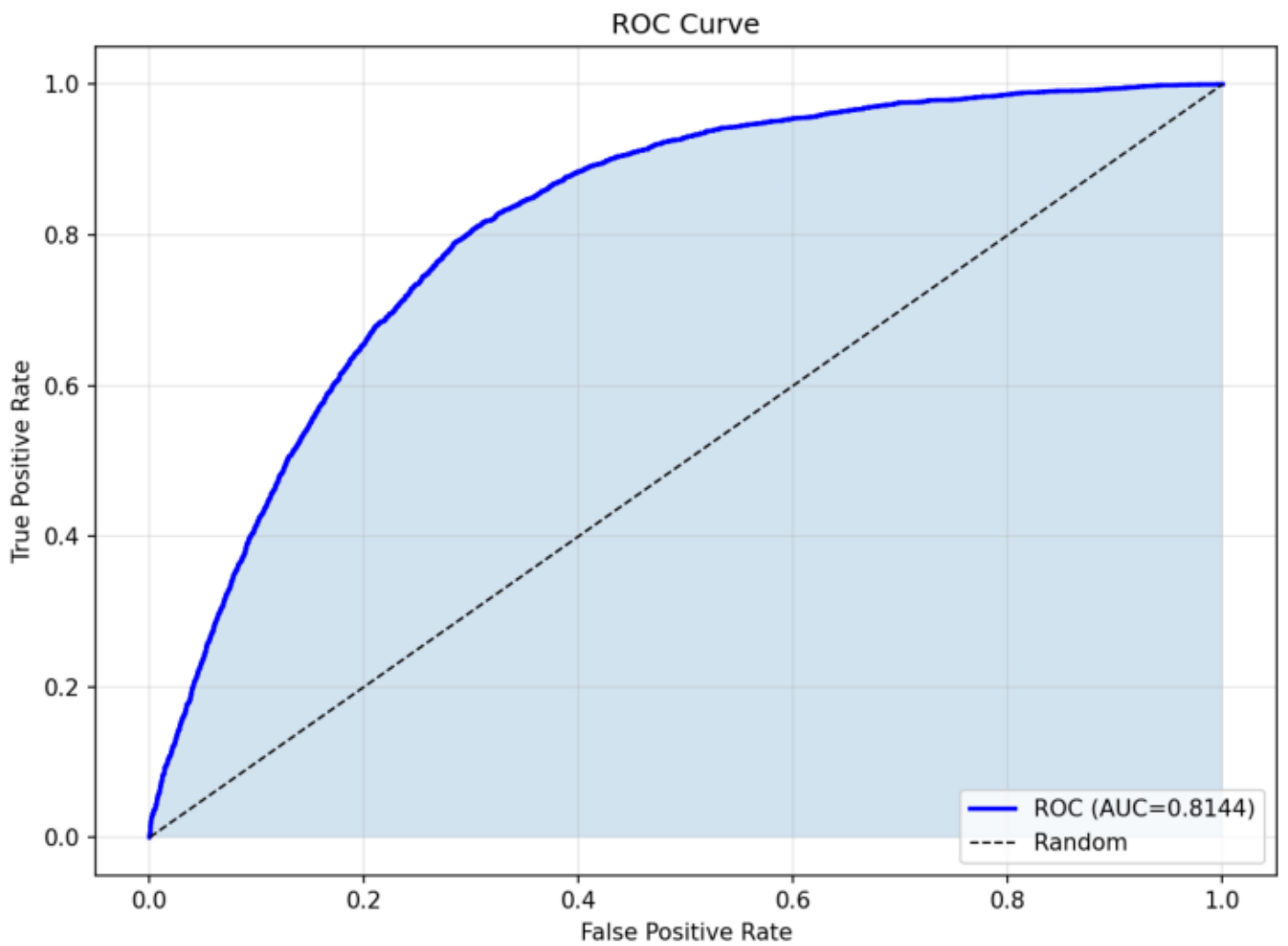
value_date_quarter=2024Q4 - 1) Model Effectiveness (2/3)

Metric	Figure	Explanation
ks_threshold	0.2300	This is the threshold that maximizes KS separation. Next: treat it as a candidate operating point, then compare it with business capacity and policy limits.
confusion_matrix[TN]	8707	True negatives correctly cleared. Next: preserve this base while tuning the alert threshold.
confusion_matrix[FP]	1252	False positives create review cost without SAR conversion. Next: reduce this count with threshold tuning or sharper risk features.
confusion_matrix[FN]	2010	False negatives are missed SAR cases. Next: if this count is unacceptable, lower the threshold or add features that capture suspicious behavior earlier.
confusion_matrix[TP]	1947	True positives are correctly escalated cases. Next: assess whether this volume is sufficient relative to expected SAR capture goals.
precision_at_k[10]	0.9000	Shows how accurate the top-10 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[50]	0.8600	Shows how accurate the top-50 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[100]	0.8700	Shows how accurate the top-100 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.

value_date_quarter=2024Q4 - 1) Model Effectiveness (3/3)

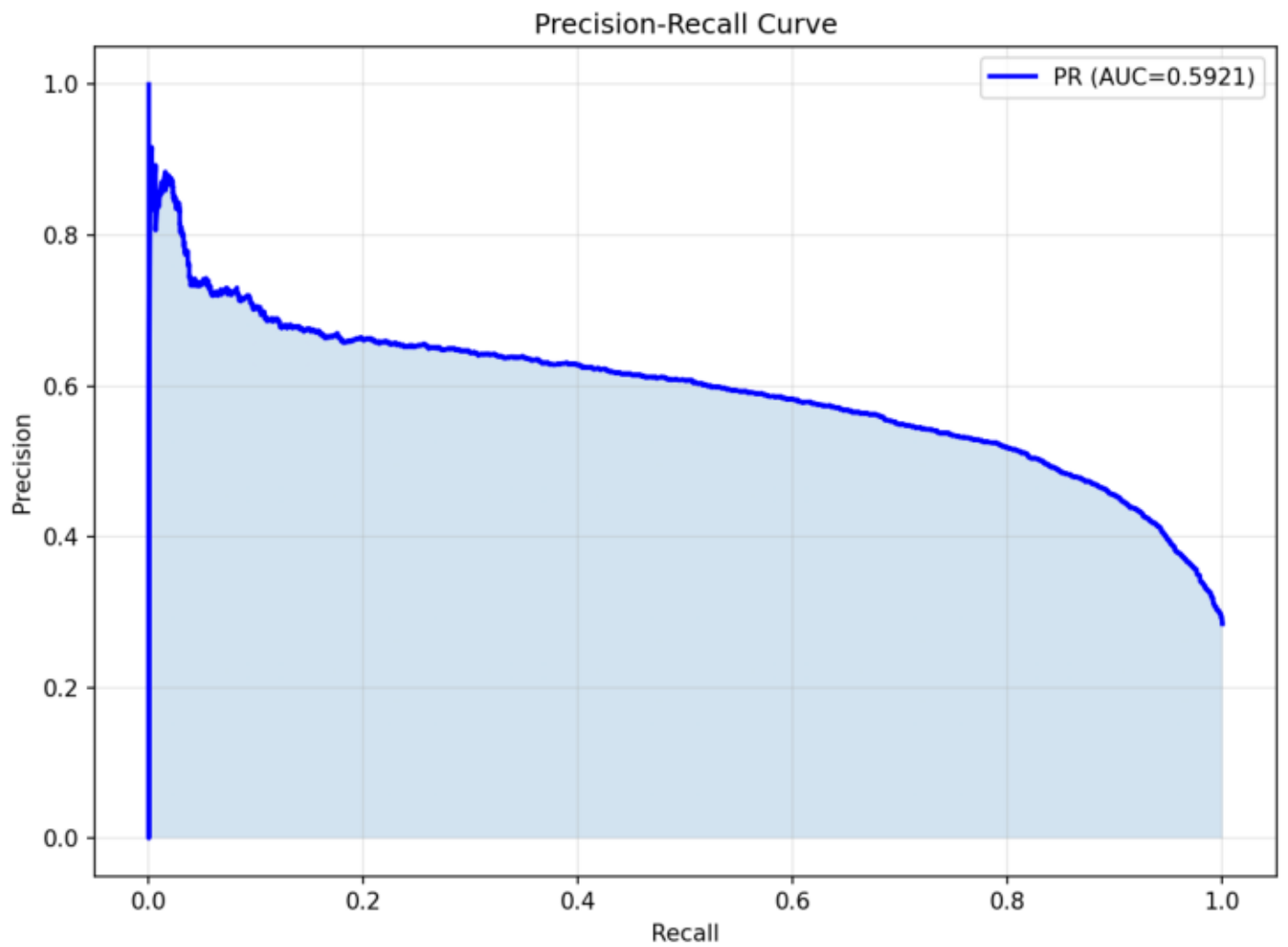
Metric	Figure	Explanation
precision_at_k[200]	0.7450	Shows how accurate the top-200 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[500]	0.7180	Shows how accurate the top-500 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
recall_at_k[10]	0.0023	Shows how much of the positive population is captured within the top-10 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[50]	0.0109	Shows how much of the positive population is captured within the top-50 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[100]	0.0220	Shows how much of the positive population is captured within the top-100 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[200]	0.0377	Shows how much of the positive population is captured within the top-200 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[500]	0.0907	Shows how much of the positive population is captured within the top-500 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.

value_date_quarter=2024Q4 - 1) Model Effectiveness



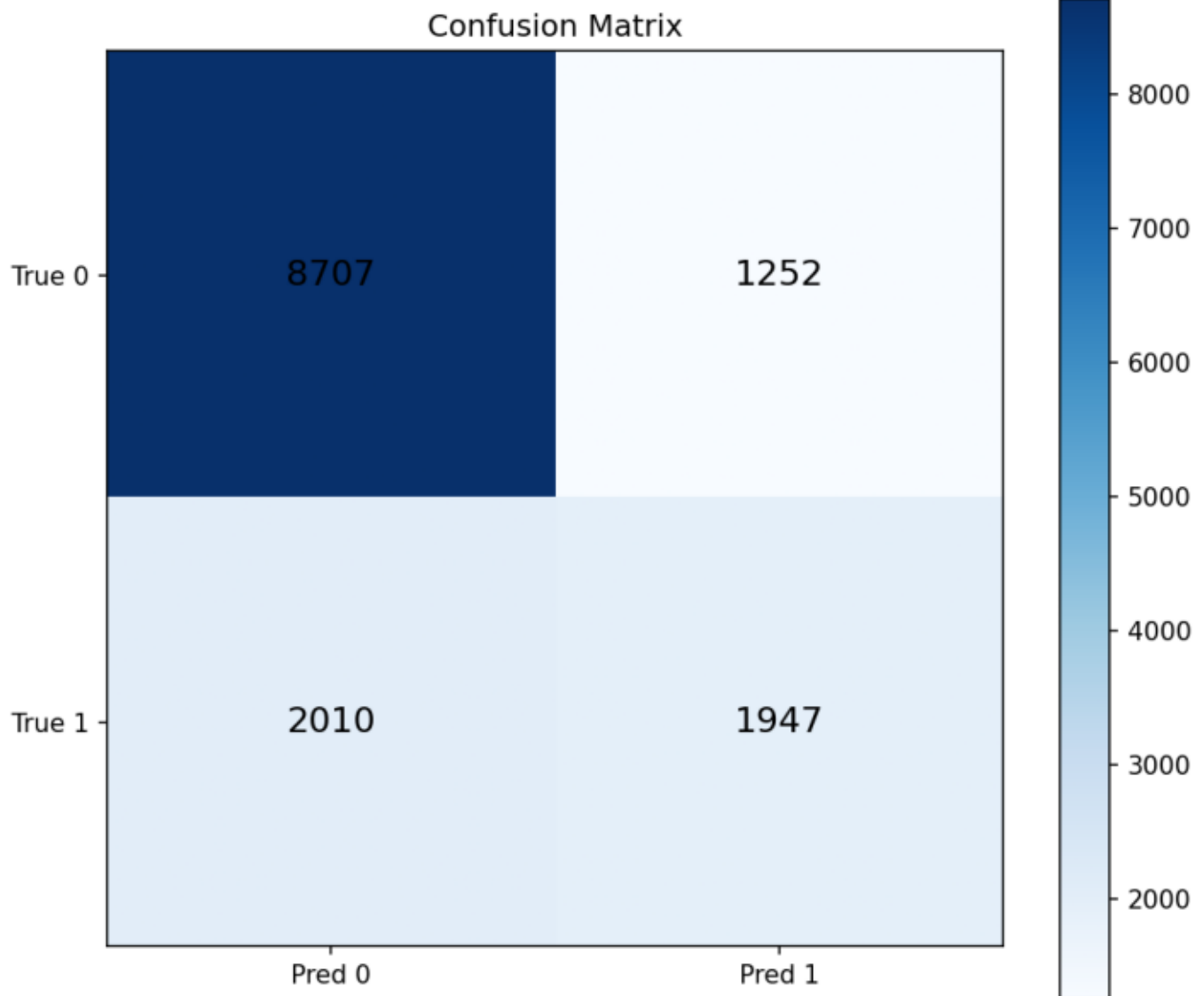
roc_curve: AUC-ROC=0.8144; curve above diagonal indicates strong separation.

value_date_quarter=2024Q4 - 1) Model Effectiveness



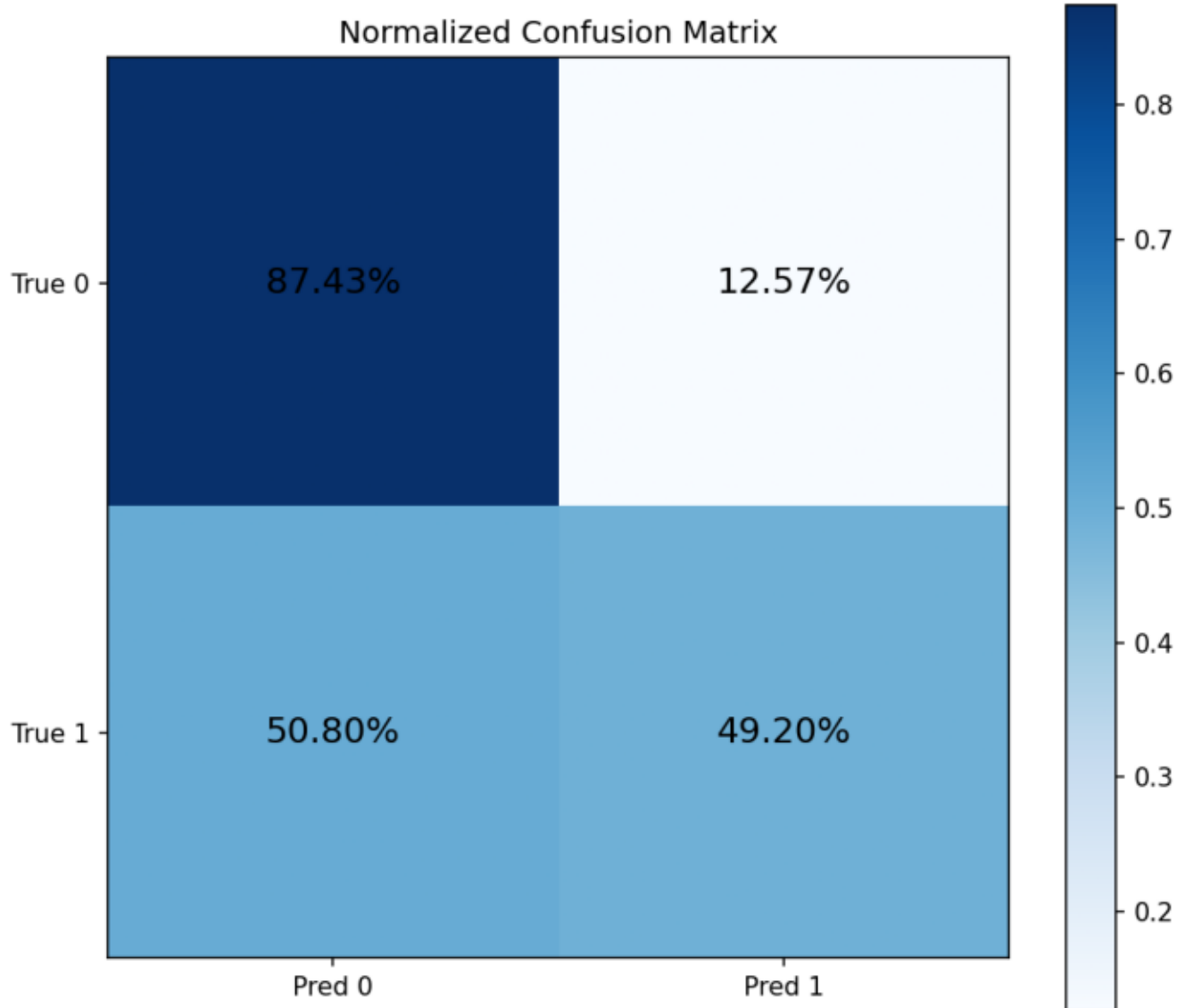
pr_curve: AUC-PR=0.5921 vs baseline 28.43%; higher curve implies better precision at recall.

value_date_quarter=2024Q4 - 1) Model Effectiveness



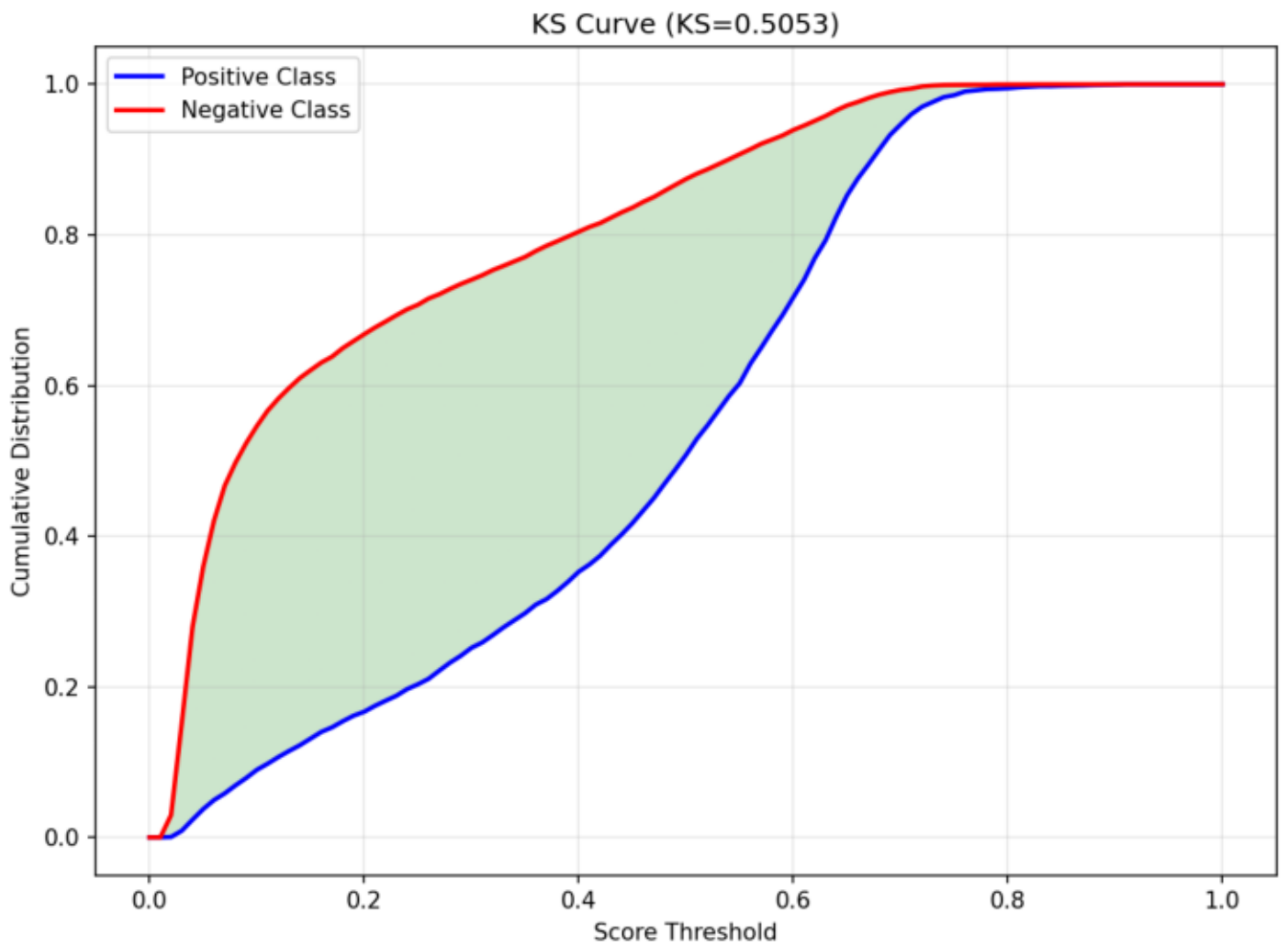
confusion_matrix: At threshold 0.50: TP=1947, FP=1252, FN=2010, TN=8707.

value_date_quarter=2024Q4 - 1) Model Effectiveness



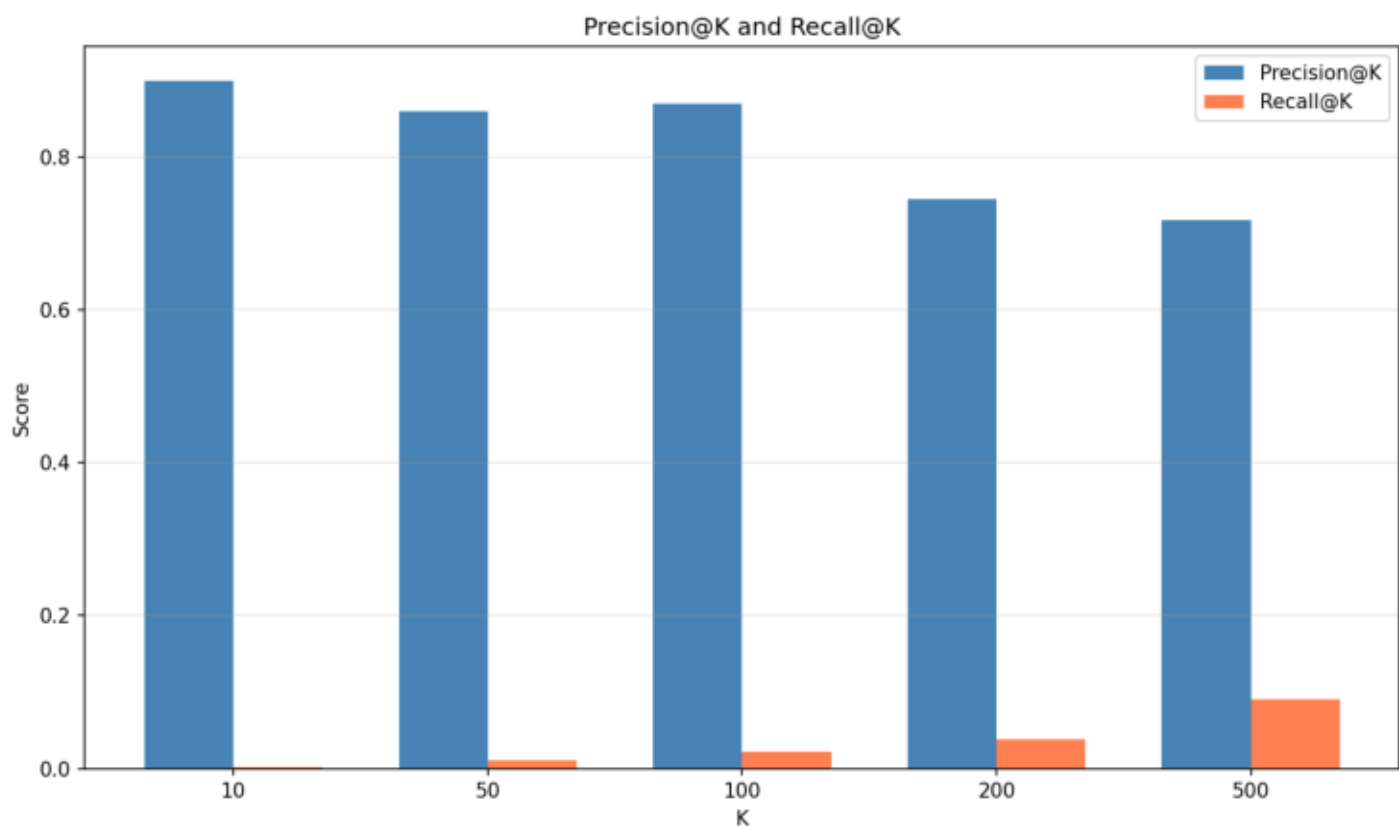
confusion_matrix_norm: Normalized rates: TPR=49.20%, TNR=87.43%.

value_date_quarter=2024Q4 - 1) Model Effectiveness



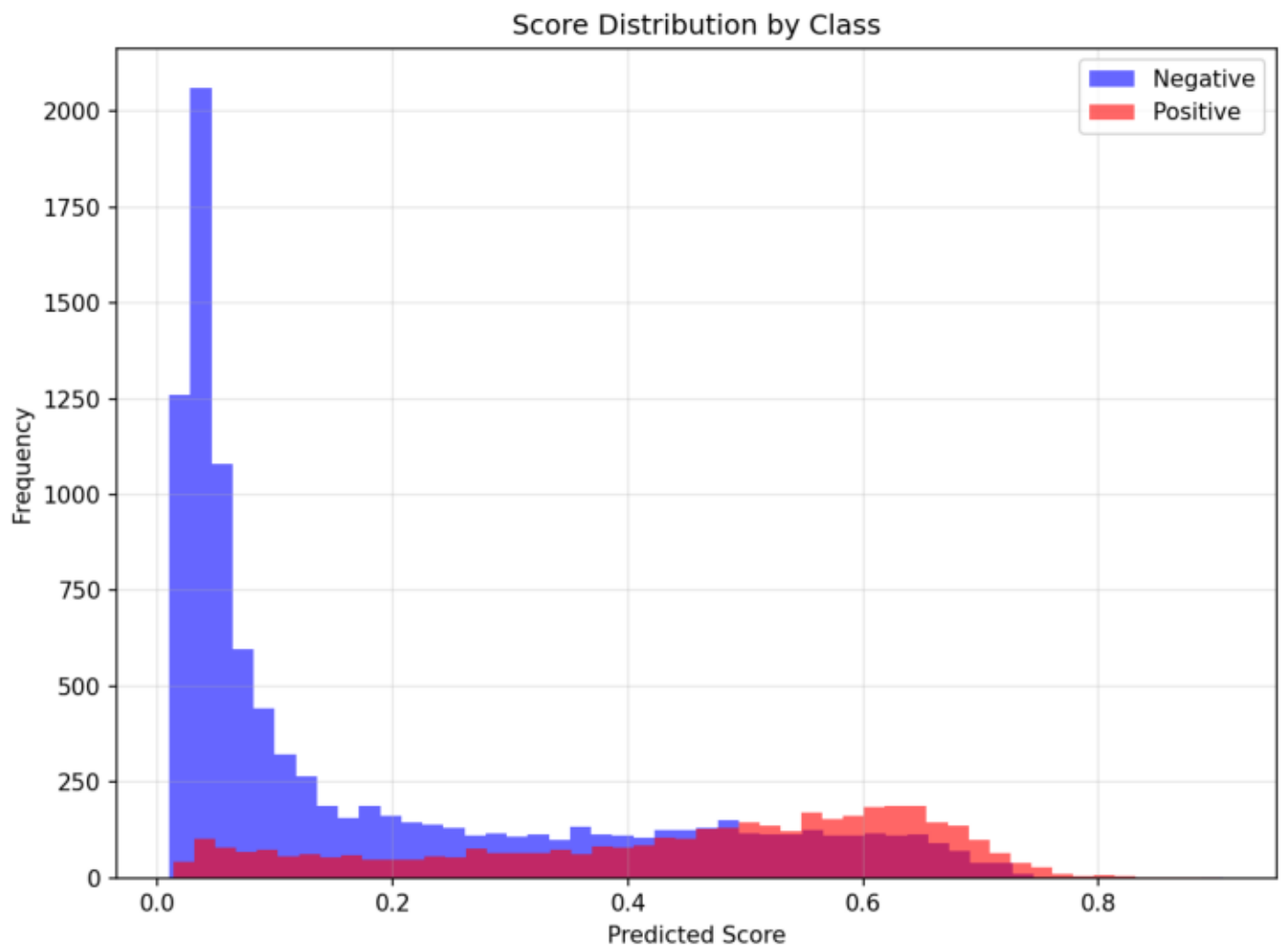
ks_curve: Maximum separation KS=0.5053 at threshold 0.23.

value_date_quarter=2024Q4 - 1) Model Effectiveness



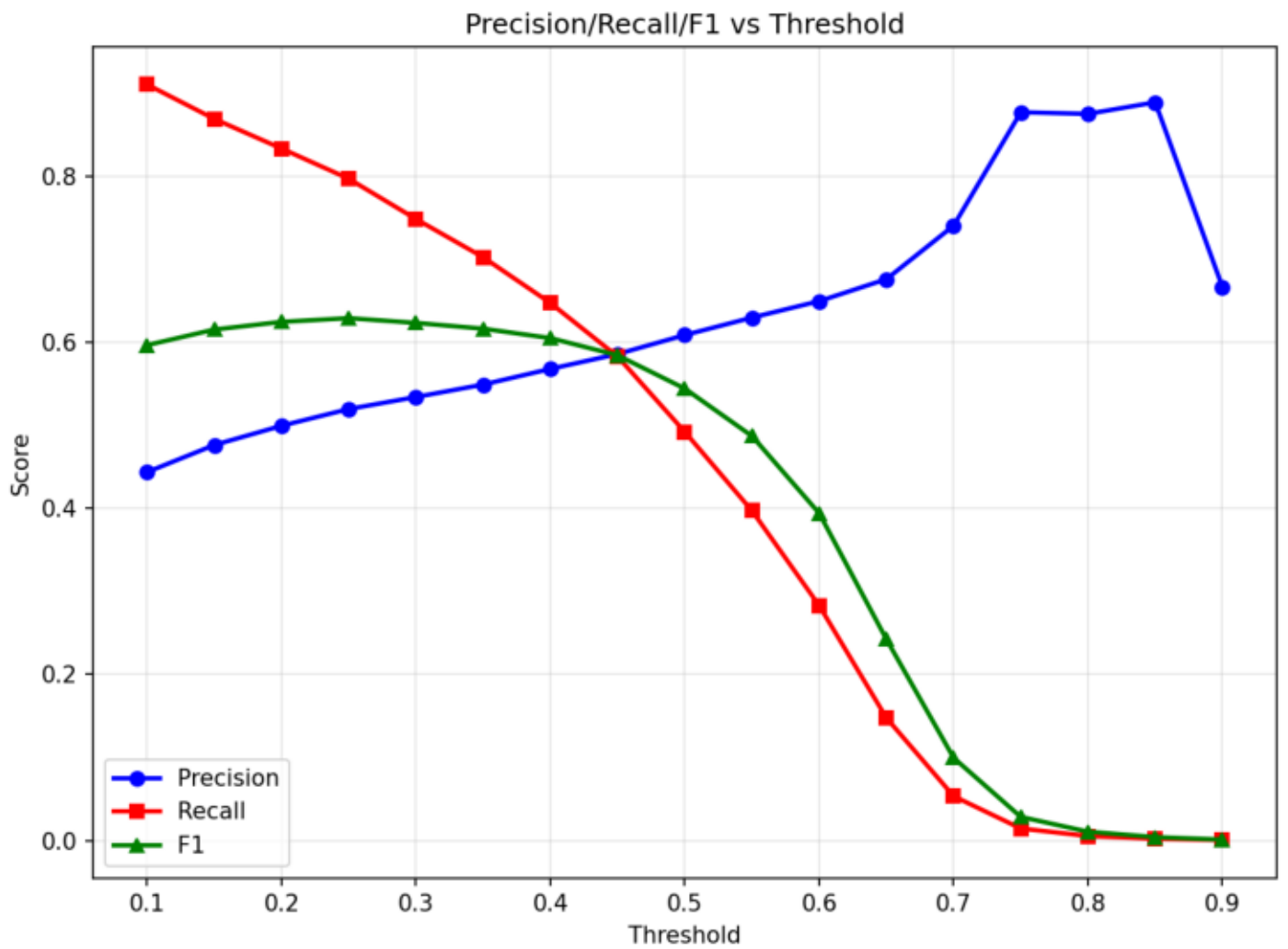
precision_recall_at_k: Bars show precision/recall as you expand the review queue.

value_date_quarter=2024Q4 - 1) Model Effectiveness



score_distribution: Mean score: positive=0.448, negative=0.189 (larger gap is better).

value_date_quarter=2024Q4 - 1) Model Effectiveness



threshold_analysis: Best F1 in tested grid is 0.6290 at threshold 0.25.

value_date_quarter=2024Q4 - 2) Model Efficiency (1/4)

Metric	Figure	Explanation
fpr	0.1257	Measures the share of negatives incorrectly flagged; 0.1257 is high. Next: if review demand is too high, raise the threshold or add features that better suppress false alerts.
tn	8707	Counts negatives correctly cleared without manual review. Next: preserve this clearance rate while checking that recall does not fall below policy expectations.
fp	1252	Counts non-events that would still be sent to review. Next: use this with staffing capacity to decide whether threshold or feature refinement is required.
threshold	0.5000	This is the active operating cutoff used to flag alerts. Next: tune it jointly with recall, precision, and review capacity rather than in isolation.
fpr_at_thresholds[thresholds.t_0.05]	0.6403	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.05. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.10]	0.4537	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.10. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q4 - 2) Model Efficiency (2/4)

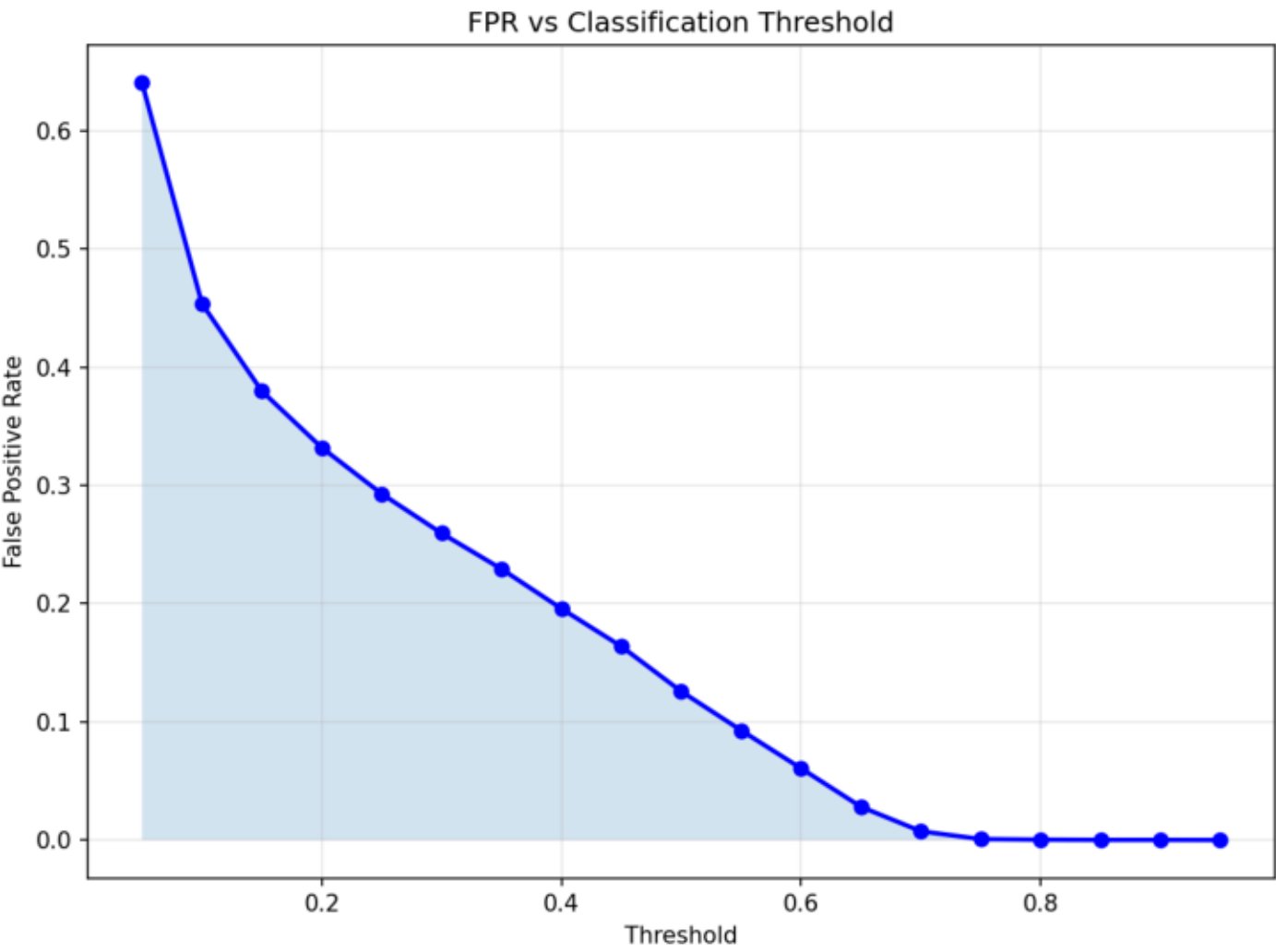
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.15]	0.3797	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.15. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.20]	0.3319	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.20. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.25]	0.2928	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.25. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.30]	0.2595	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.30. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.35]	0.2292	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.35. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.40]	0.1957	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.40. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q4 - 2) Model Efficiency (3/4)

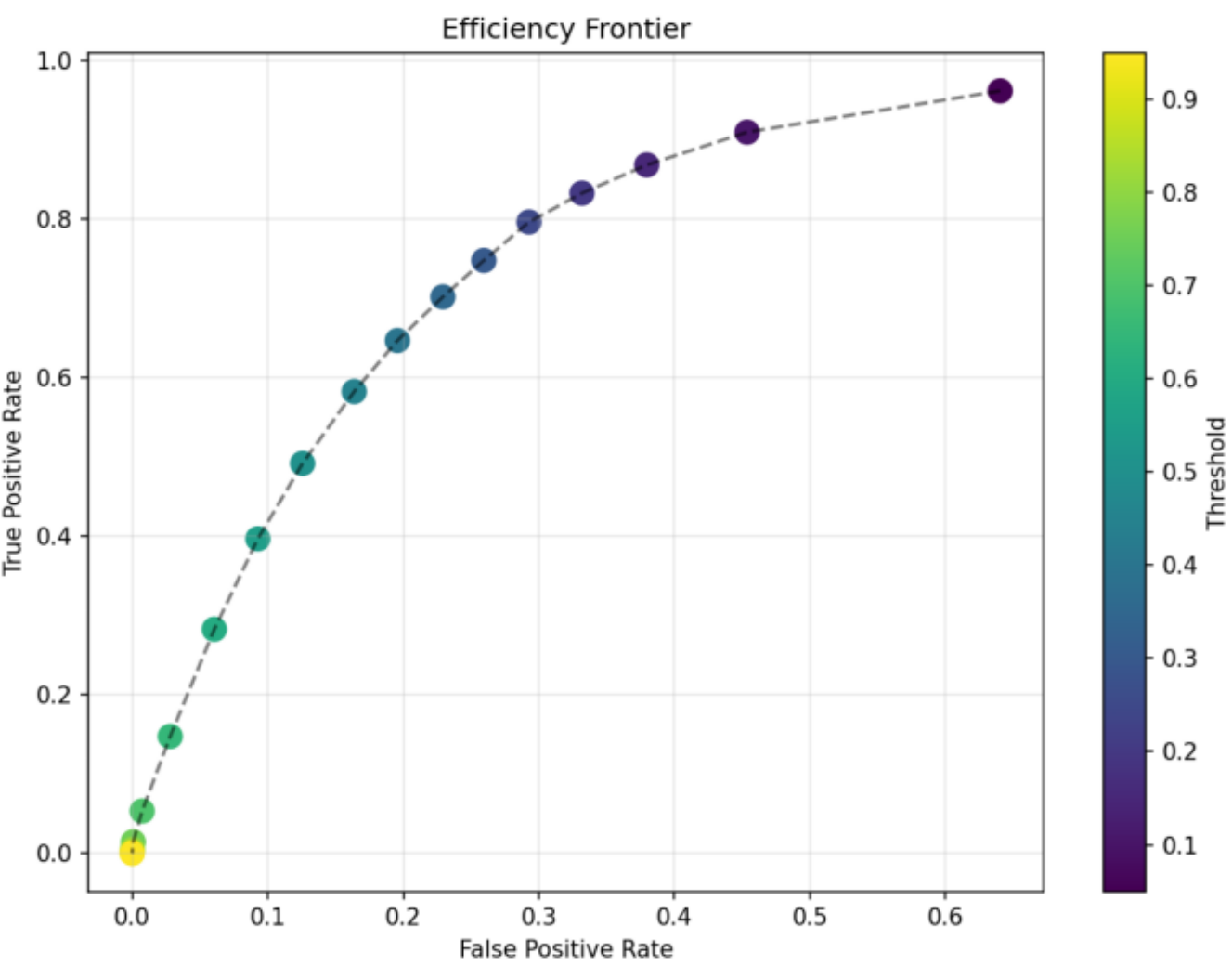
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.45]	0.1638	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.45. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.50]	0.1257	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.50. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.55]	0.0928	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.55. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.60]	0.0606	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.60. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.65]	0.0281	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.65. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.70]	0.0074	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.70. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q4 - 2) Model Efficiency (4/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.75]	0.0008	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.75. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.80]	0.0003	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.80. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.85]	0.0001	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.85. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.90]	0.0001	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.90. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.95]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.95. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

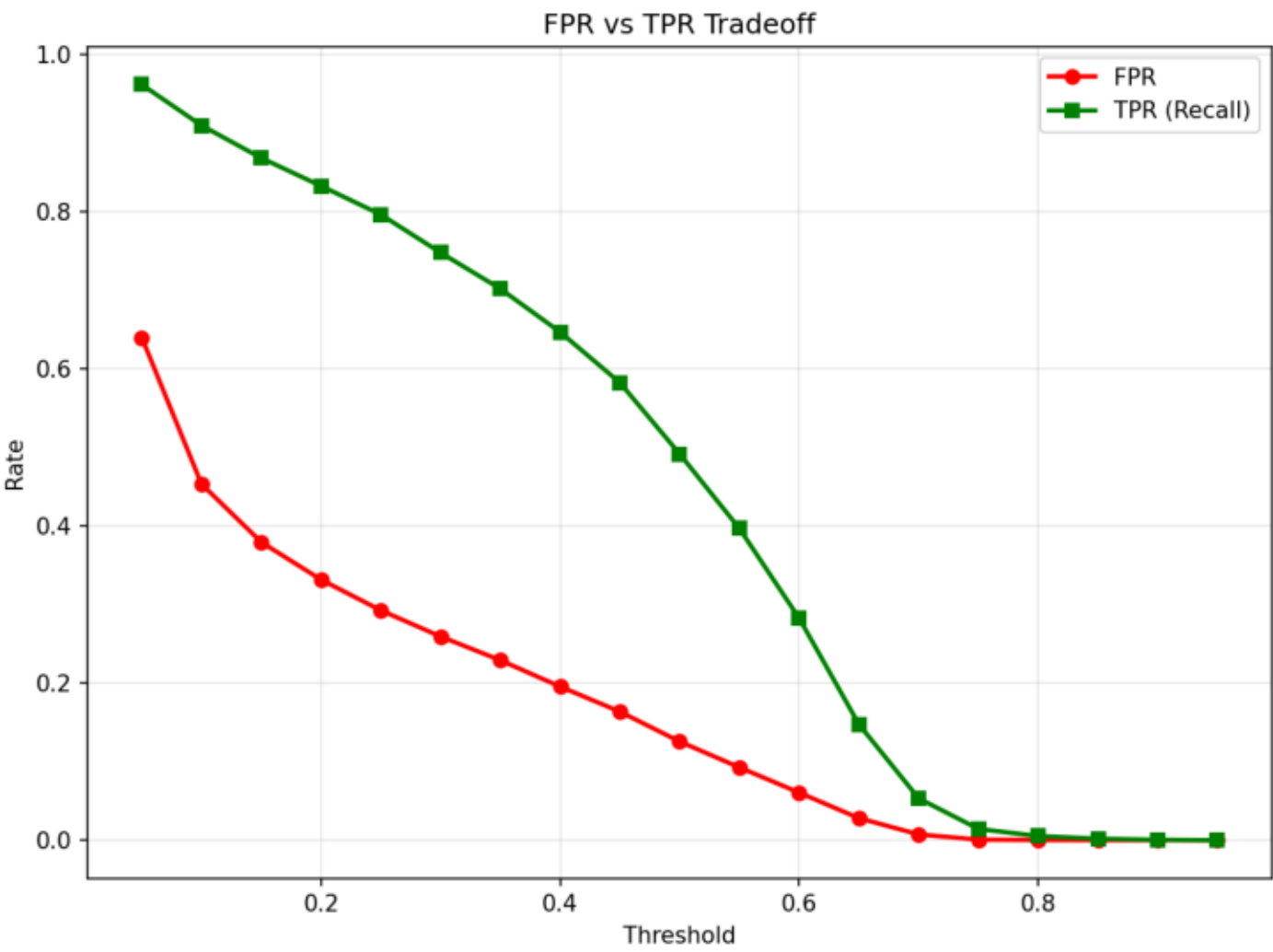


value_date_quarter=2024Q4 - 2) Model Efficiency



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

value_date_quarter=2024Q4 - 2) Model Efficiency



fpr_tpr_tradeoff: FPR and TPR curves highlight how recall drops as you reduce false positives.

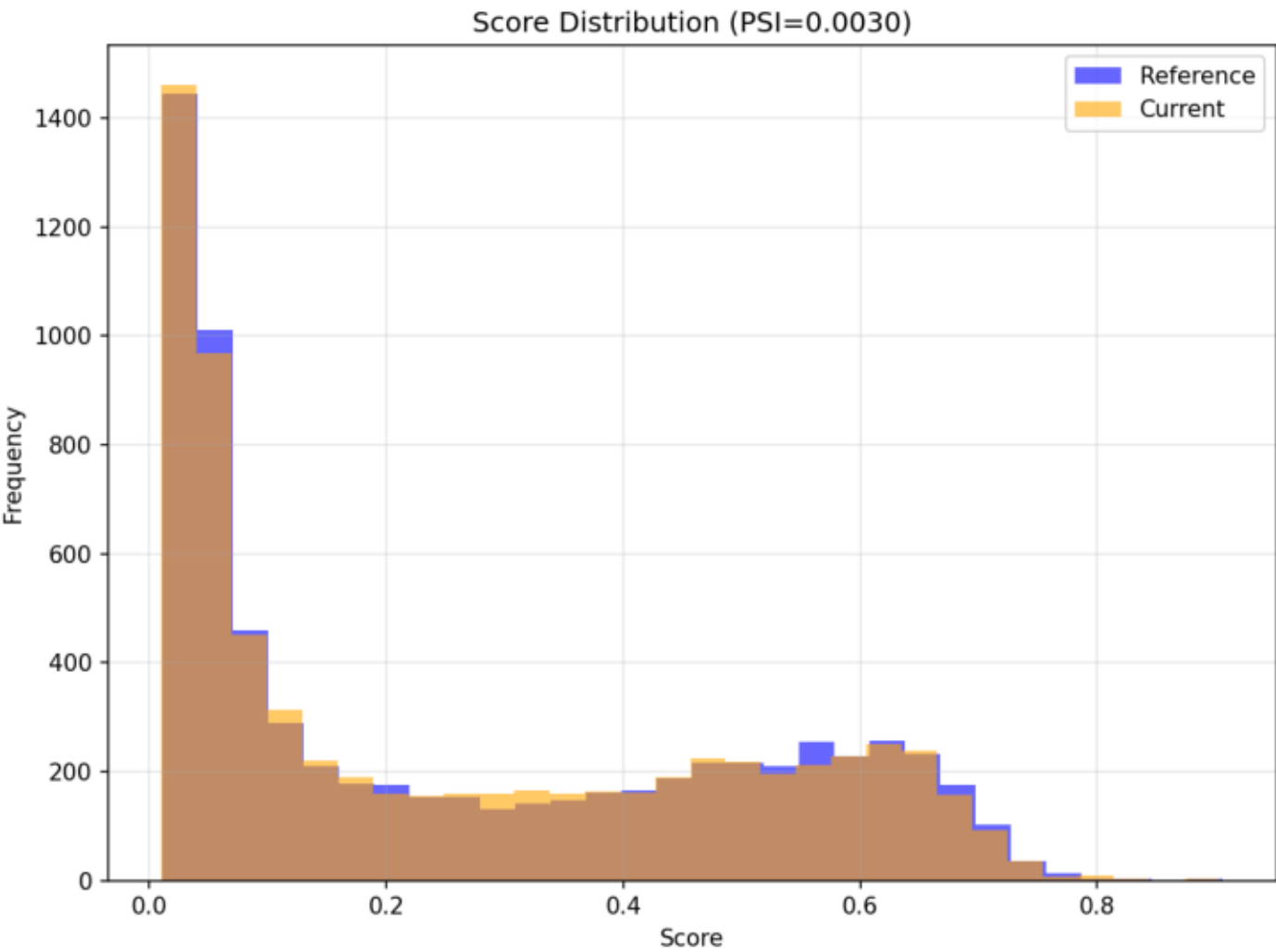
value_date_quarter=2024Q4 - 3) Model Stability (1/2)

Metric	Figure	Explanation
psi	0.0030	Measures score-distribution shift between reference and current populations; 0.0030 is negligible. Next: investigate segment mix, data drift, and recalibration needs when PSI is material.
cv_auc_roc_mean	0.8144	Average AUC-ROC across validation folds. Next: compare it with overall AUC-ROC; a large gap can indicate overfitting or unstable sampling.
cv_auc_roc_std	0.0107	Standard deviation of AUC-ROC across folds. Next: if variability is high, simplify the model or increase training stability before deployment.
cv_auc_pr_mean	0.5940	Average AUC-PR across validation folds. Next: confirm that precision-recall performance remains acceptable under resampling.
cv_auc_pr_std	0.0270	Standard deviation of AUC-PR across folds. Next: high dispersion suggests unstable alert quality and warrants deeper validation.
bootstrap_auc_roc_mean	0.8109	Bootstrap mean AUC-ROC estimates expected ranking performance under repeated sampling. Next: compare it with the point estimate to judge optimism in the current test run.
bootstrap_auc_roc_ci_lower	0.7997	Lower confidence bound for bootstrap AUC-ROC. Next: use it as a conservative performance floor in model-risk discussions.
bootstrap_auc_roc_ci_upper	0.8216	Upper confidence bound for bootstrap AUC-ROC. Next: review the interval width with the lower bound to assess uncertainty.

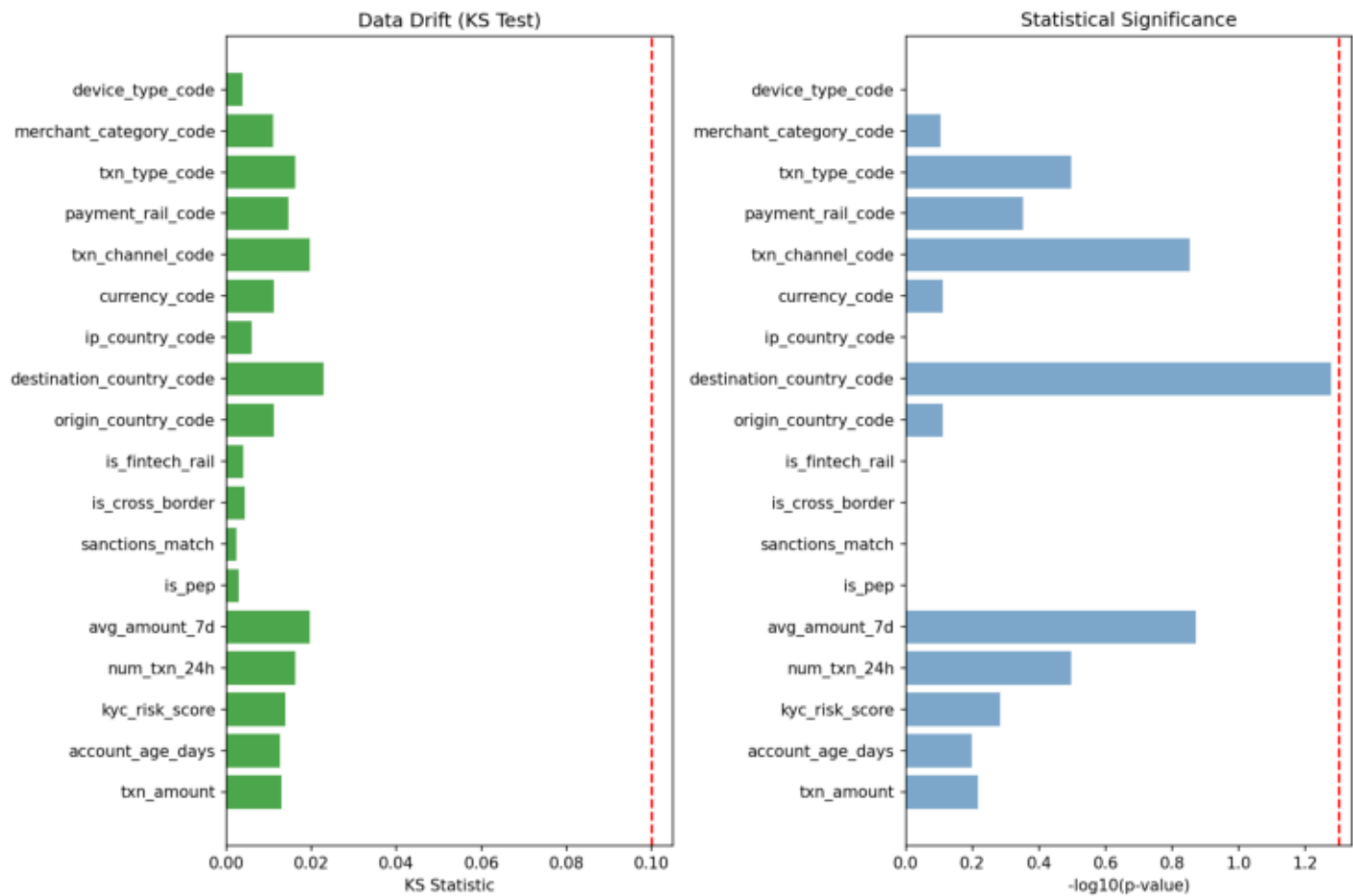
value_date_quarter=2024Q4 - 3) Model Stability (2/2)

Metric	Figure	Explanation
concept_drift_detected	False	No material concept-drift signal was detected in the evaluated slices. Next: retain ongoing monitoring because stable historical results do not guarantee future stability.
concept_drift_score	0.0076	Quantifies the magnitude of performance change across ordered slices. Next: rising scores should trigger deeper review of segment behavior and model refresh timing.

value_date_quarter=2024Q4 - 3) Model Stability

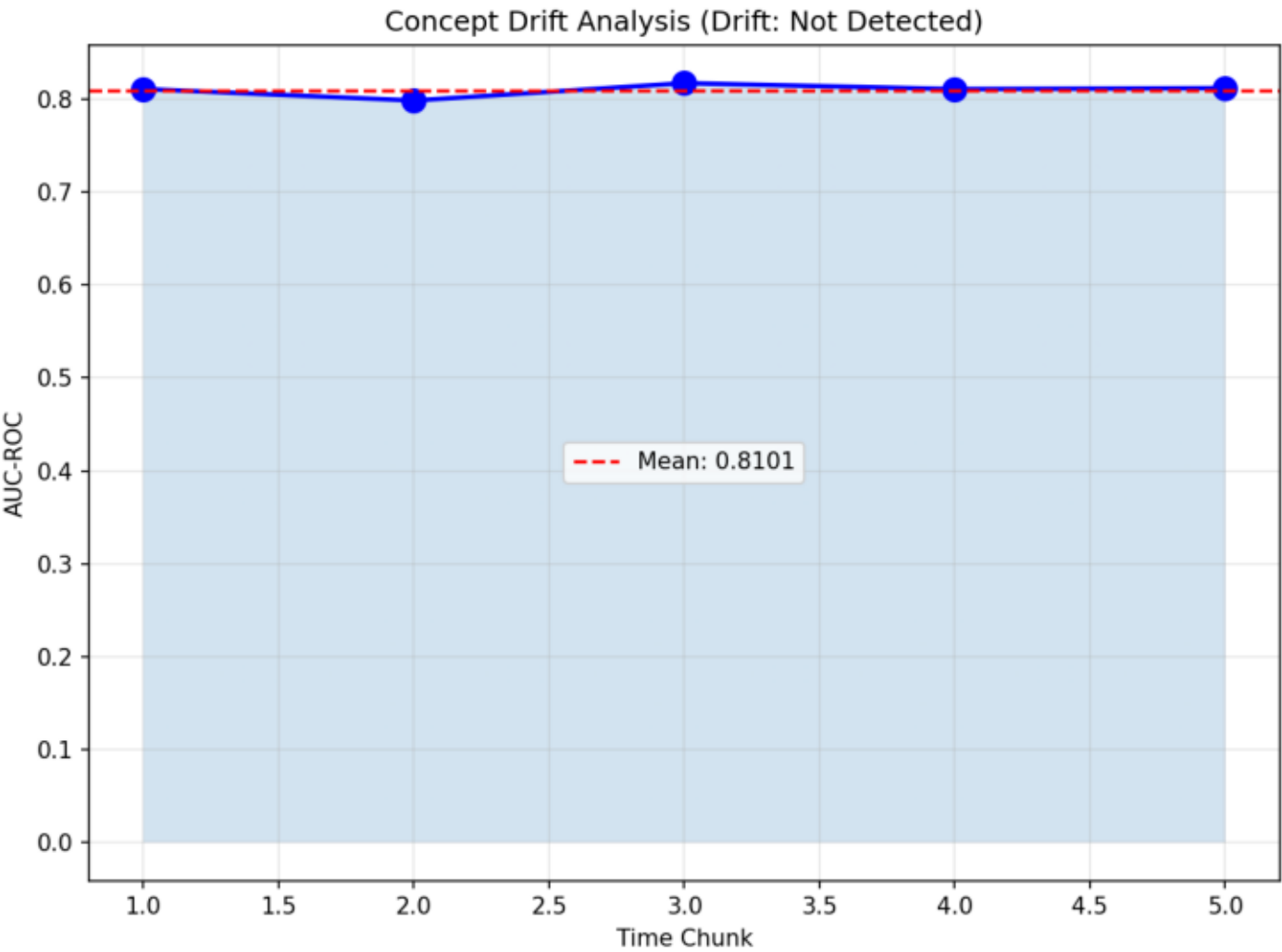


value_date_quarter=2024Q4 - 3) Model Stability



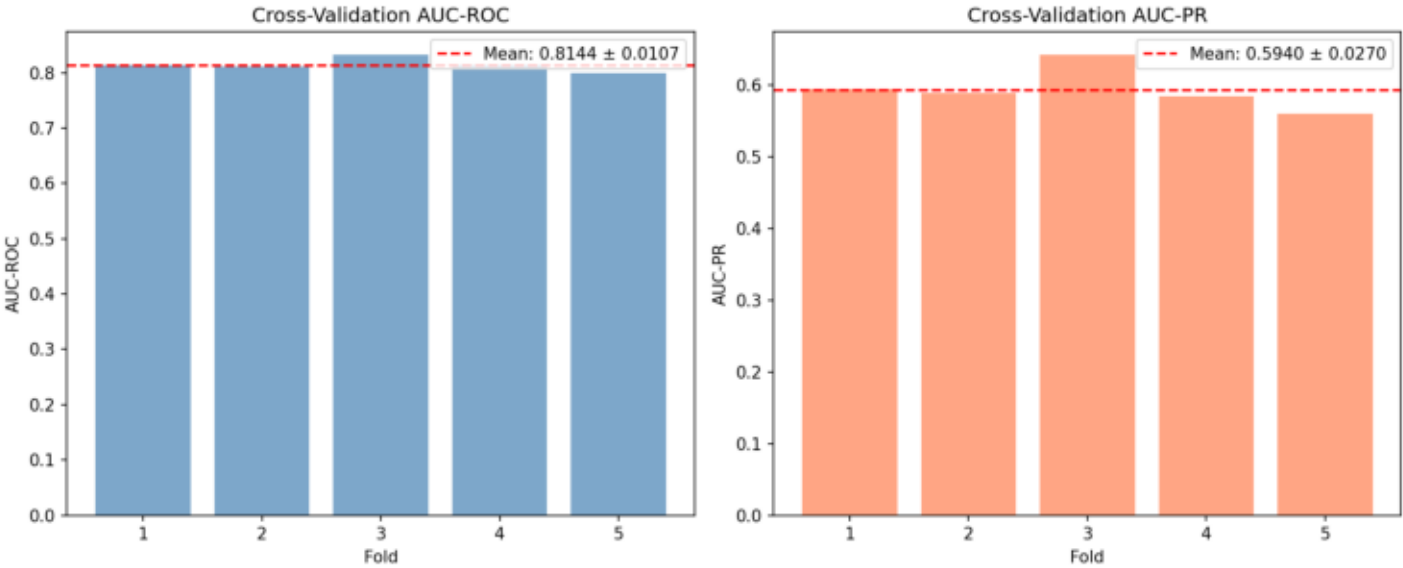
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.

value_date_quarter=2024Q4 - 3) Model Stability



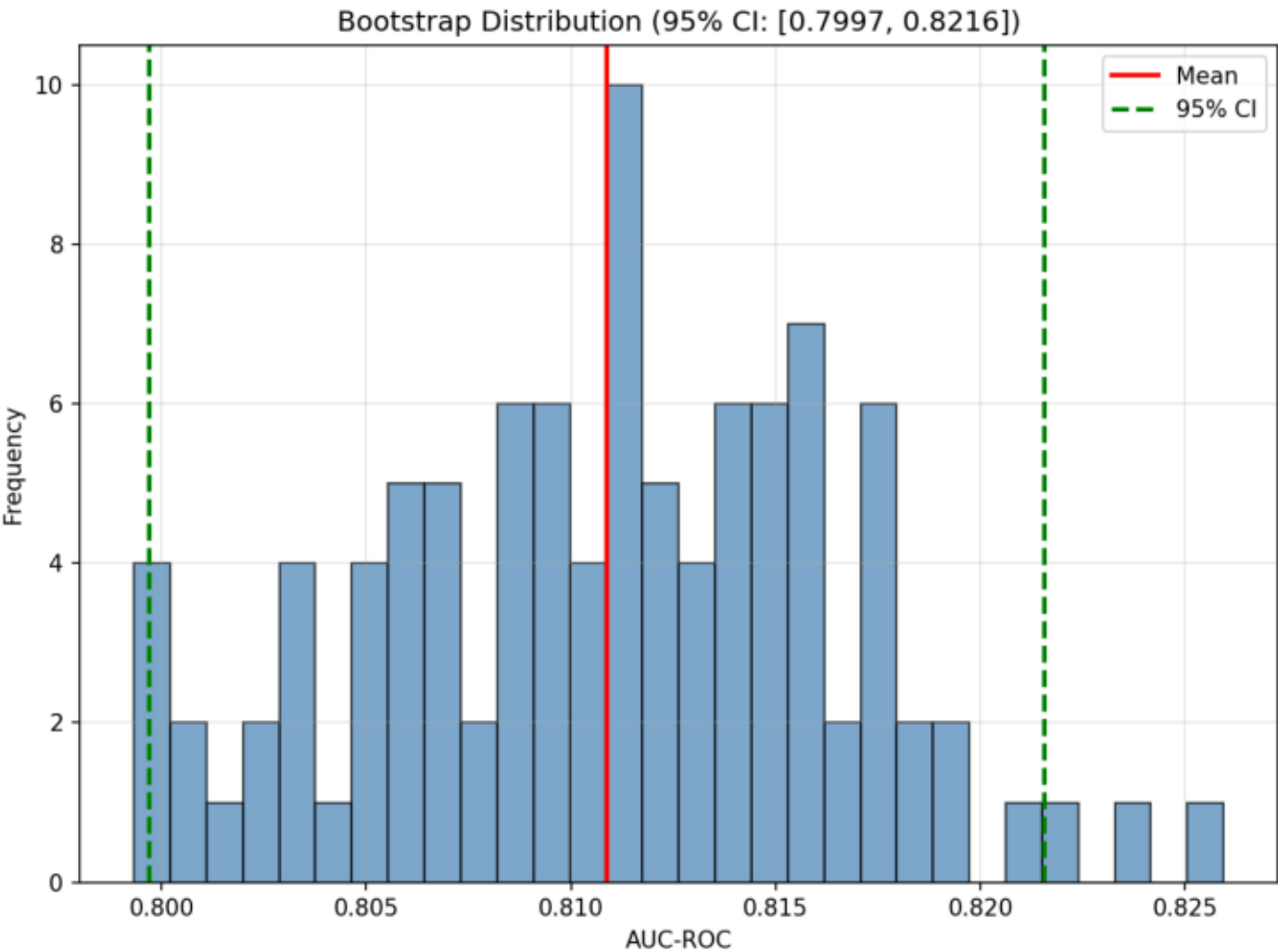
concept_drift: AUC across time chunks shows whether performance is stable over time.

value_date_quarter=2024Q4 - 3) Model Stability



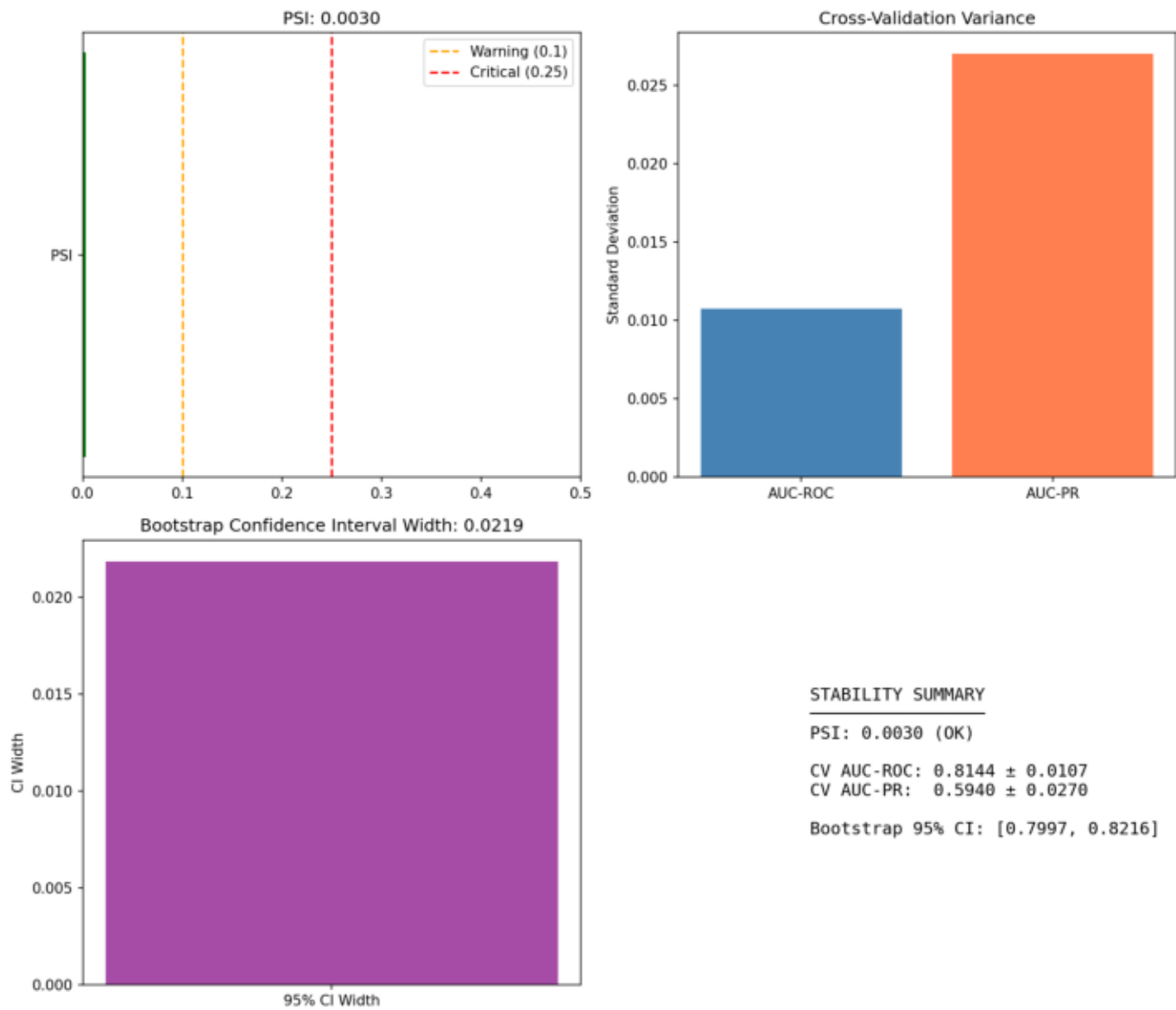
cv_results: Fold scores cluster around ROC 0.8144 and PR 0.5940.

value_date_quarter=2024Q4 - 3) Model Stability



bootstrap_distribution: Bootstrap CI [0.7997, 0.8216] reflects performance stability.

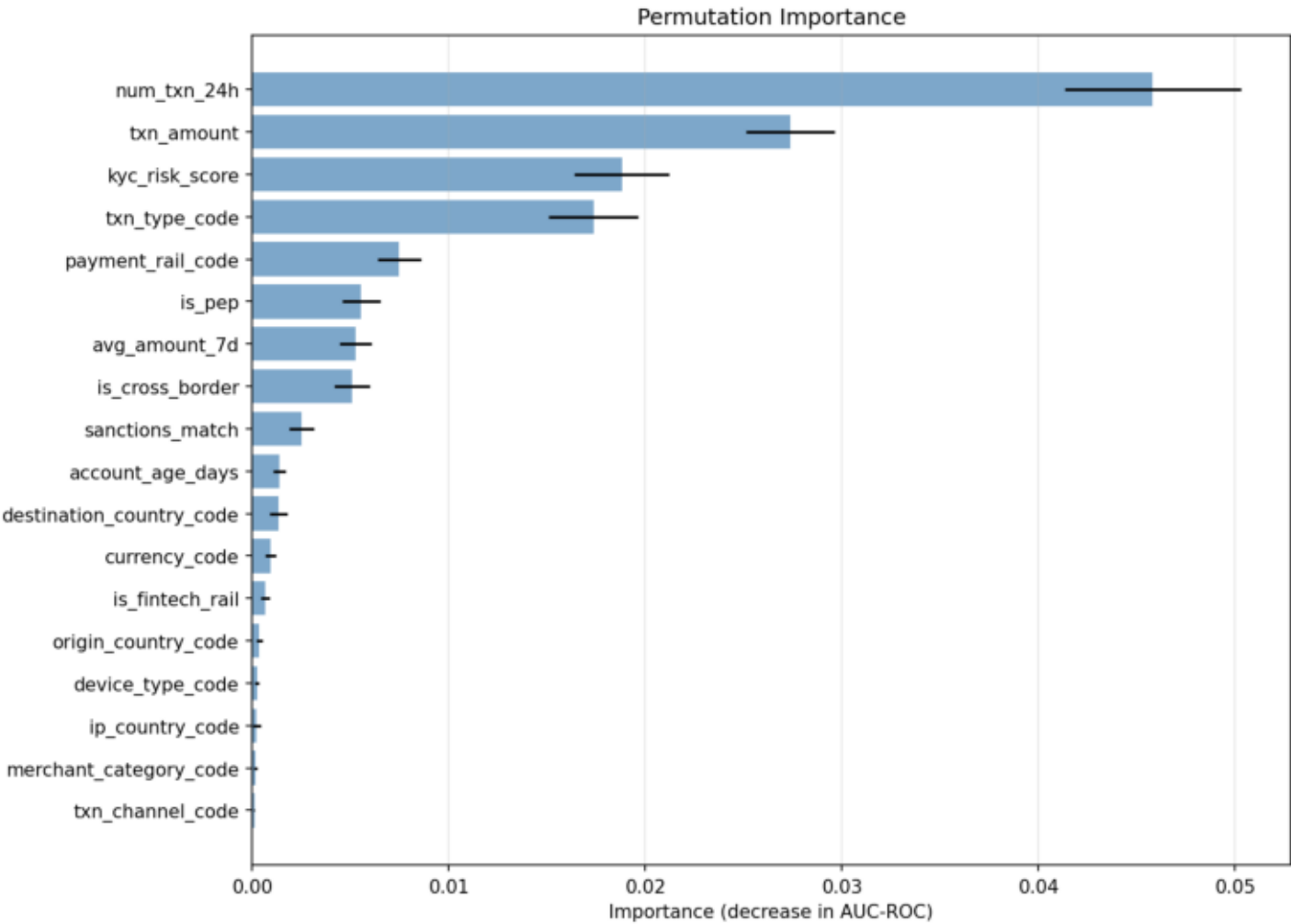
value_date_quarter=2024Q4 - 3) Model Stability



Metric	Figure	Explanation
model_type	tree	Identifies the estimator family so explanation methods can be interpreted in the right context. Next: confirm the chosen explanation methods are appropriate for this model class.
methods_used	permutation, pdp, ice, shap, lime	Lists the interpretability methods produced in this run. Next: compare signals across methods before using them in documentation or model challenge.
perm_top_features	num_txn_24h, txn_amount, kyc_risk_score, txn_type_code, payment_rail_code ...	Shows features with the largest performance drop when permuted, which highlights global importance. Next: validate that the top drivers align with business logic, feature governance, and policy expectations.
shap_top_features	num_txn_24h, txn_amount, txn_type_code, kyc_risk_score, payment_rail_code ...	Shows the most influential features under SHAP-based contribution analysis. Next: inspect whether directionality and ranking remain sensible across key segments.

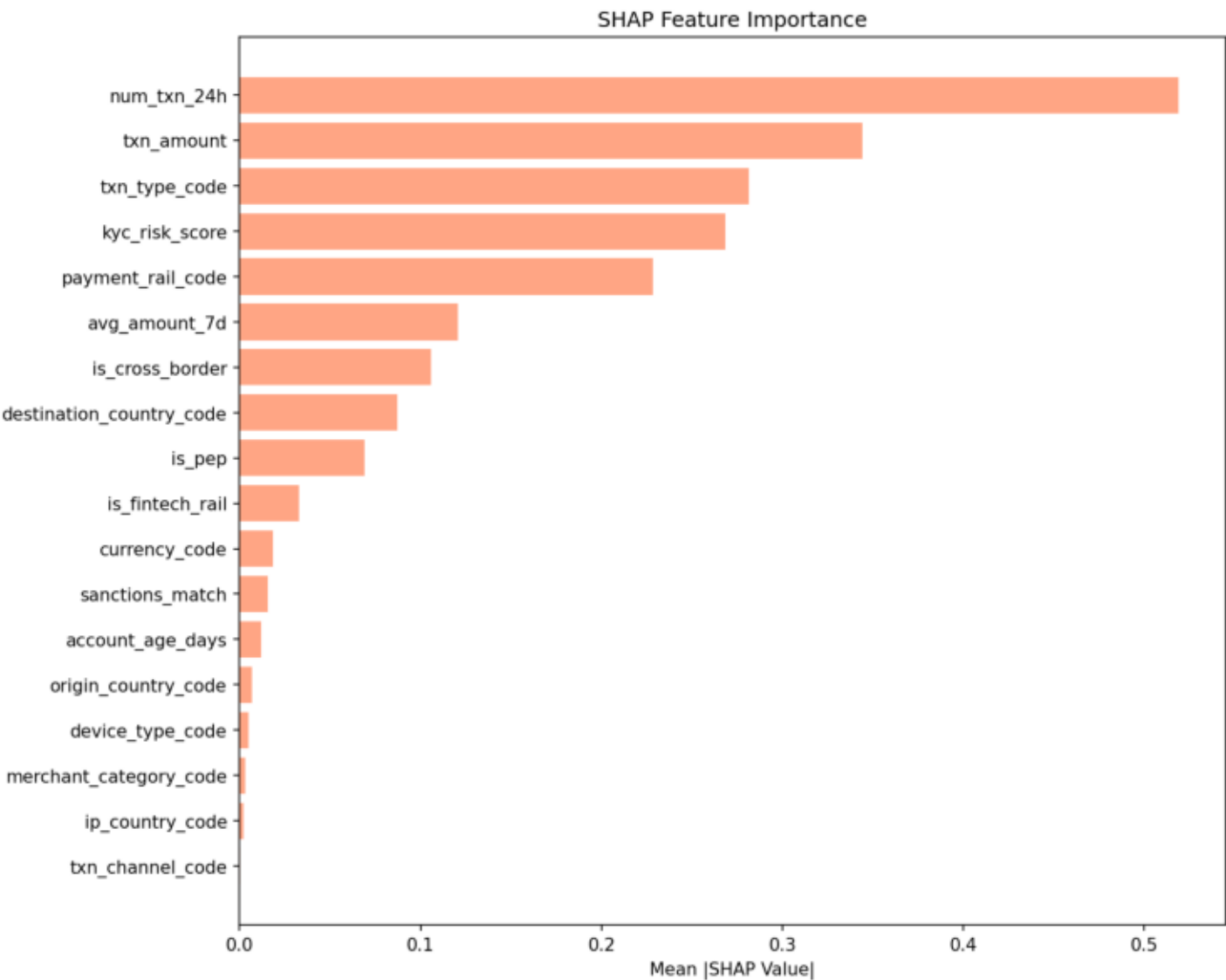
Metric	Figure	Explanation
lime_instances	3	Counts the locally explained observations reviewed with LIME. Next: examine representative true-positive and false-positive cases to understand local decision logic.
pdp_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score, num_txn_24h ...	Lists features with partial-dependence plots. Next: review whether average model response is monotonic, stable, and operationally sensible.
ice_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score	Lists features with individual conditional expectation plots. Next: inspect heterogeneity and interactions that may be hidden in average-only views.

value_date_quarter=2024Q4 - 4) Model Interpretability



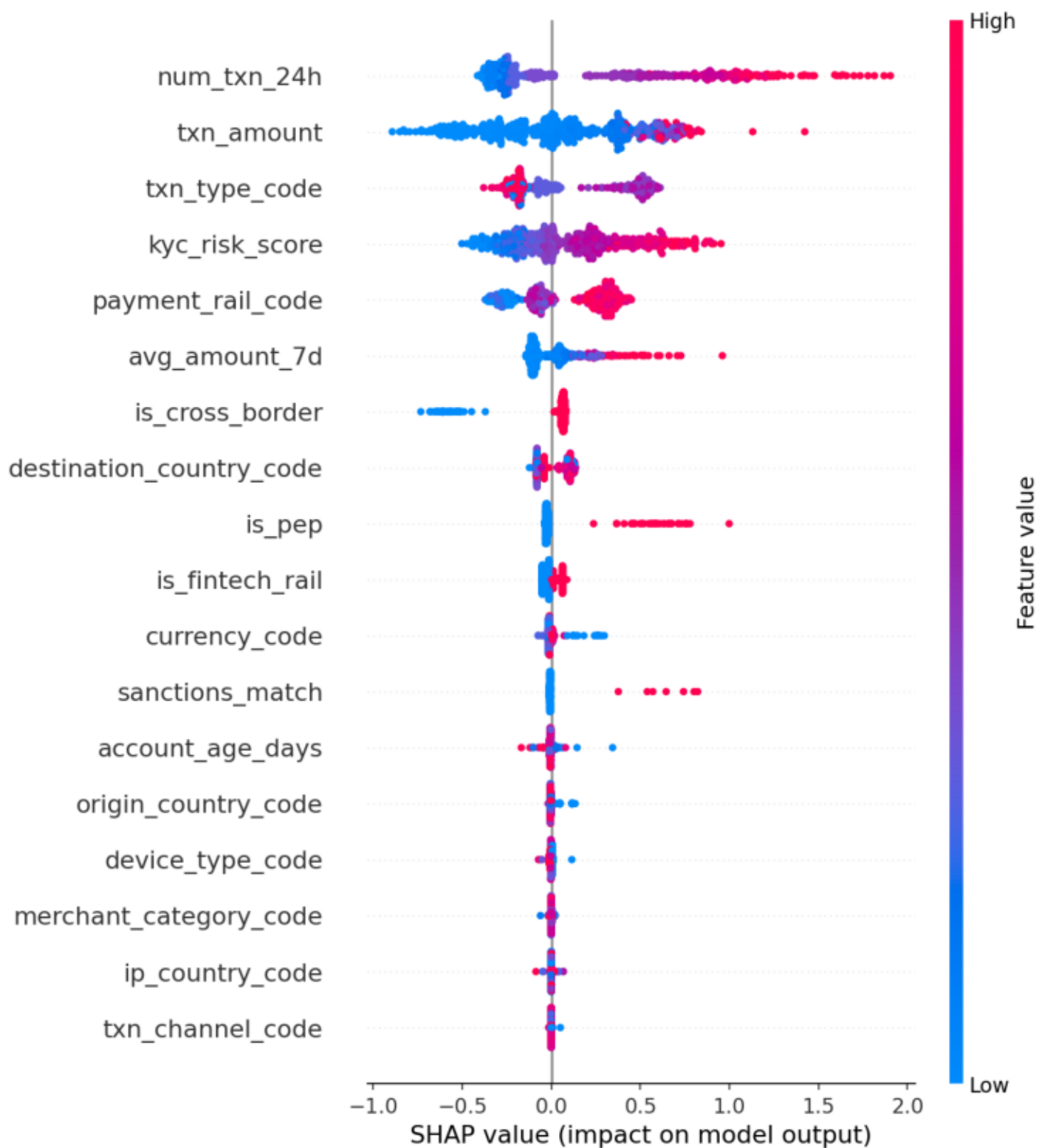
permutation_importance: Bars show how much AUC drops when each feature is permuted.

value_date_quarter=2024Q4 - 4) Model Interpretability

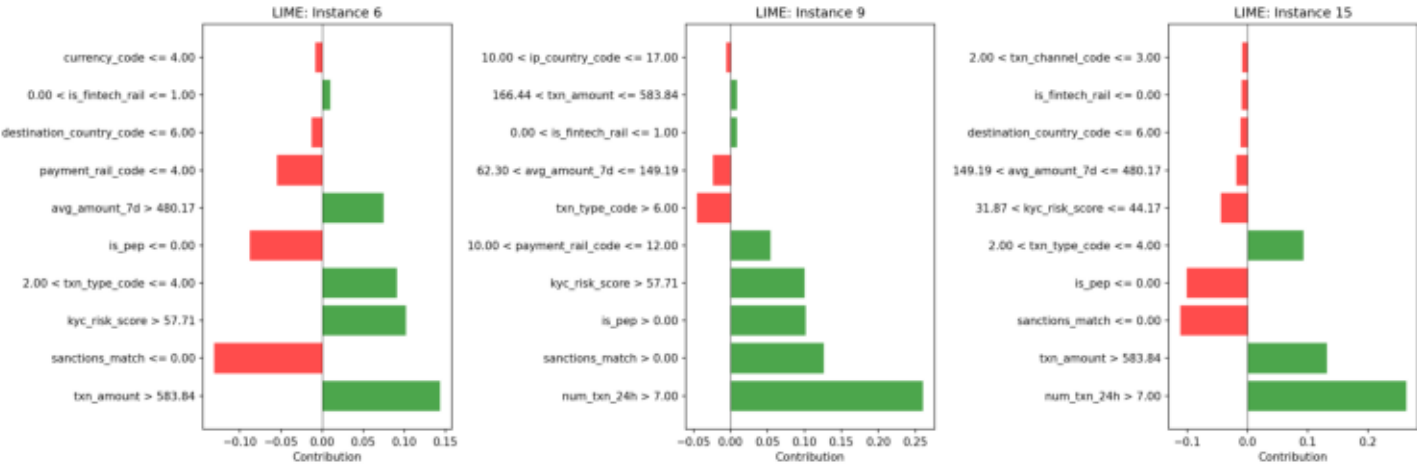


shap_bar: SHAP bar plot ranks global feature impact by mean absolute SHAP values.

value_date_quarter=2024Q4 - 4) Model Interpretability



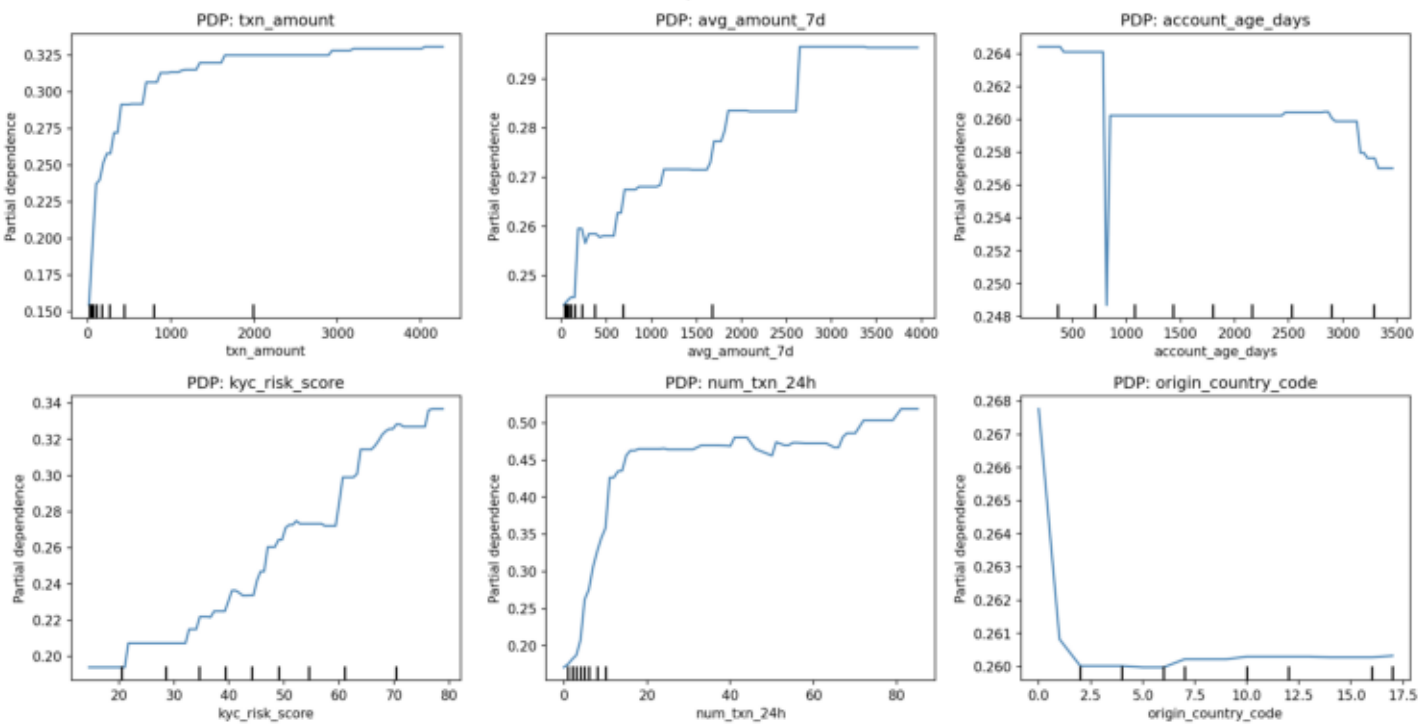
value_date_quarter=2024Q4 - 4) Model Interpretability



lime_explanation: LIME bars show local feature contributions for selected instances.

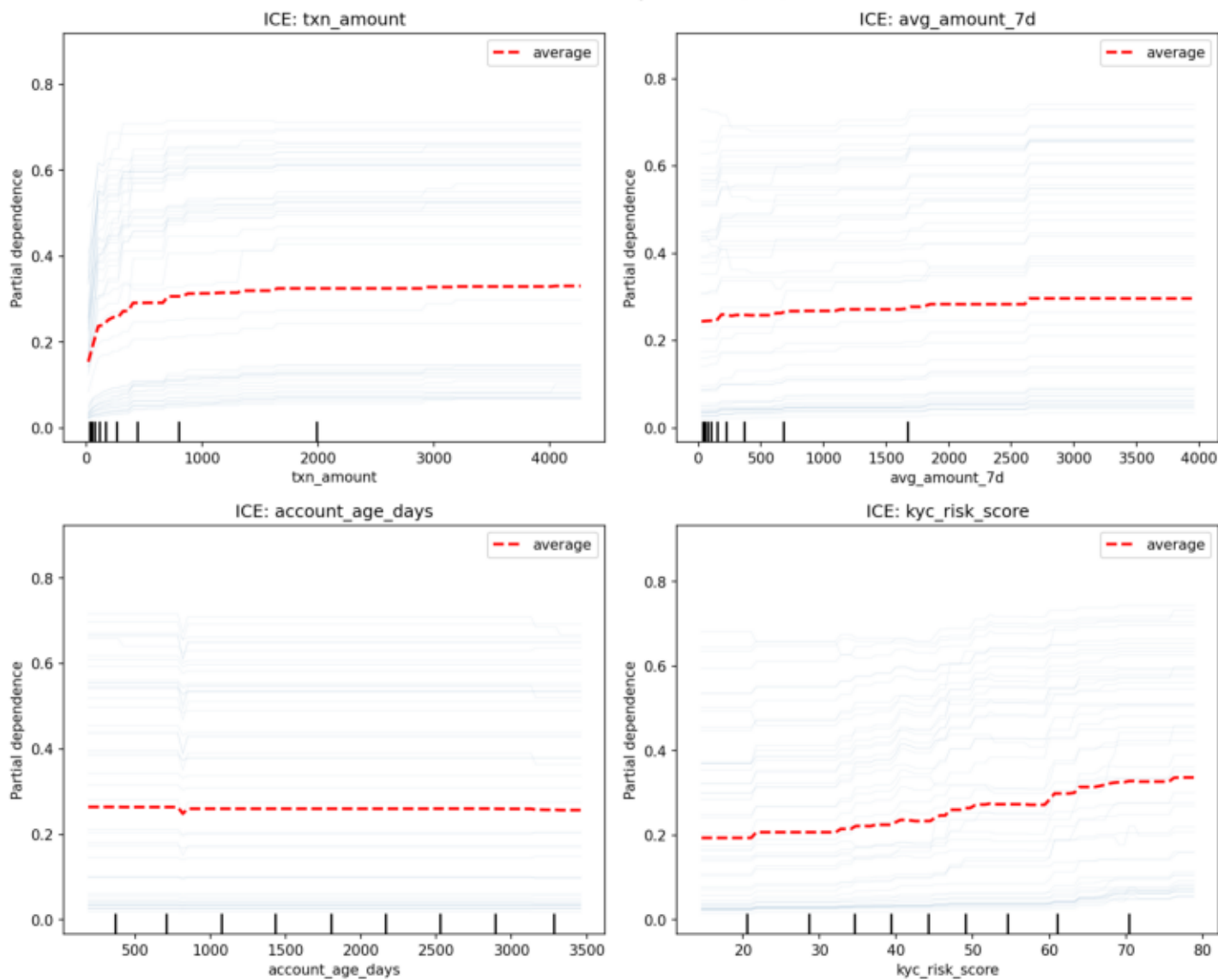
value_date_quarter=2024Q4 - 4) Model Interpretability

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.